Radio-Frequency Integrated Circuits

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1 Introduction

This is the material for an introductory radio-frequency integrated circuits course. The contents are largely based on [1], [2]; these two books are an excellent introduction to this topic and are highly recommended! For a general introduction to RF and microwave, [3] is a great read!

It is assumed that readers are familiar with the contents of this Analog Circuit Design course.

> **!** Important
>
> All course material (source code of this document, Jupyter notebooks for calculations, etc.) is made publicly available on GitHub (follow this link) and shared under the Apache-2.0 license.
>
> Please feel free to submit pull requests to fix typos or add content! If you want to discuss something that is not clear, please open an issue.
>
> The production of this document would be impossible without these (and many more) great open-source software products: `VS Code`, `Quarto`, `Pandoc`, `LaTeX`, `Typst`, `Jupyter Notebook`, `Python`, `Xschem`, `ngspice`, `CACE`, `pygmid`, `schemdraw`, `Numpy`, `Scipy`, `Matplotlib`, `Pandas`, `Git`, `Docker`, `Ubuntu`, `GNU/Linux`, ...

1.1 Wireless Transmission

In wireless transmission, we usually want to transmit data from a transmitter (TX) and its connected antenna towards a receiver (RX) using an electromagnetic (EM) wave. This arrangement is shown in Figure 1.

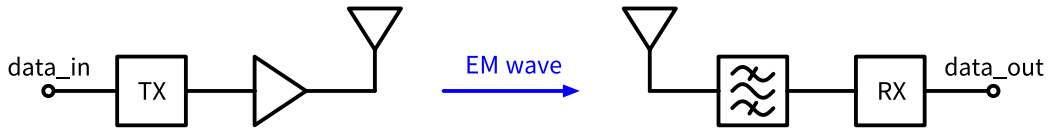


Figure 1: The block diagram of a simple wireless system.

Unfortunately, wireless transmission is hard. The wireless channel, i.e., the usage of electromagnetic waves to transmit information from a transmitter to a receiver, while tremendously useful, has quite a few undesired features:

- The wireless channel is shared between *all* users.
- As a consequence, the available bandwidth is shared; this means that bandwidth is a scarce resource.
- The wireless channel has significant losses.
- The channel is time variant, as usually the transmitter and/or the receiver move, and/or the environment changes.

In order to estimate the received power P_R from a wireless transmission, we can use **Friis' transmission formula** [3]:

$$P_\mathrm{R} = \frac{P_\mathrm{T} \cdot G_\mathrm{T}}{4\pi d^2} \cdot A_\mathrm{R} = P_\mathrm{T} \cdot \frac{A_\mathrm{R} \cdot A_\mathrm{T}}{d^2 \lambda^2} \tag{1}$$

Here, $A_\mathrm{R}/A_\mathrm{T}$ is the effective area of the receive/transmit antenna, while d is the distance (line of sight) between the two antennas, and λ is the wavelength of the EM wave. The first form of Equation 1 is simply the power density at distance d (the transmit power spread over a sphere, boosted by the transmit antenna gain G_T) multiplied by the effective area of the receive antenna. The second form follows by substituting the general relation between antenna gain and effective area,

$$G = \frac{4\pi A}{\lambda^2}, \tag{2}$$

for the transmit antenna. The effective area of an antenna depends on the type and construction, but generally we can say that

$$A \propto \lambda^2$$

For an isotropic antenna (a theoretical construct where the radiation is of equal strength in all directions) $A = \lambda^2/(4\pi)$, while for a $\lambda/2$-dipole $A = 0.13\lambda^2$. Of course, the speed of light $c = 3 \times 10^8$ m/s relates frequency f and the wavelength λ of an electromagnetic wave by

$$c = \lambda f.$$

Generally speaking, the size of an electromagnetic antenna is proportional to the wavelength of the EM wave used for transmission. For many devices, we seek antennas with dimensions on the order of a few centimeters, and this is why frequencies in the hundreds of MHz to GHz are so popular. Table 1 lists a few typical applications and their frequencies and wavelengths.

Table 1: Typical RF applications with their operating frequencies and corresponding wavelengths

Application	Frequency	Wavelength
FM Radio	88 MHz to 108 MHz	3.4 m down to 2.8 m
WiFi (lowband)	2.4 GHz	12.5 cm
WiFi (highband)	5 GHz	6 cm
Bluetooth	2.4 GHz	12.5 cm
Cellular	0.6 GHz to 5 GHz	50 cm down to 6 cm
GNSS	1.575 GHz	19 cm

As you can see in Table 1, many of these antennas would not fit into typical device form factors, i.e., often we have to use electrically small antennas. As antenna design is not part of this course but critically important for wireless systems, please refer to [4] or this website for more information on antenna design.

Note 1: Wavelength Calculation

Let's calculate the wavelength for a Bluetooth signal at 2.4 GHz. Given:

- Frequency $f = 2.4\,\text{GHz} = 2.4 \times 10^9$ Hz
- Speed of light $c = 3 \times 10^8$ m/s

Using the relationship $c = \lambda f$, we can solve for wavelength:

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8\,\text{m/s}}{2.4 \times 10^9\,\text{Hz}} = 0.125\,\text{m} = 12.5\,\text{cm}$$

This means that a quarter-wavelength monopole antenna for 2.4 GHz Bluetooth would be approximately 3.1 cm long, which easily fits into most mobile devices.

In order to get a feeling for the attenuation experienced in wireless communication, we now calculate the following example transmission. We will use the unit of dBm which is often used in RF design and is defined as

$$P|_{\mathrm{dBm}} = 10 \cdot \log_{10}\left(\frac{P|_{\mathrm{W}}}{1\,\mathrm{mW}}\right) \tag{3}$$

which is a logarithmic measure of power relative to 1 milliwatt. For example, 1 W = 30 dBm, 100 mW = 20 dBm, and 1 mW = 0 dBm.

> **i** Note 2: Wireless Transmission
>
> We use the following parameters:
>
> - Transmit power $P_{\mathrm{T}} = 1\,\mathrm{W}$
> - Frequency $f = 2.4\,\mathrm{GHz}$
> - Communication distance $d = 10\,\mathrm{km}$
> - Using $\lambda/2$ dipoles on both ends
>
> Using Equation 1 we calculate
>
> $$P_{\mathrm{R}} = P_{\mathrm{T}} \cdot \frac{0.13\lambda^2 \cdot 0.13\lambda^2}{d^2\lambda^2} = P_{\mathrm{T}} \cdot 0.13^2 \left(\frac{\lambda}{d}\right)^2 = 2.64\,\mathrm{pW} = -85.8\,\mathrm{dBm}$$
>
> With the transmit power of 1 W = 30 dBm we have an attenuation of 116 dB! This is a very large number!

The loss we calculated in Note 2 is called the **free-space path loss (FSPL)**. It is the minimum loss we can expect in a wireless communication system. In reality, the situation is often even worse. The FSPL (in dB) can be calculated as

$$\mathrm{FSPL} = 20 \cdot \log_{10}(d/\mathrm{m}) + 20 \cdot \log_{10}(f/\mathrm{Hz}) + 20 \cdot \log_{10}\left(\frac{4\pi}{c}\ \mathrm{m/s}\right). \tag{4}$$

Equation 4 can be readily derived from Equation 1 and Equation 3 assuming isotropic antennas at transmitter and receiver. Using Equation 4 we can calculate the FSPL for different distances d and frequencies f. The results are shown in Figure 2. It should be noted that the FSPL increases by 20 dB per decade of distance and 20 dB per decade of frequency, making the use of higher frequencies and longer distances very challenging.

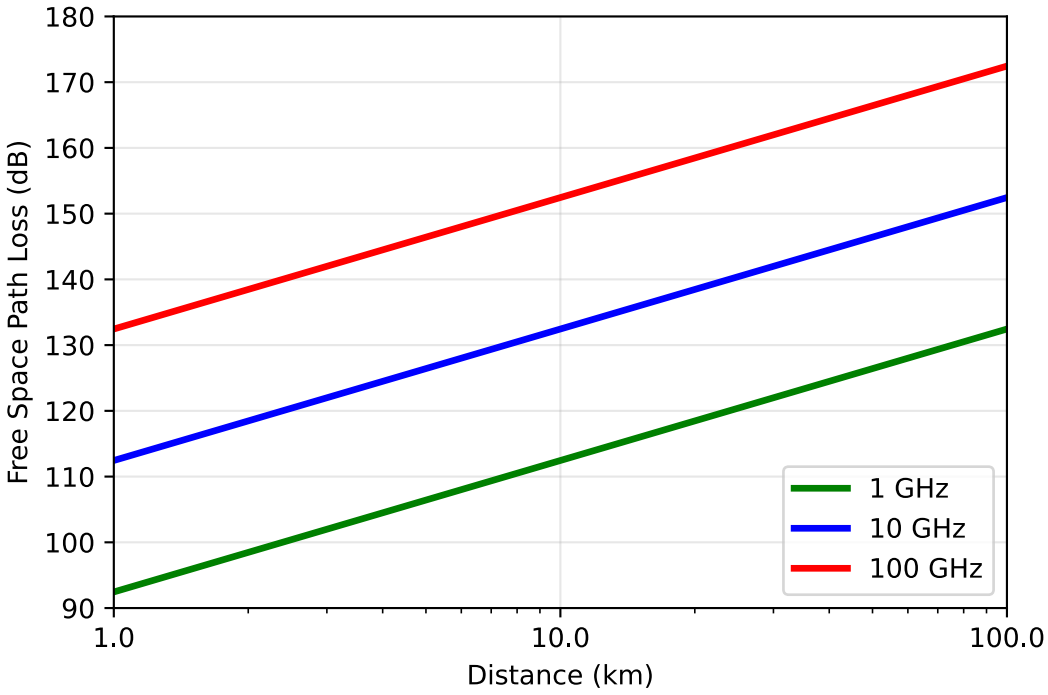


Figure 2: Free space path loss vs. distance for different frequencies (1 GHz, 10 GHz, and 100 GHz).

As dire as the situation of Figure 2 already looks, these are not even all the factors to consider:

- The given attenuation is for line-of-sight paths; often, the attenuation is significantly higher than this due to blockage by buildings, mountains, rain, or foliage.
- In the absence of a direct line-of-sight path, the EM wave is redirected by reflections, causing additional attenuation and potential destructive interference due to multi-path reception.

The consequences of this are (among others):

- The transmitter needs to generate enough **transmit power** to overcome the transmission loss; this has to be done with high **efficiency**, as the transmit device is battery operated or limited by cooling.
- The receiver has to be able to process **weak signals**, i.e., the **noise** level of the signal processing has to be very low.
- Often, the receive signal is very weak, while there are strong signals at other frequencies (i.e., other wireless transmitters are located close to the receiver). This means the receiver has to be able to process a weak signal while simultaneously tolerating **large interfering signals** (called **blockers**).
- Since the frequency spectrum is shared among many users and wireless applications, the transmit information has to be packed efficiently into a **small bandwidth**.
- Very often, wireless devices are battery-operated. This means transmit and receive functions have to be implemented using **minimum power consumption**.

As stated in the beginning, designing wireless systems is hard.

1.2 Wireless Standards

In order to allow communication between different devices, different operators, and different manufacturers, wireless communication is usually standardized. There are many different standards, each with its own characteristics. Wireless standards define every aspect of wireless communication, and can be documents of hundreds or thousands of pages. Here, we mainly focus on the radio-frequency and analog aspects of wireless standards. Summarized in Table 2 are a few popular wireless standards with their main characteristics.

Table 2: Comparison of wireless communication standards

Standard	GSM (2G)	WCDMA (3G)	LTE (4G)	**5G NR**	WiFi	Bluetooth	GNSS
Frequency range (MHz)	850, 900, 1800, 1900	850, 900, 1700, 1900, 2100	Multiple bands 450…7100 **(FR1)**, **24000…48000 (FR2)**		2400, 5000, 6000	2400	1500, 1200
Modulation	GMSK, 8PSK (EDGE)	QPSK (DL), BPSK (UL), 16QAM (HSPA), 64QAM (HSPA)	QPSK, 16QAM, 64QAM (DL+UL) 256QAM (DL**+UL**)		BPSK, QPSK, 16QAM, 64QAM, 256QAM, 1024QAM, 4096QAM	GFSK (m=0.28…0.35), π/4-DQPSK, 8DPSK	BPSK, QPSK
Transmission/ multiple access	TDMA, FDMA	DS-CDMA	OFDMA (DL**+UL**), SC-FDMA/**DFT-s-OFDM (UL)**		OFDM, CSMA/CA	FHSS	CDMA
Duplex	FDD	FDD	FDD, TDD		TDD	TDD	n/a
Channel bandwidth	200 kHz	5 MHz	1.4, 3, 5, 10, 15, 20, …, **100 MHz (FR1), 400 MHz (FR2)**		10, 20, 40, 80, 160, 320 MHz	1 MHz	16…24 MHz
Symbol rate[1]	270.833 ksym/s	3.84 Msym/s	15/**30/60** ksym/s		312.5 ksym/s (11a/g/n/ac), 78.125 ksym/s (11ax/be)	1 Msym/s	50 sym/s
Pulse shaping	Gaussian (BT=0.3)	Root Raised Cosine (α=0.22)	Rectangular		Rectangular	Gaussian (BT=0.5)	Rectangular
Transmit power	1…2 W	250 mW	200 mW (FDD), 400 mW (TDD)		100 mW	1…100 mW	n/a
PAR (UL)	0 dB (GMSK), 3 dB (8PSK)	3…8 dB	6…8 dB		Up to 12 dB	0 dB (GFSK), 3 dB (8DPSK)	n/a
MIMO	no	Not realized	DL 4x4 (up to 8x8)		2x2 (up to 8x8)	no	no
Channel bond	no	Up to 4x5 MHz (DL 2x2)	Up to 7x20/4x100 MHz		Up to 80+80+80+80 MHz	no	no

During this course, we will learn what these terms mean and how they impact the design of RF integrated circuits.

As you can see in Table 2, since LTE (4G) and 5G NR, many additional bands have been defined in the sub-6 GHz range (FR1) and also in the mm-wave range (FR2, 24.25 to 52.6 GHz). This means that modern wireless devices have to support many different frequency bands, which makes the design of RF front ends even more challenging. A good overview of the different frequency bands is given here for LTE and 5G NR.

RFIC design is a multidisciplinary field, requiring knowledge from various engineering domains, as shown in Figure 3. This makes RFIC design challenging, but also very interesting!

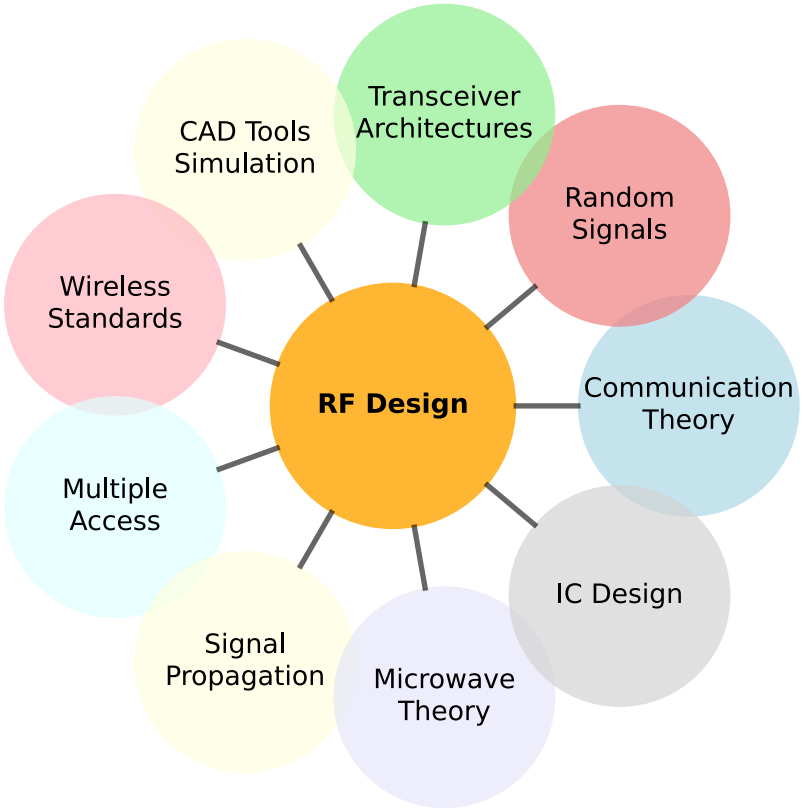


Figure 3: RF design as a multidisciplinary field requiring knowledge from various engineering domains (adapted from [1]).

Further, RFIC design requires careful consideration of many different aspects, as shown in Figure 4. Many parameters are often tightly coupled, requiring careful trade-offs during the design process.

[1]For the OFDM-based standards (LTE/5G NR, WiFi), the listed values correspond to the subcarrier spacing (numerology Δf), which equals the per-subcarrier symbol rate; they are not the aggregate symbol rate of the full signal.

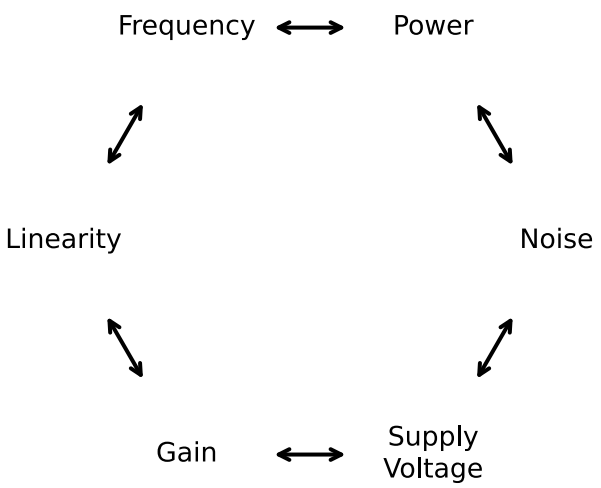


Figure 4: RFICs require careful design considerations and trade-offs (adapted from [1]).

2 Fundamentals

In this section, we will discuss a few important concepts which will be instrumental in the further study of RF circuits and systems. As the dynamic range of signals in RF circuits and systems is often limited on the top end by linearity, and on the bottom end by noise, we will discuss these two topics in some detail.

2.1 Channel Capacity

In Section 1 we have already discussed the fact that we need to pack information into a minimum bandwidth, as the available spectrum is limited. To appreciate the limits of information transfer, we need to understand how much information can be transmitted over a given bandwidth. This limit is given by the **Shannon-Hartley theorem**, which states the maximum data rate C (in bit/s) that can be transmitted over a communication channel with bandwidth B (in Hz) and signal-to-noise ratio SNR (in linear units):

$$C = B \cdot \log_2(1 + \mathrm{SNR}) \tag{5}$$

This formula provides us a theoretical upper limit on the data rate that can be achieved with a given bandwidth and SNR *under optimal conditions*. It is important to note that this limit is only achievable with ideal coding and modulation schemes, which are not practical in real-world systems. However, it provides a useful benchmark for evaluating the performance of communication systems.

i Note 3: Channel Capacity Example

Let us calculate the channel capacity for a system with an RF bandwidth of 1 MHz and an SNR of 10 dB. These are typical *assumptions* for a Bluetooth LE receiver at 1 Mbps (the Bluetooth LE specification itself does not define an SNR, but specifies the receiver sensitivity via a maximum bit error rate of 0.1 %). First, we need to convert the SNR from dB to linear units:

$$\mathrm{SNR} = 10^{10/10} = 10$$

Now we can use Equation 5 to calculate the channel capacity:

$$C = B \cdot \log_2(1 + \mathrm{SNR}) = 1\,\mathrm{MHz} \cdot \log_2(1 + 10) = 1\,\mathrm{MHz} \cdot 3.459 \approx 3.5\,\mathrm{Mbps}$$

This sounds reasonable, as the user data rate for Bluetooth LE is 1 Mbps for the given SNR, which allows for considerable overhead for coding and protocols.

2.2 Linearity

As we have already seen in Section 1.1, the transmitter has to process large signals without distorting them, while the receiver has to process small signals in the presence of large signals. Both situations mean we need metrics and models to quantify and discuss linearity properties.

We are going to use a very simple, time-invariant model to study linearity, based on a Taylor polynomial.

! Linearity and Memoryless Systems

We use a **memoryless** (also called "instantaneous") nonlinear model to simplify the mathematics. A memoryless nonlinear system has the property that the output at time t only depends on the input at time t.

In contrast, a system **with memory** will have an output at time t which depends on the input at time t and also on past inputs (e.g., at times $t - \Delta T, t - 2\Delta T, ..., t - N\Delta T$). Such a system can additionally be **time-variant**, which means its behavior changes over time, which leads to quite a few very interesting and important phenomena! Examples of systems with memory are filters, which have a frequency-dependent response, or power amplifiers with thermal memory effects.

We model a nonlinear circuit block with the input $x(t)$ and the output $y(t)$ with the following Taylor polynomial:

$$\begin{aligned} y(t) = \alpha_0 + \alpha_1 x(t) + \alpha_2 x(t)^2 + \alpha_3 x(t)^3 \\ + \alpha_4 x(t)^4 + \alpha_5 x(t)^5 + ... \end{aligned} \tag{6}$$

Usually, the blocks under study will have higher-order nonlinear terms, but we often stop at 5th order to keep things simple. For practical work, higher-order terms should be included if necessary.

Which $x(t)$ should we use to study wireless systems? Often, the bandwidth Δf_{BW} of a transmit signal is much smaller than the center frequency f_0, i.e., $\Delta f_{\mathrm{BW}} \ll f_0$. In this case, using a sinusoidal signal as a model is both simple to handle and approximately correct.

2.2.1 Single-Tone Linearity

We thus use (with A_0 being the amplitude of the input signal and $\omega_0 = 2\pi f_0$ the angular frequency)

$$x(t) = A_0 \cos(\omega_0 t)$$

and insert it into Equation 6. After some simple trigonometric manipulations we arrive at

$$\begin{aligned} y(t) = & \underbrace{\alpha_0 + \frac{1}{2}\alpha_2 A_0^2 + \frac{3}{8}\alpha_4 A_0^4}_{\text{dc component } (\omega=0)} \\ & + \underbrace{\left(\alpha_1 A_0 + \frac{3}{4}\alpha_3 A_0^3 + \frac{5}{8}\alpha_5 A_0^5\right)\cos(\omega_0 t)}_{\text{fundamental } (\omega_0)} \\ & + \underbrace{\left(\frac{1}{2}\alpha_2 A_0^2 + \frac{1}{2}\alpha_4 A_0^4\right)\cos(2\omega_0 t)}_{\text{2nd harmonic } (2\omega_0)} \\ & + \underbrace{\left(\frac{1}{4}\alpha_3 A_0^3 + \frac{5}{16}\alpha_5 A_0^5\right)\cos(3\omega_0 t)}_{\text{3rd harmonic } (3\omega_0)} \\ & + \underbrace{\frac{1}{8}\alpha_4 A_0^4 \cos(4\omega_0 t)}_{\text{4th harmonic } (4\omega_0)} \\ & + \underbrace{\frac{1}{16}\alpha_5 A_0^5 \cos(5\omega_0 t)}_{\text{5th harmonic } (5\omega_0)} + \ldots \end{aligned} \tag{7}$$

The gain H^{ω_0} of the fundamental tone at ω_0 would be

$$H^{\omega_0} = \frac{y^{\omega_0}}{x^{\omega_0}} = \alpha_1 + \frac{3}{4}\alpha_3 A_0^2 + \frac{5}{8}\alpha_5 A_0^4. \tag{8}$$

Looking at Equation 7 and Equation 8, we can make a few interesting observations:

- Even-order nonlinearities (due to α_2 and α_4) create low-frequency components; they effectively add frequency components stemming from the envelope A_0. If A_0 is a constant then this results in a dc term; if $A_0(t)$ is time-variant, it will create a squared and quadrupled version of it at low frequencies.
- The α_1-term is the gain of the circuit block.
- Odd-order nonlinearities (α_3 and α_5) can impact the gain of the fundamental term passing through the block. Often α_3 and α_5 have the same sign, which is opposite to the sign of α_1. This implies that the gain stays linear for small input powers. For larger input signals, the terms with α_3 and α_5 start to dominate, leading to gain compression or saturation effects, until the circuit block reaches a level of maximum output power. In some cases, α_3 has the same sign as α_1 but α_5 has the opposite one, which leads to gain expansion: The gain would

first demonstrate an overshoot at some input power level, until the term with α_5 becomes dominant and the gain is compressed again.

- Even- and odd-order nonlinearities create additional frequency components, so-called harmonics of the fundamental frequency. These harmonics are often unwanted, as they are far outside the wanted transmission frequency range, and need to be minimized, by either
  1. using a lowpass to filter these harmonics, or
  2. increasing the linearity, i.e., making the α_2, α_3, etc., small enough.

The created harmonics are illustrated in Figure 5. Note that measuring harmonics to quantify the nonlinearity metrics like α_2 and α_3 is often not very accurate, as these harmonics are often filtered in bandwidth-limited systems.

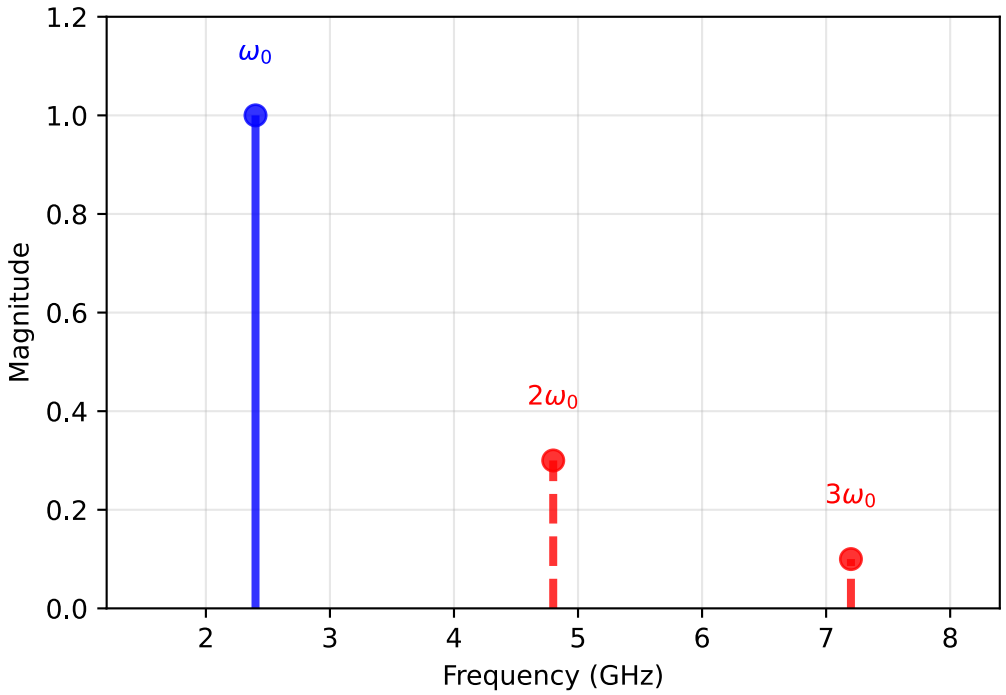


Figure 5: Single-tone test showing created harmonics at $2\omega_0$ and $3\omega_0$.

How can we quantify the nonlinearity with a one-tone test? We can sweep the input signal $x(t)$ in amplitude, and observe the output $y(t)$. If the observed gain drops by 1 dB from the small-signal value, we note the input power and call this point the **1 dB compression point** ($P_{1\mathrm{dB}}$). We should always add whether this 1-dB compression point is input- or output-referred to avoid ambiguity. The diagram in Figure 6 shows this test, evaluated directly from Equation 8 with the *voltage* gain $\alpha_1 = 10$ (i.e., a small-signal power gain of 20 dB) and $\alpha_3 = -145\,\mathrm{V}^{-2}$, referred to a 50 Ω system.

Setting the compressed gain of Equation 8 to $10^{-1/20}$ of its small-signal value α_1 gives the input amplitude at the 1-dB compression point,

$$A_{1\mathrm{dB}}^2 = \frac{1 - 10^{-1/20}}{\frac{3}{4}|\alpha_3/\alpha_1|} \approx 0.145 \cdot \left|\frac{\alpha_1}{\alpha_3}\right|, \tag{9}$$

which for the values above lands at $P_{1\mathrm{dB}} = -10$ dBm (input-referred). Note that the truncated third-order model is only meaningful up to its turnover point at $A^2 = \alpha_1/(2.25|\alpha_3|)$, which is a mere 4.9 dB above $P_{1\mathrm{dB}}$; beyond that, the higher-order terms of Equation 6 (or a saturating model, see below) are required.

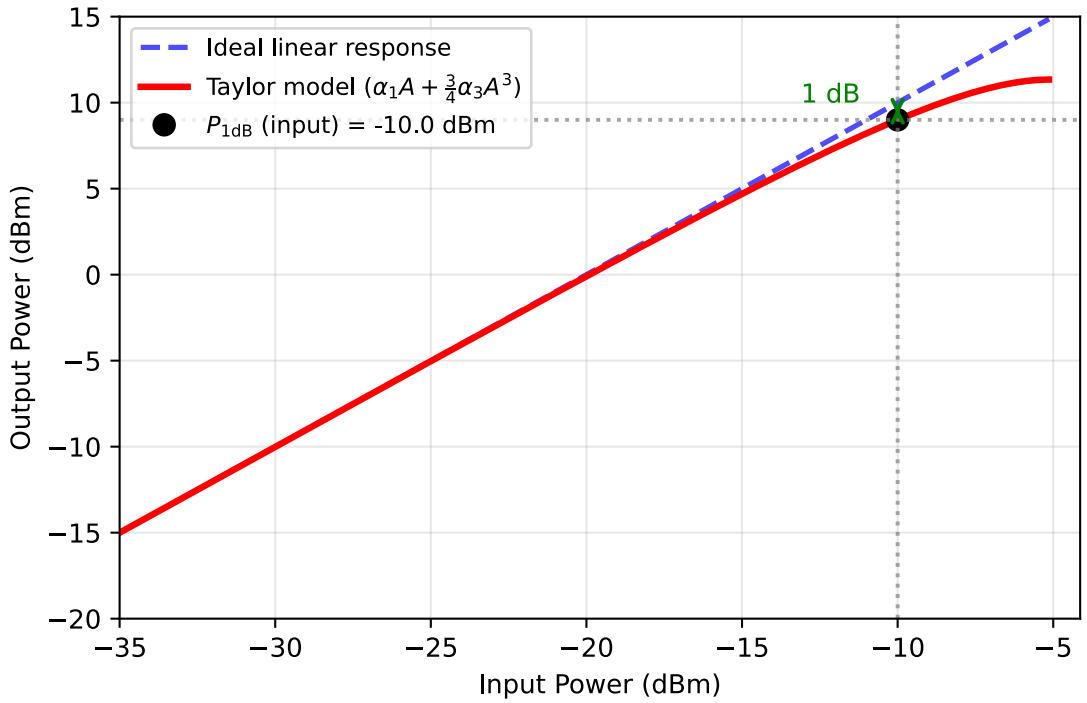


Figure 6: Single-tone 1-dB compression point test, evaluated directly from the Taylor model $y = \alpha_1 x + \alpha_3 x^3$. The curve is only drawn up to the point where the truncated third-order model turns over and stops being physically meaningful.

> ! Compressive vs. Expansive Behavior
>
> Note that for compressive behavior, α_3 and α_1 have different signs, while for expansive behavior, they have the same sign.

The following plot (Figure 7) in addition shows an expansive behavior. In contrast to the truncated Taylor-series model above, a closed-form saturating model is used here so that the output power correctly converges to the saturation power P_{sat} for large input levels. Such saturating amplitude models are widely used to describe power-amplifier nonlinearity [5]. The model uses the amplitude characteristic

$$A_{\mathrm{out}} = A_{\mathrm{sat}} \tanh(u + c \cdot u^3)$$

with the normalised amplitude $u = \sqrt{G_0 P_{\mathrm{in}}/P_{\mathrm{sat}}}$ and an expansion coefficient $c > 1/3$. At small input powers the gain equals the small-signal gain G_0; at intermediate powers the gain expands above G_0 (because the cubic term in the argument of $\tanh(\cdot)$ adds extra gain before the saturation takes effect); and at large input powers the output saturates at P_{sat} while the gain drops steeply. This (partly) expansive behavior can be used to linearize the gain of a

circuit block over a wider input power range, which is often desirable in RF power amplifiers (see Section 8.2).

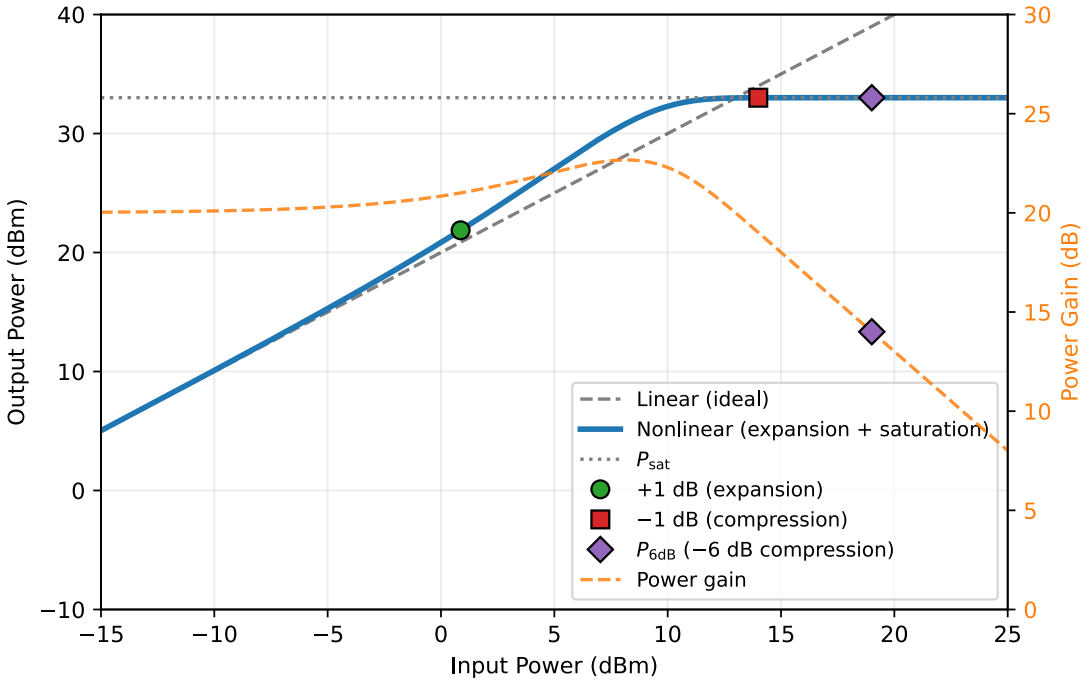


Figure 7: Output power and power gain vs. input power showing expansion, compression, and saturation towards P_{sat}. The closed-form model uses $P_{\text{out}} = P_{\text{sat}} \tanh^2(u + c \cdot u^3)$ with $u = \sqrt{G_0 P_{\text{in}}/P_{\text{sat}}}$. For (power) amplifiers the saturating power is often ill-defined, so instead of P_{sat} a compression point like P6dB is used.

2.2.2 Multi-Tone Linearity

We now extend our investigations and apply two sinusoids with different frequencies ω_1 and ω_2 and different amplitudes A_1 and A_2 and see which signals we get at the output of the nonlinear block. The two-tone test and resulting third-order intermodulation products (IM3) are illustrated in Figure 8.

$$x(t) = A_1 \cos(\omega_1 t) + A_2 \cos(\omega_2 t)$$

We apply the above stimulus to our nonlinear model described by Equation 6 and again, after some trigonometric manipulations, arrive at:

$$y(t) = y'(t) + y''(t) + y'''(t) \tag{10}$$

As many different frequency components are created by this simple two-tone test (and nonlinearity only up to 3rd order), we split the result into different equations and look at the result separately.

i What We Leave Out

To keep the following equations readable, we list only the components that are *new* compared to the single-tone case. The two-tone stimulus of course also produces the plain harmonics of each tone individually (at $2\omega_1$, $2\omega_2$, $3\omega_1$, $3\omega_2$), with exactly the coefficients we already derived in Equation 7. They are omitted below, not missing.

First, we start with the fundamental tones:

$$\begin{aligned} y'(t) = &\left(\underbrace{\alpha_1 A_1 + \frac{3}{4}\alpha_3 A_1^3}_{\text{compression/expansion}} + \underbrace{\frac{3}{2}\alpha_3 A_1 A_2^2}_{\text{cross-modulation/desens}} \right) \cos(\omega_1 t) \\ &+ \left(\underbrace{\alpha_1 A_2 + \frac{3}{4}\alpha_3 A_2^3}_{\text{compression/expansion}} + \underbrace{\frac{3}{2}\alpha_3 A_2 A_1^2}_{\text{cross-modulation/desens}} \right) \cos(\omega_2 t) \end{aligned} \tag{11}$$

As shown in Equation 11, interesting things happen:

- We (again) have the gain compression/expansion effect as already discussed in Section 2.2.1.
- In addition, we have **cross-modulation**, i.e., the squared envelope of one tone (e.g., $A_2(t)$ of the tone at ω_2) impacts the envelope of the other tone at ω_1. This can lead to unwanted signal distortion, even if there is a large frequency separation between ω_1 and ω_2!
- Further, since the sign of α_3 is usually opposite to α_1, this can also lead to **desensitization** ("desens"). If, for example, $A_2 \gg A_1$, then there would be no compression due to the tone ω_1 itself, however, the large tone at ω_2 will lead to gain compression of the tone at ω_1; this effect is called desens.

We now look at the next class of generated tones:

$$\begin{aligned} y''(t) = &\frac{1}{2}\alpha_2 A_1^2 + \frac{1}{2}\alpha_2 A_2^2 \\ &+ \alpha_2 A_1 A_2 \cos[(\omega_1 - \omega_2)t] \\ &+ \alpha_2 A_1 A_2 \cos[(\omega_1 + \omega_2)t] \end{aligned} \tag{12}$$

As we can see in Equation 12, new tones are created (besides the low frequency components we already know from the single-tone test) at the sum and difference of ω_1 and ω_2. These new frequency components are called "**intermodulation products of second order**" (IM2). These tones are created by the even-order nonlinearity (α_2). These IM2 products are far away from the wanted tones, so they are often not very problematic in amplifiers (but there can be exceptions!). However, they can be very problematic in frequency conversion blocks like mixers. We will come back to this point when discussing zero-IF receivers.

We now investigate the next couple of tones:

$$
\begin{aligned}
y'''(t) = &\frac{3}{4}\alpha_3 A_1^2 A_2 \cos[(2\omega_1 + \omega_2)t] \\
&+\frac{3}{4}\alpha_3 A_1^2 A_2 \cos[(2\omega_1 - \omega_2)t] \\
&+\frac{3}{4}\alpha_3 A_1 A_2^2 \cos[(2\omega_2 + \omega_1)t] \\
&+\frac{3}{4}\alpha_3 A_1 A_2^2 \cos[(2\omega_2 - \omega_1)t]
\end{aligned} \tag{13}
$$

The tones shown in Equation 13 are called "**intermodulation products of third order**" (IM3), and are caused by the odd nonlinearities (like α_3). While the IM3 tones located at $2\omega_1 + \omega_2$ and $\omega_1 + 2\omega_2$ are similar to the sum IM2 tone and far away from ω_1 and ω_2, the other two tones are concerning.

Expressing $\Delta\omega = \omega_2 - \omega_1$ (and assuming $\omega_1 < \omega_2$), the construction rule $2\omega_1 - \omega_2 = \omega_1 - \Delta\omega$ and $2\omega_2 - \omega_1 = \omega_2 + \Delta\omega$ results in new tones right beside ω_1 and ω_2, with a frequency separation only defined by $\Delta\omega$. This situation is illustrated in Figure 8.

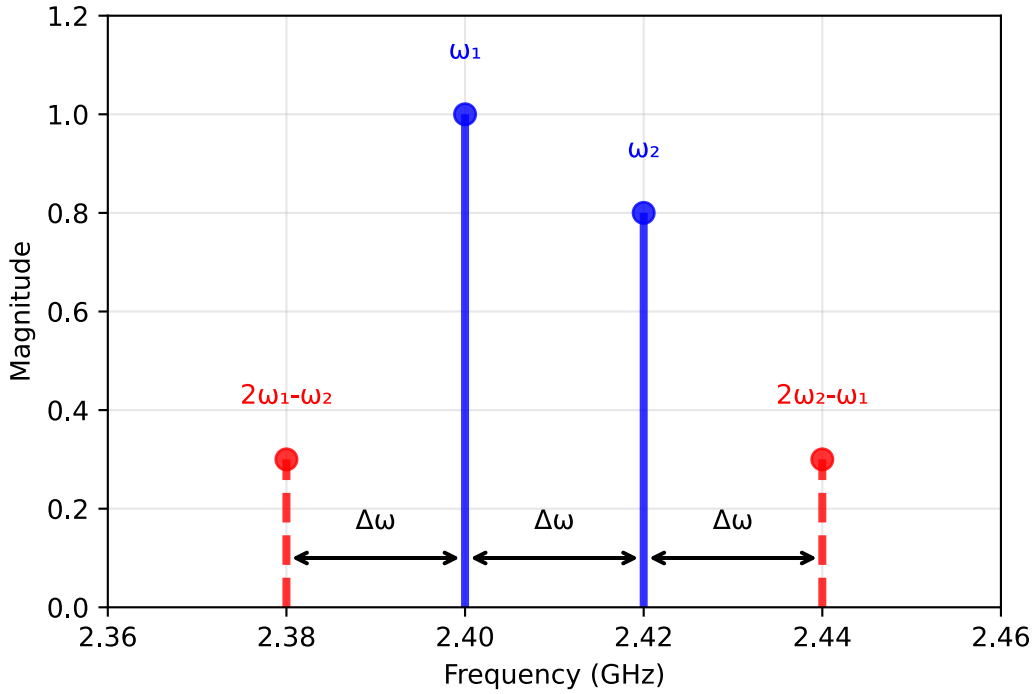


Figure 8: Two-tone test showing fundamental frequencies ω_1, ω_2 and third-order intermodulation products (IM3) at $2\omega_1$-ω_2 and $2\omega_2$-ω_1.

This close proximity of the IM3 tones can also be utilized to characterize nonlinear performance. Using gain compression or harmonic generation (H3), it can be very difficult to extract nonlinearity of third order (α_3). However, using a two-tone test, the IM3 tones can be readily measured, even if the measured signal path shows a **bandpass characteristic**! As RF systems frequently employ bandpass filters to suppress out-of-band signals, this is a very important property of the two-tone test.

The resulting test is called a two-tone test yielding the third-order intercept point (IP3). This test is widely used in RF design to characterize the linearity of amplifiers, mixers, and complete transceiver systems. The power relationship between fundamental tones and IM3 products as a function of input power is shown in Figure 9.

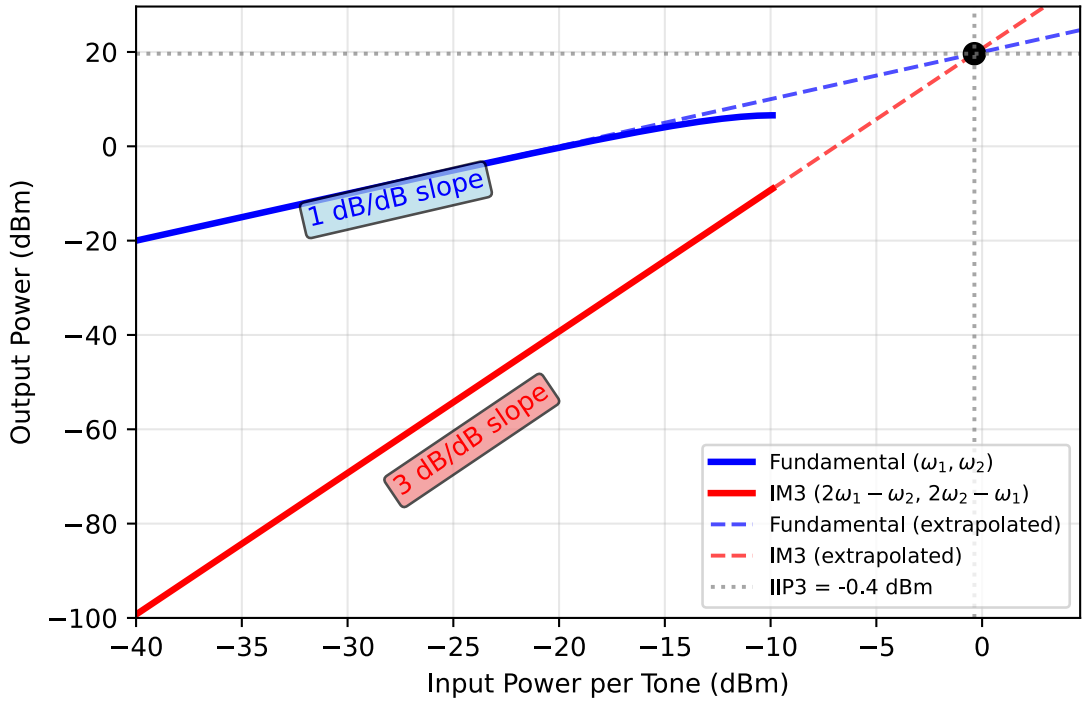


Figure 9: Two-tone IM3 test evaluated from the same Taylor model as Figure 6, showing fundamental and IM3 output power vs. input power per tone, and the extrapolated IIP3/OIP3 intercept. Equal input power per tone is assumed.

Note that, as shown in Figure 9, the IM3 products rise with a slope of 3 dB/dB, i.e., if the input power is increased by 1 dB, the IM3 products increase by 3 dB. The fundamental tones rise with a slope of 1 dB/dB (as long as we are in the linear region). The IP3 point is defined as the intersection of the **extrapolated** linear lines of fundamental and IM3 products. Equating the extrapolated fundamental $\alpha_1 A$ with the extrapolated IM3 amplitude $\frac{3}{4}|\alpha_3|A^3$ from Equation 13 gives

$$A^2_{\mathrm{IIP3}} = \frac{4}{3}\left|\frac{\alpha_1}{\alpha_3}\right|. \tag{14}$$

As both lines have different slopes, this intersection point is usually far outside the actual operating range of the circuit block under test! Figure 9 uses the same α_1 and α_3 as Figure 6, so both figures describe the *same* device: the resulting IIP3 of -0.4 dBm indeed sits 9.6 dB above the $P_{1\mathrm{dB}}$ of -10 dBm, which is precisely the rule of thumb stated in Equation 16 below.

When calculating the IIP3 (input-referred IP3), we can use the following formula, assuming equal input power per tone. It is important to always check the slope of the IM3 products to ensure that we are indeed in the third-order region! If the input power per tone is P_{in} (in dBm) and the input-referred power of one IM3 tone is P_{IM3} (in dBm), then the input-referred IP3 is given by

$$\mathrm{IIP3} = P_{\mathrm{in}} + \frac{P_{\mathrm{in}} - P_{\mathrm{IM3}}}{2} \tag{15}$$

Further, for mildly nonlinear systems (i.e., α_3 is dominant), the IIP3 can be approximated from the 1-dB compression point as [1]

$$\mathrm{IIP3}|_{\mathrm{dBm}} \approx P_{\mathrm{1dB}}|_{\mathrm{dBm}} + 9.6\,\mathrm{dB}\ . \tag{16}$$

i Further Information: The AST Linearity Test

The two-tone test assumes a spectrally pure stimulus: any distortion or quantization noise of the test signal generator (TSG) also appears at the IM3 frequencies of the system under test (SUT) and corrupts the measured IP3. In the lab, this is solved with high-quality signal generators, but for on-chip linearity monitoring (e.g., as a built-in self-test of an automotive radar receiver), such a clean TSG is very hard to integrate.

The **artificial slow time (AST) linearity test** enables accurate on-chip linearity testing with a single low-quality TSG by exploiting the TSG's repeatability. Instead of injecting a spectrally pure test signal into the SUT, the same imprecise test signal is inserted multiple times, each time scaled with a different, properly chosen analog factor. With this, all impairments affecting the test signal *before* the analog scaling (i.e., distortions of the TSG) can be separated from the SUT's nonlinear effects *after* the scaling by digital signal processing. Thus, the requirements on the integrated analog TSG are significantly reduced.

- **Application 1: IP3 with a low-quality two-tone TSG.** The AST linearity test is applied to the classic two-tone test, where the IM3 products of the SUT are recovered even if the two-tone test signal itself is heavily distorted or coarsely quantized. The concept is introduced in [6] and treated in detail in [7, ch. 4].
- **Application 2: IP3 with a low-quality single-tone TSG.** With a single tone, the in-band third-order distortion of the SUT falls onto the same frequency as the fundamental itself (it is what causes the gain compression of Section 2.2.1), so an ordinary spectrum cannot tell it apart from the fundamental. The AST linearity test separates it nevertheless, so the IP3 can be measured with an even simpler single-tone test signal. A MATLAB-automated RF demonstrator (LNA and mixer) that verifies both applications in the lab with a test signal of only 2-bit resolution is presented in [8], together with a recorded presentation [9]. In these measurements, the single-tone variant proved more sensitive to inaccurate scaling factors, so the two-tone variant is preferred for on-chip use.

If we have multiple blocks which are cascaded, and we know the gain and IIP3 of each block, we can calculate the overall IIP3 of the cascade with the following approximation. An exact calculation is very involved, as the nonlinearities of the first block (and the resulting tones) will be processed by the second block, creating even more tones; this process escalates very quickly. However, for practical purposes, the following approximation is often sufficient:

$$\frac{1}{\mathrm{IIP3}_{\mathrm{total}}} \approx \frac{1}{\mathrm{IIP3}_1} + \frac{G_1}{\mathrm{IIP3}_2} + \frac{G_1 G_2}{\mathrm{IIP3}_3} \tag{17}$$

Here G_1 is the linear gain of the first block, and $\mathrm{IIP3}_1$, $\mathrm{IIP3}_2$ are the input-referred IP3 of the first and second block, respectively. Note that all powers have to be in *linear units* when using Equation 17. An even more simplified version of Equation 17 can be used with all quantities given in dBm and dB, respectively:

$$\mathrm{IIP3}_{\mathrm{total}} \approx \min\{\mathrm{IIP3}_1, \mathrm{IIP3}_2 - G_1, \mathrm{IIP3}_3 - G_1 - G_2\} \tag{18}$$

A typical RF system cascade with multiple blocks and their individual IIP3 contributions is shown in Figure 10.

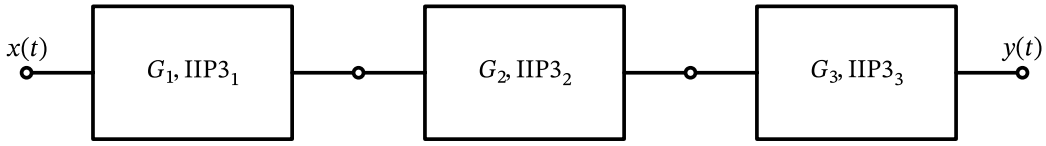


Figure 10: Block cascade for IIP3 calculation showing multiple stages with gains and individual IIP3 values.

i Note 4: Simple IIP3 Cascade Calculation

Let's calculate the overall IIP3 of two cascaded blocks. The first block is a low-noise amplifier with an IIP3 of −10 dBm and a gain of 20 dB. The second block is a mixer that has a gain of 10 dB and an IIP3 of 5 dBm. What is the value of the overall IIP3?

Using Equation 18 we can quickly estimate:

$$\mathrm{IIP3}_{\mathrm{total}} \approx \min\{-10\,\mathrm{dBm}, 5\,\mathrm{dBm} - 20\,\mathrm{dB} = -15\,\mathrm{dBm}\} = -15\,\mathrm{dBm}$$

We see that the overall IIP3 is limited by the linearity of the second block, as the first block amplifies all signals (including blockers) by 20 dB before they reach the second block.

2.3 Noise

Just as nonlinearity is a limiting factor for large signals, noise is the limiting factor for small signals. Noise is present in all electronic circuits and systems, and it is impossible to avoid it. However, we can try to minimize its impact on system performance.

Noise is usually characterized by its power spectral density (PSD) in units of Watts per Hertz (W/Hz). For example, thermal noise at room temperature has a one-sided PSD of approximately $kT = 4 \times 10^{-21}$ W/Hz, or -174 dBm/Hz (using Boltzmann's constant $k = 1.38 \times 10^{-23}$ J/K). This means that if we have a bandwidth of 1 MHz, the total thermal noise power would be

$$\begin{aligned} P_{\mathrm{thermal}}|_{\mathrm{W}} &= \mathrm{PSD}\,|_{\mathrm{W/Hz}} \cdot B \\ &= 4 \times 10^{-21}\,\mathrm{W/Hz} \cdot 1\,\mathrm{MHz} = 4 \times 10^{-15}\,\mathrm{W} \end{aligned}$$

or

$$
\begin{aligned}
P_{\text{thermal}}\big|_{\text{dBm}} &= \text{PSD}\ \big|_{\text{dBm/Hz}} + 10\log_{10}\left(\frac{B}{1\,\text{Hz}}\right) \\
&= -174\,\text{dBm/Hz} + 10\log_{10}\left(\frac{1\,\text{MHz}}{1\,\text{Hz}}\right) = -114\,\text{dBm}\ .
\end{aligned}
$$

The PSD of noise can be flat vs. frequency (which is called "white noise"), or can decrease with frequency (e.g., "flicker noise" or "1/f noise"). Further, noise can be generated by resistors (thermal noise), semiconductors (shot noise, generation-recombination noise), etc. A detailed discussion of noise sources can be found in [10], [11].

2.3.1 Types of Noise Generation

Resistors generate thermal noise (also called Johnson-Nyquist noise), which is white noise with a one-sided PSD of $4kTR$ (in V^2/Hz) when looking at the voltage across the resistor, or $4kT/R$ (in A^2/Hz) when looking at the current through the resistor. This noise is generated by the random thermal motion of charge carriers in the resistor.

> ! Thermal Noise
>
> Note that the simple approximation given above is only valid for reasonably low frequencies and typical temperatures, and is known as the Rayleigh-Jeans approximation. The full expression of Planck's blackbody radiation, accounting for quantum effects, is given by [3]
>
> $$\text{PSD} = \frac{4Rhf}{e^{hf/kT} - 1}$$
>
> where h is the Planck constant ($h = 6.626 \times 10^{-34}$ J s) and f is the frequency. The Rayleigh-Jeans approximation is valid for $f \ll kT/h$, which is approximately 6 THz at room temperature (290 K).
>
> We can integrate the above single-sided PSD over the full frequency range and show that the rms noise voltage of a resistor R is bounded to
>
> $$\overline{v_\text{n}^2} = \int_0^\infty \frac{4Rhf}{e^{hf/kT} - 1} df = \frac{2(\pi kT)^2}{3h} \cdot R$$
>
> which equates to approximately 13 mVrms noise voltage for a 1 kΩ resistor at room temperature (which is impossible to measure in practice, as there will be some form of bandwidth limitation in any real measurement setup).

MOSFETs generate several types of noise, the most important ones being the thermal noise of the channel and flicker noise.

The thermal noise of the channel can be modeled as a current noise source between the drain and source terminals with a one-sided PSD of $\overline{I_\text{n}^2} = 4kT\gamma g_{d0}$ (in A^2/Hz), where γ is a process-dependent parameter (usually between 2/3 and 2). The parameter g_{d0} is the drain-source conductance of the MOSFET at $V_\text{DS} = 0$ (i.e., in deep triode); in saturation it is commonly approximated by $g_{d0} \approx g_\text{m}$.

In saturation, it is often useful to express the thermal noise as a voltage noise source at the gate with a PSD of $\overline{V_\mathrm{n}^2} = 4kT\gamma/g_\mathrm{m}$ (in V^2/Hz). We can see that we can lower this noise of the MOSFET by increasing the transconductance g_m, which can be achieved by increasing the bias current.

In addition, at high frequencies, the MOSFET also has induced gate-current noise, which is correlated with the channel thermal noise. A detailed discussion of this noise source can be found in [11].

Flicker noise is usually modeled as a voltage noise source at the gate with a PSD of $K_f/(C'_\mathrm{ox}WLf)$ (in V^2/Hz), where K_f is a process-dependent parameter, C'_ox is the oxide capacitance per unit area, L and W are the length and width of the MOSFET, and f is the frequency. Note that we can lower the flicker noise by increasing the area of the MOSFET ($W \cdot L$), however, this increases the parasitic capacitances associated with the MOSFET, and this is often prohibitive for RF operation!

In **bipolar junction transistors (BJTs)**, the most important noise source is the shot noise due to the diffusion current in the base-emitter junction. It can be modeled as a current noise source between the collector and emitter terminals with a one-sided PSD of $2qI_\mathrm{C}$ (in A^2/Hz), where q is the elementary charge ($q = 1.6 \times 10^{-19}$ C) and I_C is the dc collector current.

> ! Equivalence of Shot and Thermal Noise
>
> Note that it has been shown in [12] that thermal noise and shot noise are actually equivalent, as both are generated by the random, thermally agitated motion of charge carriers!

Ideal **capacitors** and **inductors** do not generate noise; however, real-world capacitors and inductors have parasitic resistances which generate thermal noise.

In RF systems, additional noise sources can be present. One noteworthy example is the **cosmic microwave background** radiation, which can be modeled as a noise temperature of approximately 3 K. While this is negligible compared to thermal noise at room temperature (approximately 290 K), it can be significant in very low-noise systems, such as radio telescopes pointing to the sky. Another important noise source in RF systems is the **atmospheric noise**, which is generated by natural phenomena like lightning or in the ionosphere.

i A Note on Circuit Noise Calculations

When doing circuit noise calculations, it is instructive to keep the following points in mind:

- For circuit calculations involving noise sources, it is convenient to replace the power spectral density by equivalent sinusoidal generators in small bandwidths (typically 1 Hz).
- The noise power spectral density in a small bandwidth Δf is given by $\overline{V_\mathrm{n}^2} = \overline{v_\mathrm{n}^2}/\Delta f$ and $\overline{I_\mathrm{n}^2} = \overline{i_\mathrm{n}^2}/\Delta f$.
- The quantities $\overline{V_\mathrm{n}^2}$ and $\overline{I_\mathrm{n}^2}$ can be considered the mean-square value of sinusoidal generators. Using these values, network noise calculations reduce to familiar sinusoidal circuit-analysis calculations using V_n and I_n.
- Multiple *independent* noise sources can be calculated individually at the output, and the total noise in bandwidth Δf is calculated as a mean-square value by adding the individual mean-square contributions from each sinusoid.

2.3.2 Noise in Impedance-Matched Systems

We now want to calculate the maximum noise power that can be extracted from a noisy source. We assume the following situation as shown in Figure 11. Note that the voltage source $\overline{V_\mathrm{n,s}^2}$ models the thermal noise of the source resistor R_s resulting in a Thevenin equivalent circuit.

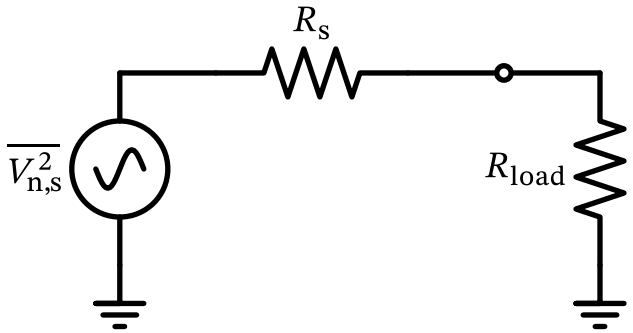


Figure 11: A noise-matched system with source and load impedances.

We know that the noise of the source resistor is given by $\overline{V_\mathrm{n,s}^2} = 4kTR_\mathrm{s}$. We assume the load resistor R_load as *noiseless* and matched to the source resistor, i.e., $R_\mathrm{load} = R_\mathrm{s}$ for maximum power transfer. The noise power spectral density delivered to the load resistor is then given by

$$P_\mathrm{n,load} = \frac{\overline{V_\mathrm{n,load}^2}}{R_\mathrm{load}} = \frac{\overline{V_\mathrm{n,s}^2}}{4R_\mathrm{s}} = kT \tag{19}$$

The calculation of Equation 19 confirms the initial statement that the maximum noise power spectral density that can be extracted from a noisy source is kT (in W/Hz). This result is independent of the actual value of the source resistance R_s.

We can further generalize the thermal noise of any impedance as

$$\overline{V_\mathrm{n}^2} = 4kT\Re\{Z\} \tag{20}$$

as for example in the complex impedance Z_ant of an antenna. Since an antenna is a reciprocal device, if we measure its radiation impedance Z_rad (for example with a vector network analyzer), we can calculate its thermal noise with Equation 20 to $\overline{V_\mathrm{n}^2} = 4kT\Re\{Z_\mathrm{rad}\}$.

2.3.3 Noise Figure

In RF systems, we often want to quantify the noise performance of a circuit block or a complete system. The most widely used metric is the **noise factor (F)**, which is defined as the ratio of the signal-to-noise ratio (SNR) at the input to the SNR at the output of a circuit block or system. If we express the noise factor in dB, we call it the **noise figure (NF)** [3]. The noise factor is given by

$$F = \frac{\mathrm{SNR}_\mathrm{in}}{\mathrm{SNR}_\mathrm{out}} = \frac{(P_\mathrm{s}/P_\mathrm{n})_\mathrm{in}}{(P_\mathrm{s}/P_\mathrm{n})_\mathrm{out}} \tag{21}$$

where P_s is the signal power and P_n is the noise power. The noise factor is always larger than or equal to 1 (or 0 dB), as no circuit can improve the SNR!

> ! SNR Improvement
>
> Note that the SNR can be improved by filtering, as filtering reduces the noise power. If the noise bandwidth is larger than the signal bandwidth, then the SNR can be improved without affecting the signal. However, this is not considered in the noise factor, as the noise factor assumes that both signal and noise pass through the same bandwidth.

Let us look at a simple model of a noisy circuit block as shown in Figure 12. The input signal S_in is accompanied by noise N_in. By definition, it is assumed that the input noise power results from a matched resistor at $T_0 = 290\,\mathrm{K}$, so that $N_\mathrm{in} = kT_0$. The circuit block has a power gain G and adds its own noise N_dut to the output signal. For simplicity, we assume that the input and output of the circuit block are **impedance matched** to avoid reflections.

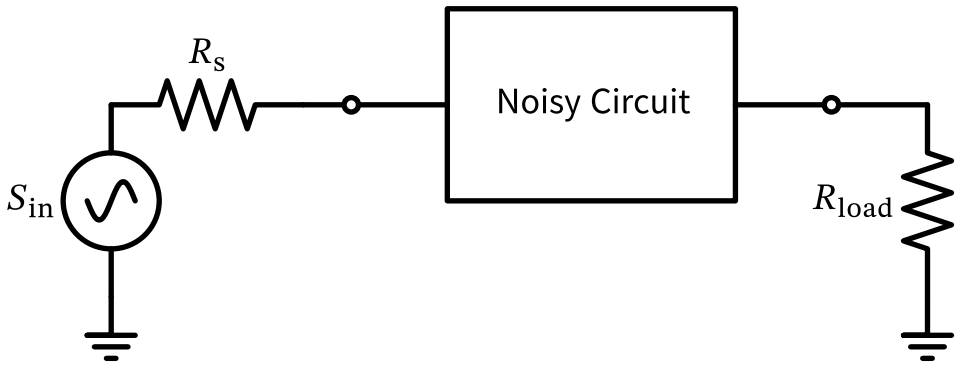


Figure 12: A noise-matched system with source and load impedances and a noisy circuit block.

The output signal and noise powers are then given by

$$S_{\text{out}} = GS_{\text{in}}$$

$$N_{\text{out}} = GN_{\text{in}} + N_{\text{dut}}$$

The resulting noise factor can then be calculated as

$$F = \frac{S_{\text{in}}/N_{\text{in}}}{S_{\text{out}}/N_{\text{out}}} = \frac{1}{G}\frac{GN_{\text{in}} + N_{\text{dut}}}{N_{\text{in}}} = 1 + \frac{N_{\text{dut}}}{GN_{\text{in}}},$$

in other words, the noise factor is 1 plus the ratio of the noise added by the device under test (DUT) to the amplified input noise.

Note that a noiseless block ($N_{\text{dut}} = 0$) has a noise factor of $F = 1$. A passive block with loss factor L (and impedance matched at input and output) has a noise factor of $F = L$ (in linear units), as it attenuates the signal and $N_{\text{out}} = N_{\text{in}} = kT$ if everything is in thermal equilibrium.

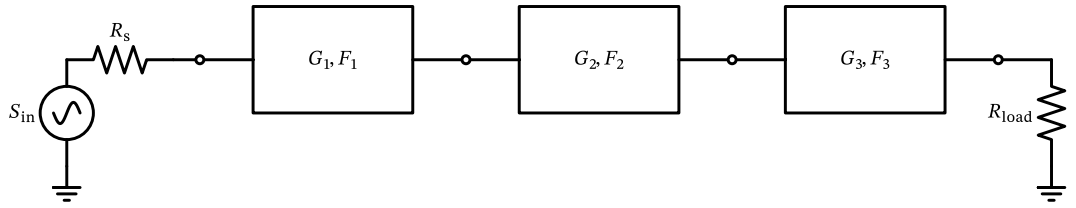


Figure 13: Block cascade for noise factor calculation showing multiple stages with gains and individual noise factors.

If we have a cascade of multiple blocks, as shown in Figure 13, we can calculate the overall noise factor with the **Friis formula** [3]

$$F_{\text{total}} = 1 + (F_1 - 1) + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} \tag{22}$$

where F_i and G_i are the noise factor and power gain of the i-th block, respectively. Note that all gains have to be in linear units (not dB) when using Equation 22. We can interpret Equation 22 as follows:

- The overall noise factor F_{total} is always larger than or equal to the noise factor of the first block (F_1).
- The noise factor of the first block is the most important one, as the noise factors of the following blocks are reduced by the gain of all preceding blocks. This is especially important in RF receivers, where the first block is usually a low-noise amplifier (LNA) with a very low noise figure (e.g., 1 dB or less) and a high gain (e.g., 10 dB or more). This ensures that the noise of the following blocks is negligible.
- The noise factor of the last block is reduced by the gain of all preceding blocks, so it is usually not very important.

Here we also see a trade-off between noise and linearity, as shown by Equation 17 and Equation 22. For low noise, we should try to maximize G_1, however, this will affect linearity (IIP3) in a negative way. As in many other situations in RF design, we have to find a good compromise between conflicting requirements.

2.3.4 Sensitivity

In RF receivers, we often want to know the minimum input signal power that can be detected with a certain SNR. This minimum input signal power is called the **sensitivity** of the receiver. The sensitivity can be calculated as

$$P_{\text{in,min}} = P_{\text{n}} \cdot \text{SNR}_{\text{min}} \cdot F \tag{23}$$

where P_{n} is the noise power at the input, SNR_{min} is the minimum detectable SNR (i.e., the SNR required to demodulate the signal with given accuracy in terms of bit error rate or signal fidelity), and F is the noise factor of the receiver. The input noise power can be calculated as

$$P_{\text{n}} = kTB$$

where k is Boltzmann's constant, T is the temperature in Kelvin, and B is the noise bandwidth of the receiver. Expressing Equation 23 in dB/dBm we get the following formula:

$$P_{\text{in,min}}|_{\text{dBm}} = -174\,\text{dBm/Hz} + \text{NF} + 10\log_{10}(B/\text{Hz}) + \text{SNR}_{\text{min}}|_{\text{dB}} \tag{24}$$

where −174 dBm/Hz is the thermal noise PSD at room temperature (290 K). We can see that the sensitivity reaches lower dBm values with lower noise figure, smaller bandwidth, and lower minimum detectable SNR.

> **i** Note 5: Sensitivity Calculation for Wi-Fi
>
> Let's calculate the sensitivity of a Wi-Fi receiver operating at 5 GHz with a noise bandwidth of $B = 80\,\text{MHz}$, a noise figure of $\text{NF} = 7\,\text{dB}$, and a minimum detectable SNR of 25 dB. This high SNR means that a high-order modulation scheme (like 64-QAM) is used for high data rates.
>
> Using Equation 24 we get:
>
> $$P_{\text{in,min}} = -174\,\text{dBm/Hz} + 7\,\text{dB} + 10\log_{10}(80 \times 10^6) + 25\,\text{dB} \approx -63\,\text{dBm}$$
>
> This means that the minimum input signal power that can be detected by the Wi-Fi receiver is approximately −63 dBm.

2.4 Modulation

In order to transmit information via an EM wave, we need to modulate the wave with the information signal. Looking at a simple sinusoidal carrier wave

$$s(t) = A\cos(\omega_0 t + \varphi)$$

we see that we can change one or more of the following parameters to encode information:

- Amplitude $A(t)$ (amplitude modulation, AM; the digital form is called amplitude-shift keying, **ASK**)
- Frequency $\omega_0(t)$ (frequency modulation, FM; the digital form is called frequency-shift keying, **FSK**)
- Phase $\varphi(t)$ (phase modulation, PM; the digital form is called phase-shift keying, **PSK**)
- Amplitude $A(t)$ and phase $\varphi(t)$ (quadrature amplitude modulation, **QAM**)

The modulation formats FM and PM have the advantage that the carrier amplitude A is constant, which makes them more robust against nonlinear distortion.

QAM is widely used in modern communication systems, as it allows to transmit more bits per symbol by combining amplitude and phase modulation. The QAM type with four different symbols is called QPSK. Higher-order modulation formats like 16-QAM, for example, use 16 different symbols, which can encode four bits per symbol (as $2^4 = 16$). Even higher-order QAM formats like 64-QAM (6 bits per symbol), 256-QAM (8 bits per symbol), 1024-QAM (10 bits per symbol), or 4096-QAM (12 bits per symbol) are also used in modern systems like Wi-Fi or LTE/5G NR.

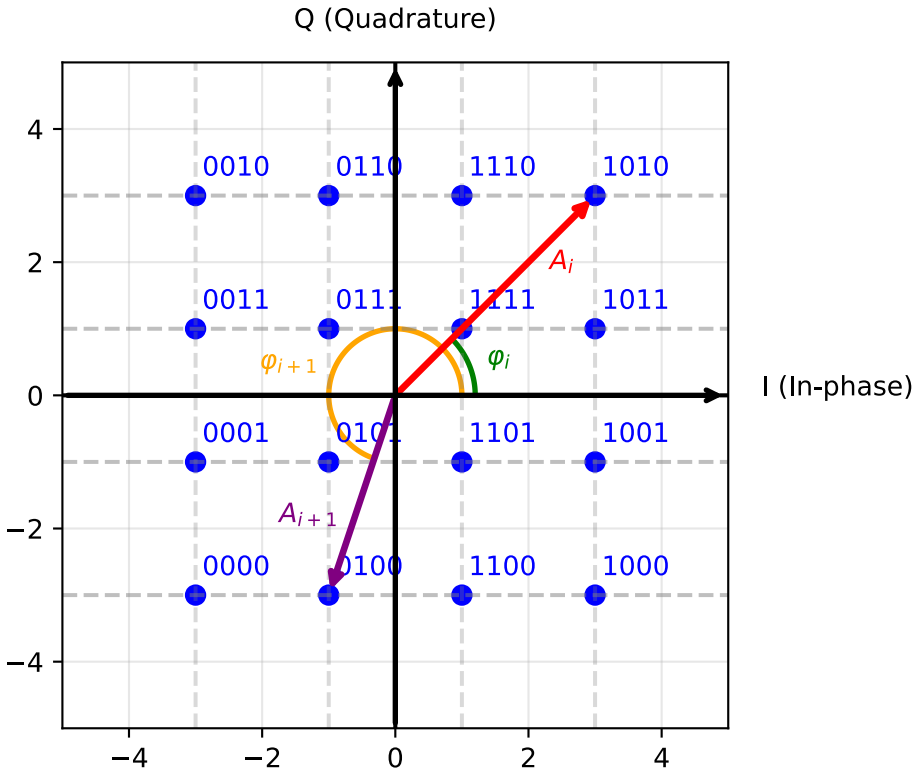


Figure 14: 16-QAM constellation diagram with Gray code labeling of constellation points.

Shown in Figure 14 is the **constellation diagram** of a 16-QAM modulation format. The constellation points are arranged in a square grid, with each point representing a unique combination of amplitude and phase. The distance between the constellation points determines the robustness against noise and interference; larger distances result in better performance, but also require more power. The mapping of bits to constellation points is called "bit mapping" or "symbol mapping". The example in Figure 14 uses a Gray code mapping, which minimizes the number of bit errors in case of a symbol error (assuming that a symbol will be wrongly decoded as one of its neighbors).

The constellation diagram can be imagined as a complex plane, where the x-axis represents the in-phase component (I) and the y-axis represents the quadrature component (Q) of the modulated signal. During transmission of a specific symbol, the RF carrier is modulated to the corresponding amplitude A_i and phase φ_i, resulting in a specific point in the constellation diagram. In Figure 14, the amplitude and phase information for two consecutive symbols, A_i/

φ_i and A_{i+1}/φ_{i+1}, is shown. If we have a bitrate with a bit duration of T_b, the symbol duration for 16-QAM (4 bits per symbol) is $T_\mathrm{s} = 4T_\mathrm{b}$. During the first T_s, the carrier is modulated to A_i/φ_i (encoding 4 bits), and during the next T_s, it is modulated to A_{i+1}/φ_{i+1} (encoding the next 4 bits).

Table 3 shows the **SNR requirements for different modulation formats** to achieve a bit error rate (BER) of 10^{-5} in an additive white Gaussian noise (AWGN) channel [13]. As we can see, higher-order modulation formats require higher SNR to achieve the same BER. Note that for the SNR values of this table **no error correction coding** (ECC) is assumed; with error correction coding the required SNR can be significantly reduced!

Table 3: Required symbol-energy-to-noise ratio E_s/N_0 and bit-energy-to-noise ratio E_b/N_0 for Gray-coded modulation schemes to achieve BER = 10^{-5} in an AWGN channel w/o error correction

Modulation	Bits/Symbol k	Required E_s/N_0 (dB)	Required E_b/N_0 (dB)
BPSK	1	9.6	9.6
QPSK	2	12.6	9.6
16-QAM	4	19.5	13.4
64-QAM	6	25.6	17.8
256-QAM	8	31.5	22.5
1024-QAM	10	37.5	27.5
4096-QAM	12	43.4	32.6

The two columns of Table 3 are related by the number of bits per symbol,

$$\left.\frac{E_\mathrm{s}}{N_0}\right|_\mathrm{dB} = \left.\frac{E_\mathrm{b}}{N_0}\right|_\mathrm{dB} + 10\log_{10}(k), \tag{25}$$

and it is worth being careful about which of the two a given data sheet or standard quotes — they differ by more than 10 dB for the high-order formats. Note also that BPSK and QPSK require the *same* E_b/N_0: QPSK transmits two bits per symbol on two orthogonal carriers, so it doubles the spectral efficiency of BPSK entirely for free in terms of energy per bit. From 16-QAM upwards, each step to the next-higher square QAM order costs roughly 6 dB in E_s/N_0, because the number of constellation points quadruples while the constellation is squeezed into the same average power.

2.5 Pulse Shaping and Spectral Efficiency

When we modulate symbols onto a carrier, the question is how we transition from one symbol A_i/φ_i to the next symbol A_{i+1}/φ_{i+1}. If we change the amplitude and/or phase of the carrier instantaneously (like with a step function), we will have a very wide spectrum, which is not desirable.

Therefore, we often apply **pulse shaping** to the symbols, which means that we smooth the transitions between symbols. This is usually done by convolving the symbol sequence with a pulse shaping filter, which has a certain impulse response. The choice of the pulse shaping filter affects the bandwidth of the transmitted signal and its spectral efficiency.

A simple pulse shape is the **rectangular pulse**, which has a sinc-shaped spectrum. However, the function $\mathrm{sinc}(x) = \sin(\pi x)/(\pi x)$ has side lobes that extend to infinity and roll off only slowly, which can cause interference with adjacent channels.

For reference, the two-sided power spectral density $S(f)$ of a random binary sequence with equal probability of -1 and 1, using rectangular pulses with a duration of T_b, is given by [13]

$$S(f) = T_\mathrm{b} \cdot \mathrm{sinc}^2(fT_\mathrm{b}) = T_\mathrm{b} \cdot \left[\frac{\sin(\pi f T_\mathrm{b})}{\pi f T_\mathrm{b}}\right]^2 \tag{26}$$

and visualized in Figure 15.

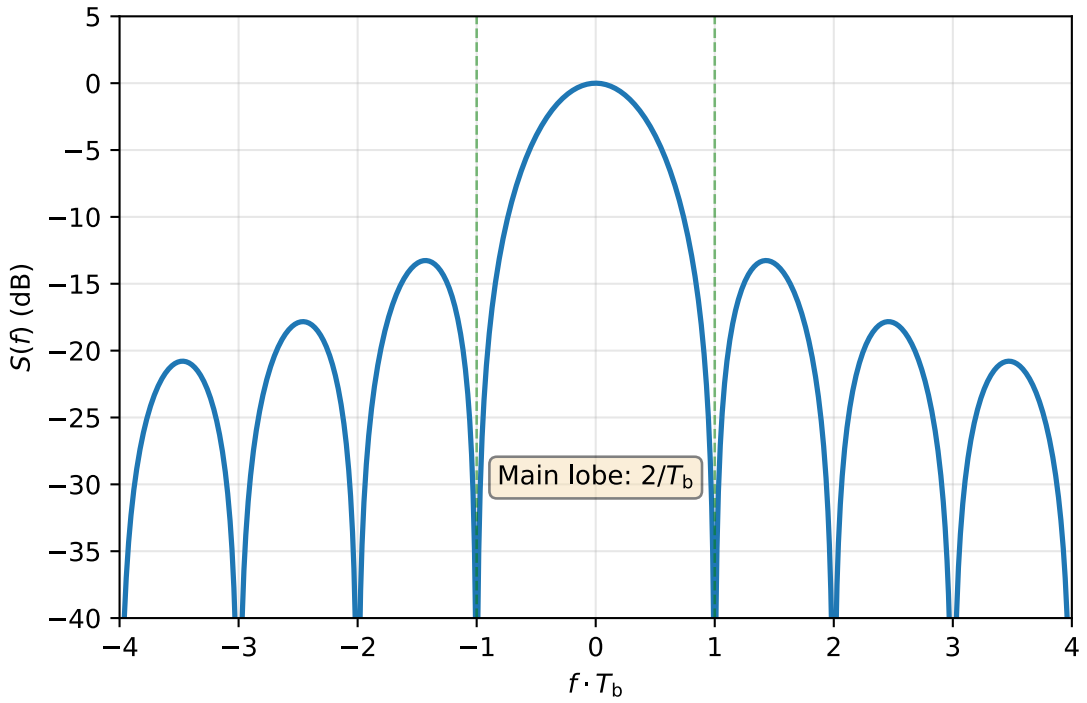


Figure 15: Power spectral density of a rectangular pulse with duration T_b. The main lobe has a bandwidth of $2/T_\mathrm{b}$, but the side lobes extend to infinity with slow roll-off, causing potential interference with adjacent channels.

As can be seen in Figure 15, the main lobe of the sinc function has a bandwidth of $2/T_\mathrm{b}$, but the side lobes are quite pronounced with a slow roll-off. This means that a significant amount of power is present outside the main lobe, which can cause interference with adjacent channels.

To avoid this, we can use pulse shapes that have better spectral properties, such as the raised cosine pulse or the root-raised cosine pulse. The **raised-cosine pulse** (see the raised-cosine filter, which satisfies the Nyquist ISI criterion) has a roll-off factor α that determines the excess bandwidth beyond the Nyquist bandwidth. The root-raised cosine (RRC) pulse is used

in practical systems (with half the pulse filter implemented at the TX, and half at the RX), as it can be implemented with a matched filter at the receiver.

The raised-cosine pulse $p(t)$ (with a spectrum shaped like a raised cosine) is defined by (T_s being the *symbol* duration, which for an n-bit-per-symbol modulation is $T_\mathrm{s} = n\,T_\mathrm{b}$):

$$p(t) = \frac{\sin(\pi t/T_\mathrm{s})}{\pi t/T_\mathrm{s}} \cdot \frac{\cos(\alpha\pi t/T_\mathrm{s})}{1-(2\alpha t/T_\mathrm{s})^2}$$

Setting $\alpha = 0$ results in a sinc pulse in the time domain (with a perfect bandwidth containment in the frequency domain), while $\alpha = 1$ results in a pulse with double the Nyquist bandwidth. The pulse shape for $\alpha = 0$ and $\alpha = 0.22$ (used in 3G) is shown in Figure 16.

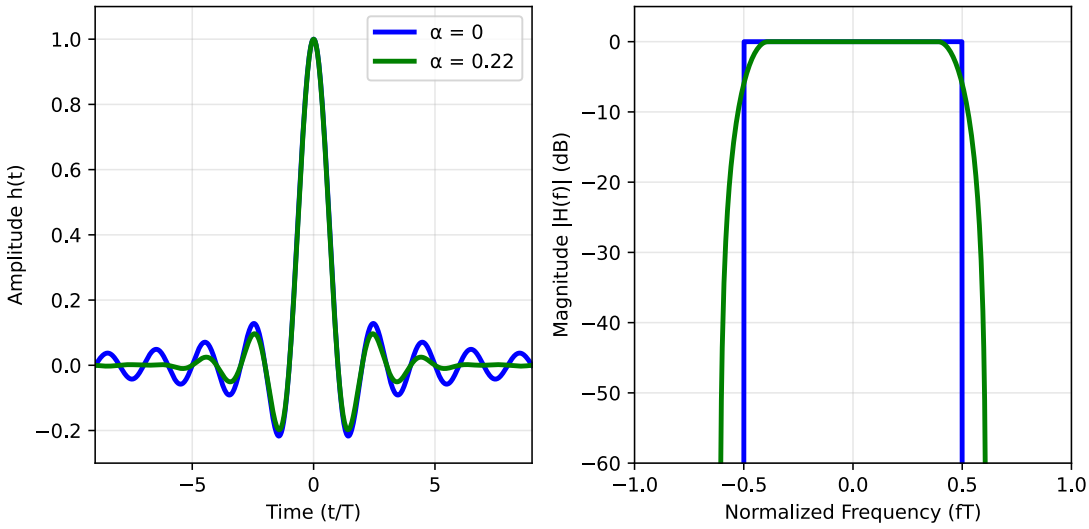


Figure 16: Raised cosine pulse shaping in time and frequency domain for different roll-off factors α.

Another often-used pulse shape is the **Gaussian pulse**, which is used in Gaussian minimum-shift keying (**GMSK**, used in 2G) modulation, or in Gaussian frequency-shift keying (**GFSK**, used in Bluetooth). The Gaussian pulse has a smooth shape and a narrow spectrum. The Gaussian pulse is defined by:

$$p(t) = \frac{\sqrt{\pi}}{\alpha} e^{-(\pi t/\alpha)^2} \quad \text{with} \quad \alpha = \frac{\sqrt{\ln 2}}{\sqrt{2}} \cdot \frac{T_\mathrm{b}}{BT_\mathrm{b}}$$

where BT_b controls the width of the pulse. The spectrum of the Gaussian pulse is also Gaussian-shaped, which helps to minimize inter-symbol interference (ISI).

The Gaussian pulse for $BT_\mathrm{b} = 0.5$ (as used in Bluetooth) is shown in Figure 17.

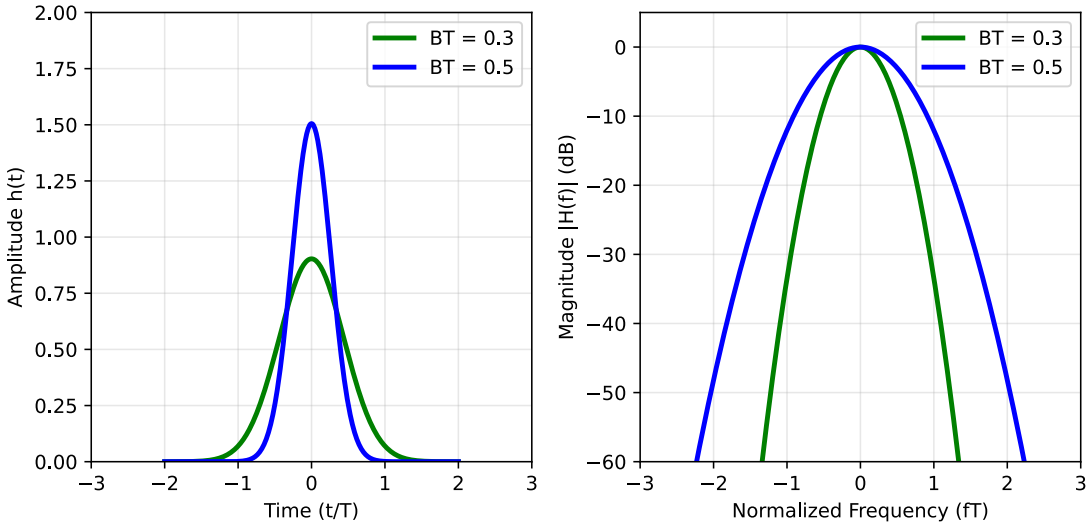


Figure 17: Gaussian pulse shaping in time and frequency domain for different bandwidth-time products BT.

For all the shown pulse shapes (rectangular, raised-cosine, and Gaussian), the trade-off between time- and frequency-domain containment is clearly visible. This is also captured in "**Küpfmüller's uncertainty principle**" [14], which states that the product of the time duration and the bandwidth of a pulse is lower-bounded by a constant. In other words, if we want to have a pulse that is very short in time, it will have a wide bandwidth, and vice versa.

2.6 Orthogonal Frequency-Division Multiplexing (OFDM)

As we have seen in the previous section, if we make the symbol rate high, we need to use short pulses with a wide bandwidth. The problem with a wide bandwidth in wireless communication is **multi-path propagation** [15], which causes frequency-selective fading. This means that some frequency parts of the channel are attenuated more than others, which can cause errors in the received signal. Equalizing such a frequency-selective channel can be very complex, especially if the channel changes rapidly (as in mobile communication).

We now face a dilemma: How can we achieve high data rates (which require high symbol rates and thus wide bandwidth) while avoiding frequency-selective fading? The key idea, implemented in **orthogonal frequency-division multiplexing (OFDM)** [15], is to split the wideband channel into multiple narrow sub-channels (subcarriers), each with a low symbol rate. This way, each subcarrier experiences essentially flat fading, which is much easier to equalize.

The key question is now how to implement this idea *efficiently*, as we now have to apply modulation to hundreds or thousands of individual subcarriers. The solution is to use the **inverse fast Fourier transform (IFFT)** at the transmitter to generate the time-domain OFDM signal from the frequency-domain symbols, and the **fast Fourier transform (FFT)** at the receiver to recover the frequency-domain symbols from the time-domain OFDM signal. This process is illustrated in Figure 18.

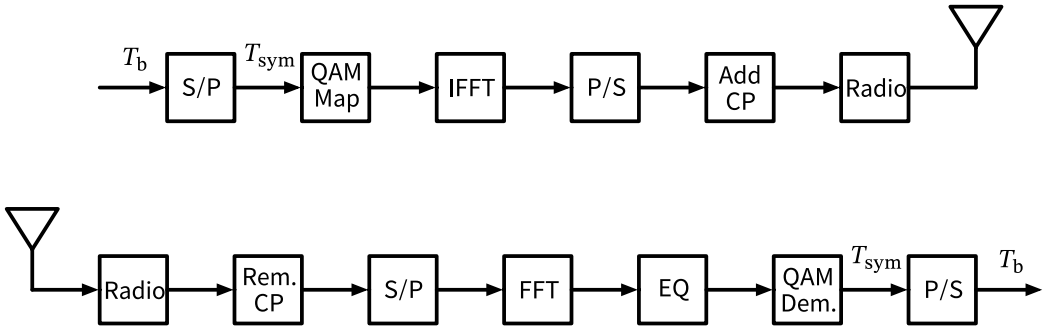


Figure 18: OFDM transmission system block diagram showing transmitter (top) and receiver (bottom) processing chains.

The OFDM transmitter takes a block of N symbols (e.g., 64-QAM symbols) and maps them onto N subcarriers. If the serial symbol stream has a symbol duration of T_s, then distributing N of these symbols over N parallel subcarriers stretches the duration seen by each individual subcarrier to $T_{sym} = N \cdot T_s$. This is precisely the point of OFDM: the per-subcarrier symbol becomes long compared to the delay spread of the channel, so each subcarrier sees essentially flat fading. The IFFT then generates the time-domain OFDM signal, which is transmitted over the wireless channel. Before transmission, the **cyclic prefix (CP)** is added to each OFDM symbol.

At the receiver, the CP is first removed, and then the FFT recovers the frequency-domain symbols, which can then be equalized (fairly simply by multiplying each subcarrier with a complex factor to correct amplitude and phase) and demodulated.

A key property of OFDM is that the subcarriers are **orthogonal** to each other, which means that they do not interfere with each other. This is achieved by choosing the subcarrier spacing Δf such that it is equal to the symbol rate $1/T_{sym}$, i.e., $\Delta f = 1/T_{sym}$. This way, the integral of the product of two different subcarriers over one symbol period is zero, which means that they are orthogonal.

To further improve the robustness against multi-path propagation, a CP is added to each OFDM symbol. The CP is a copy of the last part of the OFDM symbol, which is added to the beginning of the symbol. This way, if there are delayed copies of the OFDM symbol due to multi-path propagation, they will still fall within the CP and will not cause inter-symbol interference (ISI). The length of the CP should be longer than the maximum delay spread of the channel.

i Note 6: OFDM in LTE

In LTE OFDM is used for the downlink (base station to user equipment) with the following parameters:

- Subcarrier spacing: 15 kHz
- CP length: 5.2 µs (first symbol) resp. 4.7 µs (following symbols), and 16.67 µs as extended CP for scenarios with large delay spreads
- Number of subcarriers: 1200 (for 20 MHz bandwidth)
- Modulation: QPSK, 16-QAM, 64-QAM, 256-QAM

From the subcarrier spacing we can calculate the symbol duration as $T_{\text{sym}} = 1/\Delta f = 1/15\,\text{kHz} \approx 66.7\,\mu\text{ s}$.

We can calculate the raw bitrate for a 20 MHz LTE channel using 256-QAM modulation as

$$\text{Bitrate} = N_{\text{sc}} \cdot N_{\text{bps}} \cdot \frac{1}{T_{\text{sym}} + T_{\text{CP}}} = 1200 \cdot 8 \cdot \frac{1}{66.7\,\mu\text{ s} + 4.7\,\mu\text{ s}} \approx 134\,\text{Mbps}$$

With the overhead for control channels and error correction coding a user data rate of approximately 100 Mbps can be achieved in a 20 MHz LTE channel. This data rate can be scaled up in two independent ways: **carrier aggregation** combines several such 20 MHz channels into one logical link, while **MIMO** transmits multiple spatial layers within the *same* 20 MHz channel using several antennas. The two multiply, which is how modern devices reach gigabit rates.

2.7 Multiple Access Techniques

In wireless communication systems, multiple users need to share the same frequency spectrum. This is achieved by using **multiple access techniques**, which allow multiple users to transmit and receive data simultaneously without interfering with each other. The most common multiple access techniques are:

1. **Time division multiple access (TDMA)**: Users are assigned specific time slots for transmission, allowing multiple users to share the same frequency channel by dividing continuous time into slots.
2. **Frequency division multiple access (FDMA)**: Users are assigned specific frequency bands within the overall frequency spectrum, allowing multiple users to transmit simultaneously on different frequencies.
3. **Code division multiple access (CDMA)**: Users are assigned unique spreading codes, allowing them to transmit simultaneously over the same frequency band. The receiver uses the same code to extract the desired signal. This scheme is more precisely called direct-sequence CDMA (DS-CDMA). A sibling technique from the same spread-spectrum family (not a variant of DS-CDMA) is frequency-hopping spread spectrum (FHSS), where the carrier frequency is changed rapidly according to a pseudo-random sequence known to both the transmitter and receiver; users are separated by their hopping patterns rather than by a spreading code. This is used in Bluetooth, and has originally been co-invented by the Austrian actress Hedy Lamarr.

4. **Orthogonal frequency division multiple access (OFDMA)**: A variant of OFDM, where multiple users are assigned different subcarriers for transmission, allowing for efficient use of the frequency spectrum. This is used in 4G LTE and 5G NR.
5. **Spatial division multiple access (SDMA)**: Uses multiple antennas to create spatially separated channels, allowing multiple users to transmit simultaneously in the same frequency band.

All of the above techniques can be combined to create more efficient and flexible communication systems. For example, OFDMA can be used in conjunction with SDMA to allow multiple users to share the same frequency resources while also taking advantage of spatial diversity. Also, TDMA can be combined with FDMA to create a hybrid multiple access scheme (which has been used in 2G).

2.8 Modulation and Coding Schemes

In modern wireless communication systems several parameters can be adapted on the fly to the current channel conditions to optimize the data rate and reliability. These parameters include the modulation format (e.g., QPSK, 16-QAM, 64-QAM, etc.) and the error correction coding scheme (e.g., convolutional codes, turbo codes, LDPC codes, etc.). The combination of modulation format and coding scheme is called the **modulation and coding scheme (MCS)**. The MCS determines the number of bits per symbol and the level of error correction, which in turn affects the data rate and reliability of the communication link, as well sets the minimum required SNR.

Since there are many different MCS options available, and the receiver needs to be informed by the transmitter which MCS is used (whenever there is a change of MCS), a standardized set of MCS options is defined in most wireless communication standards. For example, the MCS options for Wi-Fi can be seen in this table, while the MCS calculation for 5G NR can be found here.

> **i** Exemplary MCS in Wi-Fi
>
> Some operating systems and wireless drivers show the MCS (and other parameters of the Wi-Fi link) during use. A typical output might be:
>
> - MCS: 7
> - RSSI: −53 dBm
> - Interference: −92 dBm
> - Bandwidth: 160 MHz
>
> Looking up MCS 7 in the Wi-Fi MCS table linked above shows that this corresponds to 64-QAM with a coding rate of 5/6. With a bandwidth of 160 MHz, 2x MIMO and a guard interval (GI) of 0.8 µs, the maximum data rate is a whopping 1441 Mbps.

3 Transceivers

Nowadays, the various small-signal RF functions for receive and transmit are integrated into so-called transceivers (TRX). A TRX is a device that can both transmit and receive signals, and is usually called an “RFIC”. While high monolithic integration is certainly the norm for radio-

frequency devices intended for standards like Bluetooth, Wi-Fi, cellular, etc., it is increasingly used also for mm-wave frequencies for applications like automotive radar and 5G cellular.

Typically, TRX include components like amplifiers, mixers, filters, oscillators, and phase-locked loops. When digital interfaces are used for the baseband I/Q data, also functions like analog-to-digital conversion (ADC) and digital-to-analog conversion (DAC) are integrated together with digital signal processing (DSP) blocks and potentially high-speed interfaces (because for wide bandwidths the I/Q sample rate can get quite high).

In these notes we will focus on the RF part of the TRX, which is responsible for the upconversion of baseband or intermediate frequency (IF) signals to the desired transmit frequency during transmission, and the downconversion of received signals from the carrier frequency to baseband or IF during reception. For filters, low-frequency amplifiers, ADCs, DACs, and DSP blocks we refer to related courses and literature, for example our analog circuit design course.

3.1 Direct-Conversion Transceiver

The following typical functions have to be performed by a TRX:

- Pulse-shaping filtering of the baseband signal (which can be implemented in analog or, in most cases, digital form).
- Modulating the baseband signal onto a carrier frequency (upconversion) in the TX or downconversion in the RX.
- Contain the RF signal in a small bandwidth (TX), or single out the wanted signal in the RX with adequate signal to noise ratio (SNR).
- Adapt gain (and linearity) to the signal strength in the RX, and to the output power in the TX.
- Generate the carrier frequency (local oscillator, LO) with low phase noise.

The dominant architecture for the TRX is the so-called **direct-conversion** or **zero-IF** architecture, where the up- and downconversion are performed in a single step [1]. This is in contrast to super-heterodyne architectures, where the signal is first converted to an intermediate frequency (IF) before being converted to baseband. The direct-conversion architecture has the advantage of reduced complexity and cost, as it requires fewer components and less filtering. However, it also has some disadvantages, such as increased susceptibility to dc offsets and I/Q imbalance. A typical TRX block diagram is shown in Figure 19.

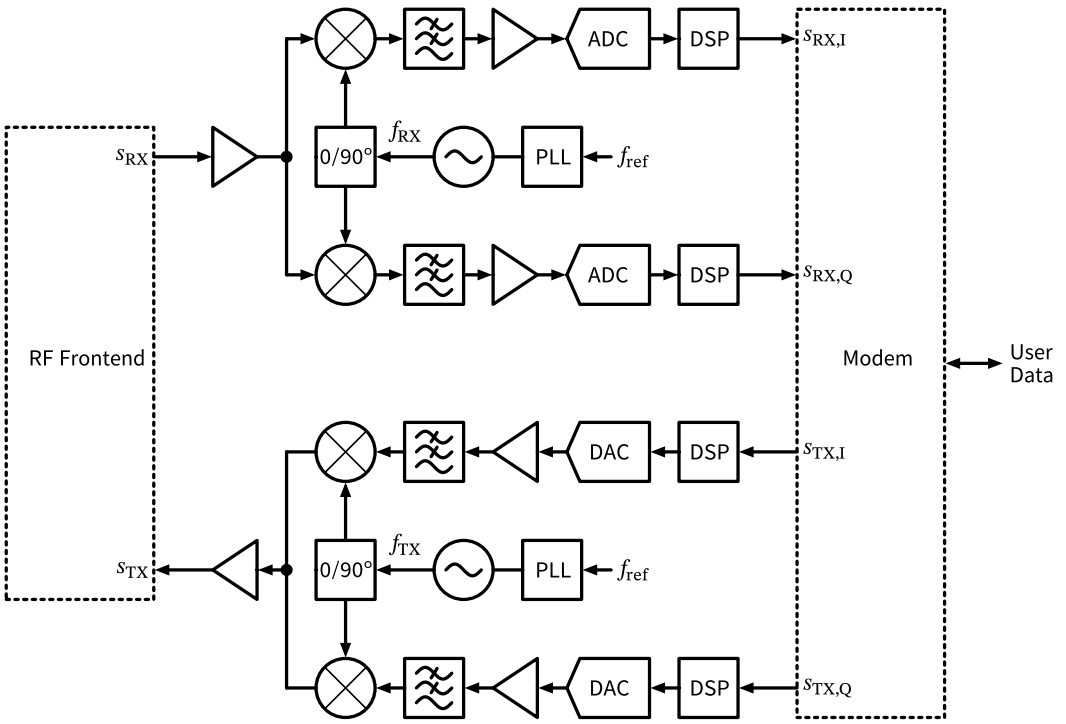


Figure 19: Block diagram of a typical transceiver (TRX) showing the main functional blocks of RX and TX. The modem provides the digital baseband processing and interfaces to the rest of the system. For implementation options of the RF front-end, see Section 3.5.

As can be seen in Figure 19, this generic architecture can be adapted in various ways. Generally, the amplifier gains are adjustable to adapt to different signal levels. If adaptive channel bandwidths are to be supported, the corner frequencies of the low-pass filters (LPF) can be adjusted, as well as (optionally) the sampling rate of the ADCs and DACs. The local oscillator (LO) frequency is generated by a phase-locked loop (PLL) synthesizer, which can be tuned to the desired carrier frequency. In case of frequency-division duplex (FDD) operation, two PLLs are used to generate the TX and RX LO frequencies, which are separated by the duplex distance f_{dpx}. In time-division duplex (TDD) operation, a single PLL is sufficient, supplying the LO signal to both RX and TX.

The modem (short for modulator-demodulator) that is shown in Figure 19 is responsible for the digital baseband processing, including functions like channel coding/decoding, modulation/demodulation, equalization, error correction, encryption/decryption, and framing/timing. The modem is usually implemented as a digital System-on-Chip (SoC) consisting of (multiple) CPUs, DSPs, and fixed-function blocks for time-critical processing. For an in-depth discussion we recommend [13] or [15].

3.2 Modulation and Demodulation

Modulation is the process of varying a carrier signal at frequency f_{c} in order to transmit information. The complex baseband signal (after converting the real-valued in-phase and quadrature signals s_{I} and s_{Q}, respectively, to analog and applying pulse-shaping filtering) is represented as ($\tilde{s}_{\mathrm{BB}} \in \mathbb{C}$; $s_{\mathrm{I}}, s_{\mathrm{Q}} \in \mathbb{R}$) [16]:

$$\tilde{s}_{\mathrm{BB}}(t) = s_{\mathrm{I}}(t) + j \cdot s_{\mathrm{Q}}(t).$$

We want to shift this signal to the carrier frequency f_{c} to create the modulated RF signal $\tilde{s}_{\mathrm{RF}} \in \mathbb{C}$, which can be done by multiplying with a complex exponential ($e^{j\omega_{\mathrm{c}}t}$, with $\omega_{\mathrm{c}} = 2\pi f_{\mathrm{c}}$):

$$\tilde{s}_{\mathrm{RF}}(t) = \tilde{s}_{\mathrm{BB}}(t) \cdot e^{j\omega_{\mathrm{c}}t} = [s_{\mathrm{I}}(t) + j \cdot s_{\mathrm{Q}}(t)] \cdot [\cos(\omega_{\mathrm{c}}t) + j \cdot \sin(\omega_{\mathrm{c}}t)].$$

The real-valued RF signal $s_{\mathrm{RF}} \in \mathbb{R}$ is obtained from $\tilde{s}_{\mathrm{RF}}$ by complex-conjugate symmetry

$$s_{\mathrm{RF}}(t) = \frac{1}{2}[\tilde{s}_{\mathrm{RF}}(t) + \tilde{s}^{*}_{\mathrm{RF}}(t)] = s_{\mathrm{I}}(t)\cos(\omega_{\mathrm{c}}t) - s_{\mathrm{Q}}(t)\sin(\omega_{\mathrm{c}}t). \tag{27}$$

The process formulated in Equation 27 is done in hardware in the TX modulator, as shown in Figure 20.

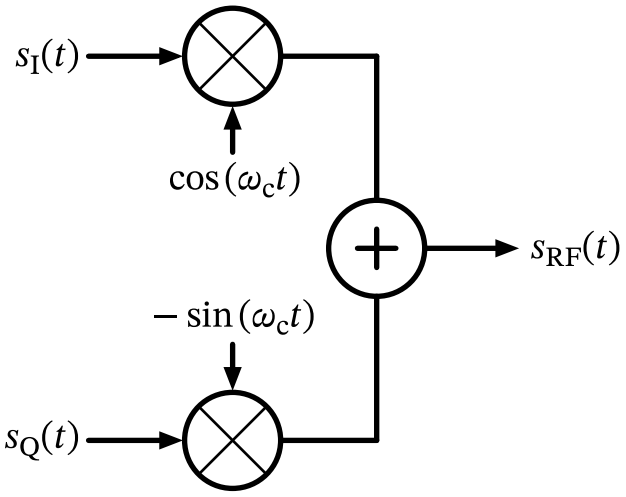


Figure 20: TX modulator.

The RF signal generation according to Equation 27 is called **quadrature modulation**. This is the modulation method used most often in modern (broadband) communication systems, as it allows one to transmit two independent signals (I and Q) in the same bandwidth. The I and Q signals are also called quadrature components, as they are 90° out of phase with each other.

Alternatively, a modulation method called **polar modulation** can be used, where the amplitude and phase of the carrier are varied according to (again using the complex-conjugate symmetry to obtain a real-valued signal):

$$\begin{aligned} s_{\mathrm{RF}}(t) &= \frac{1}{2}\left[A(t) \cdot e^{j\omega_{\mathrm{c}}t + j\varphi(t)} + A(t) \cdot e^{-j\omega_{\mathrm{c}}t - j\varphi(t)}\right] \\ &= A(t) \cdot \cos[\omega_{\mathrm{c}}t + \varphi(t)] \end{aligned}$$

This is done by converting the baseband I and Q signals (s_{I} and s_{Q}) to polar coordinates with

$$A(t) = \sqrt{s_{\mathrm{I}}^2(t) + s_{\mathrm{Q}}^2(t)}, \quad \varphi(t) = \tan^{-1}\left[\frac{s_{\mathrm{Q}}(t)}{s_{\mathrm{I}}(t)}\right].$$

As the mathematical operations required for the Cartesian to polar transformation are quite nonlinear, the $A(t)$ and $\varphi(t)$ signals are wideband. While fairly complex to implement, polar modulation can have an efficiency advantage in the TX circuits, as the signal loss associated with quadrature combination can be avoided [17].

For narrowband wireless standards the use of polar modulation can be quite efficient, as for example Bluetooth, where basically all transmitters are realized as polar modulators.

In the RX, the received RF signal is downconverted to baseband by the inverse process of Figure 20, shown in Figure 21.

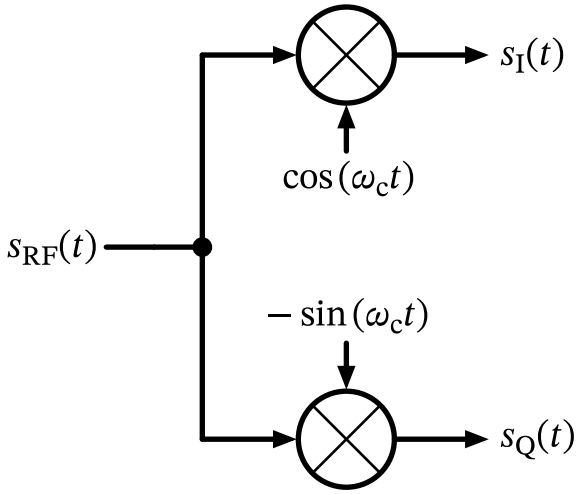


Figure 21: RX demodulator.

For demodulation we have to shift the RF signal down to baseband, which mathematically is done by multiplying with the negative complex carrier frequency $e^{-j\omega_{\mathrm{c}}t}$

$$\begin{aligned}\tilde{s}_{\mathrm{BB}}(t) &= s_{\mathrm{RF}}(t)\cdot e^{-j\omega_{\mathrm{c}}t}\\ &= s_{\mathrm{RF}}(t)\cdot\left[\cos(\omega_{\mathrm{c}}t) - j\cdot\sin(\omega_{\mathrm{c}}t)\right]\\ &\overset{\text{LPF}}{=} \frac{1}{2}\left[s_{\mathrm{I}}(t) + j\cdot s_{\mathrm{Q}}(t)\right].\end{aligned} \tag{28}$$

In hardware this is realized with two mixers, producing $s_{\mathrm{RF}}(t)\cdot\cos(\omega_{\mathrm{c}}t)$ in the I path and $-s_{\mathrm{RF}}(t)\cdot\sin(\omega_{\mathrm{c}}t)$ in the Q path. Each of these products contains, besides the wanted baseband term, a component at $2\omega_{\mathrm{c}}$, which is why Equation 28 only holds after the subsequent low-pass filter (indicated by the "LPF" above the equal sign). The factor of one-half that the recovered baseband signals carry is of no consequence, as it is simply absorbed by the RX gain.

3.3 Filtering

Filtering is an essential function in both TX and RX. In the TX, filtering is used to limit the bandwidth of the transmitted signal to the allocated channel bandwidth, and to suppress out-of-band emissions. In the RX, filtering is used to select the wanted signal from a crowded spectrum, and to suppress unwanted signals (blockers) that can cause interference or desensitization of the RX.

A typical example of filtering in the RX is shown in Figure 22, where a bandpass filter is used to attenuate strong unwanted blockers while only slightly attenuating the wanted signal.

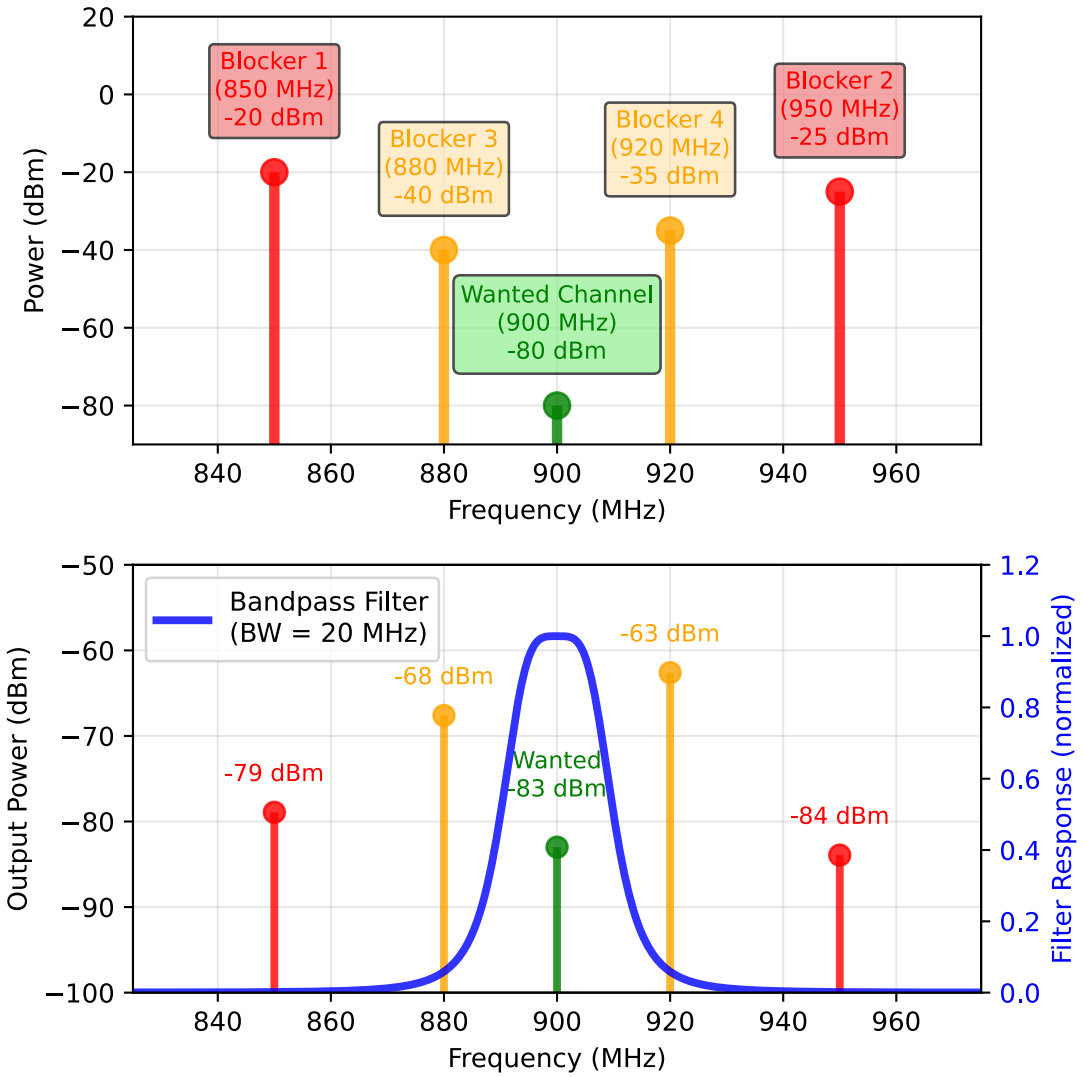


Figure 22: Filtering of wanted channel amid strong unwanted blockers. An exemplary RX scenario around 900 MHz is shown: The strong blockers (top figure) are attenuated by an RF bandpass filter (bottom figure) with a bandwidth of 20 MHz, achieving more than 40 dB rejection of the blockers while only slightly attenuating the wanted signal.

In any filter there exists a fundamental trade-off between selectivity (steepness of the filter skirts), bandwidth, and insertion loss. A very selective filter with steep skirts and large passband BW will have a high insertion loss. Conversely, a filter with low insertion loss will have a gentle roll-off and may not sufficiently suppress unwanted signals.

A useful metric to quantify the performance of a filter is the quality factor Q, defined as

$$Q = \frac{f_c}{\Delta f}$$

where f_c is the center frequency and Δf is the -3 dB bandwidth of the filter. A high Q indicates a narrow-band filter.

The achievable Q depends on the used filter technology. For example, on-chip LC filters can achieve Q values of around 10–20, while off-chip SAW or BAW/FBAR filters reach Q values into the thousands, and a crystal filter can achieve Q values of several tens of thousands.

The choice of filter technology depends on the application requirements, such as frequency range, bandwidth, insertion loss, cost, and power handling. Generally speaking, the required filtering to single out the wanted signal in the RX spectrum and decrease the power of strong blockers to a tolerable level is one of the most critical design choices, and is usually distributed at different locations in the RX chain:

- **RF filters** (between antenna and LNA) provide a first level of filtering, and are usually implemented as off-chip SAW or BAW/FBAR filters. They provide high Q and good selectivity, but have a fixed center frequency and bandwidth. They are used to pass the wanted band of interest, and to attenuate strong out-of-band blockers.
- **IF filters** (in case of a super-heterodyne receiver) provide additional filtering, and can be implemented as on-chip LC filters or off-chip SAW/BAW filters. They provide moderate Q and selectivity, and can be tuned to some extent.
- **BB filters** (after downconversion) provide the final level of filtering before entering the ADCs, and are usually implemented as on-chip active RC filters. They provide channel selection, and can be easily adjusted to different bandwidths.
- **Digital filters** (in the DSP block) provide the final level of filtering and signal processing, and can be implemented as FIR or IIR filters. They provide high flexibility and can be easily adapted to different standards and requirements. Digital filters show no variations, so they can be designed to be very selective.

It is important to note (because this dictates a lot of choices in RF design) that high-Q filters are usually fixed-frequency and fixed-bandwidth. Only baseband and digital filters can be easily adjusted to different bandwidths!

i Filter Technologies Primer

Baseband filters (analog) are usually implemented as active RC filters on-chip [18], [19]. They are very flexible and can have adjustable bandwidth by changing R and/or C. For medium/high frequencies $g_\mathrm{m}C$ filters can be used, which are also tunable by changing the transconductance g_m and/or C. For even higher bandwidths, on-chip LC filters can be used, which have a limited Q of around 10–20.

Baseband filters (digital) are implemented as FIR or IIR filters. They can be made programmable and can be easily adapted to different requirements. Digital filters show no variations, so they can be designed to be very high performance. However, a trade-off between power consumption/area (number of bits and operations) and performance exists.

Surface acoustic wave (SAW) and **bulk acoustic wave (BAW/FBAR)** filters are specialized off-chip components that can achieve high Q into the thousands. They have a fixed center frequency and bandwidth. Usually 1–2 such filters are required per supported band of interest, which can drive up implementation cost and PCB area for RF front-ends that shall support many different frequency bands.

Crystal filters can achieve very high Q values of several tens of thousands, but are usually bulky and expensive and limited to an operating frequency of tens of MHz.

LC filters can be either implemented off-chip (using discrete components) or on-chip. Off-chip LC filters can achieve higher Q values than on-chip LC filters, but are usually larger and more expensive. On-chip LC filters are limited in Q (around 10–20), but are very compact and can be integrated into the RFIC. Off-chip LC filters can achieve Q values of around 50–100, depending on the frequency and component quality. LC filters can be made tunable by changing L (difficult) and/or C (easy).

Ceramic filters are another off-chip filter technology that can achieve moderate to high Q values (up to several hundreds). They are usually smaller and less expensive than SAW or BAW/FBAR filters, but also have lower performance.

Waveguide filters are used at very high frequencies (above 10 GHz) and can achieve very high Q values (up to several thousands). They are usually bulky and expensive, and are not commonly used in mobile applications, but rather in fixed installations like base stations or satellite communication.

Optical filters can have extremely high Q values but are difficult to apply in electronic RF systems due to the need for optical components and the challenge of integrating them with electronic circuits, as well as the required electrical-optical signal conversion.

Fundamentally, the choice of filter technology is a trade-off between performance, size, cost, power handling, and flexibility. In most cases, a combination of different filter technologies is used to achieve the desired performance.

3.4 Direct-Conversion Architecture

The transceiver architecture shown in Figure 19 is called the direct-conversion or zero-IF architecture, as the downconversion in the RX and upconversion in the TX are done in a single step. This architecture has several advantages:

- Per RX- and TX-chain a single LO is required (which can even be shared between RX and TX in TDD operation).
- There are a minimum number of RF blocks, which is good for cost and power consumption.
- This architecture is very flexible and can be easily adapted to different standards and requirements, even for wideband operation. It generally shows very good performance if the disadvantages can be overcome by good design.
- This architecture allows a high integration level, as basically all blocks can be implemented on-chip.
- Direct conversion is the de facto standard architecture for cellular, Wi-Fi, and GNSS (Bluetooth receivers typically use the low-IF architecture discussed below).

However, the direct-conversion architecture also has some serious disadvantages:

- LO-RF coupling can cause self-mixing and desensitization of the RX, as well as LO leakage in the TX. This is an issue because the LO frequency is the same as the RF carrier frequency.
- Even-order distortion (characterized by a low IIP2) causes sensitivity degradation due to strong amplitude-modulated blockers [20].
- LO pulling can occur in the TX (as, again, LO and RF are at the same frequency).
- I/Q errors (amplitude and phase mismatch) of the I and Q paths can cause constellation distortion leading to increased error vector magnitude (EVM).
- Significant dc offsets can occur due to self-mixing of LO leakage or the even-order distortion products (IM2).
- Flicker noise (1/f noise) upconversion can cause increased TX noise close to the carrier, as well as increased RX noise figure.

Nowadays there exist good design techniques to mitigate these disadvantages. However, in some cases (for example very high linearity requirements, or very high frequencies) other architectures like low-IF or super-heterodyne may be preferable.

3.5 Duplexing

In the block diagram of Figure 19, we have not yet considered how to share a single antenna between RX and TX. Essentially, there are two main methods to achieve this: **frequency-division duplex (FDD)** and **time-division duplex (TDD)**.

A special case beyond FDD and TDD is **full-duplex (FD)**, where RX and TX operate *simultaneously* in the *same frequency band*. Full-duplex is technically challenging, as it requires very high isolation between RX and TX, which is difficult to achieve in practice. For some special applications, like radar, full-duplex operation is possible, but for most communication systems FDD or TDD is used.

3.5.1 Frequency-Division Duplex (FDD)

In FDD, the RX and TX operate at different frequencies, separated by a duplex distance $f_{\mathrm{dpx}} = |f_{\mathrm{RX}} - f_{\mathrm{TX}}|$. This allows simultaneous transmission and reception, which is beneficial for applications like voice communication where low latency is required. However, FDD requires

two separate frequency bands, which can be a limitation in terms of spectrum availability. Additionally, FDD requires two PLLs to generate the RX and TX LO frequencies, which increases complexity and power consumption.

The RF RX and TX paths are connected to the antenna via a duplexer filter, which is a three-port device that allows signals to pass between the antenna and the RX or TX path, while isolating the RX and TX paths from each other. A typical FDD TRX block diagram is shown in Figure 23.

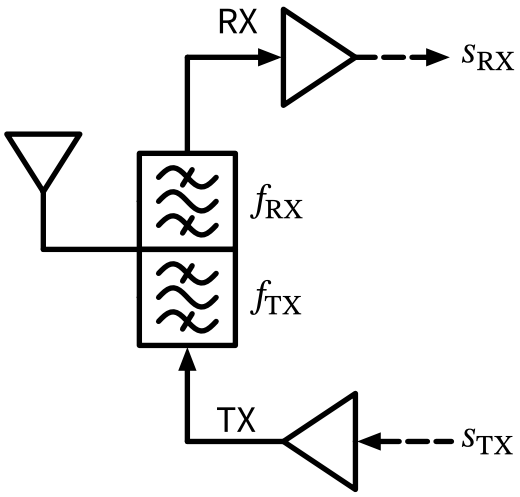


Figure 23: Block diagram of an FDD RF front-end.

Advantages of FDD:

- RX and TX can operate simultaneously, which is beneficial for low-latency applications.
- There is no need for fast switching between RX and TX, which simplifies the timing control.
- FDD allows relaxed synchronization requirements between RX and TX and different users.

Disadvantages of FDD:

- Duplexer filters are costly components with significant insertion loss depending on filtering requirements.
- It requires two separate frequency bands, which can be a limitation in terms of spectrum availability and MIMO channel estimation.
- The strong TX leakage via the duplexer filter into the RX path causes severe desensitization of the RX, which requires high linearity and good filtering (50 dB to 60 dB).
- The RX and TX chains (including their PLLs) are operating simultaneously, which increases power consumption and can cause interference between the RX and TX paths (e.g., LO pulling).

3.5.2 Time-Division Duplex (TDD)

In TDD, the RX and TX share the same frequency band but operate at different times. This allows for more efficient use of the available spectrum, as the same channel frequency can be used for both transmission and reception. TDD is particularly well-suited for applications with asymmetric traffic patterns, where the data rate in one direction is significantly higher

than in the other. However, TDD requires precise timing control to avoid interference between RX and TX periods, which can increase complexity.

In TDD, a single PLL can be used to generate the LO frequency for both RX and TX, which reduces complexity and power consumption. The RX and TX RF paths are connected to the antenna via a switch, which alternates between connecting the antenna to the RX path and the TX path. A typical TDD TRX block diagram is shown in Figure 24.

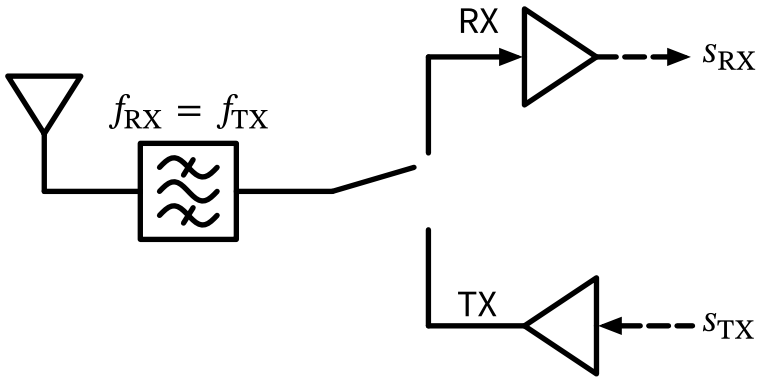


Figure 24: Block diagram of a TDD RF front-end.

Advantages of TDD:

- More efficient use of the available spectrum, as the same frequency can be used for both RX and TX.
- A single PLL can be used for both RX and TX, which reduces complexity and power consumption.
- No duplexer is required (just a single band filter), which reduces cost and insertion loss.
- No RX blocking by own TX, which relaxes linearity and filtering requirements.
- MIMO is easier to implement, as all antennas can operate in the same frequency band.

Disadvantages of TDD:

- RX and TX cannot operate simultaneously, which can be a limitation for low-latency applications.
- TDD requires precise timing control to avoid interference between RX and TX periods, which can increase complexity.
- Synchronization between RX and TX and different users is required, which can be challenging in some scenarios.

3.5.3 Comparison of FDD and TDD

Below is a summary of important wireless standards and their duplexing method as shown in Table 4.

Table 4: Comparison of duplexing methods used by major wireless standards

Wireless Standard	Duplexing Method	Comments
GSM (2G)	FDD & TDMA	TX and RX operate at different frequencies (FDD) and different times (TDMA)
UMTS (3G)	FDD	Traditional cellular standard using paired spectrum
LTE (4G)	FDD/TDD	FDD is used mostly <2.7 GHz, TDD is used >2.3 GHz
5G NR	FDD/TDD	FDD is used mostly <2.7 GHz, TDD is used >2.3 GHz
Wi-Fi (802.11)	TDD	Unlicensed spectrum operation
Bluetooth	TDD	Short-range personal area network
Zigbee	TDD	Low-power IoT applications

As you can see in Table 4, there is a tendency to use FDD for lower frequencies and long communication distances, while TDD is preferred for higher frequencies and shorter distances.

3.5.4 Full Duplex (FD)

In case of FD, neither a duplexer nor a switch can be used in the RF front-end. One popular method (which also helps to increase the RX-TX isolation) is to use two antennas, as shown in Figure 25. The TX antenna is used to transmit the signal, while the RX antenna is used to receive the signal. The two antennas are placed in such a way that they are isolated from each other, which helps to reduce the interference between the RX and TX paths. Also polarization diversity can be used to further increase the isolation.

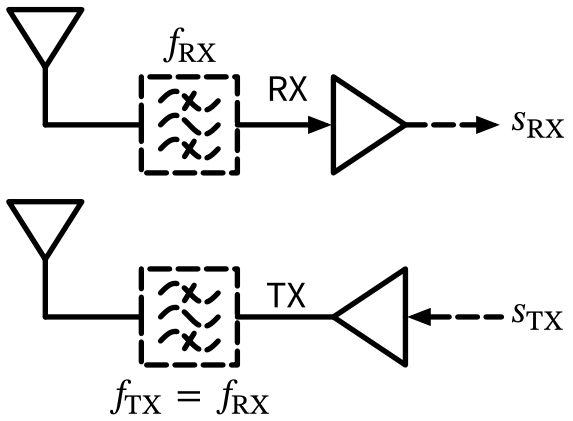


Figure 25: Block diagram of an in-band full-duplex (FD) RF front-end using two separate antennas for RX and TX, which operate simultaneously at the same frequency.

Another method to achieve FD operation is to use a circulator, which is a three-port device that allows signals to flow in one direction while isolating the other two ports. The circulator can be used to connect the RX and TX paths to the same antenna, while providing isolation between them. A typical FD TRX block diagram using a circulator is shown in Figure 26.

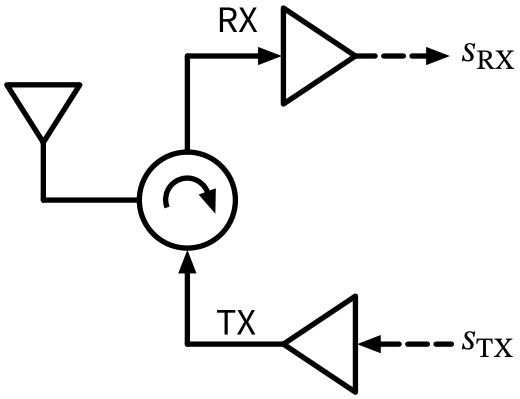


Figure 26: Block diagram of an in-band full-duplex (FD) RF front-end using a single antenna and a circulator. The circulator routes TX to the antenna and the antenna to RX, while isolating the TX from the RX port.

Regardless of whether two antennas or a circulator are used, the isolation between the TX and RX paths is always finite (typically only in the range of 15 dB to 20 dB for a circulator). As a consequence, a large fraction of the own (and strong) TX signal leaks directly into the sensitive RX port. Practical FD systems therefore require additional **self-interference cancellation** (also called leakage cancellation), performed in the RF/analog and/or digital domain, to suppress this residual TX leakage below the RX noise floor [21].

3.6 Specialty Architectures

In some cases, other architectures may be preferred over the direct-conversion architecture. During the evolution of wireless communication, many different architectures have been proposed and used. However, only a few of them are still relevant today. A few examples are shown next.

3.6.1 Super-Heterodyne Architecture

The super-heterodyne architecture is a widely used approach in radio. It works by mixing the incoming/outgoing RF signal with an LO to produce an intermediate frequency (IF) signal. This IF signal is then amplified and processed, allowing for better selectivity and sensitivity compared to direct-conversion architectures. Super-heterodyne receivers and transmitters are known for their excellent performance in terms of image rejection and dynamic range, making them suitable for a variety of applications, including traditional analog TV and radio broadcasting. A simplified block diagram of a super-heterodyne transceiver is shown in Figure 27.

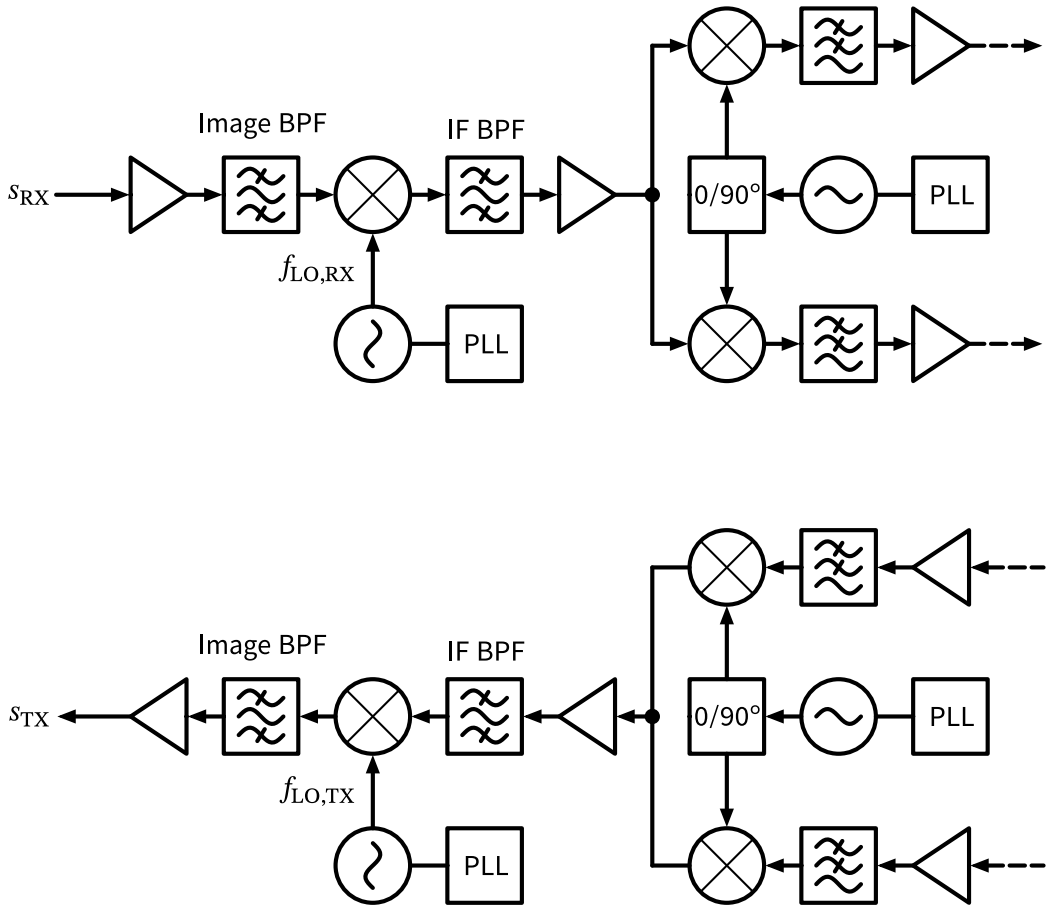


Figure 27: Block diagram of a super-heterodyne transceiver (TRX) showing the main functional blocks of RX and TX.

When you compare Figure 19 with Figure 27, you can immediately appreciate the increased complexity of the super-heterodyne architecture. It requires two PLLs to generate the RX and TX LO frequencies, as well as additional mixers, amplifiers, and filters for the IF stage. This increases cost, power consumption, and size. However, the super-heterodyne architecture can provide better performance in terms of selectivity and sensitivity, especially in challenging RF environments with strong blockers, as it allows distributed filtering at RF, IF, and baseband frequencies.

One important aspect of super-heterodyne receivers is the choice of the intermediate frequency (IF). The IF should be high enough to allow for effective filtering and high **image frequency** rejection, but low enough to avoid excessive complexity and power consumption. Common IF frequencies range from a few MHz to several hundred MHz, depending on the application and frequency band.

The image frequency is a spurious frequency that can interfere with the desired signal, and is located at $f_{\text{image}} = f_{\text{RF}} \pm 2f_{\text{IF}}$ (the sign depends on the selection of high-side or low-side mixing, which is an important design choice in superhets). To suppress the image frequency, an image-reject filter is either placed before (RX) or after (TX) the mixer. The design of

this filter is critical, as it must provide sufficient attenuation of the image frequency while maintaining low insertion loss for the desired signal.

An alternative to image filtering is the use of active image rejection techniques, such as the **Hartley** or **Weaver** architectures (see image-reject mixer) [1]. These techniques use additional mixers and phase shifters to cancel out the image frequency, allowing for improved performance without the need for a dedicated image-reject filter.

3.6.2 Low-IF Architecture

To avoid some of the issues of direct-conversion architectures (like dc offsets and flicker noise), a low-IF architecture can be used. In a low-IF architecture (which is a kind of a super-heterodyne architecture), the RX and TX signals are mixed to a low intermediate frequency (typically a few MHz to tens of MHz) in the analog domain instead of directly to baseband (the final mixing to baseband is done digitally). This allows for easier filtering of dc offsets and flicker noise, while still maintaining the benefits of a single LO and reduced complexity compared to super-heterodyne architectures. A low-IF architecture is shown in Figure 28.

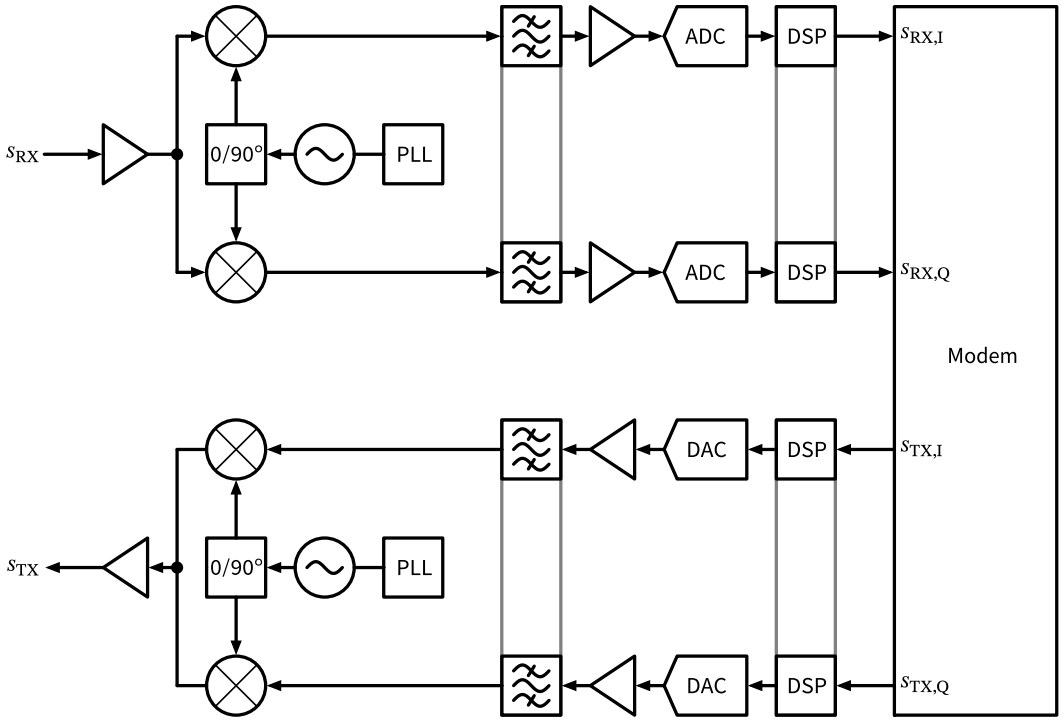


Figure 28: Block diagram of a low-IF transceiver (TRX) showing the main functional blocks of RX and TX. Note the usage of complex analog and digital baseband filters. Otherwise, the structure is similar to a zero-IF TRX as shown in Figure 19.

The low-IF architecture is the de facto standard for Bluetooth receivers. Its advantage compared to direct-conversion vanishes for larger channel bandwidths, which is why it is not used for cellular or Wi-Fi (GSM receivers might be an exception).

One noteworthy disadvantage of low-IF architectures is the required $\geq 2 \times$ BW of the analog baseband filters and the ADCs/DACs compared to direct-conversion. This might cause increased power consumption in these blocks. Additionally, the low-IF architecture still

requires careful design to mitigate issues like I/Q imbalance and LO leakage, although these issues are generally less severe than in direct-conversion architectures.

3.6.3 Direct-Modulated Oscillator and Direct-Detection Receiver (aka "Super Simple" Architecture)

For some applications with very low cost and low performance requirements, a super simple architecture can be used (for applications like a garage door opener). In this architecture, the RX and TX paths are stripped down to the bare minimum. A simple direct-detection receiver just uses an (optional) bandpass filter and a power detector, while a simple direct-modulated oscillator transmitter uses an oscillator and (optional) power amplifier. These simplified architectures are shown in Figure 29.



Figure 29: Block diagram of a direct-modulated oscillator and direct-detection receiver (aka 'Super Simple') TX and RX with optional components.

Despite the reduced architecture, digital amplitude-shift-keying (ASK) or on-off-keying (OOK) can be used. If the receiver is able to discriminate between frequencies (e.g., by using two RF filters with a power detector each), also frequency-shift-keying (FSK) can be used.

3.7 I/Q Imbalance

In direct-conversion and low-IF architectures, the I and Q paths are used to process the in-phase and quadrature components of the signal. Ideally, these paths should have identical gain and a 90° phase difference. However, in practice, there are always some mismatches between the I and Q paths, leading to **I/Q imbalance**. This imbalance can cause constellation distortion, leading to increased error vector magnitude (EVM) and degraded system performance.

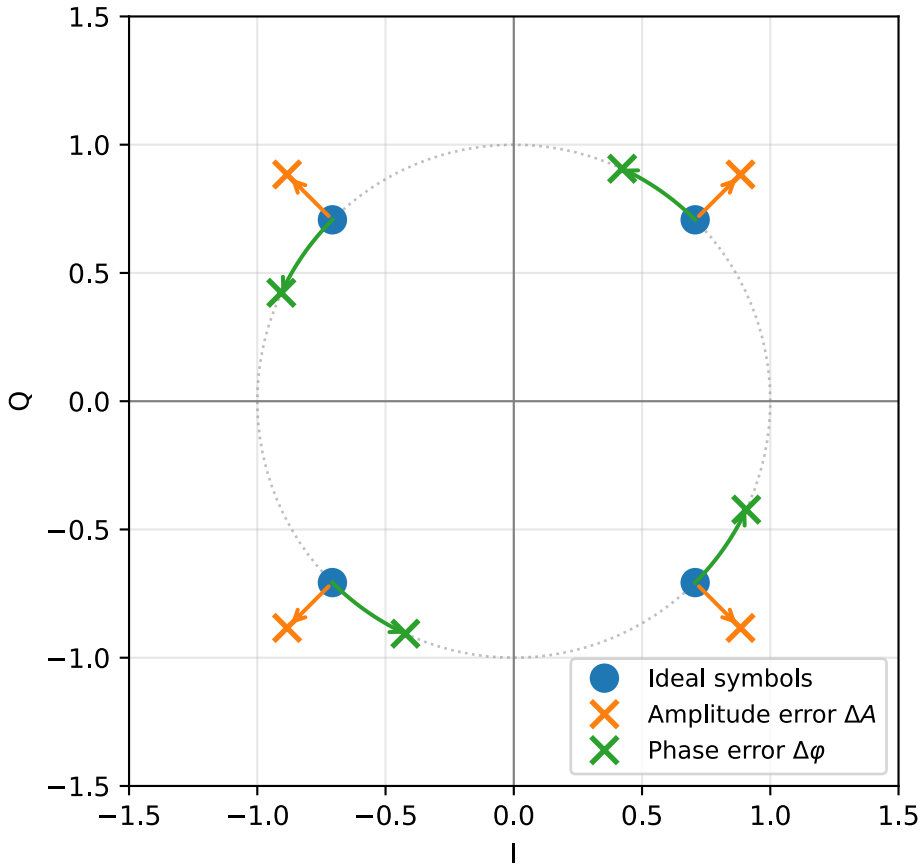


Figure 30: QPSK constellation illustrating the two error contributions. Applied to all four symbols, an amplitude error ΔA (radial displacement, orange crosses) and a phase error $\Delta\varphi$ (rotation on the circle, green crosses) are shown separately relative to the ideal symbols (blue circles).

I/Q imbalance can be characterized by two parameters: amplitude mismatch (ΔA) and phase mismatch ($\Delta\varphi$). Amplitude mismatch originates from the difference in gain between the I and Q paths, while phase mismatch refers to the deviation from the ideal 90° phase difference. The impact of I/Q imbalance on system performance depends on the modulation scheme used, with higher-order modulations being more sensitive to these impairments. Figure 30 illustrates both effects on a QPSK constellation: an amplitude mismatch displaces each symbol radially, while a phase mismatch rotates each symbol along the constellation circle.

There are two ways to quantify I/Q imbalance:

- **Image rejection ratio (IRR)**: The IRR is a measure of how well the receiver can reject the image frequency caused by I/Q imbalance. It is defined as the ratio of the power of the desired signal to the power of the image (unwanted) signal, typically expressed in dB. A higher IRR indicates better performance, with values above 30 dB to 40 dB generally considered acceptable for most applications.

The IRR can be calculated from the relative amplitude mismatch $\epsilon = \Delta A/A$ and the phase mismatch $\Delta\varphi$ (in radians) as [1]

$$\begin{aligned} \mathrm{IRR} &= \frac{1 + (1+\epsilon)^2 + 2(1+\epsilon)\cos(\Delta\varphi)}{1 + (1+\epsilon)^2 - 2(1+\epsilon)\cos(\Delta\varphi)} \\ &\approx \frac{4}{\epsilon^2 + (\Delta\varphi)^2}, \end{aligned} \tag{29}$$

where the approximation on the right holds for small amplitude and phase errors. The IRR in dB is given by

$$\mathrm{IRR}|_{\mathrm{dB}} = 10 \cdot \log_{10}(\mathrm{IRR}).$$

For example, an amplitude mismatch of 1% ($\epsilon = 0.01$) together with a phase mismatch of 1° ($\Delta\varphi \approx 0.0175$) yields an IRR of about 40 dB.

- **Error vector magnitude (EVM)**: The EVM is a measure of the difference between the ideal transmitted signal and the received signal, expressed as a percentage of the signal's magnitude. It quantifies the overall distortion in the received signal, including the effects of I/Q imbalance. Lower EVM values indicate better performance, with typical requirements ranging from 1% to 10% depending on the modulation scheme and application.

The EVM (in rms) is defined as

$$\mathrm{EVM} = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N} |s_{\mathrm{ideal},i} - s_{\mathrm{meas},i}|^2}}{\sqrt{\frac{1}{N}\sum_{i=1}^{N} |s_{\mathrm{ideal},i}|^2}} \tag{30}$$

where $s_{\mathrm{ideal},i}$ are the ideal transmitted symbols, $s_{\mathrm{meas},i}$ are the measured received symbols, and N is the number of symbols. EVM is expressed either in percent or in dB using

$$\mathrm{EVM}|_{\mathrm{dB}} = 20 \cdot \log_{10}(\mathrm{EVM}).$$

Note that the EVM (in dB) is always a negative number, with lower values indicating better performance. For example, an EVM of 1% (a very good number) corresponds to −40 dB, while an EVM of 10% corresponds to −20 dB. The SNR of the RF signal can be related to the EVM as

$$\mathrm{SNR}|_{\mathrm{dB}} \approx -20 \cdot \log_{10}(\mathrm{EVM}) = -\,\mathrm{EVM}|_{\mathrm{dB}}.$$

In order to make the I/Q mismatch sufficiently small, the following techniques can be used:

- Careful layout and matching of the components in the I and Q paths to minimize gain and phase mismatches. This usually involves good layout techniques. Further, the LO I/Q generation should be done with high accuracy.
- Calibration techniques can be used to measure and compensate for I/Q imbalance. This can be done either in the analog domain (e.g., using variable gain amplifiers and phase shifters) or in the digital domain (e.g., using digital signal processing algorithms). Digital compensation is usually preferred, as it is more flexible and can adapt to changing conditions. A CORDIC can be readily used for this purpose.

A final note on EVM: EVM is not only affected by I/Q imbalance, but also by other impairments like phase noise, nonlinearities, and noise in general. Therefore, EVM is a more comprehensive measure of the overall performance of a communication system, and often used to quantify the overall system performance in terms of SNR.

4 Low Noise Amplifiers

As shown in Section 2.3.4, the sensitivity of a receiver is determined (besides the channel bandwidth and the SNR requirement of the demodulator) mainly by the noise figure of the receiver. The noise figure is in turn determined by the noise figure of the first few components in the receive chain, as exemplified in Equation 22. Hence, as shown in Figure 23 and Figure 24, **low noise amplifiers (LNAs)** are usually the first active building block in a receiver after the antenna, switches, and initial RF filtering.

The main purpose of the LNA is to amplify the received signal with as little additional noise as possible, so that the subsequent stages in the receive chain can operate with a better signal-to-noise ratio (SNR) and do not increase the receiver NF substantially. The LNA is a critical component in the receiver design, and its performance has a significant impact on the overall sensitivity of the receiver.

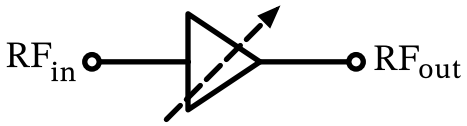


Figure 31: Block diagram of an LNA. Typically, the LNA input is impedance matched to 50 Ω, while the output is often not matched if the LNA is integrated on-chip. Often, the LNA gain is adjustable (shown with the arrow) to allow for gain control in the receiver depending on the signal conditions. The LNA also might have a low-power bypass mode to reduce the power consumption of the LNA for sufficiently strong signals.

The LNA as a building block is shown in Figure 31. It is usually designed for a specific frequency band, e.g., the 2.4 GHz ISM band or the 5 GHz WLAN band, and is typically designed for a specific impedance, e.g., 50 Ω, which is the standard impedance for RF systems. Impedance matching is usually mandated at the input; the output impedance matching is only required if the output of the LNA goes off-chip; if it is kept on-chip, impedance matching is often not needed. The LNA is also designed to be sufficiently linear, i.e., to not introduce significant distortion to the amplified signal; however, compared to the noise requirements, linearity is often less critical.

4.1 Impedance Matching

Why is impedance matching at the input of the LNA so important?

- The antenna is usually designed for a specific impedance, e.g., 50 Ω, and if the LNA input is not matched to this impedance, a significant portion of the received signal will be reflected back to the antenna, resulting in a loss of signal power.
- RF filters at the input of the LNA need to be terminated with the correct impedance to achieve the desired filter characteristics.
- Any transmission line in front of the LNA needs to be matched to avoid reflections and standing waves.

To quantify the "quality" of an impedance match, the **reflection coefficient** Γ is often used, which is defined as [3]:

$$\Gamma = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0} \tag{31}$$

where Z_0 is the characteristic impedance of the system (usually 50 Ω) and Z_{in} is the input impedance of the LNA. For a *passive* input impedance, i.e., $\Re\{Z_{\text{in}}\} \geq 0$, the reflection coefficient Γ is a complex number with a magnitude between 0 and 1 (as can be readily shown from Equation 31), where $|\Gamma| = 0$ indicates a perfect match and $|\Gamma| = 1$ indicates a complete mismatch. If the input presents a *negative* real part, then $|\Gamma| > 1$, i.e., the port reflects more power than is incident on it. This is exactly the condition we must avoid at all costs, and which we investigate in Section 4.2. This reflection coefficient can be represented on the Smith chart shown in Figure 32. A useful interactive tool to draft a matching network and visualize it on a Smith chart is this website.

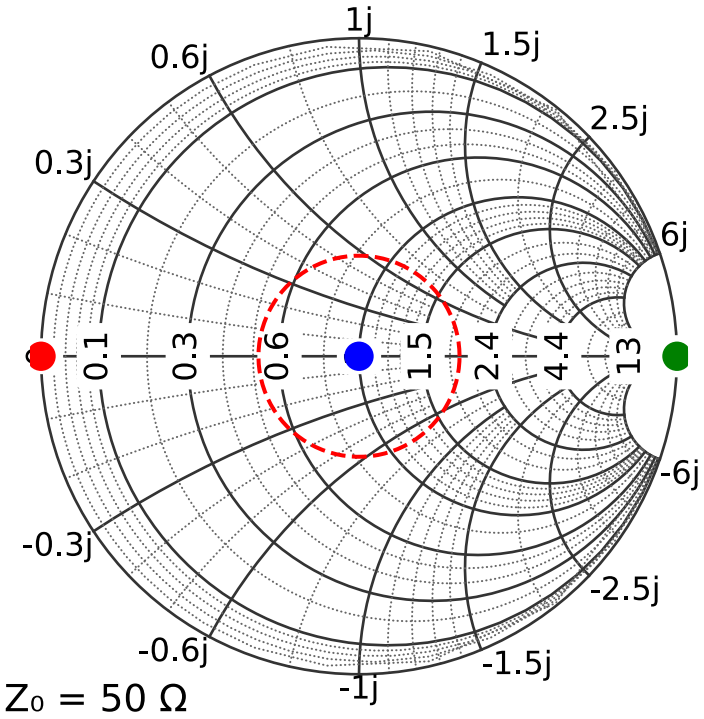


Figure 32: Smith chart showing constant resistance and reactance circles for impedance matching in RF circuits. The blue dot indicates perfect matching at 50 Ω, the green dot shows an open, and the red dot shows a short. The red dashed circle indicates a mismatch with RL = 10 dB.

However, in practice, the more commonly used metric for impedance matching is the return loss (RL), which is defined as [3]:

$$\text{RL} = -20\log_{10}|\Gamma| = 20\log_{10}\left|\frac{Z_{\text{in}} + Z_0}{Z_{\text{in}} - Z_0}\right| \tag{32}$$

A higher return loss indicates a better impedance match. A return loss of 10 dB (see Figure 32) indicates that 10% of the signal power is reflected back, while a return loss of 20 dB indicates

that only 1% of the signal power is reflected back. In practice, a return loss of at least 10 dB is desired, with higher (positive) values being better [22].

Bode-Fano Criterion

The Bode-Fano criterion states that for a given load impedance and a given bandwidth, there is a fundamental limit to how well an impedance can be matched over that bandwidth; this criterion is important in the design of LNAs! The Bode-Fano criterion can be expressed mathematically as [3, ch. 5.9], [23]

$$\int_0^\infty \ln\left(\frac{1}{|\Gamma(\omega)|}\right) d\omega \leq \frac{\pi}{RC} \tag{33}$$

where $\Gamma(\omega)$ is the frequency-dependent reflection coefficient, R is the load resistance, and C is the load capacitance (for an $R \parallel C$ load). This criterion implies that achieving a very good impedance match (i.e., a low reflection coefficient) over a wide bandwidth is fundamentally limited by the properties of the load. In practice, this means that there is a trade-off between the bandwidth of the match and the achievable return loss in an LNA design.

If we assume that we strive for match with $|\Gamma_\mathrm{m}|$ over a fractional bandwidth $\Delta\omega/\omega_0$, we can derive from the Bode-Fano criterion (and using that $Q = \omega_0 RC$ for a parallel RC circuit) that

$$\frac{\Delta\omega}{\omega_0} \leq \frac{\pi}{Q} \cdot \frac{1}{\ln(1/|\Gamma_\mathrm{m}|)}$$

For example, if we want to achieve a return loss of 20 dB (i.e., $|\Gamma_\mathrm{m}| = 0.1$) over a fractional bandwidth of 30%, we find that the quality factor of the load (i.e., the input stage of the LNA) needs to be $Q \leq 4.5$.

4.2 Amplifier Stability

When designing amplifiers, it is generally important to ensure that the amplifier is stable, i.e., that it does not oscillate! Under well-defined conditions, this is already complicated enough, but in RF design, the situation is even more complex, as the stability is usually a function of source and load impedances, which, especially when going off-chip, are often not well defined.

Hence, we have a two-port network, where we need to ensure that the amplifier is stable for all possible source and load impedances, which represent a large number of potential simulation cases. Luckily, there are well-defined criteria to check the stability of a two-port network.

One test for unconditional stability is the Rollet stability factor K, which is defined as [3]

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|} \tag{34}$$

with

$$\Delta = S_{11}S_{22} - S_{12}S_{21}. \tag{35}$$

For unconditional stability, the following conditions must be met:

$$K > 1 \quad \text{and} \quad |\Delta| < 1.$$

If these conditions are not met, the amplifier is potentially unstable, i.e., it might oscillate for some source and load impedances. In this case, further analysis is required to determine the stability of the amplifier.

Unfortunately, the K-factor test only provides a binary answer (stable/unstable) and does not provide any information about how close the amplifier is to instability. For this purpose, the so-called μ stability factor is used, which is defined as [24]

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - S_{11}^* \Delta| + |S_{12} S_{21}|}. \tag{36}$$

For unconditional stability, the following condition must be met:

$$\mu > 1$$

The μ-factor provides a measure of how close the amplifier is to instability; the larger the μ-factor, the more stable the amplifier is. A μ-factor of 1.5 or higher is usually considered sufficient for practical applications.

What we can learn from Equation 34, Equation 35, and Equation 36 is that to achieve a *stable* amplifier, we need to *minimize* the reverse gain S_{12} (also called isolation) and the input and output reflection coefficients S_{11} and S_{22}. In practice, this means that we need to design the amplifier with good reverse isolation and good input and output matching. Cascode stages are often used in RF amplifiers to achieve high gain and good isolation.

4.3 Resistively Matched Common-Source LNA

The key question is now how to design an LNA with low noise figure and an input impedance matched to 50 Ω? In order to appreciate this design challenge, we will first try a naive approach, using a common-source amplifier with resistive termination, as shown in Figure 33.

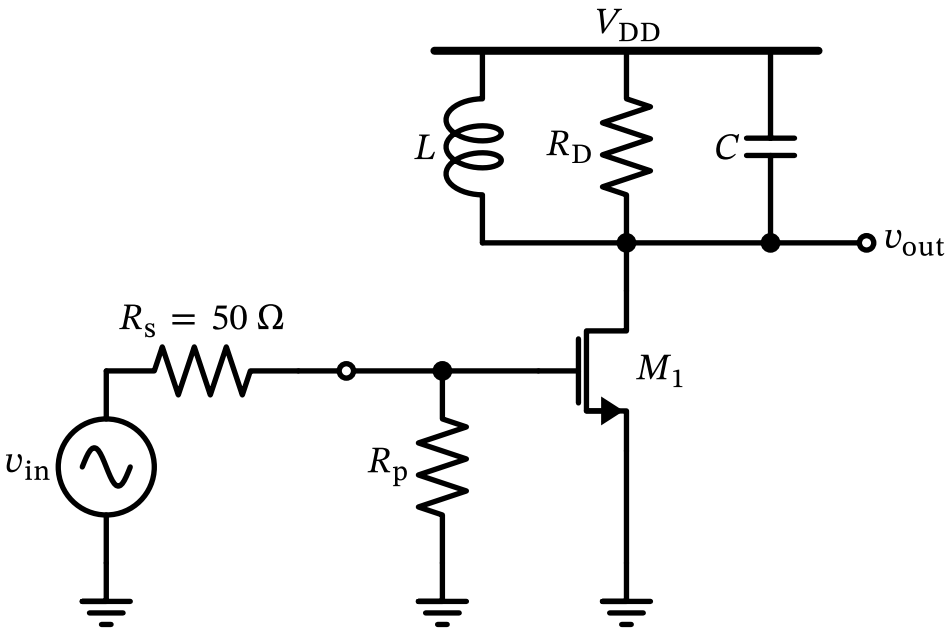


Figure 33: A simple LNA with resistive input matching and a parallel RLC tank circuit as a load (biasing details are omitted). The LNA is driven by a 50 Ω source.

If we assume the gate capacitance of M_1 is negligible, we can achieve good input impedance matching by choosing $R_p = R_s = 50\,\Omega$. The voltage gain of this simple common-source LNA is given by $A_v = -g_m R_D$, neglecting capacitances and g_{ds} of M_1 (we assume that the load tank is tuned to the desired frequency with $\omega_0 = 1/\sqrt{LC}$).

How can we calculate the noise figure of this simple LNA? We formulate

$$F = \frac{\text{total noise at output}}{\text{noise at output due to source only}}. \tag{37}$$

We derive a small-signal equivalent circuit of Figure 33, which is shown in Figure 34, to calculate the total noise at the output of the LNA.

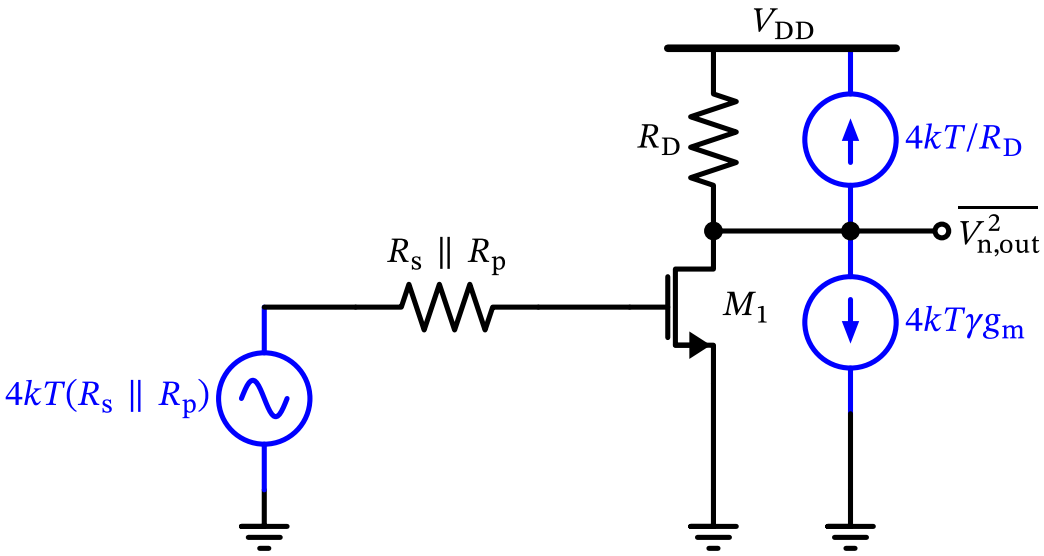


Figure 34: Equivalent circuit of resistively matched common-source LNA (noise sources shown in blue).

With the help of Figure 34, we can calculate the total output noise power spectral density as

$$\overline{V_{n,out,1}^2} = A_v^2 \cdot 4kT(R_s \parallel R_p) = (g_m R_D)^2 \cdot 4kT(R_s \parallel R_p)$$

and

$$\overline{I_{n,out}^2} = 4kT\gamma g_m + \frac{4kT}{R_D}$$

$$\overline{V_{n,out,2}^2} = R_D^2 \cdot \overline{I_{n,out}^2} = 4kT\gamma g_m R_D^2 + 4kT R_D$$

so that in total

$$\begin{aligned} \overline{V_{n,out}^2} &= \overline{V_{n,out,1}^2} + \overline{V_{n,out,2}^2} \\ &= 4kT\left[(g_m R_D)^2 (R_s \parallel R_p) + \gamma g_m R_D^2 + R_D\right]. \end{aligned} \tag{38}$$

We now need to find the output noise coming from the source only. For this we can use the equivalent circuit in Figure 35, to formulate the output noise due to the source only.

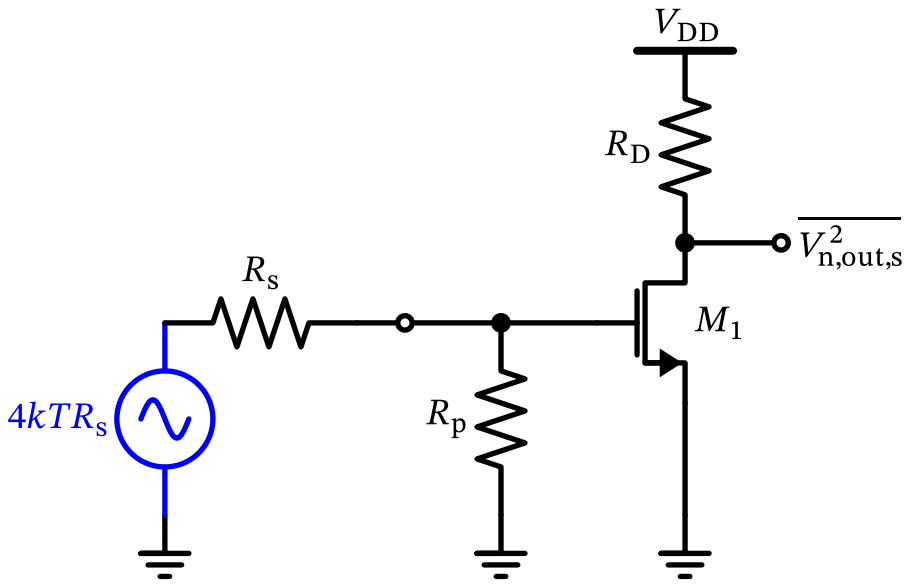


Figure 35: Equivalent circuit to calculate the output noise from the input.

We find that

$$\overline{V_{n,out,s}^2} = A_v^2 \cdot 4kTR_s \cdot \left(\frac{R_p}{R_s + R_p}\right)^2. \tag{39}$$

Finally, we can use Equation 38 and Equation 39 with Equation 37 to calculate the noise figure of the simple resistively matched common-source LNA as

$$F = \frac{\overline{V_{n,out}^2}}{\overline{V_{n,out,s}^2}} = 1 + \frac{R_s}{R_p} + \frac{\gamma R_s}{g_m (R_s \parallel R_p)^2} + \frac{R_s}{g_m^2 (R_s \parallel R_p)^2 R_D}. \tag{40}$$

i Common-source LNA with Resistive Matching

As an exercise to calculate noise behavior of circuits, derive and thus re-confirm the result of Equation 40 by yourself.

How can we interpret Equation 40? We see that we can minimize the noise factor by maximizing g_m. We then have a noise factor of

$$F \overset{g_m \gg}{=} 1 + \frac{R_s}{R_p} = 2$$

so we see that we are limited to a minimum noise figure of 3 dB, even if we spend the bias current to make g_m very large. We can go below a noise figure of 3 dB only if we give up on return loss and choose $R_p > R_s$, however, this means that the input is no longer matched to 50 Ω, so there are bounds on how far we can go. Hence, this simple resistively matched common-source LNA is not a good choice for a low-noise amplifier, with one exception: For

very wideband amplifiers, where an NF larger than 3 dB is acceptable, this configuration might be a good choice (often the solution shown in Section 4.4 is still better).

We see that we are stuck at high noise figures if we realize the real part of the input impedance with a resistor. This leaves us with the question of how to realize the real part of the input impedance otherwise. We will answer this question in the next sections (spoiler: feedback)!

4.4 Common-Gate LNA

We remember from our analog circuit design lecture that the common-gate configuration has an input impedance of $1/g_\mathrm{m}$ (conveniently neglecting parasitic capacitances). Hence, if we choose $g_\mathrm{m} = 1/50\,\Omega = 20\,\mathrm{mS}$, we can achieve input matching to 50 Ω without using a resistor at the input. This is the key idea of the common-gate LNA, which is shown in Figure 36.

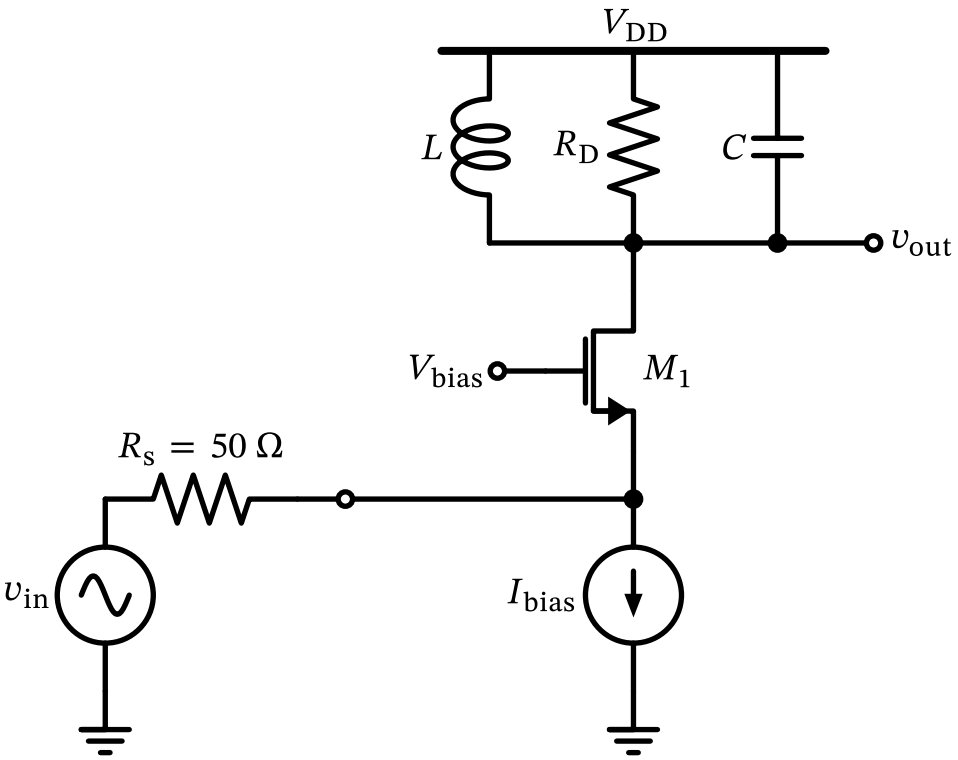


Figure 36: Circuit diagram of a common-gate LNA (biasing details are omitted).

By inspecting Figure 36 and following the practice from Section 4.3, we can directly write down the output noise voltage as (setting $g_\mathrm{m}^{-1} = R_\mathrm{s}$)

$$\begin{aligned}\overline{V_\mathrm{n,out}^2} &= kT\left[(g_\mathrm{m}R_\mathrm{D})^2 R_\mathrm{s} + \gamma g_\mathrm{m} R_\mathrm{D}^2 + 4R_\mathrm{D}\right] \\ &= kT\left(\frac{R_\mathrm{D}^2}{R_\mathrm{s}} + \gamma\frac{R_\mathrm{D}^2}{R_\mathrm{s}} + 4R_\mathrm{D}\right).\end{aligned} \tag{41}$$

The output noise due to the source only is given by

$$\overline{V_{\mathrm{n,out,s}}^2} = kT\frac{R_\mathrm{D}^2}{R_\mathrm{s}}. \tag{42}$$

Finally, we can use Equation 41 and Equation 42 with Equation 37 to calculate the noise figure of the common-gate LNA as

$$F = 1 + \gamma + \frac{4R_\mathrm{s}}{R_\mathrm{D}} \approx 1 + \gamma \quad \text{for} \quad R_\mathrm{D} \gg R_\mathrm{s}. \tag{43}$$

With a classical long-channel $\gamma = 2/3$, we can achieve a minimum noise figure of 2.2 dB, which is already better than the resistively matched common-source LNA in Section 4.3. However, using short-channel MOSFETs, γ is often larger than 1, so that the minimum noise figure of the common-gate LNA is often larger than 3 dB [1].

4.5 Inductively-Degenerated Common-Source LNA

As we have seen in Section 4.4, circuit techniques (feedback) can be used to realize the real part of an input impedance without the associated thermal noise of a resistor. We now try something different, in the hope that it will result in an even lower noise figure. We construct an LNA based on a common-source MOSFET amplifier, but we add an impedance Z_deg into the source line for series-series feedback [10]. This arrangement is shown in Figure 37.

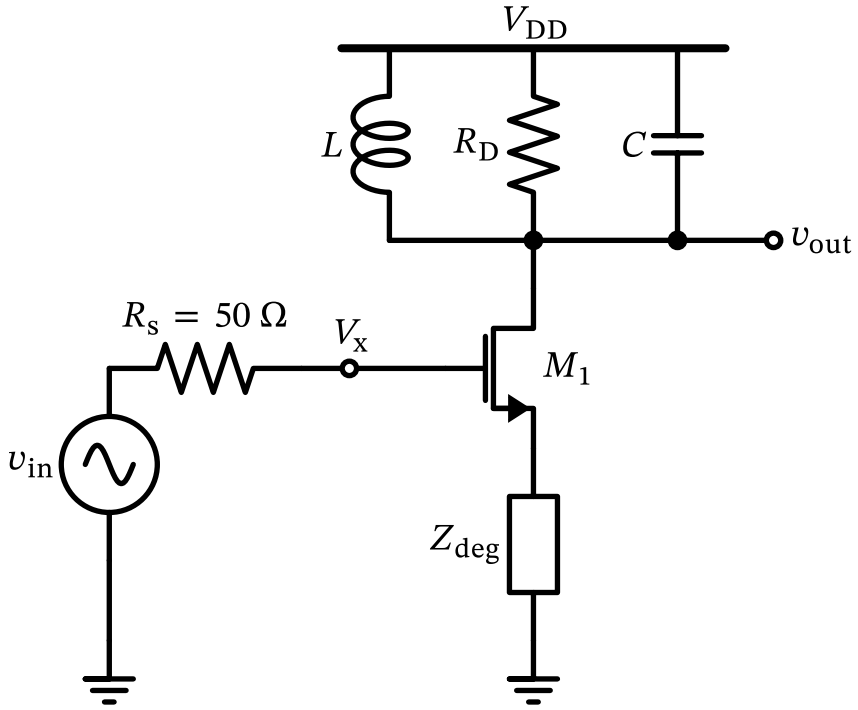


Figure 37: A common-source MOSFET stage with degeneration impedance (biasing details are omitted).

We now extract the small-signal equivalent circuit of the input stage of Figure 37, which is shown in Figure 38, to calculate the input impedance (we ignore g_ds for this analysis to simplify the derivation).

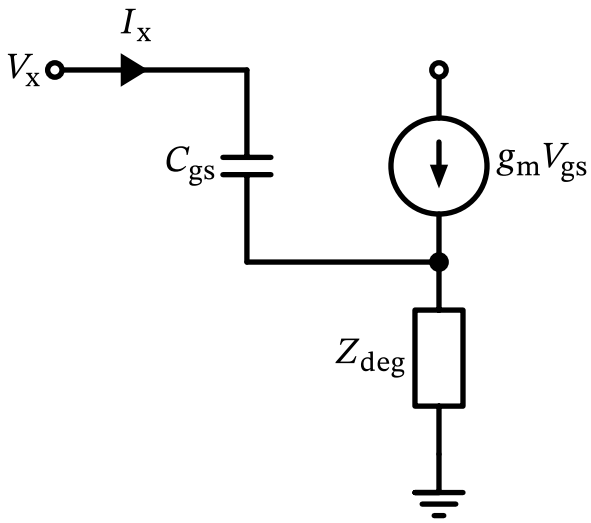


Figure 38: Equivalent small-signal circuit of the input stage around M_1.

We find that

$$V_\mathrm{x} = V_\mathrm{gs} + Z_\mathrm{deg}(I_\mathrm{x} + g_\mathrm{m}V_\mathrm{gs})$$

with $V_\mathrm{gs} = I_\mathrm{x}/sC_\mathrm{gs}$, so that we can write the input impedance as

$$Z_\mathrm{in} = \frac{V_\mathrm{x}}{I_\mathrm{x}} = \frac{1}{sC_\mathrm{gs}} + Z_\mathrm{deg} + \frac{g_\mathrm{m}Z_\mathrm{deg}}{sC_\mathrm{gs}}. \tag{44}$$

The final term in Equation 44 is the interesting one: By choosing Z_deg to be inductive (which we can do by using either an on-chip or an off-chip inductor), we can *realize a real part* of the input impedance. Choosing $Z_\mathrm{deg} = sL$, we find that

$$Z_\mathrm{in} = \frac{1}{sC_\mathrm{gs}} + sL + \frac{g_\mathrm{m}L}{C_\mathrm{gs}}.$$

By proper choice of L and C_gs, we can achieve input matching to 50 Ω at the desired frequency ω_0. We find that the real part of the input impedance is given by

$$\Re\{Z_\mathrm{in}\} = \frac{g_\mathrm{m}L}{C_\mathrm{gs}}. \tag{45}$$

Without proof (refer to [1], [2] for a derivation), we state the noise factor of this input stage (with some simplification) as ($\omega_\mathrm{T} \approx g_\mathrm{m}/C_\mathrm{gs}$ is the transition frequency of the MOSFET)

$$\begin{aligned} F &= 1 + \frac{\gamma R_\mathrm{s}\omega_0^2 C_\mathrm{gs}^2}{g_\mathrm{m}} \\ &\approx 1 + \gamma R_\mathrm{s} g_\mathrm{m}\left(\frac{\omega_0}{\omega_\mathrm{T}}\right)^2 \\ &\approx 1 + \gamma\omega_0^2 C_\mathrm{gs} L. \end{aligned} \tag{46}$$

Finally, we have an LNA input stage configuration that allows us to achieve a noise figure below 3 dB, even with $\gamma > 1$, by proper choice of a fast transistor with $\omega_T \gg \omega_0$ (and thus small C_{gs}). The inductively-degenerated common-source LNA is a widely used LNA input stage configuration in modern RFICs.

Note that the noise performance of Equation 46 cannot be lowered towards $F = 1$ because g_m, L, and C_{gs} are tied together by the input matching condition Equation 45, and we have neglected the induced gate noise of the MOSFET and other effects [1].

A bit more detailed schematic is shown in Figure 39. The inductor L_{match} is used to match the input impedance to 50 Ω at the desired frequency, L_{deg} is used to realize the real part of the input impedance, R_{bias} is used to bias the gate of M_1, M_2 is a cascode transistor which increases the output impedance and thus the gain of the stage (plus it improves the reverse isolation), and R_D, L_D, and C_D form a load tank which provides high gain at the desired frequency. A dc block is used at the input so that the bias point of M_1 is not corrupted by the input signal source. The bias voltage V_{bias2} sets the operating point of the cascode transistor M_2.

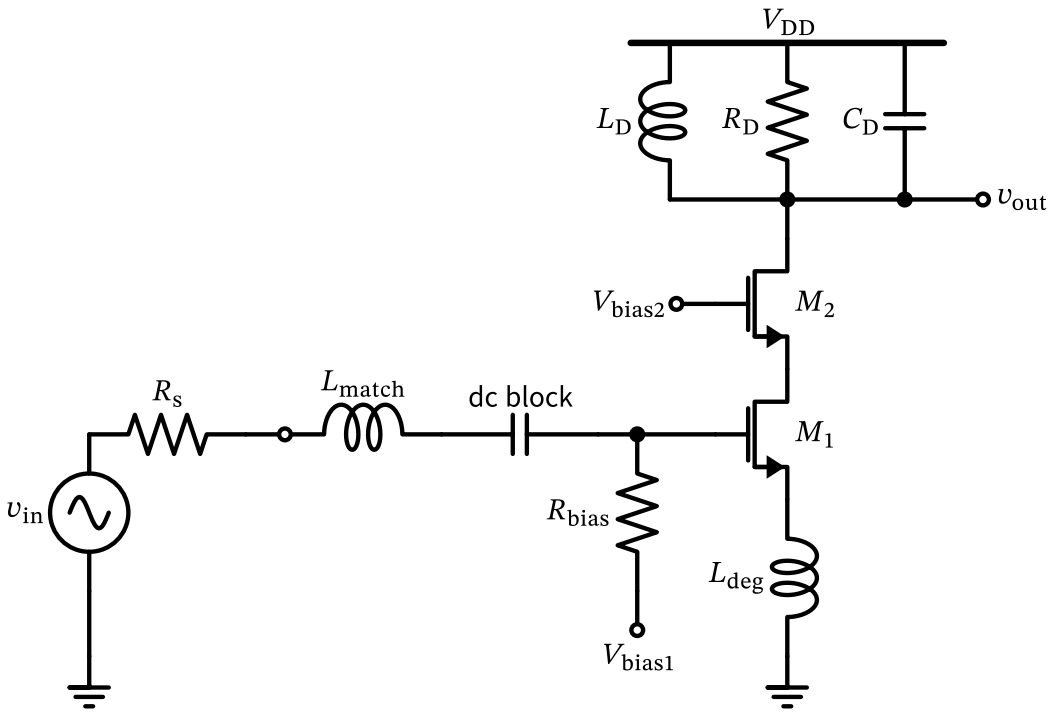


Figure 39: An (almost complete) common-source MOSFET stage with degeneration impedance and cascode.

What is missing in Figure 39 is a form of frequency tuning of the load tank to the frequency of interest, and support for different gain modes. Apart from these details, this LNA circuit is a good starting point for a practical LNA design.

4.6 Feedback LNA

One drawback of the inductively-degenerated common-source LNA (besides its limited bandwidth) is the use of at least one inductor. If the inductor is placed on-chip, it has a

comparatively large size, and if it is implemented in the package (via a bondwire) or on the PCB it adds complexity or cost to the bill-of-materials (BOM).

As an alternative, if the CMOS technology is sufficiently fast, a shunt feedback amplifier, as shown in Figure 40, might be a good choice [10].

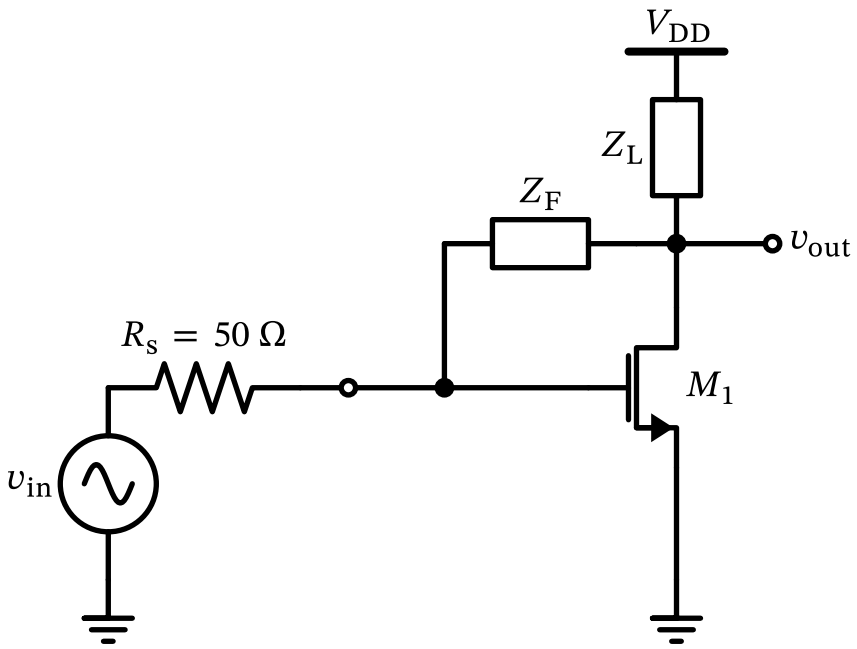


Figure 40: A shunt-feedback LNA.

The input impedance of the shunt feedback LNA is given by

$$Z_{in} = \frac{Z_F + Z_L}{1 + g_m Z_L}. \tag{47}$$

i Derivation of the Shunt-Feedback Input Impedance

We apply a test voltage V_x at the gate and find the resulting current I_x (ignoring g_{ds} and the gate capacitance for simplicity). Because the only path into the gate is through Z_{F}, we have

$$I_x = \frac{V_x - V_d}{Z_{\mathrm{F}}}.$$

Applying KCL at the drain node, where currents leave through Z_{F}, through Z_{L} to V_{DD} (ac ground), and via the transistor current source $g_{\mathrm{m}} V_x$:

$$\frac{V_d - V_x}{Z_{\mathrm{F}}} + \frac{V_d}{Z_{\mathrm{L}}} + g_{\mathrm{m}} V_x = 0$$

Rearranging for V_d and substituting back

$$\begin{aligned} I_x &= \frac{V_x - V_d}{Z_{\mathrm{F}}} \\ &= \frac{V_x}{Z_{\mathrm{F}}} \cdot \frac{Z_{\mathrm{F}} + Z_{\mathrm{L}} - Z_{\mathrm{L}}(1 - g_{\mathrm{m}} Z_{\mathrm{F}})}{Z_{\mathrm{F}} + Z_{\mathrm{L}}} \\ &= \frac{V_x \left(1 + g_{\mathrm{m}} Z_{\mathrm{L}}\right)}{Z_{\mathrm{F}} + Z_{\mathrm{L}}}, \end{aligned}$$

which gives Equation 47 directly.

The noise factor of the shunt feedback LNA is given by

$$F = 1 + \left| \frac{Z_{\mathrm{F}} + R_{\mathrm{s}}}{g_{\mathrm{m}} Z_{\mathrm{F}} + 1} \right|^2 \cdot \frac{\gamma g_{\mathrm{m}} + \Re\{Y_{\mathrm{L}}\}}{\Re\{Z_{\mathrm{in}}\}}, \tag{48}$$

which holds exactly for reactive (noiseless) Z_{F} and Z_{L} (where $\Re\{Y_F\} = 0$), and for a matched input ($R_{\mathrm{s}} = \Re\{Z_{\mathrm{in}}\}$) [2].

As you can see from Equation 48, by making g_{m} large (and spending enough bias current), the noise figure can be made arbitrarily small! Depending on the choice of Z_{F} and Z_{L}, the input impedance of this LNA can be changed in interesting ways.

By setting $Z_{\mathrm{L}} \to \infty$ (e.g., by biasing with a current source and high-impedance loading), we find that

$$Z_{\mathrm{in}} = \frac{1}{g_{\mathrm{m}}}$$

which is *independent* of Z_{F} and is a well-known result for a common-source stage. The disadvantage of this configuration is the noise factor, which (given that Z_{F} is sufficiently large) tends towards $F = 1 + \gamma$, which is the same as for the common-gate LNA.

A bit more interesting is the case when $g_{\mathrm{m}} Z_{\mathrm{L}} = A_0$ and $Z_{\mathrm{L}} \gg Z_{\mathrm{F}}$, which results in

$$Z_{\text{in}} = \frac{Z_{\text{L}}}{1 + A_0}$$

which is the well-known result that the input impedance of an amplifier with feedback is reduced by the factor $1 + A_0$, where A_0 is the open-loop gain of the amplifier. The noise factor can be made small by making g_{m} large, as we have already noted above.

A very interesting case can be achieved by choosing $Z_{\text{L}} = (sC_{\text{L}})^{-1}$ and $Z_{\text{F}} = (sC_{\text{F}})^{-1}$, which results in

$$Y_{\text{in}} = \frac{1}{Z_{\text{in}}} = \frac{g_{\text{m}} C_{\text{F}}}{C_{\text{L}} + C_{\text{F}}} + s\frac{C_{\text{L}} C_{\text{F}}}{C_{\text{L}} + C_{\text{F}}} \tag{49}$$

Looking at Equation 49, we see that the input admittance has a **real part**! By proper choice of components, we can achieve an input impedance matched to 50 Ω at the desired frequency. The noise factor can again be made small by making g_{m} large.

There is also an option, by proper choice of Z_{F} and Z_{L}, to achieve an inductive input impedance component, which can be used to resonate out parts of the input capacitance of the LNA, similar to the inductively-degenerated common-source LNA. However, in contrast to the inductively-degenerated common-source LNA, no inductor is required in this case. This configuration is called a reactance-canceling LNA [1].

5 Mixers

As we have seen in Section 3.2, we are using mixers to "upconvert" and "downconvert" signals, i.e., we shift the frequency of a signal to a higher or lower frequency with the help of an auxiliary signal, called the "**local oscillator**" (LO) signal. The block diagram of a mixer is shown in Figure 41.

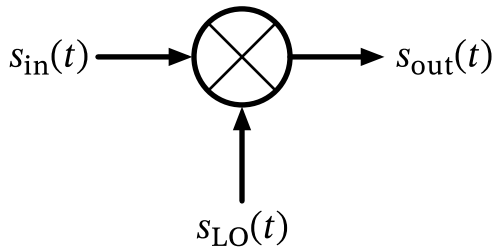


Figure 41: Mixer block diagram.

What we usually want in a mixer is that the input signal at ω_{in} is shifted by ω_{LO}, so that the output signal is located at

$$\omega_{\text{out}} = \omega_{\text{in}} \pm \omega_{\text{LO}}.$$

It should be noted that the output signal usually contains additional frequency components, which are not desired. These so-called "**spurious**" components are usually filtered out in a subsequent filtering stage.

Recalling the properties of linear time-invariant (LTI) systems, we know that in an LTI system *no new frequency components* (which are not already present in the input signal) are generated. In conclusion, in order to generate new frequency components, a mixer has to be either

- a **non-linear** system, or
- a **time-variant** system.

Next we will look at two options on how to implement a mixer.

5.1 Non-Linear Mixer

Generally speaking, we can use any nonlinear device to implement a mixer (often, this generation of new frequency components in a nonlinear system is highly undesired; in a mixer, we want this effect). As a simple example we look at the case of a square function, shown in Figure 42.

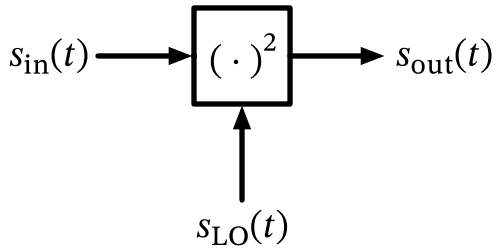


Figure 42: A squarer as a nonlinear mixer.

When we apply the sum of the signals $s_{\text{in}}(t) = \cos(\omega_{\text{in}} t)$ and $s_{\text{LO}}(t) = \cos(\omega_{\text{LO}} t)$ to the input of a squarer, we get the output signal

$$\begin{aligned} s_{\text{out}}(t) = [s_{\text{in}}(t) + s_{\text{LO}}(t)]^2 = \cos(\omega_{\text{in}} t + \omega_{\text{LO}} t) + \cos(\omega_{\text{in}} t - \omega_{\text{LO}} t) \\ +1 + \frac{1}{2}\cos(2\omega_{\text{in}} t) + \frac{1}{2}\cos(2\omega_{\text{LO}} t). \end{aligned} \tag{50}$$

We see that the output signal contains the desired frequency components at $\omega_{\text{in}} \pm \omega_{\text{LO}}$, but also additional components at dc (0 Hz), $2\omega_{\text{in}}$, and $2\omega_{\text{LO}}$. These additional components are usually unwanted and have to be filtered out in a subsequent filtering stage. As a side-note, in some circuits a **frequency doubler** is desired, which can be implemented by simply filtering out the other frequency components. In a doubler no LO signal is required, as the input signal is simply squared.

A simple example circuit is a diode, which has a non-linear current-voltage characteristic. The block diagram of a simple diode mixer is shown in Figure 43.

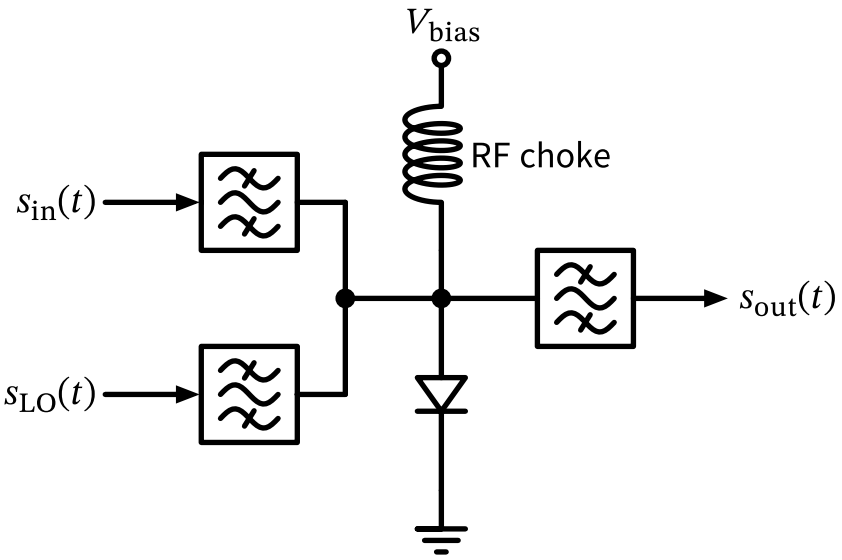


Figure 43: A diode mixer.

This structure is conceptually simple; it just requires a (fast) diode, filters to couple the desired frequency components in and out, and an RF choke to provide a dc bias voltage for the diode. However, the performance of such a simple mixer is usually not very good, as the diode is a highly non-linear device, which generates many spurious frequency components. More advanced mixer circuits use more complex non-linear devices (for example, a ring of diodes) to improve the performance. The advantage of these mixers is that they can operate up to very high frequencies (in the mm-wave range and beyond). If you can make a nonlinear device, you can make a mixer (think also of nonlinear optics, for example).

5.2 Time-Variant Mixer

In contrast to the non-linear mixer, a time-variant mixer uses an ideally **linear** device, but changes its properties over time. A simple example is a switch, which is opened and closed at the LO frequency. The block diagram of such a mixer is shown in Figure 44. This system is ideally linear from input to output, but very nonlinear when considering the LO input.

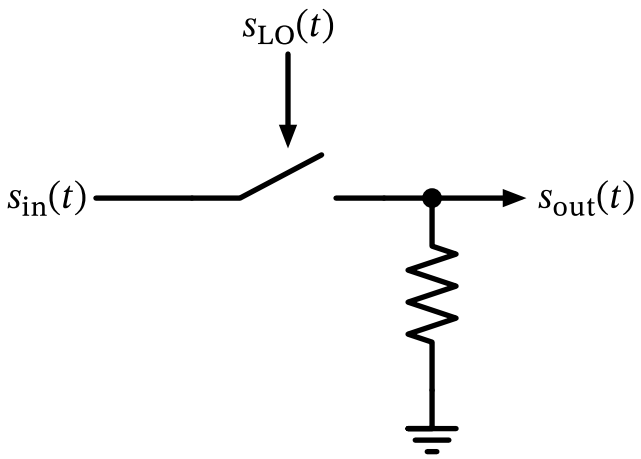


Figure 44: A switch as a time-variant mixer.

When the switch is closed ($s_{\mathrm{LO}} \geq 0$), the input signal is passed to the output, i.e., $s_{\mathrm{out}}(t) = s_{\mathrm{in}}(t)$; when the switch is open ($s_{\mathrm{LO}} < 0$), no signal is passed, i.e., $s_{\mathrm{out}}(t) = 0$. The output signal can be expressed as

$$\begin{aligned} s_{\mathrm{out}}(t) &= s_{\mathrm{in}}(t) \cdot \underbrace{\frac{1}{2}\{1 + \mathrm{sgn}[s_{\mathrm{LO}}(t)]\}}_{\text{Fourier series}} \\ &= s_{\mathrm{in}}(t) \cdot \left[\frac{1}{2} + \frac{2}{\pi}\sin(\omega_{\mathrm{LO}}t) + \frac{2}{3\pi}\sin(3\omega_{\mathrm{LO}}t) + \frac{2}{5\pi}\sin(5\omega_{\mathrm{LO}}t) + ...\right] \end{aligned}$$

where $\mathrm{sgn}(\cdot)$ is the signum function. The term in curly brackets is a square wave that switches between 0 and 1 at the LO frequency. This square wave can be expressed as a Fourier series, which contains the fundamental frequency at ω_{LO} and all odd harmonics $3\omega_{\mathrm{LO}}$, $5\omega_{\mathrm{LO}}$, and so on. In conclusion, if the input signal is expressed as $\cos(\omega_{\mathrm{in}}t)$, we get the output signal

$$s_{\mathrm{out}}(t) = \frac{1}{\pi}\cos(\omega_{\mathrm{in}}t \pm \omega_{\mathrm{LO}}t) + ... \tag{51}$$

Again, we see that the output signal contains the desired frequency components at $\omega_{\mathrm{in}} \pm \omega_{\mathrm{LO}}$, but also additional components at $3\omega_{\mathrm{LO}}$, $5\omega_{\mathrm{LO}}$, and so on. These additional components are usually unwanted and have to be filtered out in a subsequent filtering stage. The advantage of a time-variant mixer is that it can be implemented readily in CMOS, as shown in Figure 45.

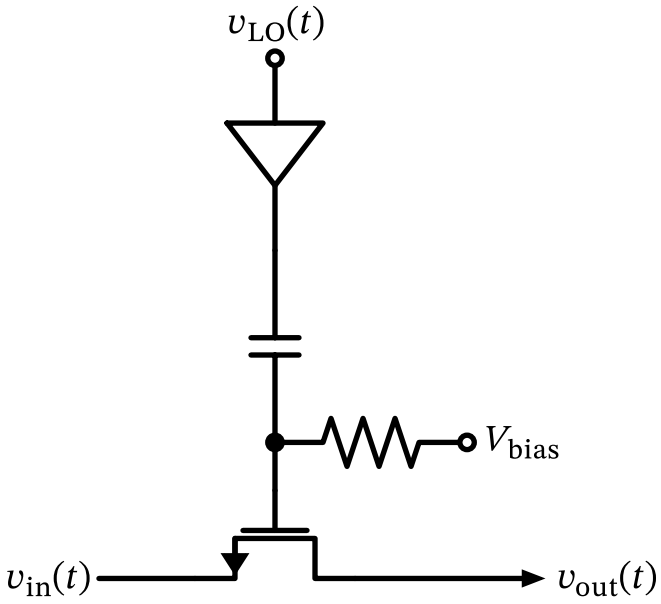


Figure 45: A MOSFET switch used as a mixer with an ac-coupled LO signal.

The implementation in Figure 45 uses a single NMOS transistor as a switch. The LO signal is ac-coupled to the gate of the transistor, so that the dc operating point is set by the bias voltage V_{bias}. When the LO signal is high enough, the transistor is switched on and the input signal $v_{\mathrm{in}}(t)$ is passed to the output; when the LO signal is low, the transistor is switched off and no signal is passed. Note that with the MOSFET we can switch voltages as well as currents, so the mixer can work in both modes, voltage mode or current mode.

A big disadvantage of this simple, **unbalanced** implementation is that the input signal is "lost" for half of the cycle when the MOSFET switch is open. A fully differential implementation (having differential ports at input, output and LO input) can alleviate this problem, as the input signal is always connected to one of the two output ports. This configuration is called a "**double-balanced mixer**" and is widely used in practice. It is shown in Figure 46.

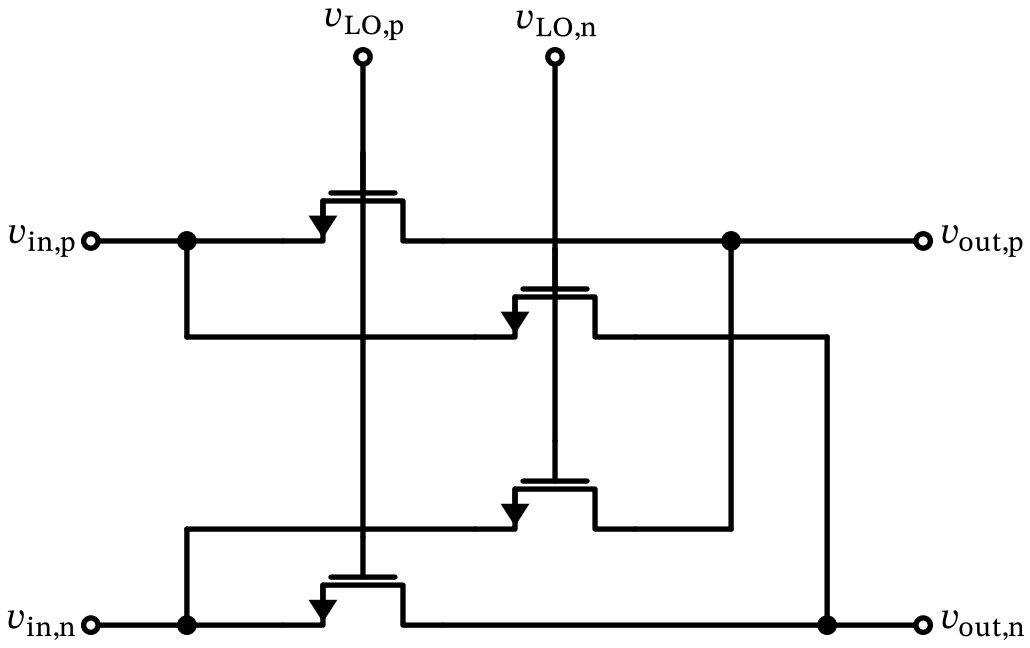


Figure 46: A fully-differential double-balanced MOSFET mixer.

The balanced structure changes the switching function from one toggling between 0 and 1 into one toggling between -1 and $+1$. The Fourier series of the latter has twice the fundamental amplitude, so the conversion gain doubles from $1/\pi$ to $2/\pi$, i.e., we gain 6 dB compared to the unbalanced mixer of Equation 51:

$$s_{\mathrm{out}}(t) = \frac{2}{\pi}\cos(\omega_{\mathrm{in}}t \pm \omega_{\mathrm{LO}}t) + \ldots \tag{52}$$

The factor $2/\pi$ corresponds to a conversion loss of 3.9 dB, which is the best an ideal balanced passive mixer can do with a 50 % duty-cycle LO driving a resistive load. A four-phase passive mixer with 25 % duty-cycle LO signals (see Figure 55) driving a capacitive (sampling) baseband load can do better and reaches $2\sqrt{2}/\pi$, i.e., a conversion loss of only 0.9 dB.

! Single-Balanced vs. Double-Balanced

Take care not to attribute these 6 dB to the *double*-balancing itself. A **single-balanced** mixer (a single-ended input driving a differential pair of switches with a differential output) already realizes the ± 1 switching function, and therefore already achieves the $2/\pi$ conversion gain. What the second level of balancing buys is *port-to-port isolation*: in the double-balanced mixer of Figure 46 both the LO feedthrough and the RF feedthrough appear as common-mode signals at the differential output and are thus suppressed. This is why the double-balanced structure is the workhorse in practice, not because of any additional conversion gain.

Note that no dc current needs to flow through the mixer structures shown in Figure 45 and Figure 46, which is a huge advantage when thinking of flicker noise. Careful, though:

this is *not* because the flicker noise of a MOSFET disappears without dc current. The gate-referred flicker noise $K_f/(C'_{\mathrm{ox}}WLf)$ of Section 2.3.1 is independent of the bias current. What matters is that this gate-referred voltage noise is converted into a *drain current* noise, $\overline{I^2_{\mathrm{n,1/f}}} = g^2_{\mathrm{m}}K_f/(C'_{\mathrm{ox}}WLf)$, and it is this current that is translated to the mixer output. In a passive mixer the switches are driven hard on and off, and with (ideally) zero dc current through them there is no bias current for the flicker noise to modulate, so the contribution at the output is drastically reduced [25]. This is in stark contrast to an active (Gilbert-type) mixer, where the switching pair carries the transconductor's bias current at all times.

Passive mixers are therefore the architecture of choice for zero-IF receivers, where the down-converted signal sits right at dc and is thus maximally exposed to 1/f noise.

In order to look at the full picture, we embed the mixer of Figure 46 in a complete RX front-end, as shown in Figure 47. An LNA (essentially a g_{m} cell) creates a current signal from the received voltage signal at the antenna. This current signal is then fed to the mixer in the current domain, and further sunk into a transimpedance amplifier (TIA), which also implements a lowpass pole. The simplified circuit is shown in Figure 47.

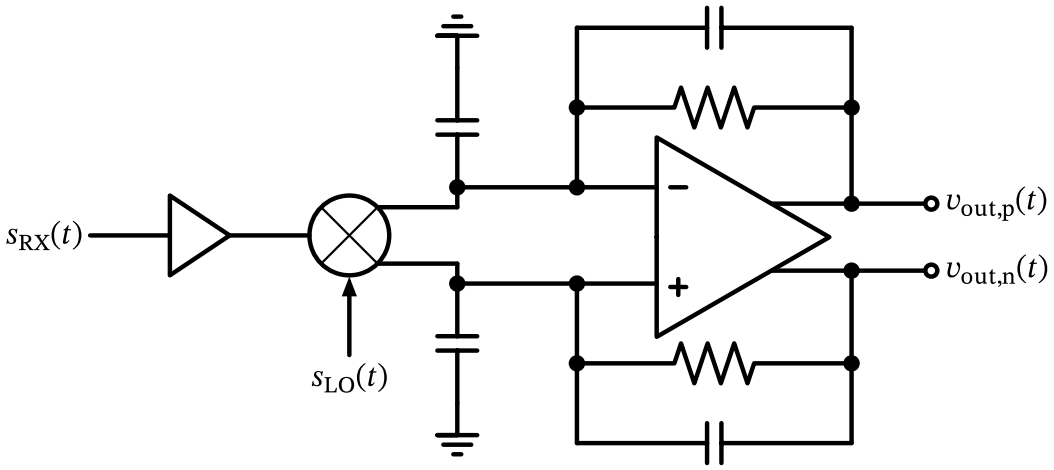


Figure 47: An RX front-end using a current-mode (passive) mixer.

The capacitors connected directly to ground at the mixer outputs are good practice, as they shunt high-frequency switching noise to ground, and in this way help the performance of the TIA, which otherwise would have to sink these currents. Note that all blocker currents originating at the LNA output and passing through the mixer are shunted by the feedback capacitors to the output of the TIA, and need to be actively driven by the TIA's output stage!

5.3 Gilbert Cell Mixer

As we have seen in Section 5.2, in CMOS we have quite a few options to implement a mixer as the MOSFET is a good current and voltage switch. This is in stark contrast to the BJT, as the bipolar transistor can only be used as an (excellent) current switch in the differential pair configuration. Hence, we need to implement a current-mode mixer in bipolar technology. The most widely used structure is the so-called "**Gilbert cell**", shown in Figure 48.

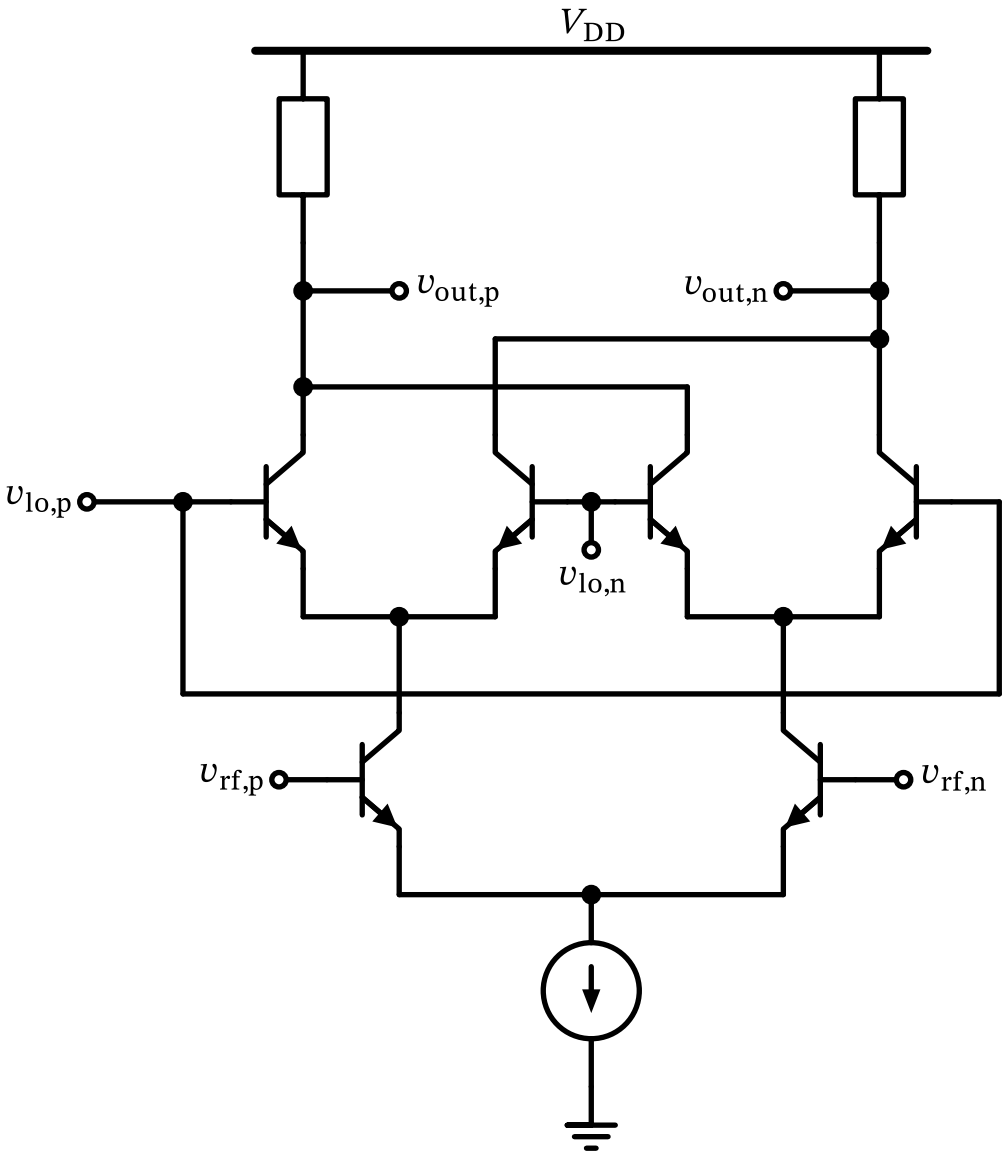


Figure 48: A Gilbert mixer based on bipolar differential pairs.

The idea is that an input transconductor creates currents, which are switched in the double differential pair and converted back to voltage in the load impedances. There are many known variations of this circuit, like skipping the current source at the bottom to increase linearity and headroom, or swapping the input transconductor for another structure. It might also be useful to "bleed" some bias current from the LO switching stage to improve the noise performance.

5.4 N-Path Filter

Without much deliberation or diving into the background, it has to be stated that passive mixers (based on MOSFET switches) have a few very interesting properties. One of them is that they can be used to implement very high-quality bandpass filters, called "**N-path filters**" [26]. The basic idea is shown in Figure 49. We use four phases of an LO signal to switch the

four capacitors to ground in a round-robin fashion. The duty cycle of each LO phase is 25% to have a non-overlapping clock.

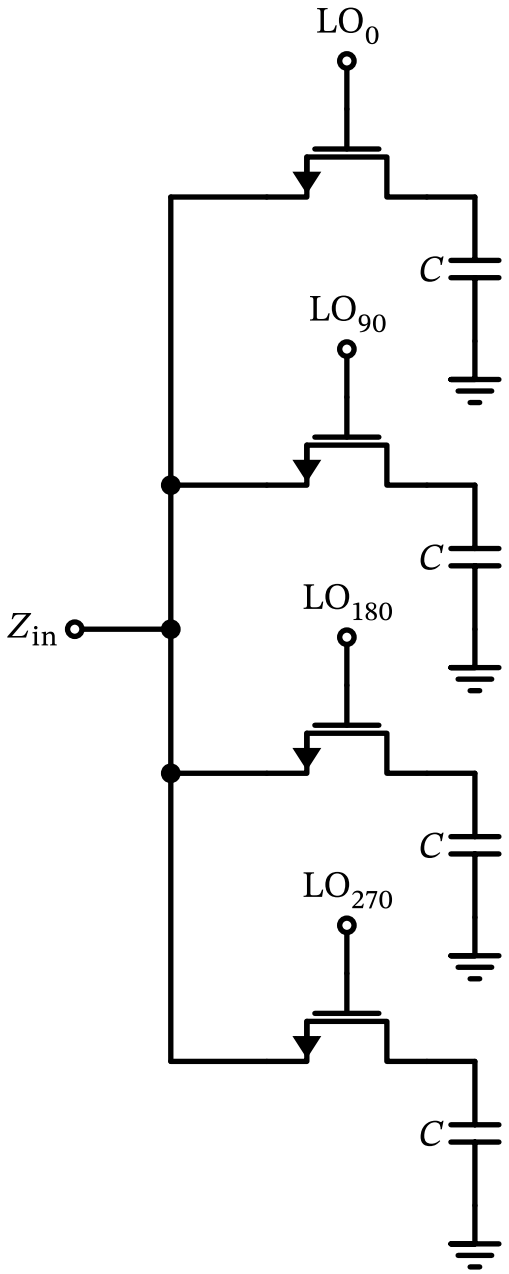


Figure 49: A 4-phase N-path filter.

The key observation to make is that a switch that is opened and closed at a certain frequency (the LO frequency) can be seen as a time-variant resistor. This time-variant resistor converts impedance seen at one end to the other end, and also changes the frequency of this apparent impedance. In other words, if we connect grounded capacitors at one end of the mixer switches and look into the other end, we see a **bandpass filter centered at the LO** frequency! In this

way we can build bandpass filters (and also bandstop filters [27]) at very high frequencies which are precisely centered (without component variations) around ω_{LO}.

The reason this works is the **transparency** of the passive mixer: the impedance connected to the baseband side of the switches appears at the RF port, frequency-shifted to $\pm\omega_{\mathrm{LO}}$. If the baseband load is a first-order RC section with impedance $Z_{\mathrm{BB}}(\Delta\omega)$, then the RF-port impedance around the LO is simply

$$Z_{\mathrm{in}}(\omega_{\mathrm{LO}} + \Delta\omega) \approx R_{\mathrm{sw}} + Z_{\mathrm{BB}}(\Delta\omega), \tag{53}$$

i.e., a *first-order* bandpass centered exactly at ω_{LO}, sitting on top of the series switch resistance R_{sw}. This is shown in Figure 50. Two properties follow directly from Equation 53 and make the N-path filter so attractive:

- The **peak impedance** is $R_{\mathrm{sw}} + R_{\mathrm{BB}}$, where R_{BB} is the effective shunt resistance at baseband. The switch resistance therefore sets the floor of what is achievable — low-R_{sw} switches are essential.
- The **bandwidth is inversely proportional to the capacitance**, $\mathrm{BW} \propto 1/C$, while the center frequency is set solely by ω_{LO}. This gives us a filter whose center frequency and bandwidth are independently and precisely programmable — something no passive LC filter can offer. The resulting effective $Q = \omega_{\mathrm{LO}}/\mathrm{BW}$ can easily reach several hundreds.

A detailed treatment, including the dependence on N and the duty cycle as well as the folding of the higher harmonics, can be found in [28].

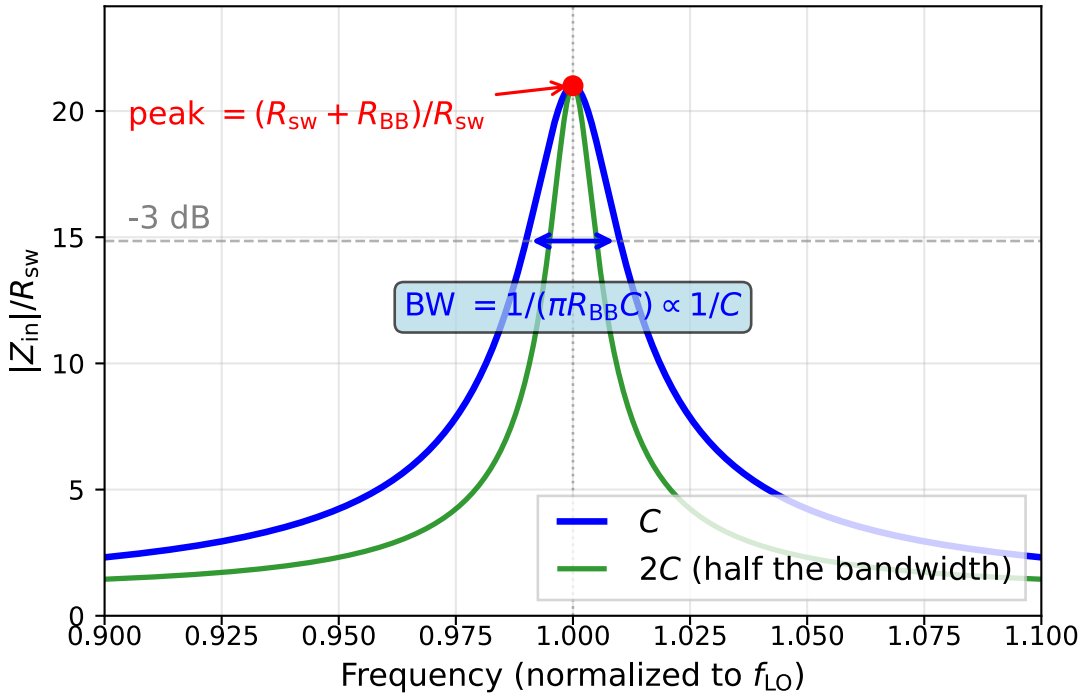


Figure 50: Input impedance of an N-path filter, obtained by frequency-translating the first-order baseband RC impedance to $\pm f_{\mathrm{LO}}$ and adding the series switch resistance. Doubling the path capacitance halves the bandwidth, while the peak impedance stays at $R_{\mathrm{sw}} + R_{\mathrm{BB}}$.

5.5 LO Generation

As you have seen in Section 3.2, for complex-valued modulation schemes (like QPSK, QAM, OFDM, and so on) we need to generate quadrature LO signals (i.e., two LO signals with a 90° phase shift). There are many ways to generate such signals, and we will now study a few of them.

5.5.1 RC/CR Phase Shift Network

Probably the simplest way to generate quadrature LO signals is to use a simple RC/CR phase shift network, as shown in Figure 51. The RC network (a first-order lowpass) generates a phase shift between 0° and −90°, while the CR network (a first-order highpass) generates a phase shift between 0° and +90°. By operating the network at $\omega_0 = 1/RC$, we can achieve a 90° phase shift between the two outputs.

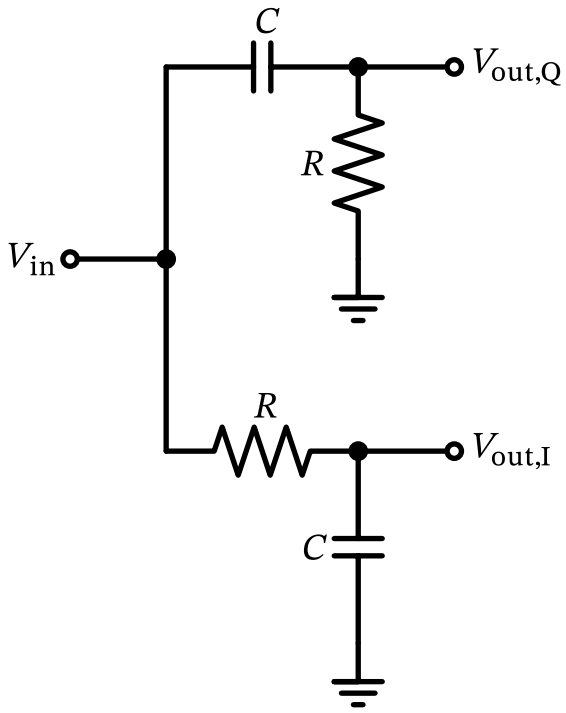


Figure 51: An RC/CR IQ generation network.

The advantage of this network is its simplicity, as it only requires a few passive components, and it can generate I and Q phases from an incoming signal at the target frequency. The disadvantage is that both the phase shift *and* the amplitude are frequency-dependent: exactly at $\omega_0 = 1/RC$ the two outputs are perfectly in quadrature *and* have equal magnitude ($1/\sqrt{2}$ each), but as soon as we move away from ω_0 the lowpass output rolls off while the highpass output rises, so we accumulate both a phase error and an amplitude imbalance. Since the resulting I/Q mismatch directly limits the image rejection (see Equation 29), this network is only usable over a narrow frequency range, and typically requires additional amplitude-limiting or gain stages. Additionally, this passive network has a 3 dB loss at ω_0, which also has to be made up by subsequent gain stages.

5.5.2 Polyphase Filter

We can create a more advanced phase shift network (following the idea of Section 5.5.1) by using a so-called "**polyphase filter**" [29]. The idea is to use multiple RC sections to create a more ideal phase shift network. An example is shown in Figure 52.

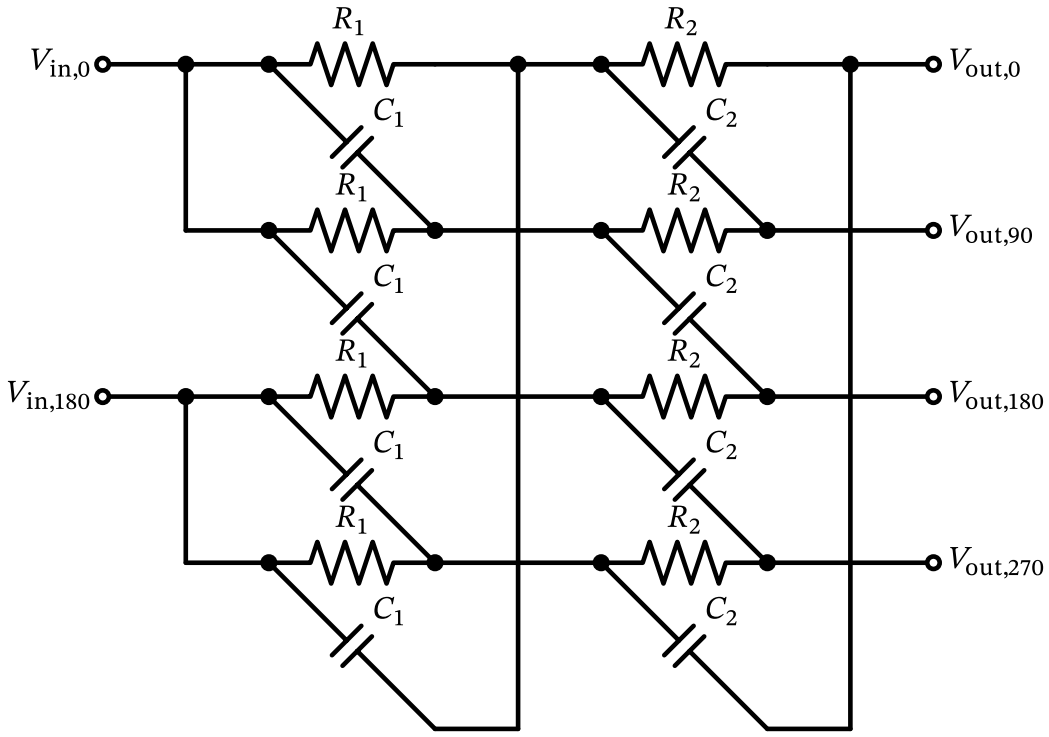


Figure 52: A two-stage polyphase network.

As shown in the figure above, we can use multiple stages (with $R_1C_1 \neq R_2C_2$) in cascade to broaden the frequency range over which we have a good 90° phase shift between the four outputs. The more stages we use, the better the performance, but also the more components are required. Note that this network is still passive, so it has an inherent loss, which also increases with the number of stages. The idea behind this network is that the network is transparent for positive frequencies, and blocks negative frequencies (or vice versa). We enter the network with real signals (having positive and negative frequency components), and at the output we get complex signals (having only positive or negative frequency components). In this sense this network is a complex filter.

Note that polyphase filters are popular when we have to create complex signals from real signals. We can use the polyphase filter in the LO path, or also in the signal path (e.g., in a receiver). The benefit is that we can work with a signal where the input frequency is equal to the output frequency. This is in strong contrast to the approach we will discuss next.

5.5.3 Flip-Flop Based Phase Generation

The most widely used approach to generate quadrature LO signals is to use digital circuits, as shown in Figure 53. The idea is to use a flip-flop (or a latch) to divide the frequency of an incoming clock signal by two. If we drive the two flip-flops (which are connected as toggle

flip-flops) by inverted clocks, then the resulting output signals are 90° phase-shifted, as shown in Figure 54.

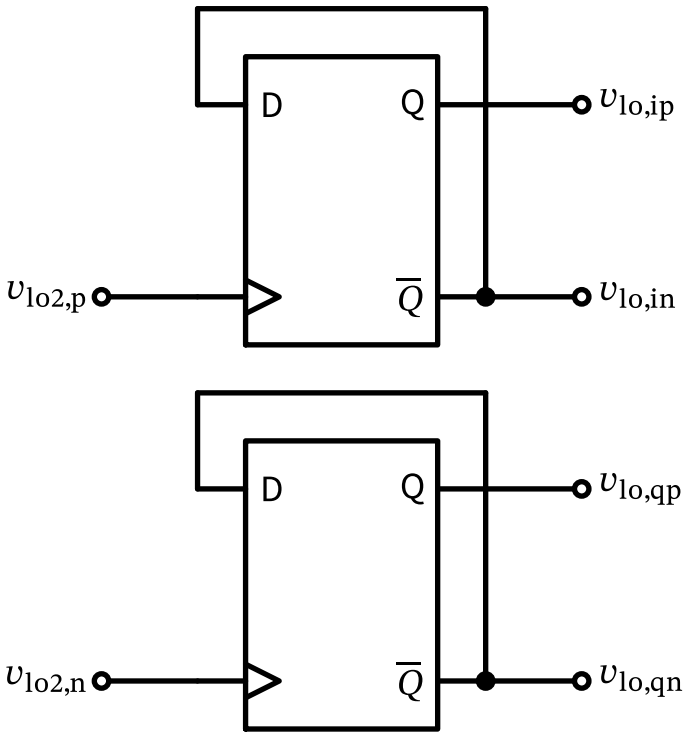


Figure 53: I/Q generation with a divide-by-2.

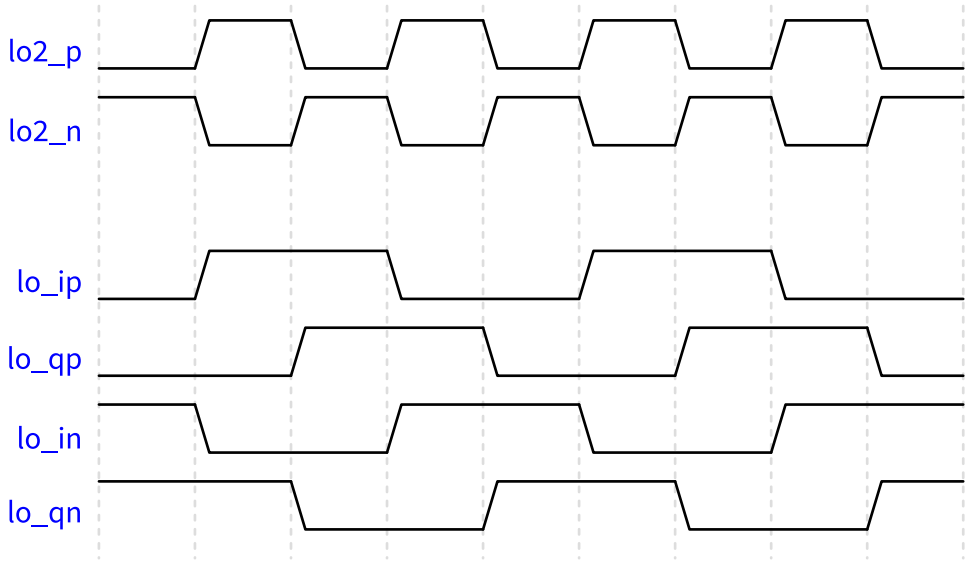


Figure 54: Input and output waveforms of I/Q generation with a divide-by-2.

It is important to note that the circuit shown in Figure 53 has to be implemented in a way so that there is no phase ambiguity, i.e., the Q-phase is lagging the I-phase by exactly 90° (as drawn in Figure 54). This can be achieved by using a reset signal to set the flip-flops to a known state at startup, or by implementing flip-flops using a differential clock input.

If the input clock is not 2x the target frequency but higher (e.g., 4x), then we can use a frequency divider approach to generate more than four phases. This could be useful in an N-path filter, where we could need more than four phases (see Section 5.4).

Often, we need four-phase LO signals with 25% duty cycle (instead of the 50% duty cycle shown in Figure 54). This can be achieved by using additional logic gates to combine the outputs of the flip-flops, as shown in Figure 55. The resulting LO waveforms are shown in Figure 56.

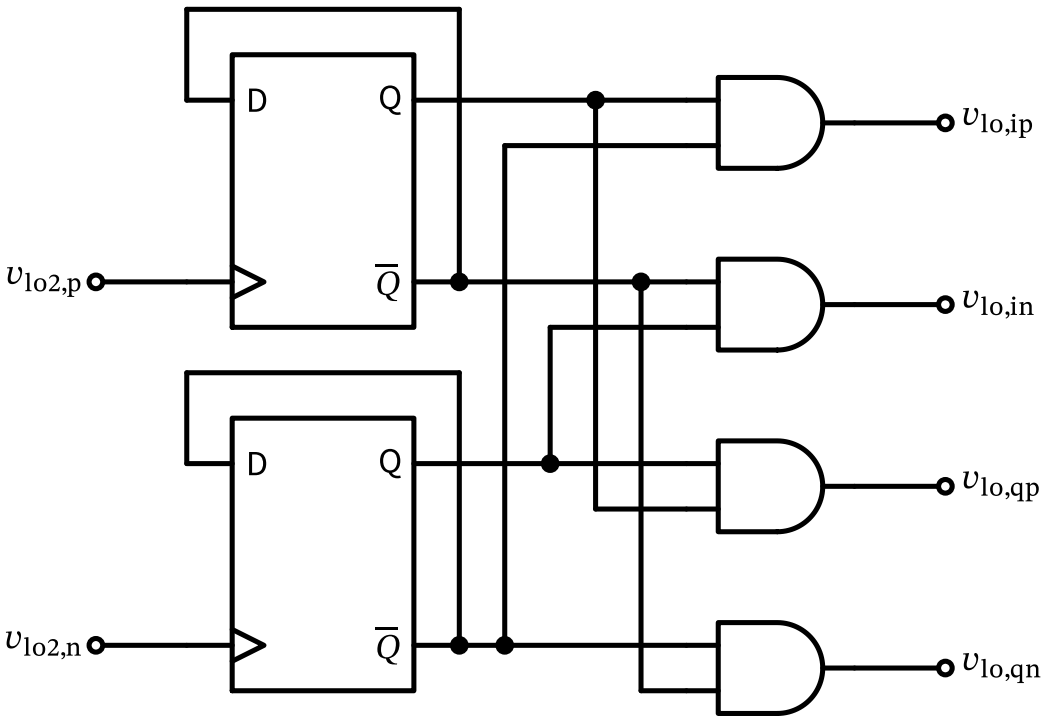


Figure 55: I/Q generation with a divide-by-2 and 25% duty cycle generation.

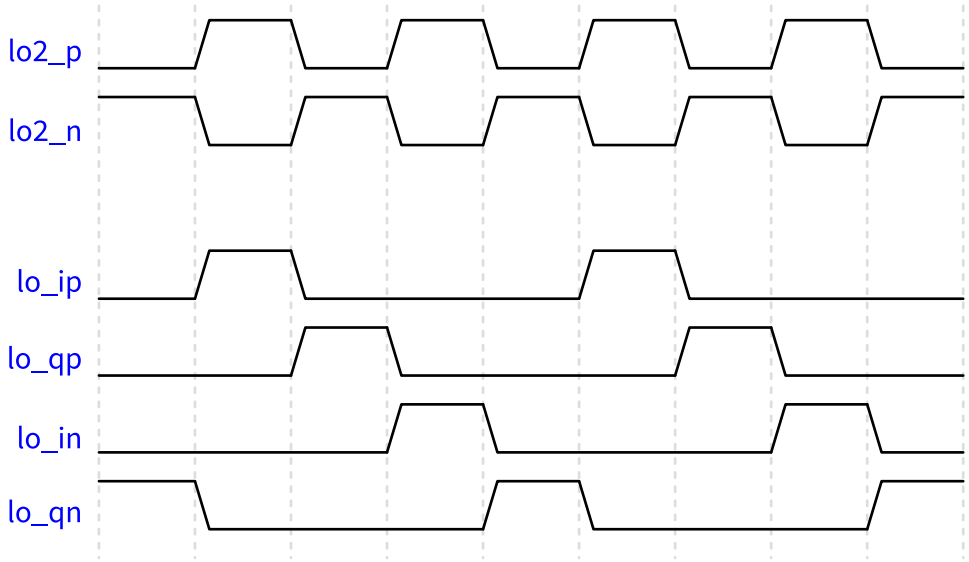


Figure 56: Input and output waveforms of I/Q generation with a divide-by-2 and 25% duty cycle generation.

The advantage of using flip-flops to generate multi-phase LO signals is that very precise phase shifts can be achieved, which can be produced over a very wide frequency range. They can be implemented in different logic styles, like CMOS, ECL, CML, and so on. The disadvantage is that we need a high-frequency clock signal above the target frequency, which might be difficult to generate. Still, I/Q generation with flip-flops is the de-facto standard approach in modern RFICs.

5.5.4 Delay-Based Phase Generation

The final approach we want to mention is to use a delay line (or multiple ones) to generate phase shifts. Using transmission lines, this approach is widely used at very high frequencies, where other approaches (like the ones shown in Section 5.5.2 or Section 5.5.3) are difficult to implement. One well-known approach is the 90° hybrid coupler, realized with $\lambda/4$ transmission lines [3]. When the frequencies are high enough (and the resulting wavelengths short enough), this approach can even be implemented on-chip. The disadvantage is that the phase shift is frequency-dependent, so the quadrature signals are only perfectly in quadrature at one frequency. Furthermore, this passive network has a 3 dB loss, so we need additional gain stages to compensate for this loss. A branch-line hybrid coupler is shown in Figure 57.

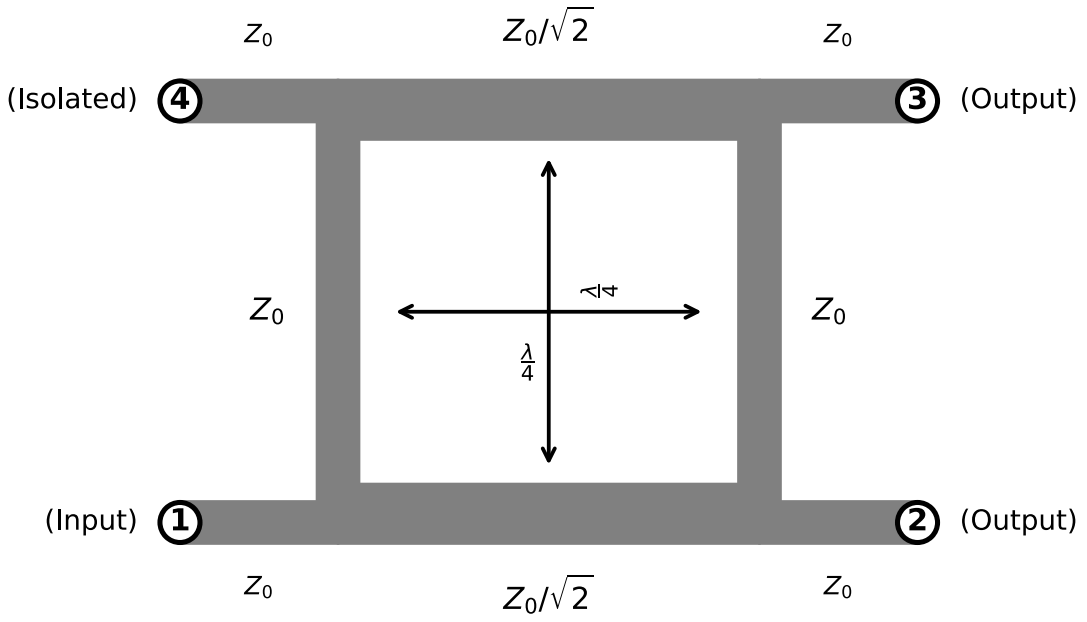


Figure 57: Branch-line hybrid coupler schematic showing the transmission line structure with characteristic impedances and λ/4 length sections.

The 3-port S-parameter matrix of an ideal branch-line coupler (assuming port 4 is properly terminated with Z_0) is given by

$$S = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -j & -1 \\ -j & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}. \tag{54}$$

A feature of the coupler in Figure 57 is that it is single-ended; a differential implementation needs either two couplers or a coupler with subsequent baluns. An alternative implementation of a 90° hybrid coupler is the Lange coupler, which uses interdigitated fingers to achieve the required coupling. The Lange coupler can be implemented in a single-ended or differential way, and it can achieve very good performance over a wide frequency range. The disadvantage is that it requires a large number of fingers, which can take up a lot of area on-chip [30].

Another approach, which is similar to the flip-flop based approach, is to use a delay line to generate the required phase shifts. This can be done with a single delay line and multiple taps, or with multiple delay lines. The idea is that we delay the input clock signal by a certain amount of time, which corresponds to a 90° phase shift at the target frequency. This approach is shown in Figure 58. Since the delay of a delay line is frequency dependent, and the delay will change with process, voltage, and temperature (PVT) variations, we have to implement a tuning mechanism to adjust the delay. This implementation is called a delay-locked loop (DLL), which is similar to a phase-locked loop (PLL, see Section 7).

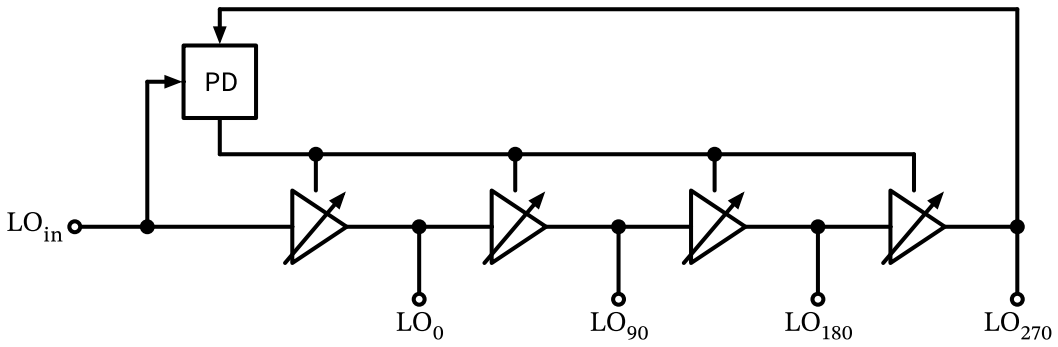


Figure 58: LO multiphase generation by delay-locked loop (DLL). An extension to more than four phases is straightforward.

With a phase detector (PD) we can compare the phase of the output signal with the input signal, and regulate the delay per stage so that after four delays the edges line up. Then we can tap the equally-spaced phases and use them as LO signals. The advantage of this approach is that we can flexibly create a required number of phases from an input clock. This feature is also used in wireline communication systems to generate multiple phases for the data sampling.

6 Oscillators

For the generation of the LO frequency to be used in a mixer for frequency conversion, oscillators are used. Ideally, oscillators produce a stable, noise-free sinusoidal signal, independent of environmental conditions like temperature or supply voltage variations. The circuit symbol of an oscillator is shown in Figure 59.

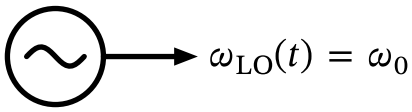


Figure 59: Oscillator symbol.

The question is how to construct an oscillator. In summary, we need to build something that oscillates, i.e., produces a sustained periodic signal with frequency ω_0. One way to achieve this is to construct a feedback loop where $|H(s = j\omega_0)| = 1$ and $\angle H(s = j\omega_0) = n \cdot 2\pi$; these conditions are called the "Barkhausen criterion" (here, $H(s)$ is the loop gain around the feedback loop). A *ring oscillator* is one example of such a feedback oscillator. For the 3-stage single-ended ring oscillator shown in Figure 60, each inverter contributes π (180°) of dc inversion plus a frequency-dependent lag caused by its output pole. At the oscillation frequency ω_0 this extra lag must be $\pi/3$ (60°) per stage, so that each inverter contributes $4\pi/3$ (240°) in total and the three inverters together provide 4π, i.e., $n \cdot 2\pi$ with $n = 2$. With a single-pole model per stage this happens at $\omega_0 = \sqrt{3}\,\omega_\mathrm{p}$, where ω_p is the pole frequency of one inverter stage. The gain condition is fulfilled by the gain of the inverters, which must be larger than unity to compensate for losses in the loop. By using an odd number of inverters, a stable locking point at dc is avoided.

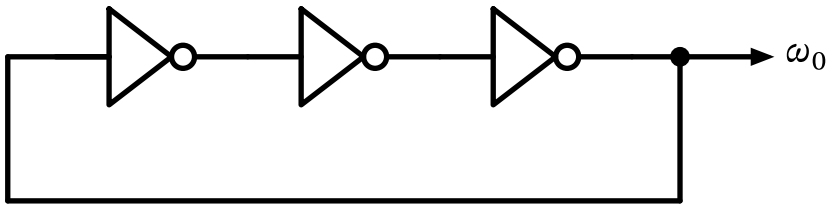


Figure 60: Three-stage single-ended ring oscillator.

Note that the output frequency of the ring oscillator shown above is ill-controlled, as it only depends on the delay (i.e., phase shift) of the inverters, which are usually dependent on process, voltage, and temperature (PVT) variations. Also, the inherent quality factor of such an oscillator is low, as exemplified by the definition of the quality factor Q as [31]

$$Q = 2\pi \frac{\text{Average energy stored}}{\text{Energy loss per cycle}} = \frac{\omega_0}{\Delta\omega} = \frac{\omega_0}{2}\sqrt{\left(\frac{dA}{d\omega}\right)^2 + \left(\frac{d\varphi}{d\omega}\right)^2} \tag{55}$$

with a ring oscillator having very low $dA/d\omega$ and $d\varphi/d\omega$ slopes ($A(\omega)$ and $\varphi(\omega)$ are the open loop gain and phase shift, respectively). It can be shown that $Q \approx 1.3$ for a 3-stage ring oscillator [31]. For some applications, this might be sufficient, but in many cases, a higher Q is desired to reduce phase noise and improve frequency stability (as we will see later in this chapter).

According to Equation 55, a high Q can be achieved by using a resonator with high energy storage capability and low energy loss per cycle. We then add an amplifier in a feedback loop to compensate for the losses of the resonator. This principle is shown in Figure 61 for a parallel LC tank circuit as the resonator. Sometimes the action of the feedback loop around the amplifier is modelled as a negative resistance $-R_{\text{amp}}$ that compensates for the losses of the tank circuit represented by the resistor R_{p}. The single parallel resistor R_{p} models the losses of the inductor L and the capacitor C.

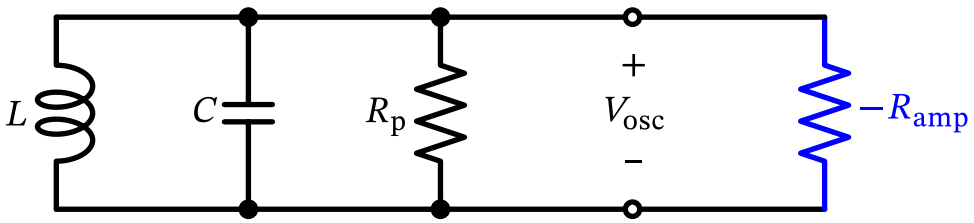


Figure 61: LC parallel tank with connected negative resistance forming an LC oscillator.

When $|-R_{\text{amp}}| = R_{\text{p}}$, the losses of the tank circuit are fully compensated, and we have a sustained oscillation at the resonance frequency $\omega_0 = 1/\sqrt{LC}$. In practice, the amplifier gain is usually set slightly higher than required for loss compensation to **start the oscillation from noise**. Then, as the oscillation amplitude increases, some **non-linear mechanism in the amplifier reduces the effective gain** until a stable oscillation amplitude is reached.

Various implementations of such LC oscillators exist, which will be discussed in the following sections.

- An alternative implementation of setting the oscillation amplitude is to use an *automatic level control* (ALC) loop, which measures the oscillation amplitude and adjusts the amplifier gain accordingly to maintain a constant output amplitude. This approach can improve the stability of the oscillation amplitude over process, voltage, and temperature variations.
- An alternative to starting the oscillation from thermal noise (which can take considerable time depending on the Q of the resonator) is to provide a known initial condition to the resonator, e.g., by pre-charging the capacitor C to a certain voltage before enabling the amplifier. This way, the oscillation can start immediately from this initial energy stored in the resonator.

As a first hint on how to optimize an LC oscillator for low phase noise, we can try to maximize Q according to Equation 55. The energy stored in the capacitor at peak voltage is

$$E_{\mathrm{stored}} = \frac{C V_{\mathrm{osc,p}}^2}{2}$$

so to maximize this we should maximize both C and the peak oscillation voltage $V_{\mathrm{osc,p}}$. On the other hand, the energy loss per cycle is related to the power dissipated in the resistor R_{p} by

$$E_{\mathrm{loss}} = \frac{V_{\mathrm{osc,p}}^2}{2R_{\mathrm{p}}} \cdot \frac{1}{f_0}$$

which we can minimize by maximizing R_{p}. In summary, to maximize Q we should use a large capacitance C, a high oscillation amplitude $V_{\mathrm{osc,p}}$, and a high tank resistance R_{p}. Note that increasing C will reduce L for a given ω_0. Calculating Q from these expressions yields

$$Q = 2\pi \frac{E_{\mathrm{stored}}}{E_{\mathrm{loss}}} = \omega_0 C R_{\mathrm{p}}$$

which confirms the above observations.

Fundamentally, if we describe the output voltage of the oscillator as

$$v_{\mathrm{osc}}(t) = V_{\mathrm{osc,p}}(t) \cdot \cos[\omega_0(t) t + \varphi(t)]$$

we want to keep the amplitude variations small, i.e., $V_{\mathrm{osc,p}}(t) = V_{\mathrm{osc,p}}$, and also the frequency variations small, i.e., $\omega_0(t) = \omega_0$. We further want to minimize any phase fluctuations, i.e., $\varphi(t) = \varphi_0$.

As a side note, the definition of Q in Equation 55 can also be understood as describing how fast the energy stored in the resonator is dissipated once the excitation is removed. Per cycle, a fraction of $2\pi/Q$ of the stored energy is lost, so the energy decays as $E(t) = E_0\, e^{-\omega_0 t/Q}$. At this rate, the stored energy would be used up after $Q/(2\pi)$ cycles, which is also the time after which it has fallen to $1/e$. For example, if $Q = 1000$ at $f_0 = 1\,\mathrm{GHz}$, this takes about 159 cycles, which is 159 ns. After $Q = 1000$ cycles (1 µs), the energy is down to $e^{-2\pi} \approx 0.19\,\%$ and the amplitude to about 4 %, so the resonator has practically rung down. Both durations are sometimes called the “ring-down time” of the resonator, so it is worth stating which one is meant.

6.1 Oscillator Noise

For calculating the noise of an oscillator, we assume that an LC-based oscillator as shown in Figure 61 is used. We assume that the oscillator is in steady-state operation, i.e., $-R_{\mathrm{amp}} = R_{\mathrm{p}}$. We can then simplify the circuit to the one shown in Figure 62.

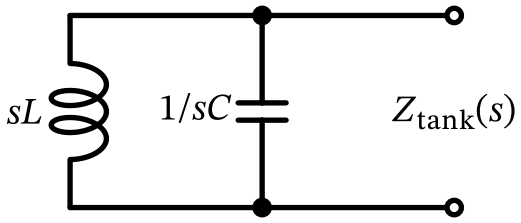


Figure 62: LC parallel tank with oscillator in steady-state operation.

We can calculate

$$Z_{\mathrm{tank}}(s) = \frac{sL\frac{1}{sC}}{sL + \frac{1}{sC}} = \frac{sL}{1 + s^2LC}.$$

At $s = j\omega_0$, we have $Z_{\mathrm{tank}}(s) \to \infty$, so let's approximate around ω_0:

$$Z_{\mathrm{tank}}(j\omega_0 + j\Delta\omega) = \frac{j(\omega_0 + \Delta\omega)L}{1 - (\omega_0 + \Delta\omega)^2LC}$$

For $\Delta\omega \ll \omega_0$, we can approximate $(\omega_0 + \Delta\omega)^2 \approx \omega_0^2 + 2\omega_0\Delta\omega$ and $\omega_0 + \Delta\omega \approx \omega_0$, which yields (using $\omega_0^2LC = 1$)

$$Z_{\mathrm{tank}}(j\omega_0 + j\Delta\omega) = -\frac{j}{2\Delta\omega C}.$$

We now use the correspondence $C = 1/\omega_0^2L$ and express $L = R_{\mathrm{p}}/\omega_0Q$ to get

$$|Z_{\mathrm{tank}}(j\omega_0 + j\Delta\omega)| = \frac{R_{\mathrm{p}}}{2Q}\left(\frac{\omega_0}{\Delta\omega}\right)$$

which provides us with an expression for the magnitude of the tank impedance around resonance. We now calculate the noise power if a noise current is injected into this impedance. We use the single-sided noise current of R_{p} (see Section 2.3.1), increased by a factor F contributed by the active circuit providing $-R_{\mathrm{amp}}$:

$$\overline{I_n^2} = \frac{4kTF}{R_{\mathrm{p}}}$$

The noise voltage across the tank at a frequency offset $\Delta\omega$ from ω_0 is then

$$\overline{V_n^2}(\Delta\omega) = |Z_{\mathrm{tank}}(j\omega_0 + j\Delta\omega)|^2 \cdot \frac{4kTF}{R_{\mathrm{p}}} = \frac{kTFR_{\mathrm{p}}}{Q^2}\left(\frac{\omega_0}{\Delta\omega}\right)^2$$

This absolute noise voltage is not of much interest per se. We normalize it to the oscillation amplitude V_{p}, and only account for 1/2 of the noise, as the total noise power is split equally

into amplitude noise and phase noise [31], and we are only interested in the **phase noise**. This is because we assume the amplitude noise is removed by amplitude clipping in the LO chain routing the oscillator signal to the mixer. This is often a valid assumption. We introduce the symbol $\mathcal{L}\{\cdot\}$ to denote this normalized phase noise of the oscillator:

$$\mathcal{L}\{\Delta\omega\} = \frac{\frac{1}{2}\frac{kTFR_\mathrm{p}}{Q^2}\left(\frac{\omega_0}{\Delta\omega}\right)^2}{\left(\frac{V_\mathrm{p}}{\sqrt{2}}\right)^2} = \frac{kTFR_\mathrm{p}}{V_\mathrm{p}^2} \cdot \frac{1}{Q^2} \cdot \left(\frac{\omega_0}{\Delta\omega}\right)^2 \tag{56}$$

This equation is known as "**Leeson's equation**" [32], and it provides us with important insights on how to design low phase noise oscillators.

At first glance Equation 56 seems to say that a *large* tank resistance R_p is bad, since it sits in the numerator. This is misleading, because Q in the denominator itself depends on R_p. Substituting $Q = \omega_0 C R_\mathrm{p}$ makes the true dependence explicit:

$$\mathcal{L}\{\Delta\omega\} = \frac{kTF}{V_\mathrm{p}^2\,\omega_0^2 C^2 R_\mathrm{p}} \cdot \left(\frac{\omega_0}{\Delta\omega}\right)^2 \tag{57}$$

so for a given tank capacitance C the phase noise indeed *falls* with increasing R_p, i.e., with a lower-loss tank. With this in mind, the design guidelines are:

1. Use a resonator with high quality factor Q to reduce phase noise. On-chip LC tanks usually provide moderate Q values (e.g., 10 to 30), while off-chip crystals can provide very high Q values (e.g., 10,000 to 100,000).
2. Maximize the oscillation amplitude V_p to increase the stored energy in the resonator. In CMOS implementations, this is often limited by the supply voltage and/or device breakdown voltages.
3. Use an active circuit with low noise factor F to minimize the noise contribution of the negative resistance $-R_\mathrm{amp}$. A good value to aim for is $F \approx 2$.
4. Maximize the tank resistance R_p to minimize the thermal noise contribution. This can be achieved by using high-Q inductors and low-loss capacitors.

The phase noise $\mathcal{L}\{\Delta\omega\}$ is expressed in the unit of dBc/Hz, i.e., in decibels of phase noise relative to the carrier power per 1 Hz of bandwidth. When expressing the phase noise of an oscillator in dBc/Hz it is important to state both the oscillation frequency ω_0 and the offset frequency $\Delta\omega$ at which the phase noise is evaluated. For example, we could say that an oscillator has a phase noise of –137 dBc/Hz at 3 MHz offset from a carrier frequency of 2 GHz.

The phase noise expressed with Equation 56 describes an important region of the total oscillator phase noise where the phase noise decreases with $1/(\Delta\omega)^2$, i.e., with 20 dB per decade. This region is often called the "thermal noise region" as it is dominated by thermal noise from the tank resistor and the active circuit. However, at lower offset frequencies, other noise mechanisms (like flicker noise) can dominate, leading to different slopes of the phase noise vs. offset frequency curve. At larger offset frequencies, the phase noise can flatten out due to thermal noise floor limitation of buffer amplifiers following the oscillator. A typical phase noise plot of an LC oscillator is shown in Figure 63.

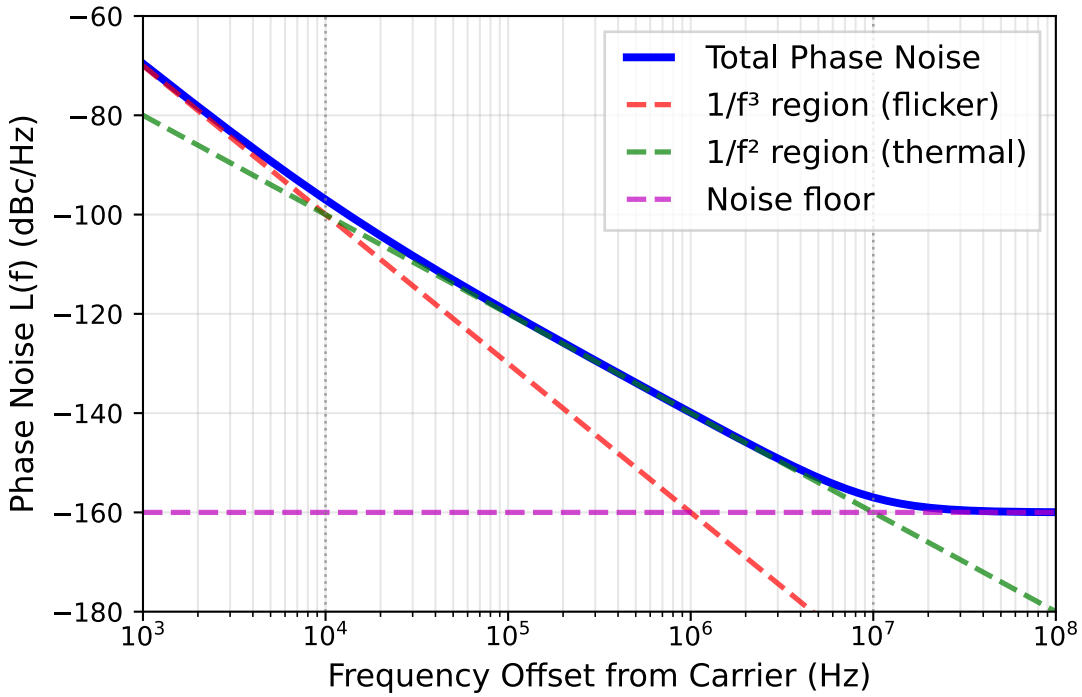


Figure 63: Phase noise spectrum of a typical LC oscillator showing characteristic $1/f^3$ and $1/f^2$ slopes vs. frequency offset from carrier. The flicker noise corner (where the $1/f^3$ region transitions to the $1/f^2$ region) is around 10 kHz, and the thermal noise floor onset is marked at 10 MHz.

It is important to note that the phase noise does not go to infinity as $\Delta\omega$ approaches zero, as Equation 56 might suggest. This is an artifact of the simplified model used to derive Leeson's equation. In reality, the phase noise has the form of a Lorentzian spectrum around the carrier frequency, which means that it flattens out at very low offset frequencies with a trend described by

$$\mathcal{L}\{\Delta\omega \ll\} \propto \frac{1}{\omega_{\mathrm{B}}^2 + \Delta\omega^2}$$

with ω_{B} being the half-power bandwidth of the oscillator spectrum.

Why is phase noise important?

1. In RX and TX, the phase noise of the LO creates jitter, which degrades signal quality and contributes to EVM degradation.
2. In TX, phase noise of the LO creates noise sidebands around the carrier, which can lead to adjacent channel interference together with spectral regrowth due to nonlinearities and thermal noise.
3. In RX, reciprocal mixing of the phase noise sidebands of the LO with strong adjacent channel signals creates in-band noise, which degrades the SNR and sensitivity of the RX.
4. In systems using multiple carriers or carriers with high-order modulation schemes (like in OFDM), phase noise can lead to inter-carrier interference (ICI) and symbol misinterpretation, further degrading system performance.

> **Note 7: Phase Noise and Frequency Division**
>
> When using a frequency divider to generate a lower-frequency LO signal from a higher-frequency oscillator (e.g., to generate IQ phases), the phase noise of the divided signal is affected by the division ratio N [33]. Specifically, the phase noise of the divided signal improves by $20\log_{10}(N)$ dB compared to the original oscillator phase noise. This means that if you divide the frequency by a factor of 2, the phase noise will be lowered by 6 dB!
>
> The statement above is related to the phase noise at the output coming from the input of the divider. Of course, the frequency divider itself also contributes some additional phase noise, which can degrade the overall phase noise performance. However, in many practical cases, the improvement due to frequency division outweighs the additional noise introduced by the divider.

6.2 Jitter

Jitter is a time-domain representation of phase noise. It quantifies the timing variations of a periodic signal, such as the zero-crossings or rising edges of a clock signal. Jitter is typically measured in seconds (or fractions thereof) and can be expressed as root mean square (RMS) jitter or peak-to-peak jitter. The relationship between phase noise and RMS jitter can be approximated by the following equation [33]:

$$\sigma_{\mathrm{a}} = \frac{1}{2\pi f_0}\sqrt{2\int_{f_1}^{f_2}\mathcal{L}\{f\}df} \tag{58}$$

where σ_{a} is the absolute RMS jitter, f_0 is the carrier frequency, and the integral is taken over the frequency offset range from f_1 to f_2. This equation shows that higher phase noise levels lead to increased jitter, which can adversely affect the performance of digital communication systems by causing timing errors and signal integrity issues. Note that the phase noise $\mathcal{L}\{f\}$ in Equation 58 must be expressed in linear scale (not dBc/Hz) when using this equation.

6.3 Reciprocal Mixing

Reciprocal mixing is a phenomenon that occurs in receivers when the phase noise sidebands of the local oscillator mix with strong adjacent channel signals, resulting in in-band noise that degrades the signal-to-noise ratio of the desired signal. This effect is particularly pronounced in systems with high-order modulation schemes or closely spaced channels, where even small amounts of phase noise can lead to significant performance degradation.

To analyze reciprocal mixing, we consider a scenario where a strong interferer is present at a frequency offset Δf from the desired signal. The phase noise of the local oscillator at this offset frequency can be characterized by its power spectral density $\mathcal{L}\{\Delta f\}$. When the local oscillator mixes with the interferer, the phase noise sidebands effectively "fold" into the desired signal band, creating additional noise. This noise level can be estimated by multiplying the power of the interferer by the phase noise level at the offset frequency and considering the channel bandwidth B of the RX

$$P_{\mathrm{RM}} = P_{\mathrm{interferer}} + \mathcal{L}\{\Delta f\} + 10\log_{10}(B) \tag{59}$$

where P_{RM} is the power of the reciprocal mixing noise introduced into the desired signal band.

Note 8: Reciprocal Mixing Example

Let us assume the following example from a Bluetooth LE receiver: The sensitivity target for the RX is −70 dBm, and we assume a required SNR of 10 dB in a 1 MHz bandwidth (the same assumptions as in Note 3; the Bluetooth LE specification itself defines the sensitivity via a bit error rate of 0.1 %, not via an SNR). An adjacent channel interferer is present at −27 dB channel to interferer ratio at an offset of 3 MHz. How much phase noise can the LO have to meet the sensitivity target?

From the sensitivity target and the SNR requirement, we can calculate the maximum allowable noise floor in the RX. We add a margin of 3 dB to account for implementation losses:

$$\begin{aligned} P_{\text{noise,max}} &= P_{\text{sens}} - \text{SNR} - P_{\text{margin}} \\ &= -70\,\text{dBm} - 10\,\text{dB} - 3\,\text{dB} = -83\,\text{dBm} \end{aligned}$$

The interferer power is:

$$\begin{aligned} P_{\text{interferer}} &= P_{\text{sens}} - \text{C/I} \\ &= -70\,\text{dBm} + 27\,\text{dB} = -43\,\text{dBm} \end{aligned}$$

Using Equation 59, we can express the maximum allowable phase noise at 3 MHz offset at 2.4 GHz as:

$$\begin{aligned} \mathcal{L}\{3\,\text{MHz}\} &= P_{\text{noise,max}} - P_{\text{interferer}} - 10\log_{10}(B/\text{Hz}) \\ &= -83\,\text{dBm} - (-43\,\text{dBm}) - 60\,\text{dB} = -100\,\text{dBc/Hz} \end{aligned}$$

The reciprocal mixing scenario described in Note 8 is visualized in Figure 64.

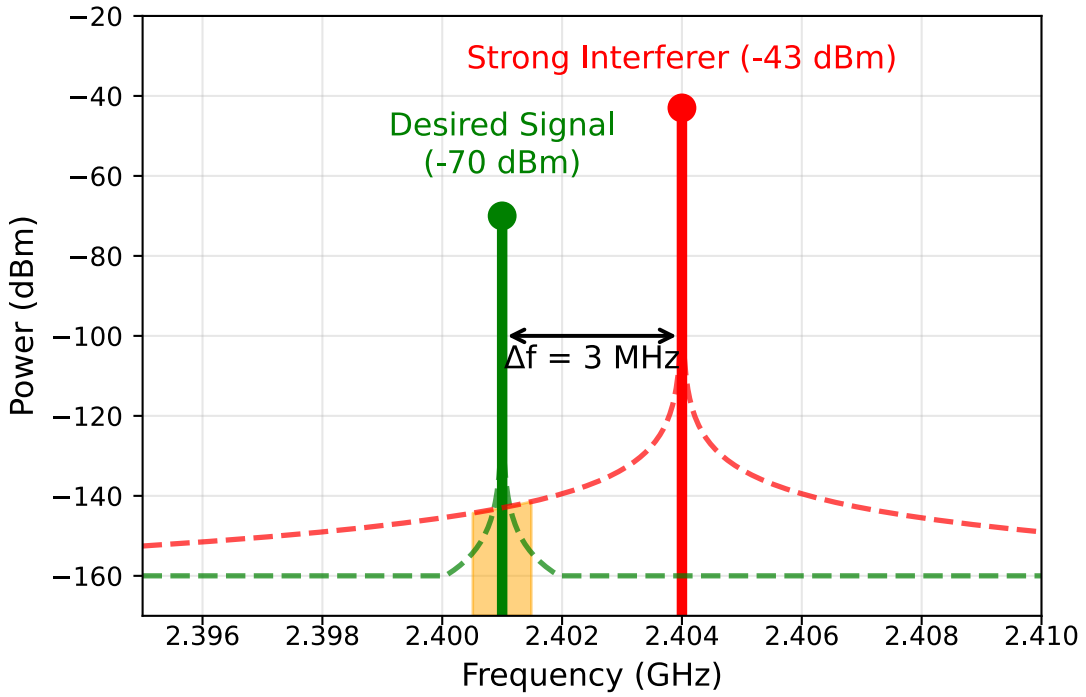


Figure 64: Reciprocal mixing in a receiver: A strong interferer at offset frequency mixes with LO phase noise sidebands, creating in-band noise that degrades the desired signal SNR. The desired signal shows the same, scaled phase noise profile around it as the interferer, as both mix with the same LO phase noise.

6.4 Single-Ended Oscillators

Single-ended oscillators are commonly used in RF applications due to their simplicity and ease of integration, especially in quartz oscillators. A negative resistance is implemented using a single transistor amplifier as is shown in Figure 65. The transistor is configured with two capacitors C_1 and C_2 to provide a phase-shifted feedback path.

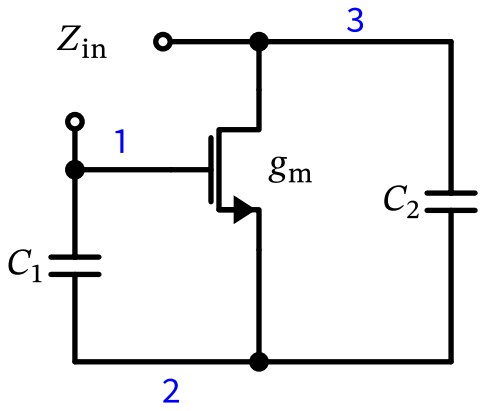


Figure 65: Circuit diagram of a single-ended negative resistance implementation.

It can be shown that the differential input impedance looking into the transistor gate and drain is given by

$$Z_{\text{in}}(s) = -\frac{g_{\text{m}}}{\omega^2 C_1 C_2} + \frac{C_1 + C_2}{s C_1 C_2} = -R_{\text{amp}} + \frac{1}{s C_{\text{amp}}}$$

which consists of the series combination of a negative resistance $-R_{\text{amp}}$ and a capacitance C_{amp}. Note that the circuit in Figure 65 does not show a ground symbol. In fact, any of the three nodes marked with blue numbers can be used as a reference node (ground), and this results in the following well-known oscillator topologies, summarized in Table 5.

Table 5: The three "three-point" oscillator topologies obtained by grounding one node of Figure 65

Reference Node	Amplifier Configuration	Common Name
Node 1 (Gate)	common-gate	grounded-gate (also called common-base Colpitts)
Node 2 (Source)	common-source	Pierce oscillator
Node 3 (Drain)	common-drain	Colpitts oscillator

The important insight is that the negative resistance $-R_{\text{amp}}$ is *identical* in all three cases: choosing a different reference node does not change the oscillation mechanism at all. What it does change is which node serves as the output, how the transistor is biased, and how supply and substrate noise couple into the resonator — which in practice is what decides the choice.

> **i** A Note on Oscillator Naming
>
> The naming in Table 5 is not used consistently across the literature. The grounded-source case is universally called the Pierce oscillator, but *both* the grounded-gate and the grounded-drain variants are referred to as "Colpitts" by different authors (they differ only in which of the two capacitors provides the feedback), and some crystal-oscillator literature treats the grounded-drain circuit as just another implementation of the Pierce oscillator [34].
>
> What is *not* a grounding variant is the **Clapp oscillator**: it is a Colpitts with an additional capacitor placed **in series with the inductor**, introduced to keep the transistor's parasitic capacitances from pulling the frequency [35]. Interestingly, a quartz crystal oscillator gets this for free: as we will see in Figure 66, the motional branch of a crystal is L_{m} in series with a very small C_{m}, so C_{m} plays exactly the role of Clapp's series capacitor. This is the very reason why changes of the surrounding capacitances (e.g., parasitic or transistor capacitances) move the frequency by only a few ppm (see Equation 61), and thus why quartz oscillators are so stable.

When we investigate the equivalent electrical circuit of a quartz crystal, we find that it has inductive behavior between its series resonance frequency and its parallel resonance frequency, which are very close together. The quartz crystal equivalent circuit is shown in Figure 66.

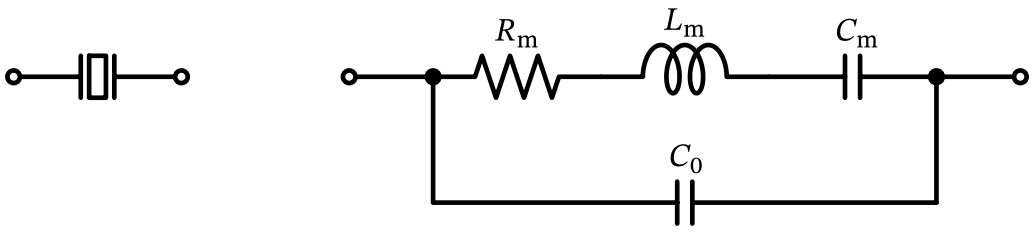


Figure 66: Quartz crystal equivalent circuit.

This means we can operate the quartz crystal as a high-Q inductor in an oscillator circuit, which results in the single-ended quartz crystal oscillator shown in Figure 67, which is a very popular choice for high-performance crystal oscillators. For an in-depth treatment of this circuit, including its start-up behavior and amplitude regulation, we recommend [34]. Note that in a simple implementation the current-bias MOSFET can be replaced by an inverter stage biased in the linear region.

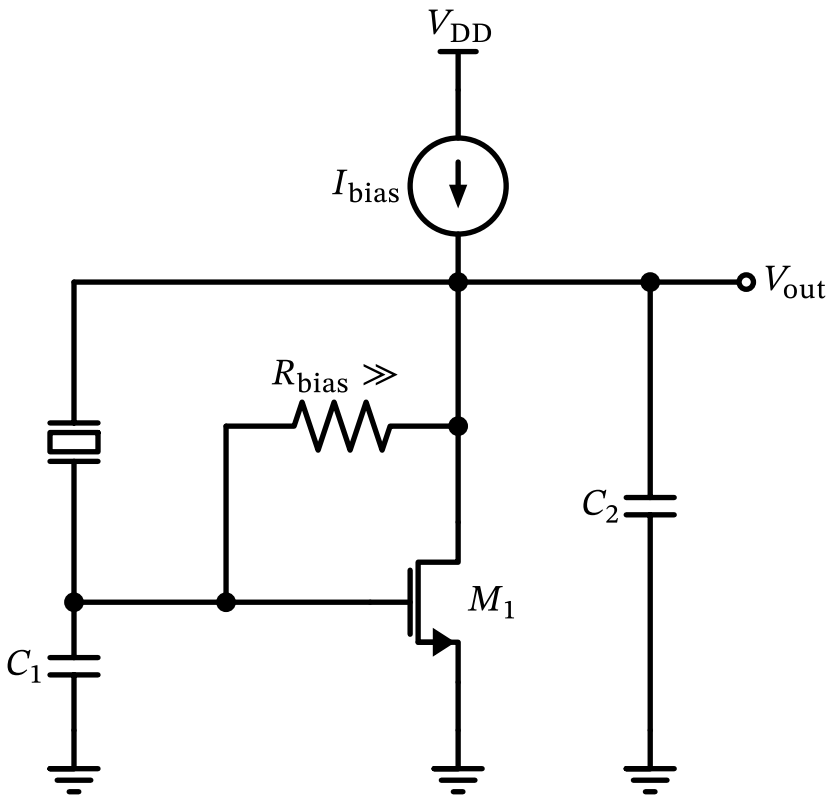


Figure 67: Circuit diagram of a single-ended Pierce crystal oscillator operating the quartz between series and parallel resonance where it acts as a large high-Q inductor. Note that the quartz crystal has no dc path, hence we need a high-ohmic bias resistor to connect M1 into a diode configuration.

The series combination of C_1 and C_2 provides the load capacitance required by the crystal to oscillate at its specified frequency. Note that the oscillation frequency is *not* set by C_load

resonating with the motional inductance L_{m}: since the motional capacitance is tiny (C_{m} is in the femtofarad range, while C_{load} is in the picofarad range), the frequency is set almost entirely by the crystal itself. The oscillator runs slightly *above* the series resonance frequency

$$f_{\mathrm{s}} = \frac{1}{2\pi\sqrt{L_{\mathrm{m}} C_{\mathrm{m}}}}, \tag{60}$$

pulled up by the load capacitance according to ($C_{\mathrm{load}}^{-1} = 1/C_1 + 1/C_2$, and C_0 being the parallel plate capacitance of the crystal)

$$f_0 \approx f_{\mathrm{s}}\left[1 + \frac{C_{\mathrm{m}}}{2(C_0 + C_{\mathrm{load}})}\right]. \tag{61}$$

The resulting total *pulling* away from f_{s} is only in the range of some tens to hundreds of ppm, while a small change of the load capacitance moves the frequency according to the sensitivity

$$\frac{1}{f_{\mathrm{s}}}\frac{\partial f_0}{\partial C_{\mathrm{load}}} \approx -\frac{C_{\mathrm{m}}}{2(C_0 + C_{\mathrm{load}})^2}.$$

For typical values of $C_{\mathrm{m}} = 5\,\mathrm{fF}$, $C_0 = 2\,\mathrm{pF}$, and $C_{\mathrm{load}} = 10\,\mathrm{pF}$, the oscillator runs about 208 ppm above f_{s}, but a change of C_{load} moves it by only about 17 ppm/pF, so 0.1 pF of parasitic capacitance shifts the frequency by less than 2 ppm. Since the total pulling is nevertheless significant, crystal data sheets specify a nominal load capacitance: the designer trims C_1 and C_2 (often with an on-chip switched capacitor bank) to hit the specified frequency, and can use the same knob to compensate for aging and temperature drift.

6.5 Differential Oscillators

After discussing single-ended oscillators in Section 6.4, we now turn to differential oscillator topologies, which are widely used in integrated LC oscillators due to their superior common-mode noise rejection and reduced even-order harmonics. A popular way to create a differential negative resistance is to use a cross-coupled pair of transistors as shown in Figure 68.

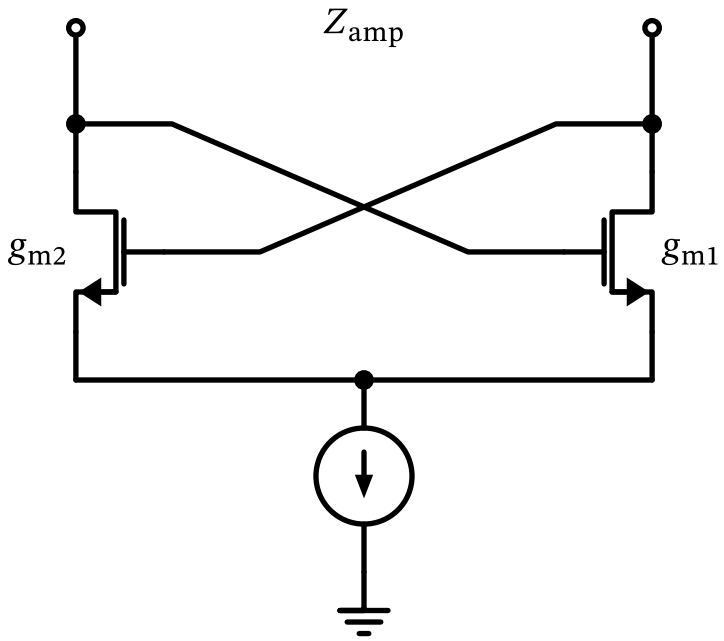


Figure 68: A cross-coupled differential pair.

We assume a symmetrical circuit by setting $g_{\text{m1}} = g_{\text{m2}} = g_{\text{m}}$. By analyzing the small-signal equivalent circuit, we can derive the differential input impedance looking into the gates of the transistors:

$$Z_{\text{in}} = -\frac{2}{g_{\text{m}}}$$

A practical implementation of a differential LC oscillator using the cross-coupled pair is shown in Figure 69 using a so-called “NMOS core”, as it is using an NMOS-based differential pair.

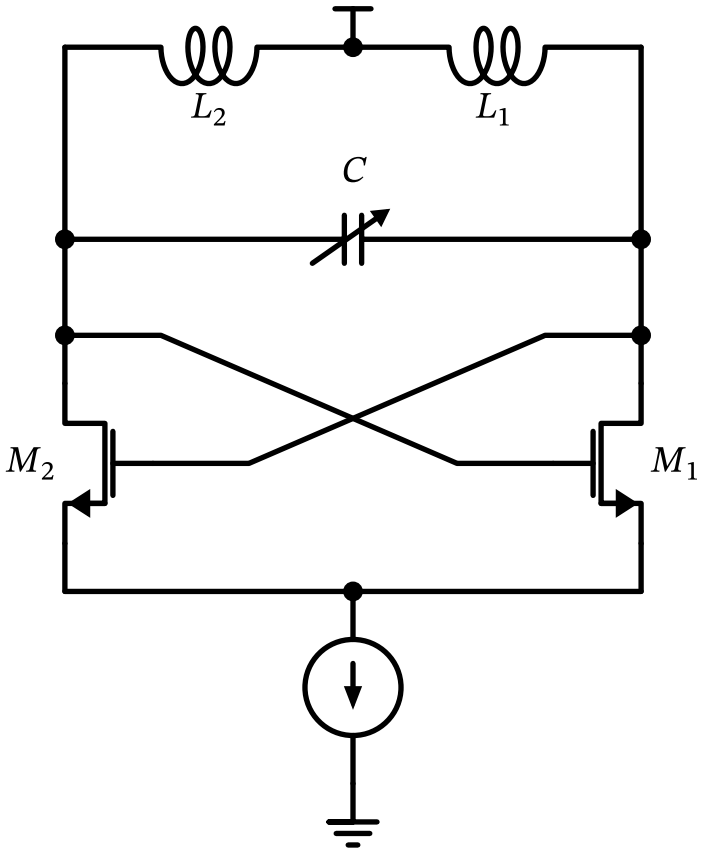


Figure 69: An LC differential oscillator using an NMOS cross-coupled differential pair. L_1 and L_2 are usually implemented as a single on-chip spiral inductor with a center tap.

By putting the circuit in Figure 69 on its head, we obtain a "PMOS core" oscillator, which is also widely used in integrated LC oscillators. This configuration is shown in Figure 70.

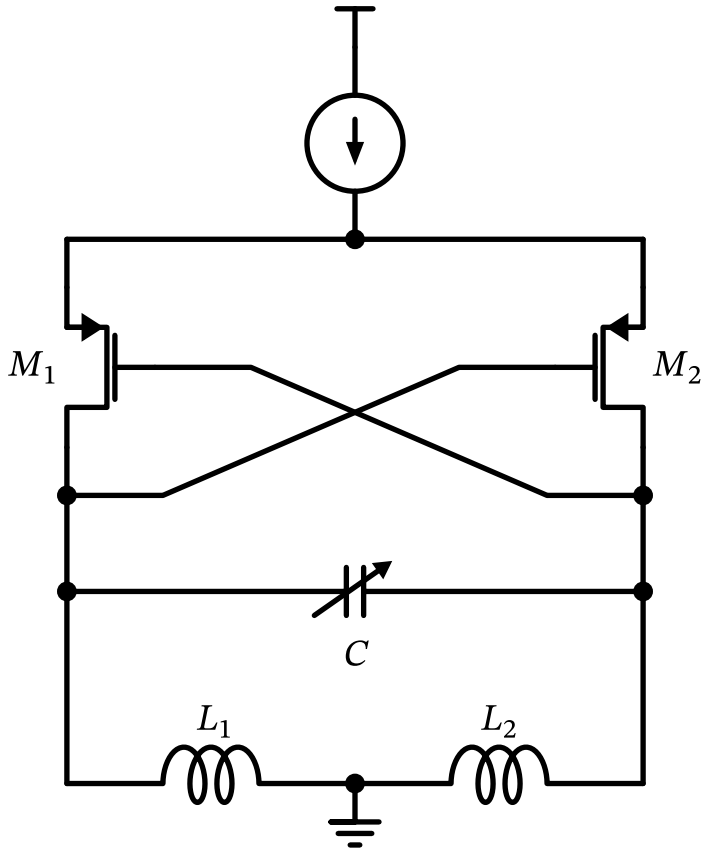


Figure 70: An LC differential oscillator using a PMOS cross-coupled differential pair. L_1 and L_2 are usually implemented as a single on-chip spiral inductor with a center tap.

In both oscillator topologies, the oscillation frequency is determined by the LC tank circuit formed by $L = L_1 + L_2$ and C. The oscillation frequency can be approximated by

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

As both topologies have their inductor center tap tied to a supply rail (V_{DD} for the NMOS core, V_{SS} for the PMOS core), the voltage swing across the tank can go well beyond the supply voltage, which is an advantage of these topologies. However, note that the maximum voltage swing is still limited by the device breakdown voltages of M_1 and M_2, which can be critical in advanced CMOS technologies with low breakdown voltages. To utilize both an NMOS and a PMOS cross-coupled pair in parallel, a so-called "complementary (or CMOS) LC oscillator" can be used, which is shown in Figure 71. Here, the voltage swing across the tank is limited to the supply voltage, so it is inherently safe regarding device breakdown. Also, the transconductance of both NMOS and PMOS devices contributes to the negative resistance, which can reduce power consumption for a given oscillation amplitude.

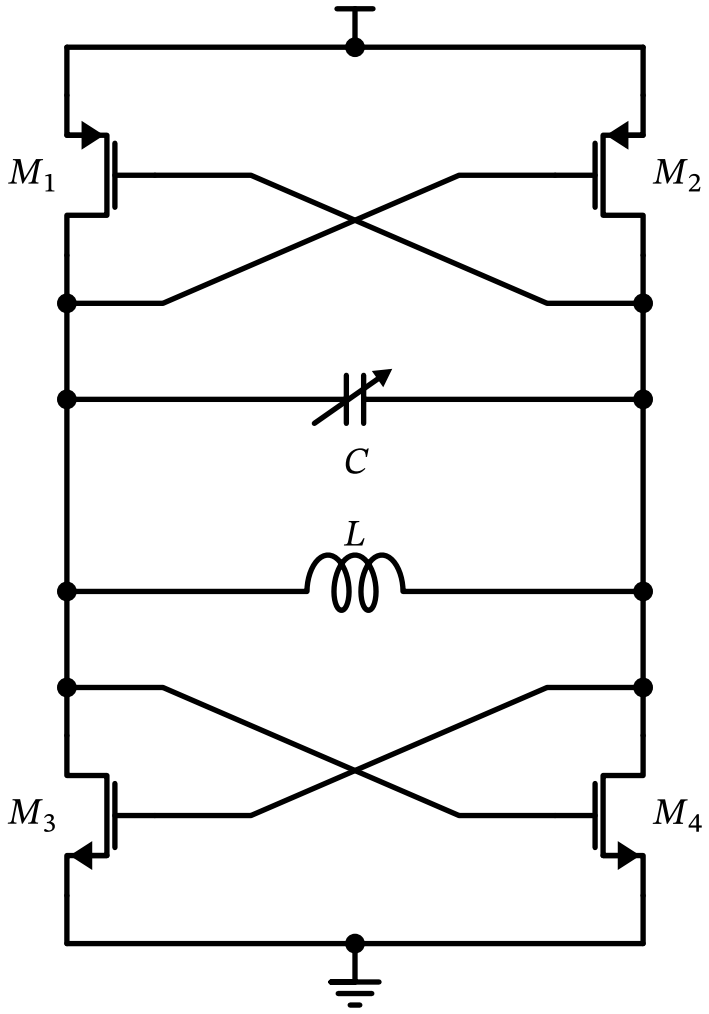


Figure 71: An LC differential oscillator using a PMOS and an NMOS cross-coupled differential pair.

Note that no bias current sources are shown in Figure 71, but can be added in practice. However, they require some voltage headroom, which lowers the maximum voltage swing across the tank, which is negative for phase noise performance according to Leeson's equation in Equation 56.

The inductor L used in LC tanks is often implemented as a differential spiral inductor, which can be realized on-chip using the top metal layers of the CMOS process. If higher Q is sought, then off-chip inductors must be used, which is more expensive and uses package pins to connect the inductor. Alternatively, bondwire inductors can be used, which utilize the bondwires connecting the die to the package leads as inductors. Bondwire inductors can provide high Q values and are a cost-effective solution for improving oscillator performance without requiring off-chip components. For mm-wave frequencies, transmission line stubs can be used as inductors, which can be implemented on-chip using coplanar waveguide (CPW) or microstrip structures.

6.6 Frequency Tuning of Oscillators

Note that all three LC oscillator topologies shown in Section 6.5 have a fixed oscillation frequency determined by the LC tank circuit. In practice, it is often desirable to be able to tune the oscillation frequency over a certain range. There are only two ways to achieve this, either by tuning the inductance L or the capacitance C of the tank circuit. In integrated LC oscillators, it is common to use a fixed inductor and a tunable capacitor, which is shown in Figure 69, Figure 70, and Figure 71.

The need to change the oscillator frequency can arise from various requirements:

1. We need to precisely tune the oscillator frequency to match a desired carrier frequency.
2. The oscillator frequency needs to be adjusted to compensate for process, voltage, and temperature (PVT) variations that can affect the resonant frequency of the LC tank.

In order to achieve a tunable capacitance, a so-called "varactor" (variable capacitor) is often used. A varactor is a semiconductor device that exhibits a capacitance that varies with the applied voltage. Common ways to implement a varactor in integrated circuits are:

1. Use the gate-to-channel capacitance of a MOSFET operated in reverse bias. By changing the voltage applied to the gate, the depletion region width in the channel changes, which in turn changes the capacitance.
2. Use a PN-junction diode operated in reverse bias. The depletion region width of the diode changes with the applied voltage, leading to a change in capacitance.
3. Use the gate-to-channel capacitance of a special MOSFET structure in accumulation mode, which can be achieved by using an NMOS situated in an n-well (in contrast to the usual operation in a p-well). Changing the gate-to-channel voltage changes the accumulation capacitance [36].
4. Use a switched capacitor bank, where multiple capacitors are connected in parallel or series using switches (usually MOSFETs) to achieve discrete capacitance values.

In order to characterize the tuning sensitivity of an oscillator, the metric K_{VCO} (voltage-controlled oscillator gain) is often used, which is defined as

$$K_{\mathrm{VCO}} = \frac{df_0}{dV_{\mathrm{tune}}}$$

with V_{tune} being the control voltage applied to the varactor. The unit of K_{VCO} is usually expressed in MHz/V or GHz/V. A high K_{VCO} means that a small change in tuning voltage results in a large change in oscillation frequency, which can be beneficial for wide tuning ranges but can also make the oscillator more sensitive to noise on the tuning voltage line.

! Watch the Units of K_{VCO}!

Data sheets and measurements state K_{VCO} as a *frequency* sensitivity in MHz/V or GHz/V, as defined above. All loop equations that follow (in Section 6.7 and throughout Section 7) instead need the *angular* frequency sensitivity in rad/(s·V), which is larger by a factor of 2π:

$$K_{\mathrm{VCO}}\big|_{\mathrm{rad/(s\cdot V)}} = 2\pi \cdot K_{\mathrm{VCO}}\big|_{\mathrm{Hz/V}}$$

Forgetting this factor is one of the most common sources of error in PLL loop design: since the natural frequency scales as $\omega_{\mathrm{n}} \propto \sqrt{K_{\mathrm{VCO}}}$, a missing 2π moves ω_{n} by a factor of $\sqrt{2\pi} \approx 2.5$, and it moves the damping factor by the same factor (in one direction or the other, depending on the loop filter used). Since both change together, the $-3\,\mathrm{dB}$ loop bandwidth moves by even more: for a type-II charge-pump PLL (Section 7.2) designed for $\zeta = 0.707$, a K_{VCO} that is 2π larger than assumed increases the loop bandwidth by about $4.7\times$. In the remainder of these notes, K_{VCO} inside loop equations is always understood to be in rad/(s·V).

Note that the tuning sensitivity K_{VCO} is usually quite nonlinear over the tuning range, so it is common to specify K_{VCO} at a certain operating point or as an average value over the tuning range.

In order to achieve a wide tuning range while maintaining a sufficiently small K_{VCO} for phase noise reasons, a combination of coarse and fine tuning mechanisms can be used [37]. For example, a switched capacitor bank can provide coarse tuning steps, while a varactor can provide fine tuning within each step. It is important to ensure that the overall tuning range is free of dead zones, where the oscillator cannot be tuned to certain frequencies due to non-overlapping tuning ranges of the coarse and fine tuning elements.

While there are many ways to implement a switched capacitor for use in an oscillator tuning circuit, one popular way is shown in Figure 72. Many such switched capacitors with different values of C (often binary weighted) can be combined to form a capacitor bank for coarse frequency tuning.

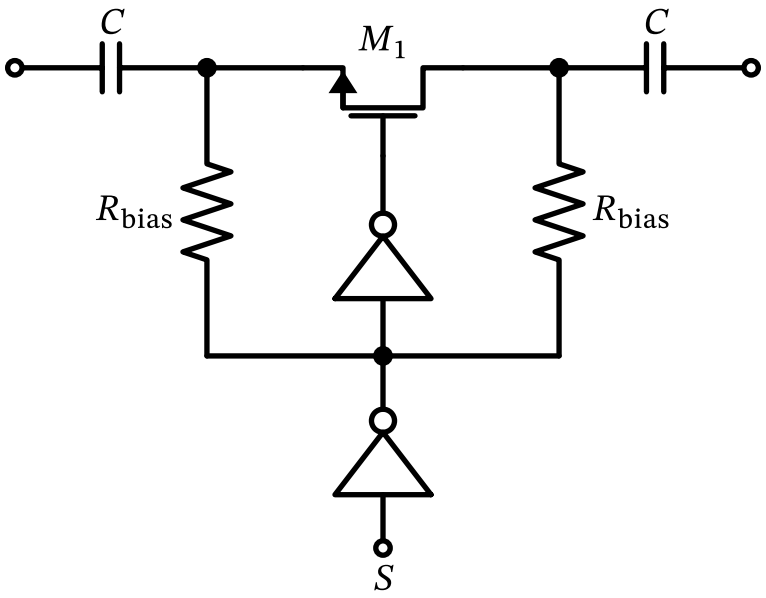


Figure 72: A switched capacitor for use in an oscillator's switched-capacitor tuning bank. The bias resistors tie the drain/source nodes to ground during turn-on of M_1 (for low on resistance), while they tie the drain/source nodes to VDD during turn-off of M_1 to prevent accidental turn-on of the drain/source to bulk diodes of M_1.

Here, when the control signal S is high, the effective capacitance is $C/2$, with a parasitic series resistance R_{on} due to the switch. When S is low, the effective capacitance is C_{off}, which is the parasitic drain-source capacitance of the MOSFET switch.

Note that there exists a trade-off when designing the switched capacitor: A larger switch (with increased W) reduces R_{on}, but increases C_{off}. A large R_{on} leads to increased losses in the tank circuit, which degrades the quality factor Q and increases phase noise according to Equation 56. On the other hand, a large C_{off} reduces the effective tuning range of the switched capacitor, which can be detrimental if a wide tuning range is required.

In **digitally-controlled oscillators (DCOs)**, the tuning voltage V_{tune} is replaced by a digital control word that selects different capacitance values from a capacitor bank. This approach allows for precise and repeatable frequency tuning, which is beneficial in applications requiring frequency synthesis or channel selection. These fine tuning steps can also be implemented according to Figure 72; however, the value of C must be sufficiently small.

6.7 Oscillator Modelling

The instantaneous frequency of an oscillator can be expressed as

$$\omega_{\text{VCO}}(t) = \omega_0 + K_{\text{VCO}} \cdot V_{\text{tune}}(t)$$

where ω_0 is the nominal oscillation frequency, K_{VCO} is the tuning sensitivity, and $V_{\text{tune}}(t)$ is the tuning voltage applied to the varactor. Looking at the instantaneous phase $\varphi_{\text{VCO}}(t)$ of the oscillator, we can integrate the instantaneous frequency $\omega_{\text{VCO}}(t)$ to obtain

$$\varphi_{\mathrm{VCO}}(t) = \int_0^t \omega_{\mathrm{VCO}}(\tau)d\tau = \omega_0 t + K_{\mathrm{VCO}} \int_0^t V_{\mathrm{tune}}(\tau)d\tau.$$

Inspecting this equation, we see that with respect to phase, an oscillator is a **perfect** integrator of the tuning voltage over time! For simulation purposes, we can therefore model an oscillator as an integrator block as shown in Figure 73.

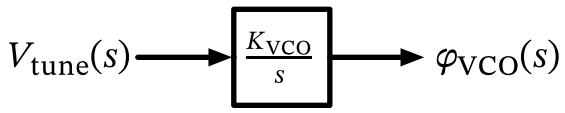


Figure 73: A model of a VCO as a perfect integrator for the excess phase in the s-domain.

In this model we look at the phase output of the oscillator as an *excess phase* with respect to the nominal phase $\omega_0 t$. This delta phase is obtained by integrating the tuning voltage $V_{\mathrm{tune}}(t)$ scaled by K_{VCO}.

7 Phase-Locked Loops

Thinking about oscillators, we face a dilemma:

- Good phase noise performance is reached with high to very high Q in the resonator (see Equation 56).
- Oscillator tunability requires a tunable resonator, which usually results in a low-to-moderate Q, unfortunately.
- Oscillators have inherent frequency stability issues due to temperature variations, device aging, and supply voltage fluctuations. However, for wireless communication systems, precise frequency control and a very stable LO (down to a few ppm of accuracy) are required.

7.1 Basic PLL Architecture

We solve the issues above by using a **phase-locked loop (PLL)** to stabilize and control the frequency of a tunable oscillator, mostly in the form of a VCO. A PLL is a feedback control system that locks the phase of the VCO output to the phase of a reference signal, typically generated by a crystal oscillator with excellent frequency stability. The block diagram of a basic PLL is shown in Figure 74.

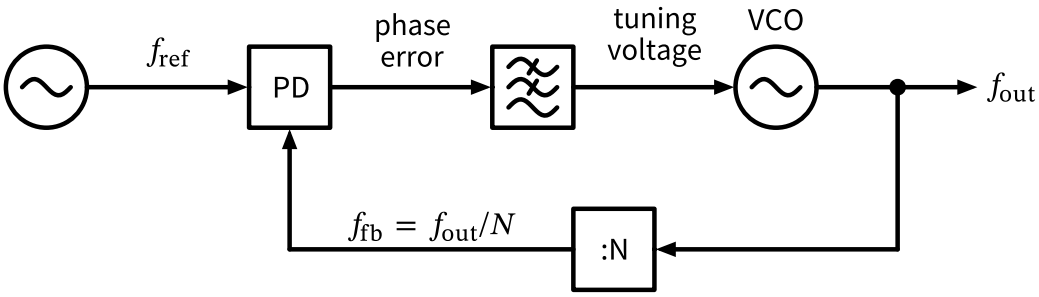


Figure 74: Block diagram of a PLL. A reference phase signal is compared to the phase of a VCO output signal in a phase detector. The phase error is low-pass filtered and used to tune the VCO frequency.

The phase of a reference signal (often coming from a high-Q crystal oscillator) is compared to the phase of a VCO output signal in a phase detector (PD). The phase error is low-pass filtered and used to tune the VCO frequency. By continuously adjusting the VCO based on the phase difference, the PLL ensures that the VCO output remains synchronized with the reference signal, effectively combining the stability of the reference with the tunability of the VCO. When the output signal of the VCO is fed back to the phase detector through a frequency divider (with division ratio N), the PLL can generate output frequencies that are integer multiples of the reference frequency, given by $f_{\text{out}} = N \cdot f_{\text{ref}}$.

The PD compares the phase of the reference signal with the phase of the VCO output signal. It works in the time domain and produces an output voltage proportional to the arrival time difference (phase difference) between the two input signals. This output voltage, known as the phase error signal, indicates whether the VCO is leading or lagging the reference signal in phase, expressed as

$$\Delta\varphi = 2\pi \frac{\Delta t}{T_{\text{ref}}}$$

and

$$V_{\text{error}} = K_{\text{PD}} \cdot \Delta\varphi$$

where $\Delta\varphi$ is the phase difference (caused by the arrival time difference) in rad, K_{PD} is the phase detector gain in V/rad, Δt is the time difference between the two signals, and T_{ref} is the period of the reference signal.

An important consideration of the PD is its linear operating range. The PD can only provide a linear output voltage over a limited window of phase differences. For an XOR gate (or a multiplier used as a PD) this window is π wide: the characteristic rises linearly from $\Delta\varphi = 0$ to $\Delta\varphi = \pi$ and then folds back, so the loop operates around a quiescent point of $\Delta\varphi = \pi/2$ (quadrature) with a usable range of only ±90° around it. If the phase difference exceeds this range, the PD characteristic "wraps around," leading to ambiguity in the phase error signal. This can cause the PLL to lose lock or behave unpredictably. Therefore, the PLL design must ensure that the phase difference remains within the linear range of the PD during normal operation, which is especially troublesome if the output frequency f_{out} is not yet close enough

to its final steady-state value. Note that a simple PD can be implemented by using an XOR gate for digital signals or a mixer for analog signals.

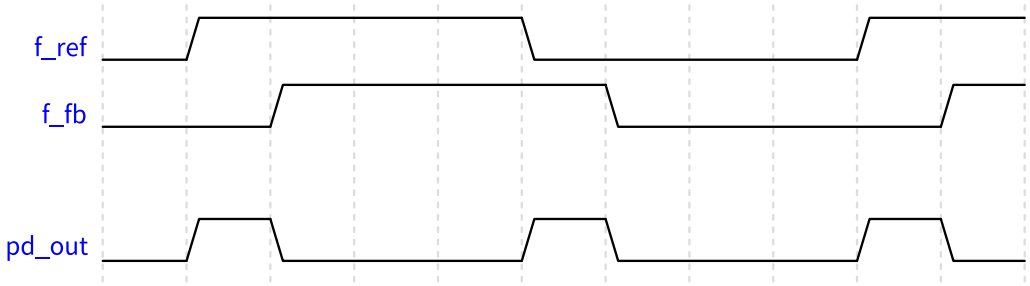


Figure 75: Input and output waveforms of an XOR-based phase detector.

We can see in Figure 75 that the output of the XOR PD is high when the two input signals differ and low when they are the same. The output duty cycle is proportional to the phase difference between the two input signals. Note that this PD has no notion of frequency, only phase! Still, it can be used to lock the frequency of the VCO to the reference frequency, as a constant phase difference implies equal frequencies; however, this works only if the initial frequency difference is small enough. Also, this XOR-based PD does not evaluate the edges of the input signals, only their logic states. Therefore, it is crucial that both input signals are square waves with a 50% duty cycle to ensure proper operation of the PD.

i JK Flip-Flop Phase Detector

Instead of an XOR gate, which requires 50% duty cycle input signals, an edge-triggered JK flip-flop can be used. While the overall behavior is similar, the JK flip-flop-based PD is insensitive to duty cycle variations of the input signals, as it only evaluates the rising edges of the reference and feedback signals [38].

As the output signal of the PD contains high-frequency components (at least at twice the reference frequency), a low-pass filter (LPF) is used to smooth the phase error signal before it is applied to the VCO tuning input. The LPF also determines the dynamic response of the PLL, affecting its stability and transient behavior. A well-designed LPF ensures that the PLL can quickly respond to changes in the reference signal or disturbances while maintaining stability and minimizing overshoot or oscillations in the output frequency. Often, the LPF has voltage-mode inputs and outputs to connect to the PD and VCO, respectively.

Using a digital divider in the feedback path to only pass every N-th cycle of the VCO output to the PD allows the PLL to generate output frequencies that are integer multiples of the reference frequency. This is particularly useful in applications such as frequency synthesis, where a wide range of frequencies is required from a single stable reference source. By adjusting the division ratio $N \in \mathbb{N}$, the PLL can produce various output frequencies while maintaining phase lock with the reference signal. However, note that the frequency resolution of the output frequencies Δf_{out} is limited to integer multiples of the reference frequency!

Putting everything together into an s-domain block diagram, the PLL can be modeled as shown in Figure 76.

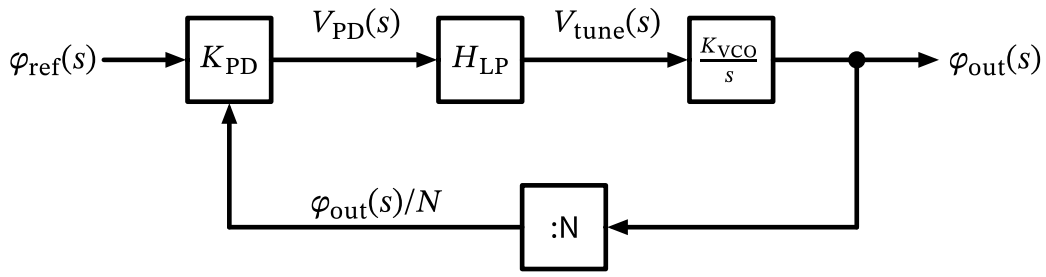


Figure 76: Laplace domain model of a PLL. Note that input and output signals are (excess) phase in rad.

For the loop filter transfer function $H(s)$, we assume a simple first-order low-pass filter with a cutoff frequency of $\omega_{\text{LP}} = 1/T_{\text{LP}}$ given by

$$H_{\text{LP}}(s) = \frac{1}{1 + sT_{\text{LP}}}.$$

! Why Regulate Phase Instead of Frequency (PLL vs. FLL)?

Before we proceed, it is worth discussing why PLLs regulate the phase of the VCO output rather than its frequency directly. The reason lies in the relationship between phase and frequency: Frequency is the time derivative of phase. By controlling the phase, the PLL inherently controls the frequency as well. Additionally, even if the phase regulation has a steady-state phase error φ_{err}, the frequency error in steady state is **zero**, as can be appreciated when inspecting

$$\omega_{\text{out}}(t) = \frac{d\varphi_{\text{out}}(t)}{dt} = \frac{d}{dt}[N \cdot \varphi_{\text{ref}}(t) + \varphi_{\text{err}}] = N \cdot \frac{d\varphi_{\text{ref}}(t)}{dt} = N \cdot \omega_{\text{ref}}(t).$$

In principle, one could also design a frequency-locked loop (FLL) that compares the frequencies of the reference and VCO output signals directly (by, e.g., using counters); however, this usually results in non-zero frequency error in steady state. Nevertheless, FLLs are often used in combination with PLLs to improve acquisition time and robustness, especially in scenarios with large initial frequency offsets.

We now derive the closed-loop transfer function of the PLL from reference phase $\varphi_{\text{ref}}(s)$ to output phase $\varphi_{\text{out}}(s)$. First, we note that

$$V_{\text{PD}}(s) = K_{\text{PD}}\left[\varphi_{\text{ref}}(s) - \frac{\varphi_{\text{out}}(s)}{N}\right]$$

and

$$\varphi_{\text{out}}(s) = \frac{K_{\text{VCO}}}{s} \cdot H_{\text{LP}}(s) \cdot V_{\text{PD}}(s) = \frac{K_{\text{VCO}}}{s} \cdot \frac{1}{1 + sT_{\text{LP}}} \cdot K_{\text{PD}}\left[\varphi_{\text{ref}}(s) - \frac{\varphi_{\text{out}}(s)}{N}\right].$$

Note that this equation has one integrator $1/s$ (which we can also see in Figure 76). We call PLLs with one integrator in the loop a **Type-I PLL**. We can now rearrange to find the closed-loop transfer function $H(s)$ as

$$H(s) = \frac{\varphi_{\text{out}}(s)}{\varphi_{\text{ref}}(s)} = N \cdot \frac{K_{\text{VCO}} \cdot K_{\text{PD}} \cdot \omega_{\text{LP}}/N}{s^2 + s \cdot \omega_{\text{LP}} + K_{\text{VCO}} \cdot K_{\text{PD}} \cdot \omega_{\text{LP}}/N}. \tag{62}$$

We can use all techniques known from control theory to analyze the stability and dynamic response of the PLL based on Equation 62, however, we have to keep the following in mind:

- $H(s)$ is a small-signal model of the PLL around its locked operating point, only valid for small perturbations. Large-signal behavior, such as acquisition and lock range, is not captured by this model.
- A PLL is a sampled system, working at instants of T_{ref}. $H(s)$ is a continuous-time approximation, which is valid only if the loop bandwidth is much smaller (typically $1/10$) than the reference frequency f_{ref}. If this approximation is not valid, a discrete-time model of the PLL must be used, deriving the z-domain transfer function $H(z)$.

If we compare the canonical form of a second-order system given by

$$H(s) = N \cdot \frac{\omega_{\text{n}}^2}{s^2 + s \cdot 2\zeta\omega_{\text{n}} + \omega_{\text{n}}^2},$$

with Equation 62, we can identify the natural frequency ω_n and the damping factor ζ of the PLL as

$$\omega_{\text{n}} = \sqrt{\frac{K_{\text{VCO}} \cdot K_{\text{PD}} \cdot \omega_{\text{LP}}}{N}} \tag{63}$$

and

$$\zeta = \frac{1}{2}\sqrt{\frac{\omega_{\text{LP}} \cdot N}{K_{\text{VCO}} \cdot K_{\text{PD}}}}. \tag{64}$$

This allows us to use standard control theory results to design the PLL parameters K_{PD}, K_{VCO}, ω_{LP}, and N to achieve the desired dynamic response and stability margins. Remember that K_{PD} has to be inserted in V/rad and K_{VCO} in rad/(s·V), i.e., 2π times the value quoted in Hz/V (see the callout in Section 6.6). Note that ω_{n} and ζ are not independent, as they both depend on the same set of PLL parameters! This seriously limits the design of the loop dynamics and means that we need a more complex loop filter $H_{\text{LP}}(s)$ to achieve more freedom in selecting ω_{n} and ζ independently. Often, we want to choose $\zeta = 1/\sqrt{2} \approx 0.707$ to achieve a maximally-flat Butterworth response with minimal overshoot.

The poles of a second-order system are given by

$$p_{1,2} = -\omega_{\text{n}}\left(\zeta \pm \sqrt{\zeta^2 - 1}\right)$$

with $\zeta = 1/\sqrt{2}$ resulting in complex conjugate poles at

$$p_{1,2} = -\frac{\omega_{\text{n}}}{\sqrt{2}} \pm j\frac{\omega_{\text{n}}}{\sqrt{2}}.$$

To summarize: We can make a simple PLL using an XOR for a PD, use a first-order LPF, plus a feedback divider to get frequency multiplication of $f_{\text{out}} = N \cdot f_{\text{ref}}$. However, the control over the loop dynamics is very limited, and the locking procedure of the PLL is tricky, as the PD has a limited linear range of only ±90° around its quiescent point, requiring that the VCO frequency at the start of the locking procedure is already close enough to the desired output frequency. In addition, N is an integer, limiting the frequency resolution of the output frequencies to integer multiples of the reference frequency.

The question is how to improve the PLL design to overcome these limitations. We will explore advanced PLL architectures and techniques in the following section.

7.2 Charge-Pump PLL

The fundamental limitations of the simple PLL architecture presented in Section 7.1 can be addressed by using a more sophisticated phase detector known as a **phase-frequency detector (PFD)** in combination with a **charge pump (CP)** [39]. This combination allows for better control over the loop dynamics and improved locking behavior.

In order to drive the CP, we envision a PFD with two digital outputs: an "up" signal that indicates when the VCO phase lags behind the reference phase, and a "down" signal that indicates when the VCO phase leads the reference phase [40]. The PFD compares both the phase and frequency of the reference and feedback signals, providing a more robust locking mechanism. The typical implementation of a PFD is shown in Figure 77.

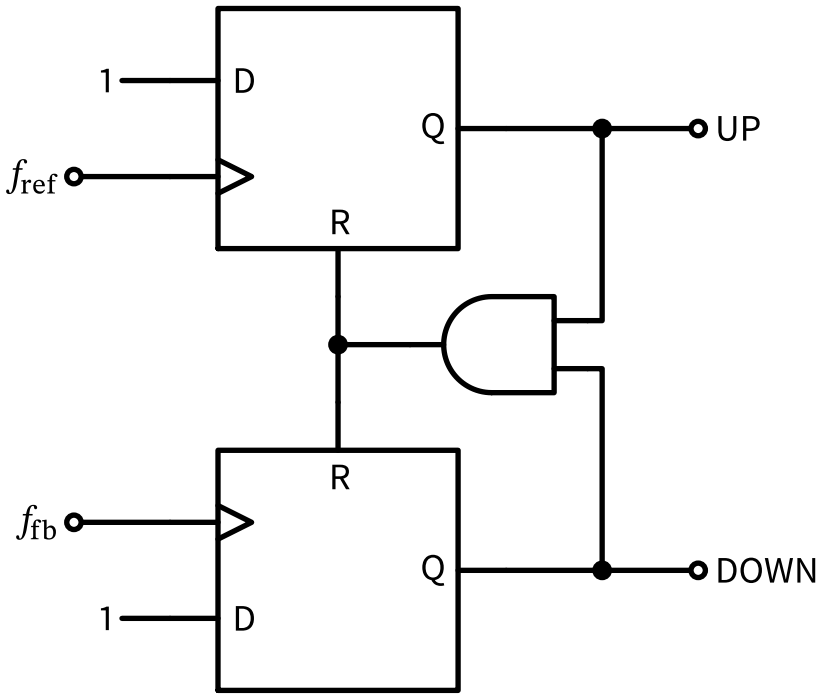


Figure 77: Implementation of a phase-frequency detector.

The PFD operates as follows:

- If the reference signal leads the feedback signal (its rising edge arrives first), the “UP” output is activated, causing the CP to source current into the integration capacitor, *increasing* the VCO tuning voltage and thus the VCO frequency.
- If the feedback signal leads the reference signal (its rising edge arrives first), the “DOWN” output is activated, causing the CP to sink current from the integration capacitor, *decreasing* the VCO tuning voltage and thus the VCO frequency.
- If both signals are aligned in phase and frequency, both outputs are only activated for a short time (the gate delays of the DFF and the AND), and the CP does not source or sink current; the integration capacitor in the loop filter holds its voltage, maintaining the VCO frequency.
- The rising edge that arrives last resets both outputs, ensuring that the CP only sources or sinks current for a duration proportional to the phase difference between the two signals.

In the following, we will look at four different cases of PFD operation to illustrate its behavior.

1. *Reference leads feedback*: The “UP” output is activated, and the CP sources current to the loop filter, increasing the VCO frequency.

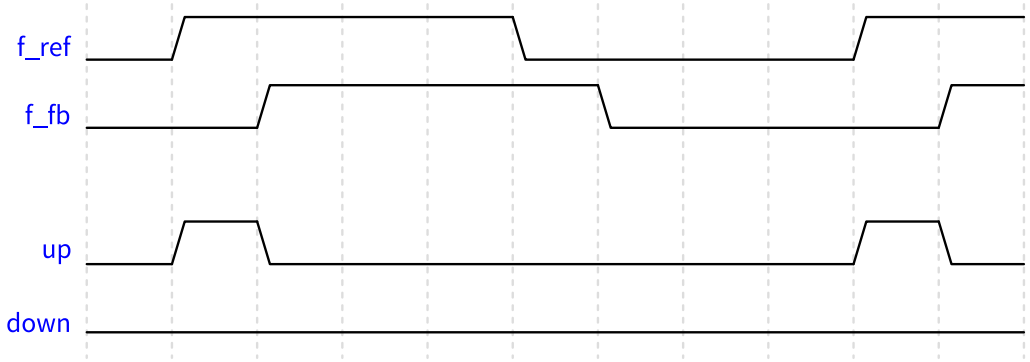


Figure 78: Reference leading the VCO feedback signal, frequencies are already aligned.

2. *Feedback leads reference*: The “DOWN” output is activated, and the CP sinks current from the loop filter, decreasing the VCO frequency.

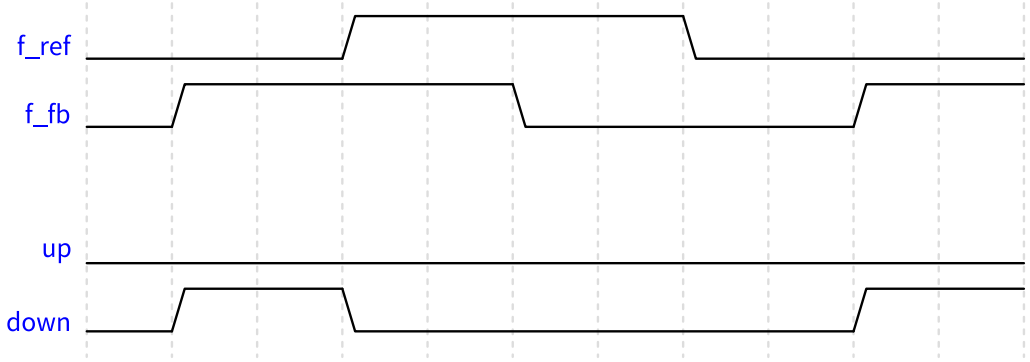


Figure 79: VCO feedback signal leading the reference, frequencies are already aligned.

3. *Both signals aligned*: Both outputs are briefly activated, but the CP does not source or sink current, maintaining the VCO frequency.

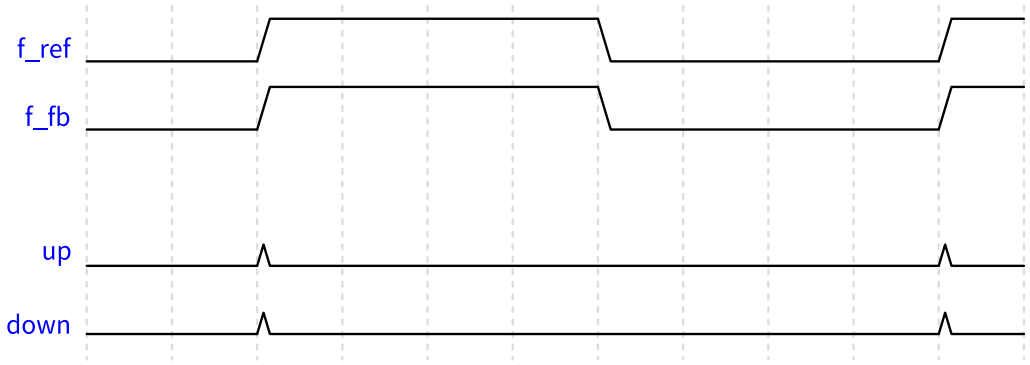


Figure 80: VCO feedback and reference signal are aligned in phase and frequency.

4. *Large frequency difference*: The PFD continues to source or sink current until the phases align, allowing the PLL to acquire lock even with large initial frequency offsets.

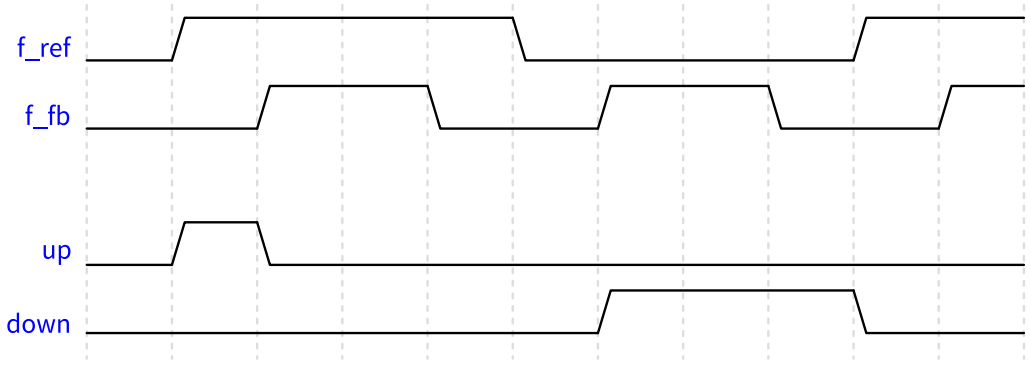


Figure 81: VCO feedback and reference signal have different frequencies (the VCO frequency is too high). There will be some periods with a wrong output (as shown here in the first cycle), but on average the phase-frequency detector output will indicate that the VCO frequency must be decreased.

Let us now investigate the circuit of the CP. The UP and DOWN signals from the PFD control two current sources/sinks connected to the loop filter capacitor, as shown in Figure 82.

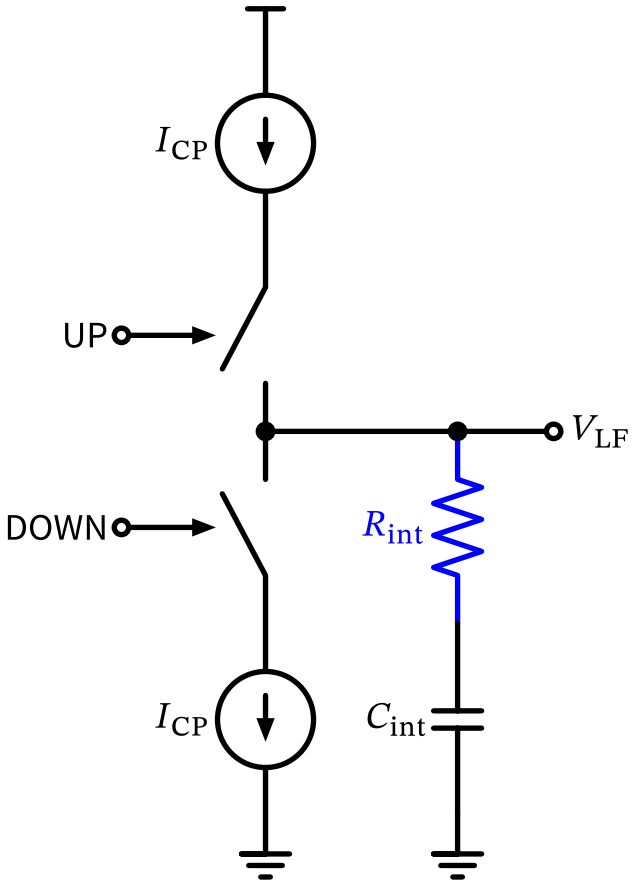


Figure 82: Charge pump consisting of two matched current sources and switches. The resistor in series with the capacitor introduces a zero.

When the UP signal is high, the CP *sources* a constant current I_{CP} into the loop filter capacitor C_{int}, increasing its voltage linearly over time according to the phase error. Conversely, when the DOWN signal is high, the CP *sinks* a constant current I_{CP} from the capacitor, decreasing its voltage linearly.

Note that the top and bottom current sources must be well matched to ensure symmetric operation of the CP and should have a wide operating range. The output impedance of the CP is ideally very high so that the charge pump current does not depend on the voltage across the loop filter capacitor. Further, the switches introduce charge injection and clock feedthrough, which can cause unwanted ripples on the loop filter voltage, leading to spurious signals in the PLL output. Careful design of the switches and careful layout techniques are necessary to minimize these effects.

The voltage change on the capacitor can be expressed as

$$\Delta V_{LF} = \frac{1}{C_{int}} \int_0^{t_{on}} I_{CP} \cdot dt \tag{65}$$

where t_{on} is the duration for which the UP or DOWN signal is active, proportional to the phase difference between the reference and feedback signals.

Taking the Laplace transform of Equation 65, we find the transfer function of the CP from the UP/DOWN control signals to the loop filter voltage as

$$H_{\mathrm{CP}}(s) = \frac{I_{\mathrm{CP}}}{2\pi}\left(\frac{1}{sC_{\mathrm{int}}} + R_{\mathrm{int}}\right) \tag{66}$$

where we have added a resistor R_{int} in series with the capacitor C_{int} to introduce a zero in the loop filter transfer function. The need for this zero will become apparent when we analyze the PLL loop dynamics next.

The resulting PLL using a PFD-CP combination is shown in Figure 83.

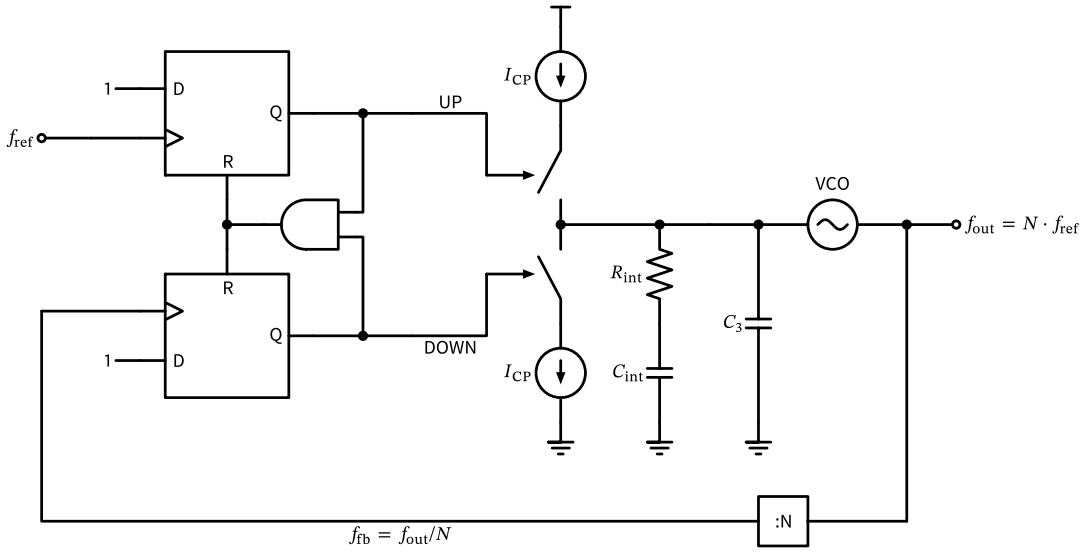


Figure 83: Diagram of a PFD-CP PLL. Note that the capacitor C_3 has been added to reduce the CP switching ripple, and its presence makes the loop filter 3rd order.

Neglecting C_3 for now, we can derive the closed-loop transfer function of the PLL from reference phase $\varphi_{\mathrm{ref}}(s)$ to output phase $\varphi_{\mathrm{out}}(s)$ as

$$H(s) = \frac{\varphi_{\mathrm{out}}(s)}{\varphi_{\mathrm{ref}}(s)} = N \cdot \frac{\frac{I_{\mathrm{CP}}K_{\mathrm{VCO}}}{2\pi C_{\mathrm{int}}N}(1 + sC_{\mathrm{int}}R_{\mathrm{int}})}{s^2 + s\frac{I_{\mathrm{CP}}K_{\mathrm{VCO}}R_{\mathrm{int}}}{2\pi N} + \frac{I_{\mathrm{CP}}K_{\mathrm{VCO}}}{2\pi C_{\mathrm{int}}N}} = N \cdot \frac{s \cdot 2\zeta\omega_{\mathrm{n}} + \omega_{\mathrm{n}}^2}{s^2 + s \cdot 2\zeta\omega_{\mathrm{n}} + \omega_{\mathrm{n}}^2} \tag{67}$$

with

$$\omega_{\mathrm{n}} = \sqrt{\frac{I_{\mathrm{CP}}K_{\mathrm{VCO}}}{2\pi C_{\mathrm{int}}N}} \tag{68}$$

and

$$\zeta = \frac{R_{\mathrm{int}}}{2}\sqrt{\frac{I_{\mathrm{CP}}K_{\mathrm{VCO}}C_{\mathrm{int}}}{2\pi N}}. \tag{69}$$

Looking at Equation 68 and Equation 69, we can see that we now have one additional *independent* parameter (R_{int}) to set the natural frequency ω_{n} and the damping factor ζ of the

PLL independently (compare this to Equation 63 and Equation 64). This allows us to design the loop dynamics more flexibly to achieve the desired performance.

With Equation 69, we can now appreciate the need for the resistor R_{int} in series with the capacitor C_{int} in the loop filter. Without this resistor, the damping factor ζ would be zero, leading to an undamped system with oscillatory behavior and poor transient response. The resistor introduces a zero in the loop filter transfer function, providing the necessary damping to stabilize the PLL and improve its dynamic response.

Looking at Figure 83 and Equation 66, we can also see that the PFD-CP introduces another integrator in the loop. This integrator, together with the integrator from the VCO, makes the PLL a **Type-II PLL**. A Type-II PLL achieves zero steady-state phase error not only for a step in the reference phase (which a Type-I PLL manages as well), but also for a *step in the reference frequency*, i.e., a ramp in the reference phase. This is the practically important case, as it means the loop settles to exactly $\Delta\varphi = 0$ after a channel change, without the static phase offset that a Type-I PLL would exhibit. Only for a frequency *ramp* (a parabolic phase input) does a Type-II PLL retain a finite static phase error.

In conclusion:

- The PLL regulates the phase of a VCO to match the phase of a reference signal, using feedback control. Even with a steady-state phase error, the frequency error is zero (Type-I PLL). In a Type-II PLL, both phase and frequency errors are zero in steady state.
- The lowpass behavior of the loop filter determines the dynamic response and stability of the PLL.
- Further, the PLL transfer function from reference phase to output phase has the form of a lowpass, which passes the reference oscillator phase noise to the output inside the passband. The phase noise of the VCO is suppressed inside the loop bandwidth, as the PLL corrects for phase deviations.
- Outside the PLL loop bandwidth, the VCO phase noise dominates, as the PLL cannot correct for phase deviations fast enough.

This phase noise shaping behavior is illustrated in Figure 84, which shows how the reference and VCO phase noise contributions combine to form the total PLL output phase noise.

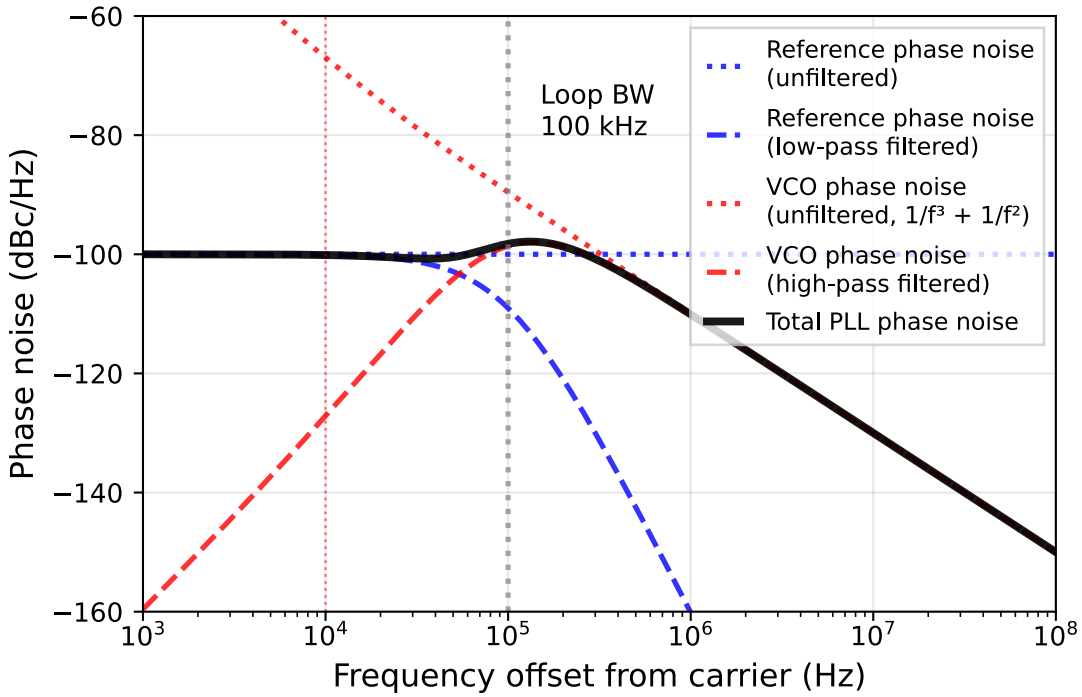


Figure 84: PLL phase noise contributions at the PLL output showing reference phase noise (increased by 20*log(N) and low-pass shaped), VCO phase noise (high-pass shaped), and total phase noise. The loop bandwidth determines the crossover frequency between reference and VCO phase noise contributions. For effective rejection of the 1/f³ VCO phase noise, a higher-order loop filter (shown is 3rd order) is beneficial.

Note that inside the PLL loop bandwidth the reference phase noise is multiplied by N^2 in power, i.e., $20 \log_{10}(N)$ in dB, as the PLL output frequency is N times the reference frequency. In Figure 84, $N = 100$ and a reference phase noise of −140 dBc/Hz is used. Large values of N can significantly increase the reference phase noise contribution at the PLL output, which is why low phase noise references are essential for high-performance PLLs.

7.3 All-Digital PLL

In the PLL implementations discussed so far, we have used analog components such as the VCO, loop filter, and charge pump. While these analog PLLs can achieve excellent performance, they also come with challenges such as component variability, temperature sensitivity, and integration complexity in modern digital processes. The question arises: Can we implement a PLL using only digital components to overcome these challenges? The answer is yes, leading to the concept of the **all-digital phase-locked loop (ADPLL)** [41], whose block diagram is shown in Figure 85.

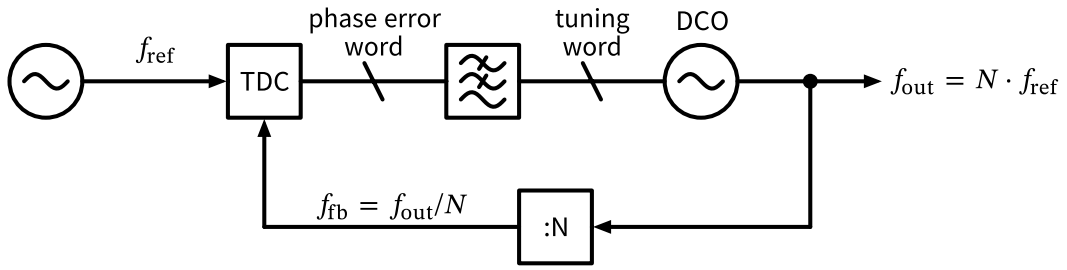


Figure 85: Block diagram of an all-digital PLL. A reference phase signal is compared to the phase of a digitally-controlled oscillator (DCO) output signal in a time-to-digital converter (TDC). The phase error word is digitally low-pass filtered and used to tune the DCO frequency via a frequency control word.

Conceptually, the ADPLL in Figure 85 is similar to the analog PLL in Figure 74, with the key difference being that all components are implemented digitally. The phase detector is replaced by a time-to-digital converter (TDC), which measures the time difference between the reference and feedback signal edges and converts it into a digital phase error word. The loop filter is implemented as a digital filter (FIR, IIR, or similar), processing the phase error word to generate a tuning word for the digitally controlled oscillator (DCO). The DCO generates the output frequency based on the tuning word, completing the feedback loop.

The advantages of the ADPLL are that it can be fully integrated in a digital nm CMOS process, benefiting from scalability, low power consumption, small chip area, and immunity to analog component variations. Additionally, the digital nature of the ADPLL allows for easy programmability and adaptability to different applications. However, the ADPLL also faces challenges such as quantization noise from the TDC and DCO, nonlinearity (DNL and INL) of TDC and DCO, as well as coupling of digital switching noise into the oscillator output signal.

7.3.1 Time-to-Digital Converter

The TDC is a crucial component of the ADPLL, responsible for measuring the time difference between the reference and feedback signal edges and converting it into a digital phase error word. The TDC operates by sampling the time interval between two events (the rising edges of the reference and feedback signals) and quantizing this interval into discrete levels. TDCs can be implemented using various techniques. One popular method is using a digital delay line. An exemplary implementation is shown in Figure 86.

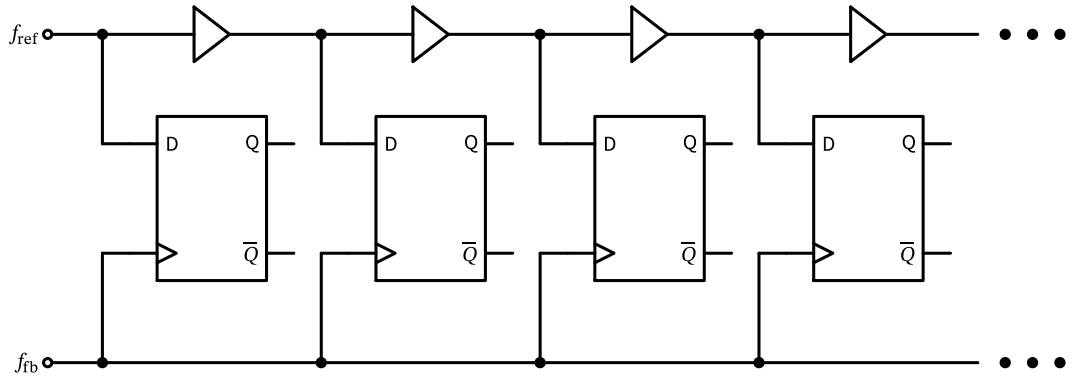


Figure 86: Basic TDC implementation as a delay line with parallel capture flip-flops. The TDC delay resolution is limited to two inverter delays.

In Figure 86, the reference signal edge runs through a series of buffers (delay elements) connected in a delay line configuration. The feedback signal edge is used to sample the state of the delay line at the moment it arrives. The outputs of the DFFs represent the time difference between the reference and feedback signals, which can be encoded into a digital phase error word. The resolution of the TDC is determined by the delay between each stage in the delay line, with finer delays providing higher resolution. However, increasing the resolution also increases the complexity and power consumption of the TDC.

The phase noise of the TDC (at the output of the ADPLL) arising from this time measurement quantization can be modeled as

$$\mathcal{L}\{f\} = 10 \cdot \log\left[\frac{(2\pi)^2}{12 f_{\text{ref}}} \cdot \left(\frac{\Delta T_{\text{TDC}}}{T_{\text{DCO}}}\right)^2\right]$$

where ΔT_{TDC} is the time resolution of the TDC, T_{DCO} is the period of the DCO output frequency, and f_{ref} is the reference frequency. This equation shows that improving the TDC resolution (reducing ΔT_{TDC}) and increasing the reference frequency can help reduce the phase noise contribution from the TDC.

i TDC Resolution Example

To get an idea of the performance requirements for the TDC, we assume a DCO output frequency of 2.4 GHz (e.g., for a Bluetooth application) and a reference frequency of 40 MHz. If we aim for a TDC phase noise contribution of −100 dBc/Hz at the DCO output, we can rearrange the above equation to find the required TDC time resolution:

$$\Delta T_{\text{TDC}} = T_{\text{DCO}} \cdot \sqrt{\frac{12 f_{\text{ref}} \cdot 10^{\mathcal{L}\{f\}/10}}{(2\pi)^2}} = \frac{1}{2.4 \times 10^9} \cdot \sqrt{\frac{12 \cdot 40 \times 10^6 \cdot 10^{-10}}{(2\pi)^2}} \approx 15\,\text{ps}$$

This number is challenging but achievable with modern TDC designs in nm CMOS.

7.3.2 Digitally Controlled Oscillator

With the implementation of the TDC clarified in Section 7.3.1, we now turn our attention to the digitally controlled oscillator (DCO), which generates the output frequency of the ADPLL based on a digital tuning word. The DCO is a digital equivalent of the VCO used in analog PLLs, and its frequency is adjusted by changing the digital input value. Fundamentally, two possible implementations exist:

1. An analog-controlled oscillator (e.g., a VCO) is combined with a digital-to-analog converter (DAC) to convert the digital tuning word into an analog control voltage. This approach is shown in Figure 87.

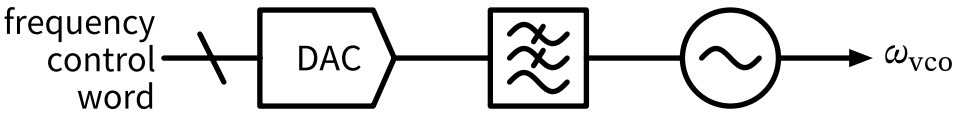


Figure 87: Digitally controlled oscillator using a DAC.

2. A digitally-controlled oscillator uses a large number of small varactors or switched capacitor banks to adjust the oscillation frequency directly based on the digital tuning word. An example implementation of a switched varactor is shown in Figure 88, or a switched capacitor like shown in Figure 72 is used.

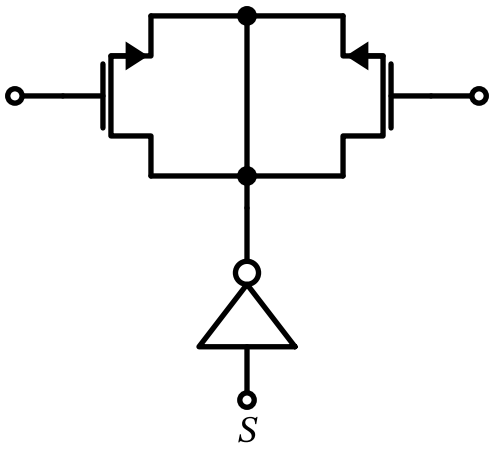


Figure 88: A switched differential varactor used for fine frequency control in a DCO.

It has to be noted that while an ADPLL seems conceptually simpler than an analog PLL, the design of high-performance TDCs and DCOs can be quite complex and requires careful consideration of quantization noise, linearity, and power consumption. Coupling effects between different circuit blocks can also introduce unwanted spurs and phase noise folding in the ADPLL output. Therefore, while ADPLLs offer many advantages, they also present unique design challenges that must be addressed to achieve adequate performance.

7.4 Fractional-N PLL

With the PLL architectures discussed so far, the output frequency resolution is limited to integer multiples of the reference frequency, i.e., $f_{\text{out}} = N \cdot f_{\text{ref}}$ with $N \in \mathbb{N}$. For a fine output frequency resolution, this requires a very low reference frequency. With the rule of thumb that the PLL loop bandwidth should be less than $f_{\text{ref}}/10$ to ensure stable operation, a low reference frequency also limits the achievable loop bandwidth, which is detrimental to phase noise performance and settling time. Further, we have learned in Section 7.2 that large values of N increase the reference phase noise contribution at the PLL output by $20\log_{10}(N)$ dB, making it challenging to achieve low phase noise performance with high frequency resolution.

How to break these fundamental limitations? The answer lies in the concept of the **fractional-N PLL**, which allows for non-integer frequency multiplication.

Assume we want to generate an output frequency of $f_{\text{out}} = 2.45\,\text{GHz}$ from a reference frequency of $f_{\text{ref}} = 40\,\text{MHz}$. This requires a frequency multiplication factor of $N = 61.25$, which is not an integer. To achieve this, the fractional-N PLL employs a feedback divider that can switch between two integer division ratios, N_1 and N_2, such that the average division ratio over time equals the desired fractional value (the averaging is performed by the low-pass loop filter, providing the averaged tuning voltage to the VCO / the tuning word to the DCO). In this case, we can choose $N_1 = 61$ and $N_2 = 62$. By alternating between these two division ratios in a controlled manner, the PLL can effectively achieve an average division ratio of 61.25, allowing it to generate the desired output frequency of 2.45 GHz. The following sequence illustrates this concept over four reference cycles:

$$N = N_1, N_1, N_1, N_2 \Longrightarrow \text{Average} \quad N = \frac{61 + 61 + 61 + 62}{4} = 61.25$$

A simple sequence like the one above works, but would lead to significant spurs in the PLL output spectrum due to the periodic nature of the division ratio switching, especially when the fractionality is close to an integer, and thus the sequence is long. Consider the case of $N = 61.01$, where we would need to switch to $N_2 = 62$ only once every 100 cycles of $N_1 = 61$:

$$N = \frac{99 \cdot 61 + 1 \cdot 62}{100} = 61.01$$

This periodic switching introduces spurious tones in the PLL output spectrum at multiples of $f_{\text{ref}}/100$, which can potentially pass through the loop filter and appear in the output spectrum, degrading the signal quality.

To minimize these spurs, more sophisticated techniques such as *delta-sigma modulation* are employed to *randomize* the switching sequence, spreading the quantization noise over a wider frequency range and reducing its impact on the PLL output. Additionally, the quantization noise introduced by the fractional division is *pushed to higher frequencies* where it can be more easily filtered out by the loop filter.

7.4.1 Delta-Sigma Modulator

For an in-depth understanding of delta-sigma modulation, we refer to [42]. Here, we briefly summarize the key concepts relevant to fractional-N PLLs. In a nutshell, a delta-sigma modulator (DSM) is a feedback system that shapes quantization noise to higher frequencies,

allowing for high-resolution digital-to-analog conversion or frequency synthesis. The basic structure of a first-order DSM is shown in Figure 89.

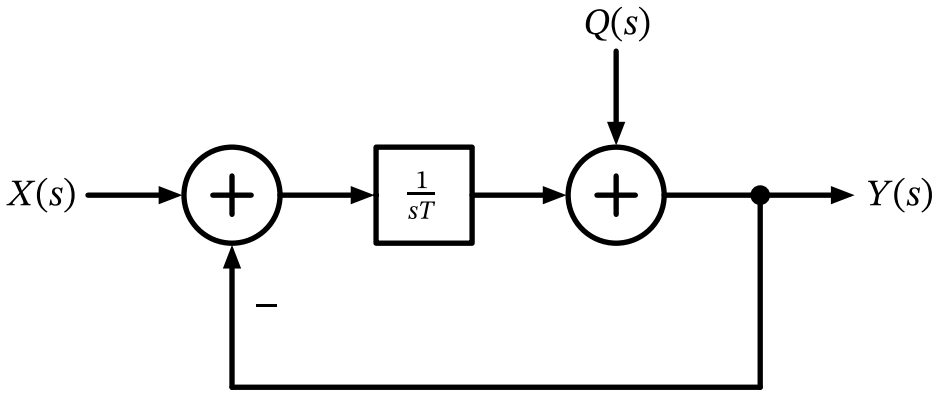


Figure 89: A first-order continuous-time delta-sigma modulator.

In Figure 89, the input signal $X(s)$ is transferred to the output $Y(s)$ with the following transfer function (we set $Q = 0$):

$$Y(s) = X(s) \cdot \frac{1}{1 + sT} = X(s) \cdot H_{\mathrm{LP}}(s)$$

The quantization noise $Q(s)$ introduced by the quantizer is shaped by the feedback loop, resulting in the following output contribution:

$$Y(s) = Q(s) \cdot \frac{sT}{1 + sT} = Q(s) \cdot H_{\mathrm{HP}}(s)$$

We can see that the quantization noise is highpass filtered, pushing it to higher frequencies, where it can be more easily filtered out by a subsequent lowpass filter.

The digital implementation of a first-order DSM is straightforward and one possible form is shown in Figure 90. At the point of truncation from $N + 2$ bits to 1 bit (the MSB) we imagine quantization noise $q[n]$ being added to the signal.

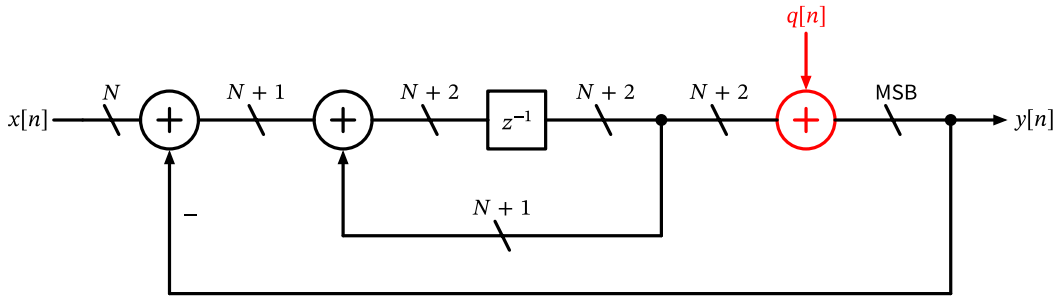


Figure 90: A first-order digital delta-sigma modulator.

Calculating the z-domain transfer functions from input $X(z)$ to the output $Y(z)$ (the signal transfer function, STF) and quantization noise $Q(z)$ to $Y(z)$ (the noise transfer function, NTF), we find

$$\frac{Y(z)}{X(z)} = \mathrm{STF}(z) = z^{-1}$$

and

$$\frac{Y(z)}{Q(z)} = \mathrm{NTF}(z) = (1 - z^{-1}) \Longrightarrow \mathrm{NTF} = 1 - z^{-1}.$$

We see that the STF is a simple delay, while the NTF has a zero at DC, pushing the quantization noise to higher frequencies. Evaluating the frequency response of the NTF to find the frequency-shaped PSD of the quantization noise, we find

$$S_{\mathrm{y}}(f) = S_{\mathrm{q}}(f)|H(f)|^2 = S_{\mathrm{q}}(f) \cdot 2|1 - \cos(2\pi f T_{\mathrm{s}})| \Longrightarrow H(f) = \sqrt{2|1 - \cos(2\pi f T_{\mathrm{s}})|}.$$

with T_{s} being the sampling period of the DSM. The resulting $H(f)$ is shown in Figure 91. With the assumed PLL loop bandwidth of 100 kHz and a reference frequency of 40 MHz, we can see that the quantization noise is significantly attenuated inside the PLL loop bandwidth, minimizing its impact on the PLL output phase noise. The general trend is that higher-order DSMs and faster sampling frequencies lead to better noise shaping and lower in-band quantization noise.

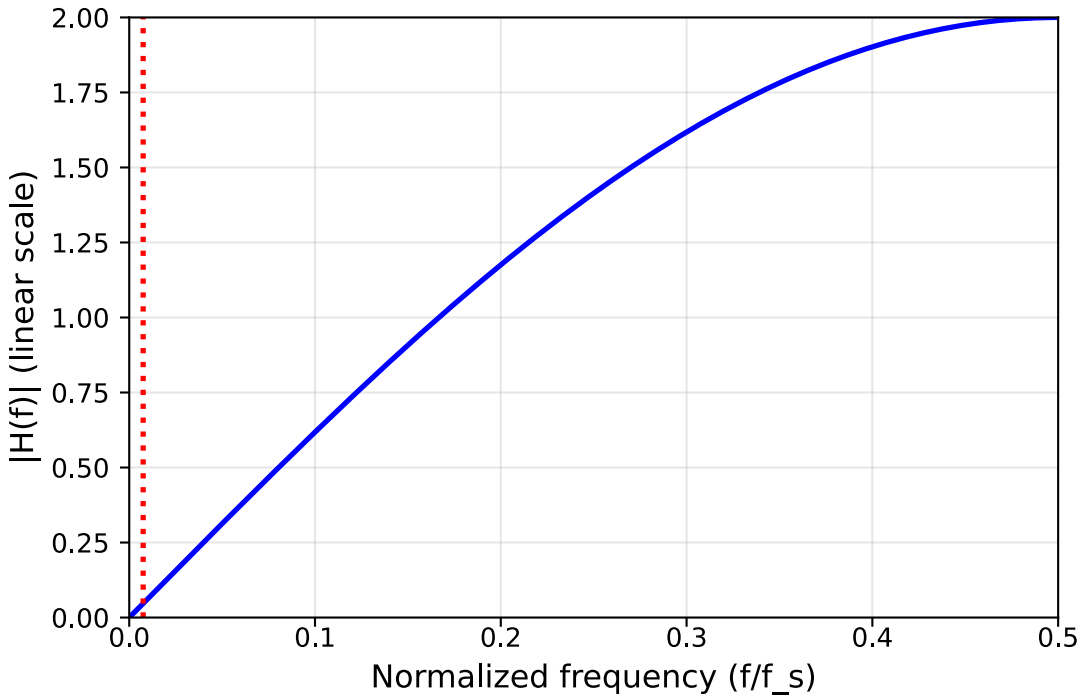


Figure 91: NTF H(f) of a first-order delta-sigma modulator showing the high-pass noise shaping characteristic. The quantization noise is suppressed at low frequencies and increases towards the Nyquist frequency. An exemplary loop filter bandwidth of 300 kHz for a 40 MHz reference frequency is indicated.

If we analyze the signal properties of the number sequence generated by the first-order DSM in Figure 90, we find that its randomization properties are not ideal, leading to spurs in the PLL output spectrum. To improve the randomization properties, higher-order DSMs are used in fractional-N PLLs, especially 2nd or 3rd order. A consequence of a higher-order DSM is that the produced sequence can take more values than just N and $N + 1$. The range of produced values is shown in Table 6.

Table 6: DSM output ranges and characteristics for different orders

DSM Order	Output Range	Noise Shaping
1	$N, N + 1$	20 dB/decade
2	$N - 1 ... N + 2$	40 dB/decade
3	$N - 3 ... N + 4$	60 dB/decade

A standard 3rd-order DSM implementation has issues with stability due to the phase shift of three integrators in the feedback loop. To mitigate this, a well-known architecture called the **MASH (multi-stage noise shaping) DSM** is often used [43], which consists of multiple first-order DSM stages cascaded together. An example Python implementation of a 3rd-order MASH DSM is shown in Figure 92, showing a snippet of the time-domain output and the resulting frequency spectrum, showing the noise shaping. As can be clearly seen, even using a 3rd-order MASH, there are strong spurious signals showing up in the spectrum. The time series shows that for some static input values (like here 0.125) patterns with a short period can exist.

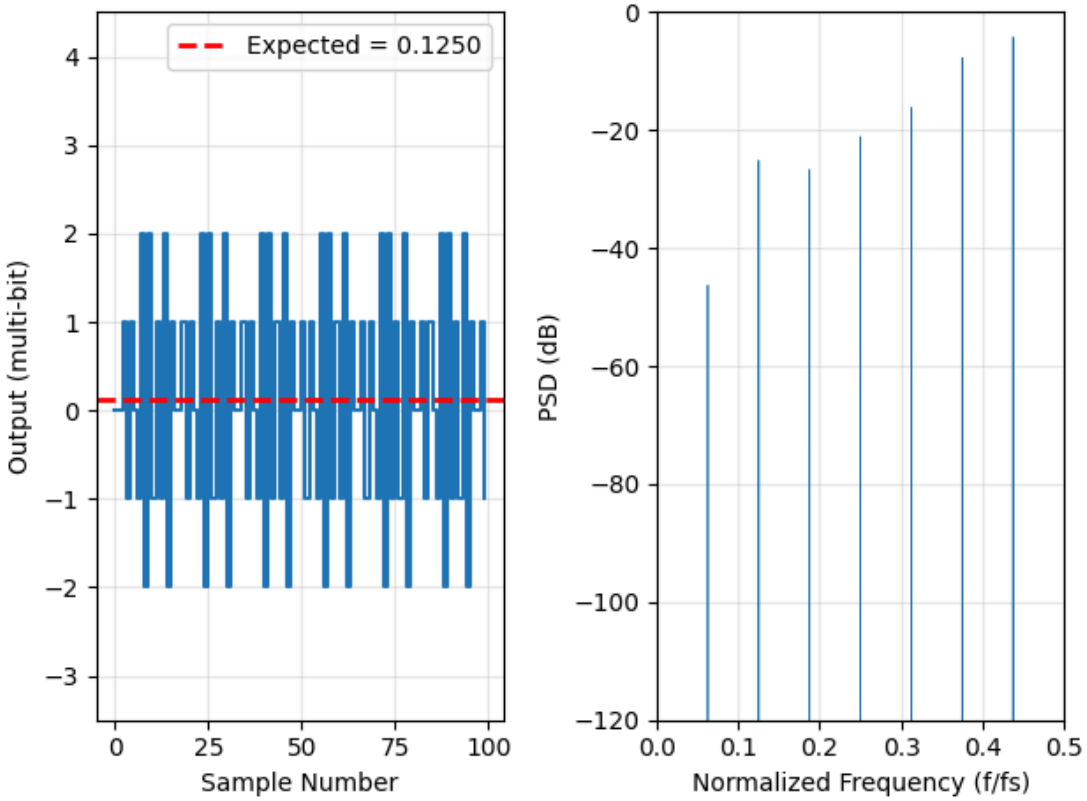


Figure 92: Time series and the spectrum of a MASH 3rd-order modulator.

As a potential solution, we can add a very small dithering signal (i.e., adding a random signal to the input), which breaks these patterns. As shown in Figure 93, dithering indeed causes enough randomization so that the noise shaping can be clearly seen without noticeable spurs (this is evident when comparing the time series of Figure 92 and Figure 93). Care has to be taken that the added random noise is small enough to not cause issues with in-band noise.

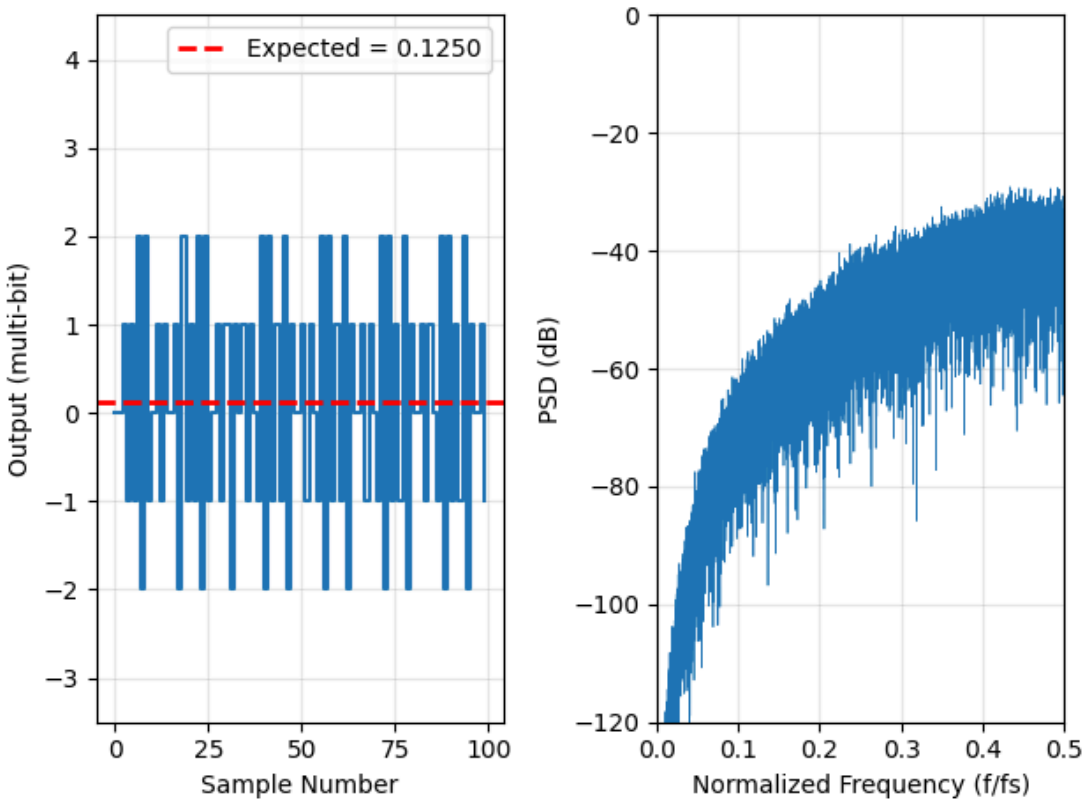


Figure 93: Time series and the spectrum of a MASH 3rd-order modulator with added dither.

A block diagram of a 3rd-order MASH 1-1-1 implementation including dither injection (shown in blue) is depicted in Figure 94. The dither is typically implemented digitally as a pseudo-random bit sequence (PRBS) generator, often in the form of a linear-feedback shift register (LFSR).

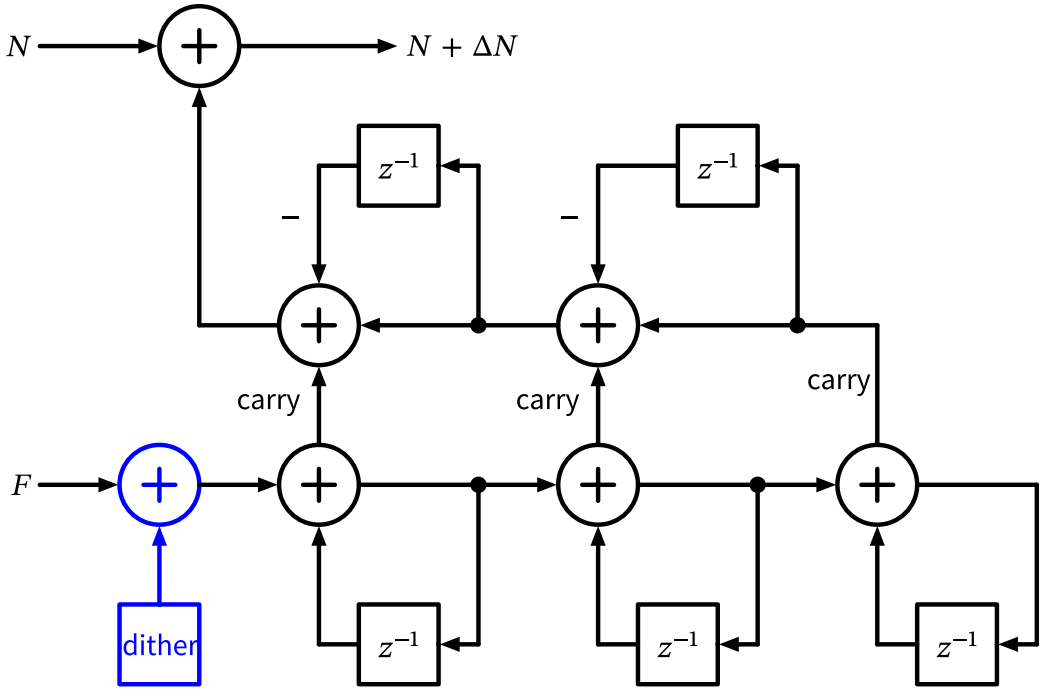


Figure 94: Block diagram of a 3rd-order MASH 1-1-1 delta-sigma modulator with three cascaded first-order stages and digital noise cancellation logic including dither injection.

7.4.2 Fractional-N PLL Implementation

A block diagram of a fractional-N PLL using a DSM and a multi-modulus divider (MMD) is shown in Figure 95. In order to ease the timing loop of the jittered clock coming from the MMD, the DSM is typically clocked by the MMD's output.

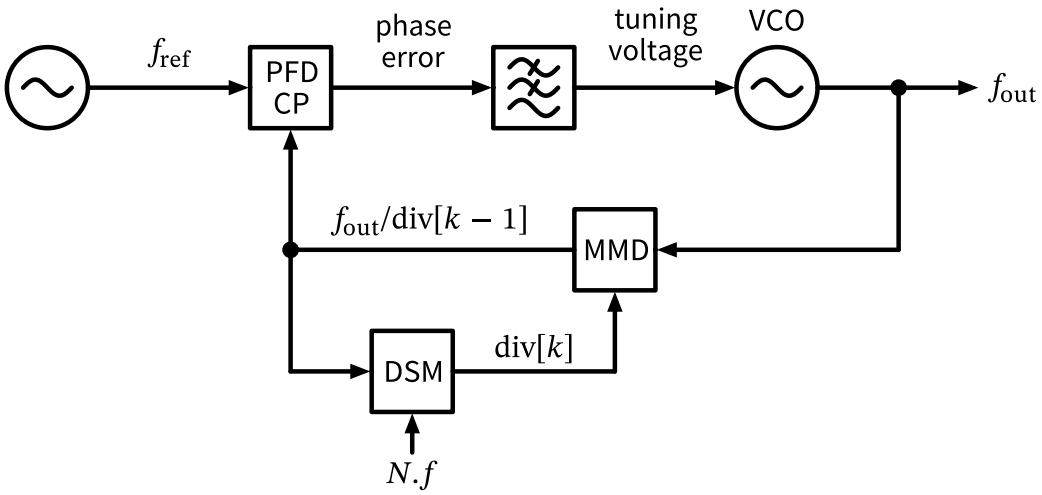


Figure 95: Block diagram of a fractional-N PLL. The VCO signal is divided by a sequence of integer divider values div[k] whose average value is given by $N.f$, where N is the integer part of the divider value and f is the fractional part of the divider.

For a practical implementation of a fractional-N PLL, it is of utmost importance that the path from the MMD output to the loop filter output is linear, as only the averaging/lowpass filtering of the noise-shaped quantization noise reduces the noise. Any *nonlinearity* in this path will lead to distortion of the quantization noise, causing quantization noise to fold back into the PLL loop bandwidth, degrading the PLL output phase noise!

One prominent source of nonlinearity is the difference between the source and sink currents of the charge pump, as well as any nonlinearity present in the PFD/CP operating point of $\Delta\varphi \approx 0$ (like a deadzone). Since a Type-II PLL ideally operates at exactly this point, with random excursions around it due to the DSM-induced jitter, any nonlinearity here will seriously degrade the PLL performance. There are predominantly two options to mitigate this issue:

1. Use a Type-I PLL architecture with $\Delta\varphi \neq 0$.
2. Introduce a phase offset in a Type-II PLL to operate away from $\Delta\varphi = 0$.

A simple way to introduce a phase offset is to add a fixed current source/sink to the charge pump, as shown in Figure 96.

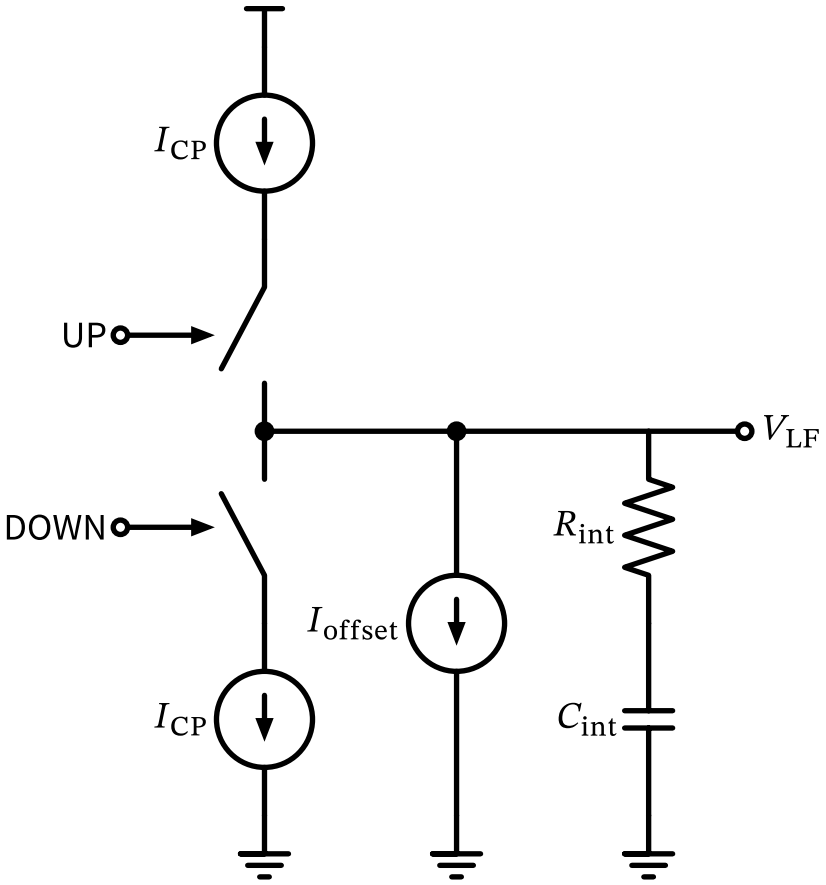


Figure 96: Charge pump with offset current for use in a fractional-N Type-II PLL.

As we have seen multiple times in the PLL discussion, using a higher f_{ref} is beneficial for phase noise performance, loop bandwidth, and TDC resolution (in an ADPLL). However, in many applications the choice of reference frequency is constrained by other factors. One way to

achieve a higher reference frequency is by using a **frequency doubler**. As shown in Figure 97, a frequency doubler can be implemented using an XOR gate and a delay line. The delay line is set to a quarter of the period of the reference frequency, i.e., $T_\mathrm{d} = T_\mathrm{ref}/4$. The XOR gate then produces an output frequency of $f_\mathrm{double} = 2 \cdot f_\mathrm{ref}$. For stable operation, this delay line must be designed to be relatively insensitive to process, voltage, and temperature (PVT) variations, or must be adjusted with a control loop.

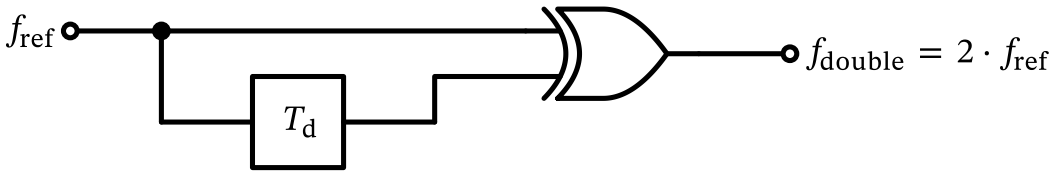


Figure 97: Implementation of reference frequency doubler. Ideally, T_d should be set to $T_\mathrm{ref}/4$.

As a final remark, we note that the output of the MMD in Figure 95 is a jittered clock due to the changing division ratio and the multiple gates and flip-flops that make up the MMD. This jitter degrades the PLL phase noise performance. Fortunately, there is an easy fix for this issue by using a retiming flip-flop after the MMD, clocked by the oscillator output, as shown in Figure 98.

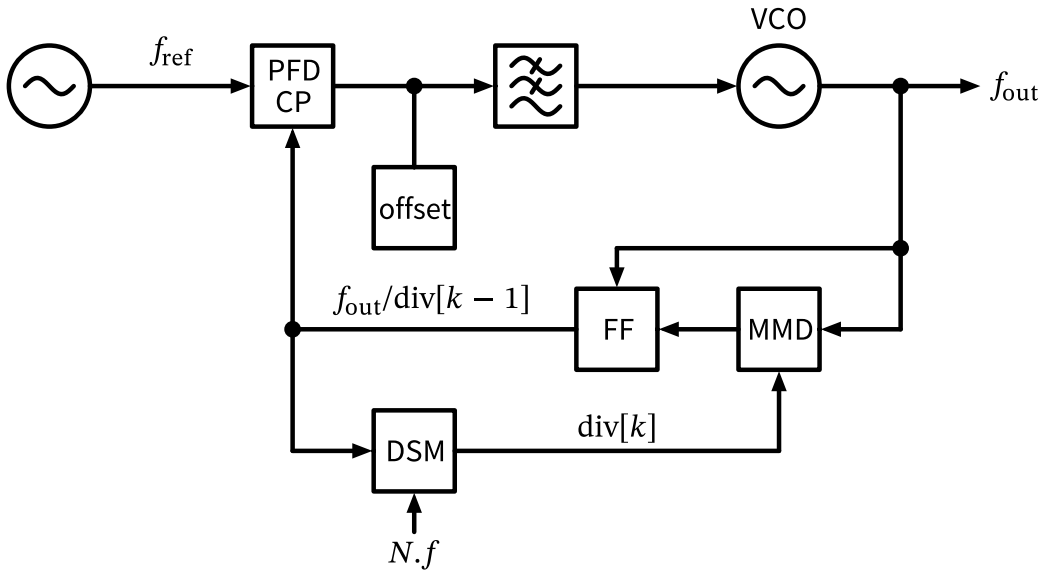


Figure 98: Block diagram of a fractional-N PLL including a retiming flip-flop to reduce the MMD-related jitter.

This retiming flip-flop effectively synchronizes the MMD output to the oscillator clock domain, significantly reducing the jitter seen by the loop filter and improving the overall phase noise performance of the fractional-N PLL.

8 Power Amplifiers

Power amplifiers (PAs) are essential components in wireless communication systems, as they boost the signal power from (usually) a few milliwatts up to several watts to ensure reliable transmission over the air. While PAs are simple in concept, their design and operation involve several trade-offs between efficiency, linearity, bandwidth, and cost.

A basic PA based on a single common-source stage is shown in Figure 99. The input signal is applied to the gate of the transistor via an input matching network. The transistor amplifies the signal at the drain terminal, and the output matching network transforms the $50\,\Omega$ load to the optimal load impedance for the desired output power under linearity and efficiency constraints.

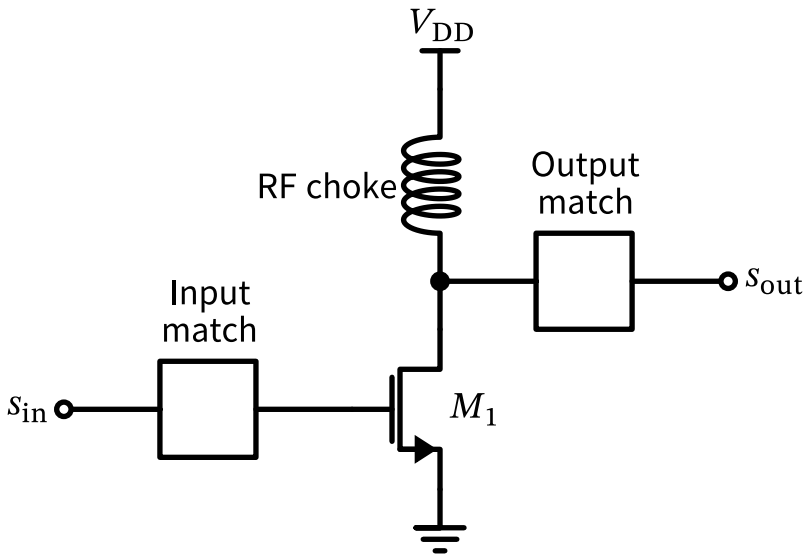


Figure 99: A basic single stage single-ended common-source power amplifier with input and output matching.

The RF choke at the drain allows the drain node to swing above the supply voltage, enabling higher output power. However, the transistor M_1 must be able to handle the high voltages and currents involved. Unfortunately, in silicon technologies the Johnson limit, which is the product of the breakdown voltage and the transit frequency of the transistor, is limited to around 200 GHz·V. This often requires the use of different semiconductor technologies for the PAs, such as GaAs, InP, or GaN. Implementing the PA in a different technology than the rest of the RFIC means implementing it off-chip.

Assuming the output voltage swing is limited only by the supply voltage V_{DD}, the maximum output power delivered to a load R_{L} is given by

$$P_{\mathrm{out,max}} = \frac{V_{\mathrm{DD}}^2}{2R_{\mathrm{L}}}.$$

For a typical supply voltage of $V_{\mathrm{DD}} = 3\,\mathrm{V}$ and a load of $R_{\mathrm{L}} = 50\,\Omega$, this results in a maximum output power of $P_{\mathrm{out,max}} = 90\,\mathrm{mW}$ or $19.5\,\mathrm{dBm}$. Higher output powers require either higher

supply voltages (which may not be available in battery-powered systems) or lower load impedances. The output matching network is thus designed in such a way as to present a lower impedance to the PA, e.g., $R_\mathrm{L} = 5\,\Omega$, which increases the maximum output power to $P_{\mathrm{out,max}} = 900\,\mathrm{mW}$ or $29.5\,\mathrm{dBm}$. The matching networks are usually implemented as LC networks or transmission line sections, often using transformers to achieve the required impedance transformation.

In order to drive the large output transistor M_1 in Figure 99, a driver stage is often required. This driver stage provides the necessary voltage and current gain to ensure that the PA operates in its optimal region. If needed, even another pre-driver stage can be added, but higher gain might cause stability issues and increased output noise. A two-stage power amplifier is shown in Figure 100.

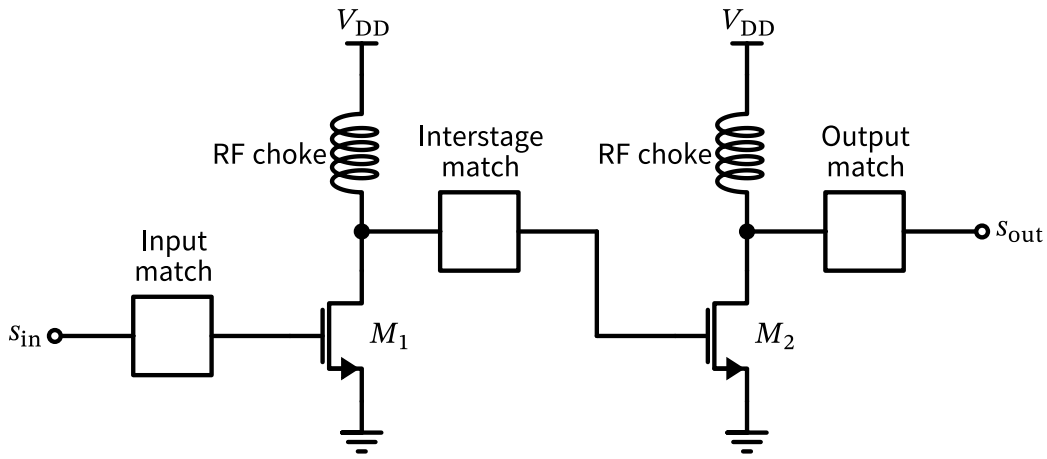


Figure 100: A basic two-stage single-ended common-source power amplifier with input and output matching.

8.1 PA Efficiency

One of the main characteristics of a PA is its **drain efficiency**, which is defined as the ratio of the output RF power to the total DC power consumed from the supply:

$$\eta = \frac{P_\mathrm{out}}{P_\mathrm{dc}}$$

Since η does not take into account the required input power to drive the PA, a more comprehensive metric is the **power-added efficiency (PAE)**:

$$\mathrm{PAE} = \frac{P_\mathrm{out} - P_\mathrm{in}}{P_\mathrm{dc}}$$

Ideally, we want both efficiency metrics as high as possible, but in practice there is a trade-off between efficiency and linearity (for linearity metrics refer to Section 2.2). High efficiency is often achieved by operating the PA in saturation, which leads to signal distortion and spectral regrowth. Various techniques exist to improve the linearity of PAs, such as **predistortion**, **feedback**, and **envelope tracking**.

8.2 PA Nonlinearity

For describing PA nonlinearity, various models can be used. One simple model is the **Saleh model** (originally developed for traveling-wave tube amplifiers), which describes the AM/AM and AM/PM conversion of the PA using a smooth transition from linear operation to saturation [5]. This model is defined by the following equations in the polar domain ($r(t)$ and $\psi(t)$ are the modulated input signal amplitude and phase, respectively; $y(t)$ is the output signal):

$$y(t) = A[r(t)] \cos\{\omega_0 t + \psi(t) + \varphi[r(t)]\}$$

with

$$A(r) = \frac{\alpha_1 r}{1 + \beta_1 r^2} \quad \text{and} \quad \varphi(r) = \frac{\alpha_2 r^2}{1 + \beta_2 r^2}$$

where α_1, β_1, α_2, and β_2 are the Saleh model parameters that define the PA characteristics. The AM/AM conversion $A(r)$ describes how the output amplitude saturates with increasing input amplitude, while the AM/PM conversion $\varphi(r)$ describes the phase distortion introduced by the PA. While this simple model does not capture memory effects or frequency-dependent behavior, it provides a good starting point for understanding PA nonlinearity.

Note that most discrete PAs are implemented similarly to Figure 100. For RFIC-integrated PAs, a single-ended implementation is problematic due to power supply parasitics and coupling issues. Therefore, integrated PAs are very often implemented in a differential configuration, which improves power supply rejection and reduces even-order harmonics [44], [45]. A generic differential PA implementation is shown in Figure 101.

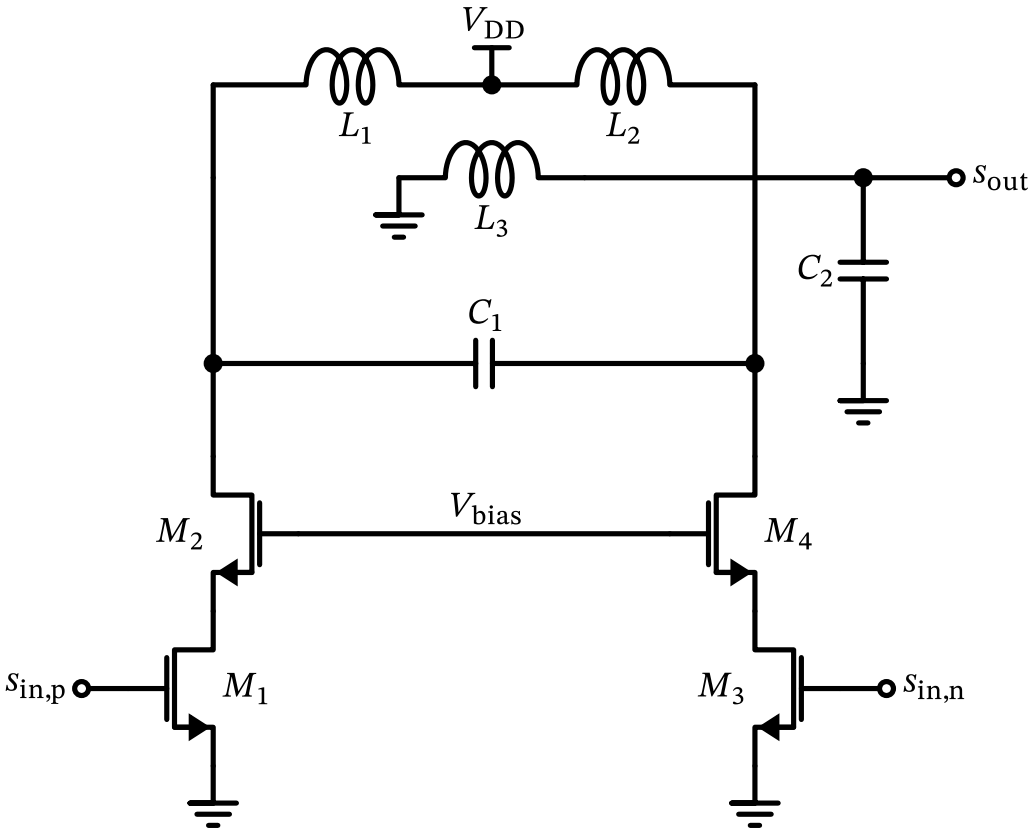


Figure 101: A generic differential power amplifier with cascode stages for peak voltage handling. Inductors L_1, L_2, and L_3 are implemented as a transformer, and form the load and output matching network together with C_1 and C_2. Note that the load transformer allows a differential to single-ended conversion, as well as a dc block.

The cascode transistors M_2 and M_4 allow higher output voltage swings by stacking the transistors, thus reducing the voltage stress on each individual transistor. They can be implemented as thick-oxide transistors to further increase the breakdown voltage.

While the power amplifier circuits look simple in principle, their design and optimization require large-signal models and extensive electromagnetic (EM) simulations to accurately capture the behavior of the matching networks and parasitic effects. Additionally, thermal management is crucial in PA design, as high power dissipation can lead to device degradation and failure.

In order to accurately capture and model large-signal effects, load-pull measurements are often performed during PA design. In a **load-pull measurement** setup, the PA is driven with a known input signal while the output load impedance is varied systematically using impedance tuners. By measuring the output power, efficiency, and linearity metrics at different load impedances, the optimal load conditions for the PA can be determined. This information is then used to design the output matching network to achieve the desired performance.

8.3 Amplifier Classes

Power amplifiers can be classified into different classes based on their conduction angle, which is the portion of the input signal cycle during which the transistor conducts current. The most common classes are Class A, Class B, Class AB, Class C, Class D, Class E, and Class F. Each class has its own trade-offs between efficiency, linearity, and complexity. The following table summarizes the main characteristics of each amplifier class:

Table 7: Amplifier Classes

Class	Conduction Angle	Efficiency	Linearity	Description
A	360°	Low	High	Bias current is larger than peak output current
B	180°	Medium	Medium	Conducts half the cycle (zero bias current)
AB	180°-360°	Medium	Medium-high	Between A and B (a little bit of bias current)
C	<180°	High	Low	Conducts less than half the cycle
D	N/A	Very high	Low-high	Switching amplifier, uses PWM or similar
E	N/A	Very high	Low	Switching amplifier with tuned load
F	N/A	Very high	Low	Uses harmonic tuning for efficiency
G	N/A	Very high	Medium	Multi-level supply voltage for efficiency
H	N/A	Very high	Medium	Adaptive supply voltage for efficiency

- **Class A** amplifiers are the most linear but also the least efficient, as they conduct current throughout the entire input signal cycle. They are typically used in applications where linearity is critical, such as low power analog signal processing. Drain efficiency is limited to $\eta \leq 50\%$.
- **Class B** amplifiers are more efficient than Class A, as they conduct current only during half of the input signal cycle. However, they suffer from crossover distortion at the zero-crossing point of the input signal. The upper limit for drain efficiency is $\eta \leq 78.5\%$.
- **Class AB** amplifiers are a compromise between Class A and Class B, providing better linearity than Class B while still being more efficient than Class A. They are widely used in small-signal applications and linear RF power amplifiers, where better efficiency than Class A is desired.
- **Class C** amplifiers are highly efficient but exhibit significant distortion, as they conduct current for only a part of the input signal cycle. They are typically used in applications where efficiency is more important than linearity, such as RF transmitters for constant-envelope modulation schemes (e.g., FM, FSK). Drain efficiency can theoretically approach $\eta = 100\%$, but only as the output power falls to zero.

- **Class D, E, and F** amplifiers are switching amplifiers that achieve very high efficiencies by operating the transistor as a switch, either fully on or fully off. They use various techniques such as pulse-width modulation (PWM), tuned loads, and harmonic tuning to optimize efficiency. These classes are commonly used in high-power applications, such as audio amplifiers and RF transmitters.
- **Class G and H** amplifiers use multiple supply voltages or adaptive supply voltages to improve efficiency, and can be combined with the other classes. They are often used in applications where power consumption is a concern, such as battery-powered devices. Class AB combined with Class G or H techniques is state-of-the-art in mobile device PAs. Class G and H are sometimes also referred to as "**envelope tracking**" techniques.

8.4 PA Stability and Neutralization

Looking at the basic PA circuit in Figure 99, it becomes apparent that a PA can easily become unstable due to the high gain and large signal swings involved. The Miller capacitance C_{m} between the gate and drain of the transistor creates a feedback path that can lead to oscillations, as shown in Figure 102. Additionally, parasitic capacitances and inductances in the layout and packaging can exacerbate the problem.

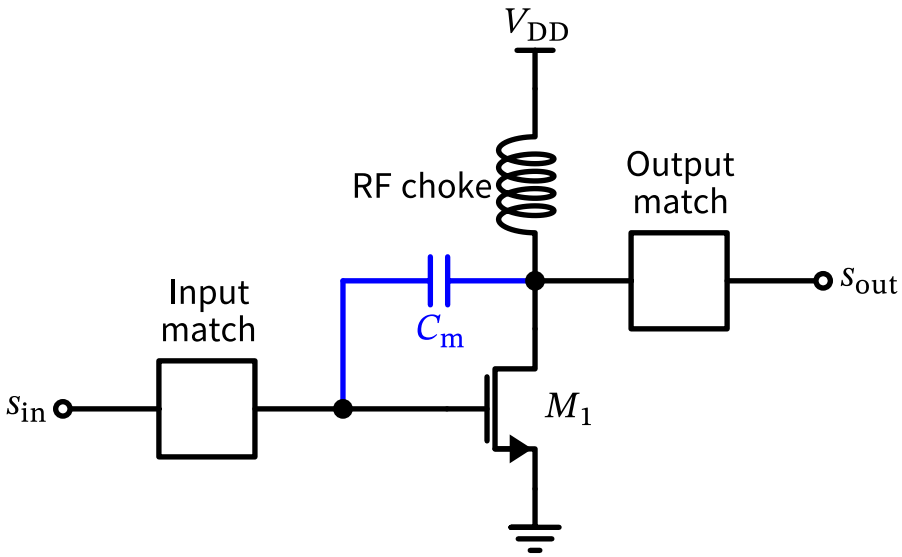


Figure 102: A basic single stage single-ended common-source power amplifier showing the Miller capacitance C_{m} between input and output (in blue).

The Miller capacitance effectively creates a feedback loop from the output to the input, which can lead to instability, as also shown in Section 4.2. Lowering S_{12} (and thus getting rid of the Miller effect due to C_{m}) is crucial for PA stability. This can be mainly achieved in two ways:

- Adding a cascode stage, which isolates the input from the output and reduces the effect of C_{m} significantly (see Figure 101).
- Adding a **neutralization** capacitor between the gate and drain, which cancels out the effect of C_{m}. Alternatively, feedback networks or additional circuitry can be used to stabilize the PA.

In order to implement neutralization, we add in parallel to the (intrinsic) Miller capacitance C_{m} a capacitor C_{n}, driven by an inverted version of the output signal. The neutralization capacitor C_{n} is chosen such that it cancels out the effect of C_{m}, i.e., $C_{\mathrm{n}} = C_{\mathrm{m}}$. This effectively eliminates the feedback path from the output to the input, improving stability. However, implementing neutralization requires careful design and tuning, as any mismatch between C_{m} and C_{n} can lead to residual feedback and potential instability. This arrangement is shown in Figure 103.

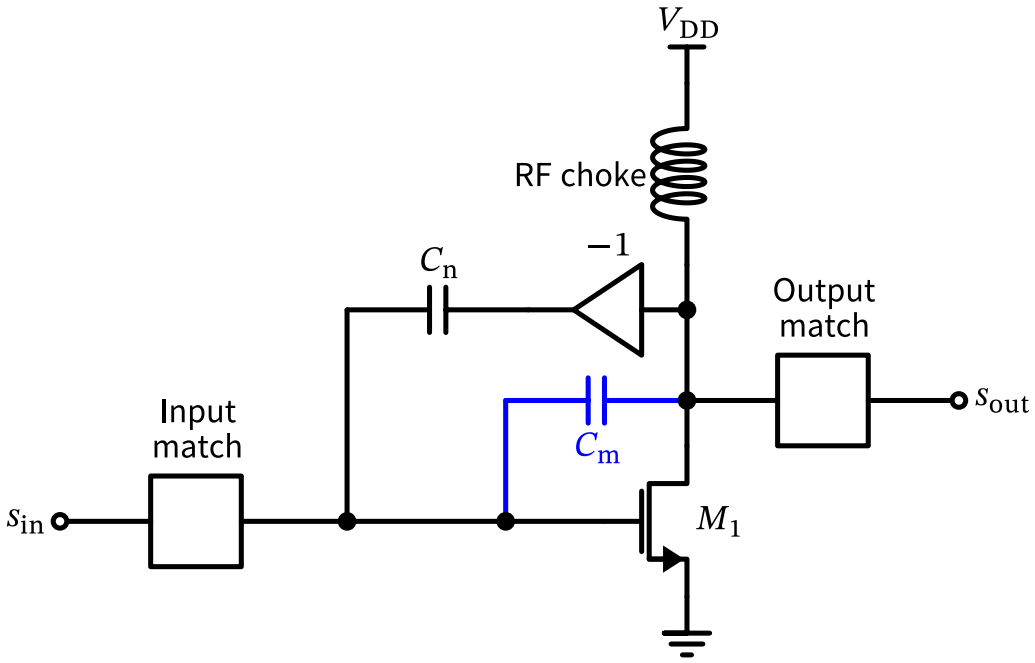


Figure 103: A basic single stage single-ended common-source power amplifier showing the Miller capacitance C_{m} between input and output (in blue) and the neutralization capacitor C_{n}.

In a differential PA implementation, the neutralization capacitors can be connected cross-coupled between the drains and gates of the two transistors, further improving stability, as shown in Figure 104. The cancellation of the Miller effect also leads to improved bandwidth, which is beneficial for wideband applications.

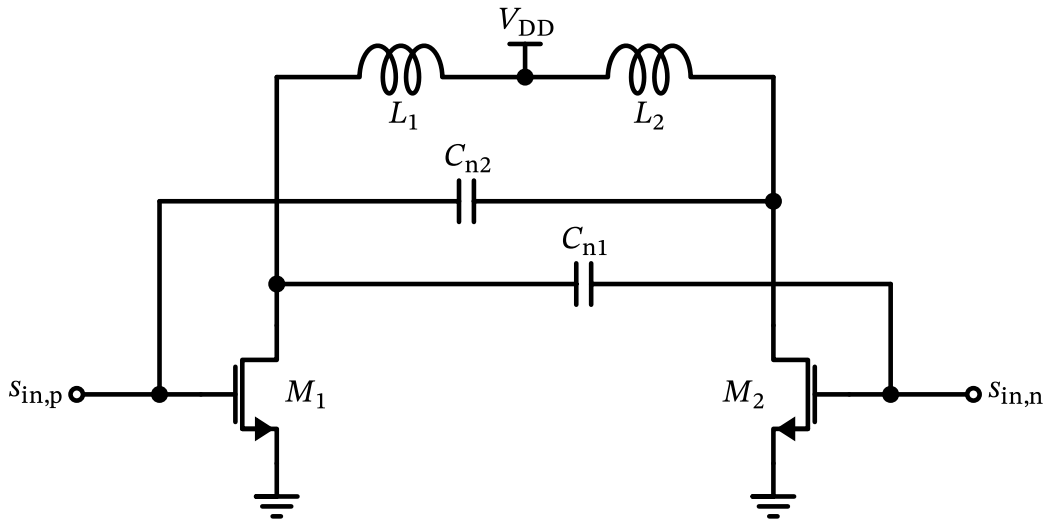


Figure 104: A generic differential power amplifier with neutralization for wideband operation and improved stability.

8.5 PA Advanced Techniques

To further improve the performance of power amplifiers (but outside the scope of these lecture notes), several advanced techniques can be employed:

- **Predistortion**: This technique involves applying an inverse distortion to the input signal before it enters the PA, effectively canceling out the nonlinearity of the PA. Digital predistortion (DPD) is commonly used in modern communication systems to improve linearity and reduce spectral regrowth.
- **Envelope Tracking**: This technique dynamically adjusts the supply voltage of the PA based on the envelope of the input signal, improving efficiency by reducing power consumption during low signal levels (see Class G and Class H).
- **Load Modulation**: This technique involves varying the load impedance seen by the PA to optimize performance for different signal conditions. This can be achieved using tunable matching networks or switched load networks. An active method is the **Doherty amplifier** architecture, which uses two amplifiers (a main and a peaking amplifier) to improve efficiency at back-off power levels.

9 Spurs, Crosstalk and Self-Interference

In practical RFICs, unwanted tones and unintended coupling paths often limit the achievable dynamic range (expressed for example as SNR or EVM) or cause performance issues (like out-of-band emissions in TX or RX blocking) that render standards-compliant operation impossible. In modern carrier-aggregation transceivers, the integration of several TX and RX paths together with multiple local oscillators on a single die turns this into a major source of concern [21].

A *spur* (short for “spurious tone”) is a discrete spectral line at an unwanted frequency. A common metric is the spur level relative to the wanted carrier, expressed in dBc:

$$L_{\mathrm{spur}} = 20\log_{10}\left(\frac{V_{\mathrm{spur}}}{V_{\mathrm{carrier}}}\right)\mathrm{dBc}\ .$$

Crosstalk is undesired signal transfer from an aggressor node to a victim node through electric, magnetic, substrate, or supply coupling [46]. Coupling mainly happens through parasitic capacitances from high voltage-swing nodes, or inductively from high-current loops. Shared traces (like supply and ground connections) can also lead to ohmic coupling.

A first-order estimate for capacitive crosstalk from aggressor to victim is

$$\left|\frac{V_{\mathrm{victim}}}{V_{\mathrm{aggressor}}}\right| \approx \omega C_{\mathrm{c}}|Z_{\mathrm{victim}}|,$$

showing that coupling tends to worsen at higher frequency and higher victim impedance. In high-performance RFICs, crosstalk and spur levels on the order of -80 dBc or lower are often required to meet system-level specifications.

9.1 Coupling Paths

Figure 105 shows the main on-chip coupling paths in a typical TRX architecture. The following discussion uses the crosstalk nomenclature introduced in [21].

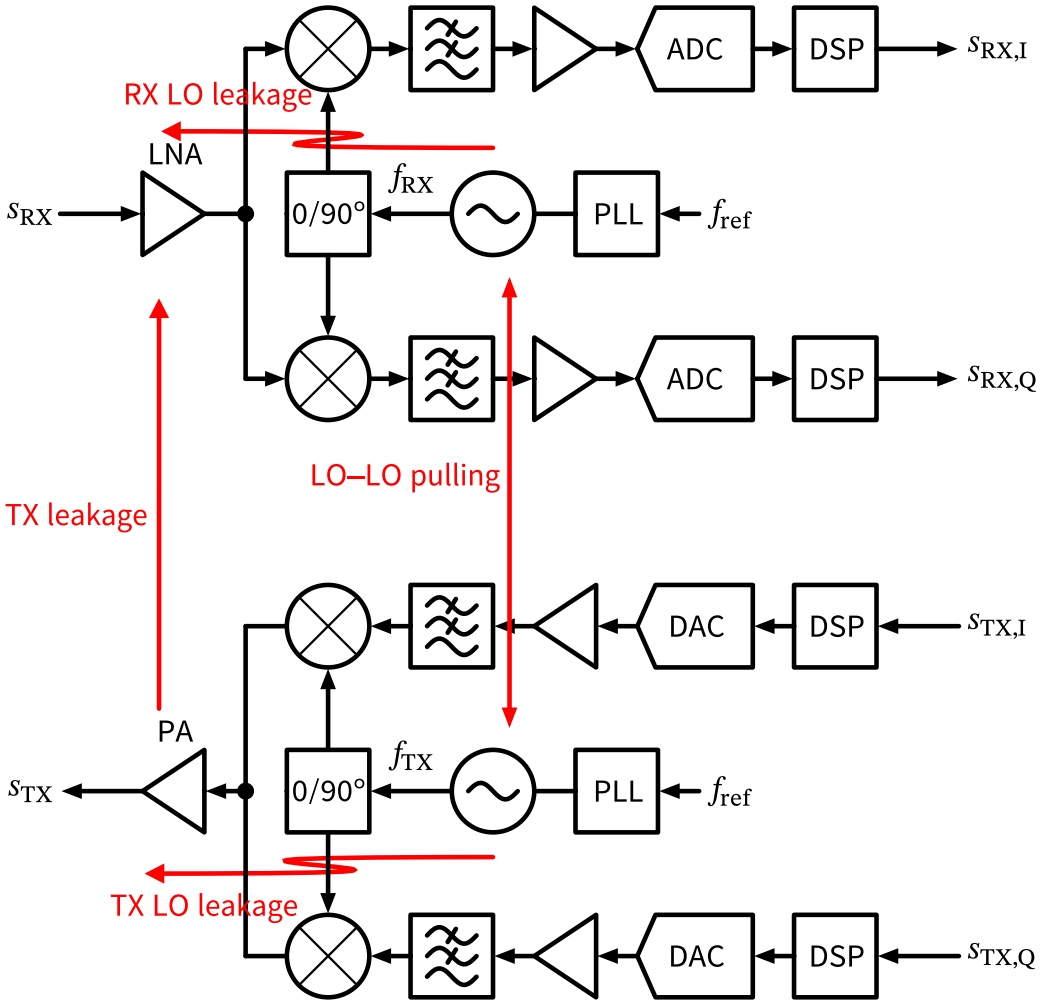


Figure 105: Block diagram of a typical transceiver (TRX) showing the main coupling paths.

- TX-to-RX coupling through the RFIC, duplexer and shared front-end elements ("TX leakage") can cause issues such as gain desensitization, reciprocal mixing, receive-band noise from the TX, and IM2/IM3 distortion in the RX [47].
- RX VCO coupling ("RX LO leakage") from the RX VCO/PLL into the LNA/mixer can cause increased IM2 products and LO radiation.
- LO-to-LO coupling through the LO distribution network can cause reciprocal mixing due to LO spurs in RX, or OOB emissions on TX. When the two oscillators run at nearby frequencies, injection pulling between the VCOs can also degrade the spectral purity of both LOs [48].
- TX VCO coupling ("TX LO leakage") from the TX VCO/PLL into the PA can cause increased LO leakage and EVM degradation.

9.2 Synthesizer Spurs

In addition to the inter-block coupling discussed above, the frequency synthesizer itself is a notorious source of *intra-PLL* spurs (see also Section 7). The most relevant mechanisms are:

- **Reference spurs** at integer multiples of the reference frequency, $f_{\text{spur}} = f_{\text{LO}} \pm k\, f_{\text{ref}}, k \in \mathbb{N}^{+}$. They are caused by charge-pump mismatch and leakage, finite isolation of the loop-filter node, and supply or substrate coupling at f_{ref}. Their amplitude is set by how strongly the loop filter rejects components at f_{ref} relative to the loop bandwidth.
- **Fractional-N spurs** generated by the periodicity and quantization noise of the $\Delta\Sigma$ modulator that controls the divider (see Section 7.4.1). They typically appear as families of tones around f_{LO} at offsets related to the fractional ratio. Higher-order modulators and dithering reduce their level at the cost of increased close-in noise.
- **Divider and digital-switching spurs** at sub-harmonics of f_{LO} and at digital clock frequencies, coupling into the VCO through the supply and substrate.
- **Digital spurs** originating from TDC or DCO nonlinearities in all-digital PLLs, or from digital blocks in the vicinity of the VCO.

Once these spurs appear on the LO, they translate directly into the receive band through reciprocal mixing of a strong blocker, or into the transmit spectrum through the upconversion mixer. If these spurs are inside the channel bandwidth, they cause EVM degradation and may violate spectral mask requirements.

9.3 Spur and Crosstalk Mitigation

Mitigation of spurs and crosstalk is a complex topic that requires careful design of the entire system, including the architecture, floorplan, circuit design, layout, and packaging. Some common techniques are summarized here. In addition, inductor-to-inductor coupling has to be considered and may require quadrupole inductors [49] or patterned ground shields [50].

Substrate and supply coupling deserve special attention because they couple every block to every other block through a common low-impedance node [46]. Effective measures include placing sensitive analog and RF blocks in dedicated deep-n-well or triple-well pockets, surrounding aggressors and victims with substrate guard rings, separating supply domains for digital, analog, RF, and LO blocks, and using on-chip and on-package decoupling so that switching currents do not modulate the supply seen by the VCO or LNA. A careful floorplan that keeps high-swing TX nodes, LO distribution, and digital clocks physically far from sensitive RX inputs and VCO inductors is usually more effective than any post-layout fix.

If supply decoupling is insufficient, series or shunt supply regulators and additional low-pass filters for supply and reference lines can help to reduce supply-bound coupling [51]. However, these measures increase the power consumption and area of the RFIC, and may not be feasible in all applications.

Where possible, mitigation should also be performed by digital signal processing of the RX or TX baseband signals, for example digital predistortion, digital IM2 cancellation [20], or digital self-interference cancellation [21]. However, these techniques require additional hardware and power, and are not always feasible.

10 Appendix: RF Circuit Simulation

There is no breadboard for an integrated circuit: every design decision in an RFIC is checked by circuit simulation before tapeout. A SPICE-type simulator [52] offers a small set of **fundamental analyses** (dc operating point, small-signal ac and noise, and transient), which are

perfectly adequate for amplifiers, filters, and bias circuits. RF circuits, however, push these analyses to their limits, for several reasons:

- **Widely separated time scales.** An LO at several GHz carries information at kHz to MHz offsets, and flicker noise matters down to a few Hz. A time-domain simulation must resolve the fastest signal for as long as the slowest one needs.
- **Large periodic signals.** The LO of a mixer, the clock of a switched-capacitor filter, or the carrier of a PA moves the operating point periodically. With respect to the small signal, such a circuit is *time-variant* (see Section 5.2), so an analysis linearized around a constant operating point cannot capture its behavior.
- **Frequency translation of noise.** In such time-variant circuits, noise is modulated by the large signal and folded from many frequencies onto the wanted output frequency.
- **High-Q resonators and oscillators.** High-Q circuits take many periods to settle, and an oscillator has no stable dc operating point at all. Its frequency and phase noise are results of the large-signal operation.

These challenges led to a set of **advanced RF analyses**: transient noise, periodic steady-state analysis by shooting (PSS) and by harmonic balance (HB), and the small-signal (PAC, HBAC) and noise (PNOISE, HBNOISE) analyses built on top of a periodic steady state [53]. For a long time, the complete set was only available in commercial simulators, while open-source simulators offered parts of it (e.g., harmonic balance in Xyce, and transient-noise sources and an experimental oscillator PSS in ngspice). Since September 2026, the development version of the open-source simulator VACASK [54] supports all of them. The IIC-OSIC-TOOLS container ships them from its release 2026.10 on, so that the complete RF verification flow of this appendix can be run with open-source tools. The analysis names used in this appendix follow VACASK, which largely uses the names of Cadence Spectre (Spectre calls the operating point `dc`, the small-signal analysis `ac` and the S-parameter analysis `sp`, where VACASK uses `op`, `ac` and `acsp`). Section 10.4 lists the corresponding ngspice commands.

Figure 106 shows how the analyses build on each other. All of them start from the dc operating point. The small-signal analyses linearize the circuit either around this constant operating point, which gives a linear time-invariant (LTI) system, or around a periodic steady state, which gives a linear periodically time-varying (LPTV) system.

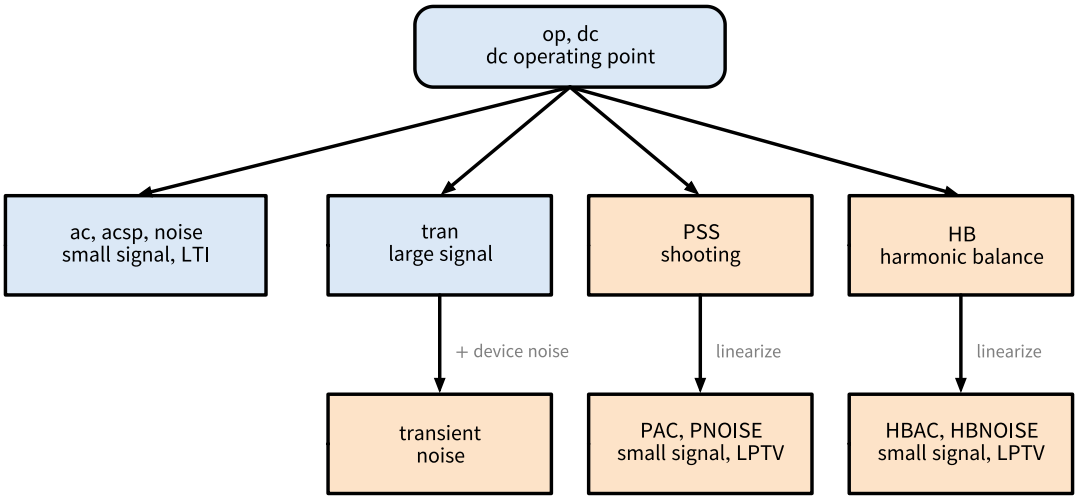


Figure 106: Family tree of circuit simulation analyses. Every analysis starts from the dc operating point. The small-signal analyses linearize around a constant operating point (LTI) or around a periodic steady state (LPTV). The periodic steady state is found either in the time domain (shooting PSS) or in the frequency domain (harmonic balance, HB). The fundamental analyses are shown in blue, the advanced RF analyses in orange.

All analyses solve the same circuit equations. With modified nodal analysis (MNA), a circuit is described by

$$\mathbf{f}(\mathbf{x}(t)) + \frac{\mathrm{d}}{\mathrm{d}t}\mathbf{q}(\mathbf{x}(t)) = \mathbf{u}(t), \tag{70}$$

where the vector $\mathbf{x}$ contains the unknown node voltages and selected branch currents (of inductors and voltage sources), $\mathbf{f}$ the currents of the resistive elements (summed per node according to Kirchhoff's current law) together with the branch equations of voltage sources and inductors, $\mathbf{q}$ the charges of capacitors and fluxes of inductors, and $\mathbf{u}$ the independent sources. The analyses differ only in the assumption they make about $\mathbf{x}(t)$:

- **dc operating point**: all time derivatives vanish, $\mathbf{f}(\mathbf{x}_0) = \mathbf{u}_{\mathrm{DC}}$.
- **ac and noise**: a small deviation from $\mathbf{x}_0$, so that Equation 70 can be linearized around $\mathbf{x}_0$ (LTI).
- **transient**: no assumption at all. The time derivative is discretized, and Equation 70 is integrated step by step.
- **PSS and HB**: the solution is periodic, $\mathbf{x}(t+T) = \mathbf{x}(t)$ (or quasi-periodic for several incommensurate tones in HB).
- **PAC, PNOISE, HBAC, HBNOISE**: a small deviation from a periodic steady state $\mathbf{x}_{\mathrm{s}}(t)$, so that Equation 70 can be linearized around $\mathbf{x}_{\mathrm{s}}(t)$ (LPTV).

10.1 Fundamental Analyses

10.1.1 DC Operating Point and DC Sweep

The **operating point** analysis (`op`) sets all time derivatives in Equation 70 to zero (capacitors become open circuits, inductors short circuits) and solves the remaining nonlinear algebraic

equations $\mathbf{f}(\mathbf{x}_0) = \mathbf{u}_{\mathrm{DC}}$ with the Newton-Raphson method. Starting from an initial guess $\mathbf{x}^{(0)}$, it iterates

$$\mathbf{G}(\mathbf{x}^{(i)})\,\Delta\mathbf{x}^{(i)} = \mathbf{u}_{\mathrm{DC}} - \mathbf{f}(\mathbf{x}^{(i)}), \qquad \mathbf{x}^{(i+1)} = \mathbf{x}^{(i)} + \Delta\mathbf{x}^{(i)}, \tag{71}$$

where $\mathbf{G} = \partial\mathbf{f}/\partial\mathbf{x}$ is the Jacobian (the conductance matrix of the linearized circuit). Each iteration requires the solution of a sparse linear system. Newton-Raphson converges very fast close to the solution but can fail from a poor starting point. Simulators therefore fall back to continuation methods such as gmin stepping (temporarily adding conductances to ground) or source stepping (ramping up the supplies). The same Newton iteration is the inner engine of every other analysis in this appendix.

A **dc sweep** (`.dc` in ngspice, a `sweep` around `op` in VACASK) repeats the operating point analysis while stepping a source or a parameter, using each solution as the starting point of the next one. It yields static transfer characteristics, e.g., the dc output of a power detector versus a bias voltage, or the multiplier characteristic of a Gilbert cell (Section 5.3).

> **!** Always Check the Operating Point First
>
> Every small-signal analysis (ac, acsp, noise) linearizes around the operating point and inherits all its errors. Check bias currents, $g_{\mathrm{m}}/I_{\mathrm{D}}$ values, and saturation margins before looking at any other result. Note also that circuits with positive feedback (latches, oscillators) have an operating point that is an *unstable* equilibrium: it is a valid solution of the equations, but the real circuit never stays there. For bistable circuits, the operating point may not converge at all, and a defined state (a reset or a nodeset) helps.

10.1.2 Small-Signal AC Analysis

The **ac analysis** (`ac`) assumes a small sinusoidal excitation, $\mathbf{x}(t) = \mathbf{x}_0 + \mathrm{Re}\{\mathbf{X}e^{j\omega t}\}$, and linearizes Equation 70 around the operating point. In the frequency domain, this gives one complex linear system per frequency:

$$(\mathbf{G} + j\omega\mathbf{C})\mathbf{X}(j\omega) = \mathbf{U}(j\omega), \qquad \mathbf{G} = \frac{\partial\mathbf{f}}{\partial\mathbf{x}}, \quad \mathbf{C} = \frac{\partial\mathbf{q}}{\partial\mathbf{x}}, \tag{72}$$

with both Jacobians evaluated at $\mathbf{x}_0$. As the system is linear, the amplitude of the source is arbitrary (usually 1 V), and the results are transfer functions: gain, phase, bandwidth, input and output impedances, and the loop gain of feedback circuits (for stability). An ac sweep is very fast, as it requires only one sparse linear solve per frequency point.

By construction, the ac analysis cannot show any distortion, compression, or frequency conversion: the response appears only at the excitation frequency.

10.1.3 S-Parameter Analysis

The **S-parameter analysis** (`acsp` in VACASK, `.sp` in ngspice) is an ac analysis with *ports*. Each port is an excitation source with a reference impedance Z_0 (usually 50 Ω). With the voltage V_i and the current I_i flowing into port i, the incident and reflected power waves and the scattering parameters are [3]

$$a_i = \frac{V_i + Z_0 I_i}{2\sqrt{Z_0}}, \quad b_i = \frac{V_i - Z_0 I_i}{2\sqrt{Z_0}}, \quad S_{ij} = \frac{b_i}{a_j} \quad (a_k = 0 \ \text{ for } \ k \neq j). \tag{73}$$

For an amplifier, S_{11} describes the input match (return loss, see Equation 32), S_{21} the gain, S_{12} the reverse isolation, and S_{22} the output match. From the same data, the stability factors K and μ follow directly (Equation 34 and Equation 36). S-parameters are also the natural description of passive structures like inductors, transformers, couplers, and filters, which are usually characterized by electromagnetic (EM) field simulation and enter the circuit simulation as Touchstone data or as fitted lumped-element models.

10.1.4 Small-Signal Noise Analysis

The **noise analysis** (`noise`) models every noise source of the circuit (thermal noise of resistors and channels, shot noise of junctions, flicker noise, see Section 2.3.1) as a stationary random source, whose power spectral density (PSD) $S_i(f)$ is evaluated at the dc operating point. For uncorrelated sources, the contributions add in power at the output

$$S_{\text{out}}(f) = \sum_i |H_i(f)|^2 S_i(f), \tag{74}$$

where $H_i(f)$ is the small-signal transfer function from source i to the output. Correlated sources, such as the induced gate noise and the channel noise of a MOSFET (see Section 2.3.1), add cross terms, which the simulator takes into account. Simulators either solve the linearized circuit once per noise source, or obtain all $H_i(f)$ at once from a single adjoint (transposed) solve per frequency. The results are the output noise PSD, the input-referred noise (dividing by the gain), and, most valuable for design, a ranking of the **contributions** of every device and every noise mechanism. The noise factor (Equation 21) follows as the ratio of the total output noise to the part contributed by the source resistance alone. A load resistor that only models the next stage must be excluded from the output noise, as its noise is not part of the circuit under test.

The noise analysis is the standard tool for LNAs, baseband amplifiers, TIAs, and bias and reference circuits. Its key assumption, however, is a *constant* operating point. In a circuit driven by a large periodic signal, the noise sources are modulated by the signal, and noise from many frequencies is translated to the output frequency. A plain noise analysis of a mixer, a sampling circuit, or an oscillator is therefore wrong, as we will see in Section 10.2.5.

10.1.5 Transient Analysis

The **transient analysis** (`tran`) makes no assumption about the waveforms. It discretizes time, $t_{n+1} = t_n + h_n$, and replaces the time derivative in Equation 70 with a numerical integration formula. With the simplest one, the backward Euler method, the equations at the new time point become

$$\mathbf{f}(\mathbf{x}_{n+1}) + \frac{\mathbf{q}(\mathbf{x}_{n+1}) - \mathbf{q}(\mathbf{x}_n)}{h_n} = \mathbf{u}(t_{n+1}), \tag{75}$$

which is solved for $\mathbf{x}_{n+1}$ with the Newton iteration of Equation 71. Practical simulators use the trapezoidal rule or the Gear (backward differentiation) formulas of second order, and adapt the time step h_n by estimating the local truncation error.

The transient analysis is the most general one: it handles any nonlinearity, any waveform, start-up, settling, and digital control. It is therefore the reference that all other analyses can be checked against. Its drawbacks in RF circuits are:

- **Cost.** The number of time points is the simulated time divided by the step size. The step size is set by the fastest signal (the LO or RF carrier and its edges), the simulated time by the slowest effect (settling of high-Q networks and bias circuits, the IF signal, and the frequency resolution $1/T_{\mathrm{sim}}$ of a subsequent FFT).
- **Numerical artifacts.** The trapezoidal rule can show spurious ringing, while the Gear formulas add numerical damping, which can make an oscillator die out or attenuate a high-frequency carrier. A maximum time step that resolves the carrier is therefore mandatory. In a 160 GHz receiver testbench, for example, omitting it let the Gear integrator damp the carrier, and the IF output came out twelve times too small in a run that finished quickly and looked healthy [55].
- **Spectral analysis.** Distortion and spurs require an FFT of a waveform in steady state, with coherent sampling or windowing, resampled to a uniform time grid, and with simulator tolerances tight enough that the numerical noise floor lies well below the products of interest.

i Note 9: Transient Simulation Effort of a Mixer

We want to simulate the conversion gain of a downconversion mixer with $f_{\mathrm{LO}} = 2.4\,\mathrm{GHz}$ and an IF around 1 MHz, with a frequency resolution of 10 kHz. The simulated time must be at least

$$T_{\mathrm{sim}} \geq \frac{1}{10\,\mathrm{kHz}} = 100\,\mu\ \mathrm{s}\ .$$

With 50 time points per LO period, the step size is $h = 1/(50 \cdot 2.4\,\mathrm{GHz}) \approx 8.3\,\mathrm{ps}$, and we need

$$N = \frac{T_{\mathrm{sim}}}{h} = 100\,\mu\ \mathrm{s} \cdot 50 \cdot 2.4\,\mathrm{GHz} = 1.2 \cdot 10^{7}$$

time points, each requiring a Newton solve of the full circuit, and this excludes the settling time. A PSS analysis of the same mixer integrates only a few tens of LO periods (10^3 to 10^4 time points, depending on the time resolution per period), and the conversion gain then follows from a PAC analysis with one linear solve per IF frequency (Section 10.2.4).

10.1.6 Comparison of the Fundamental Analyses

Table 8: Comparison of the fundamental analyses.

Analysis	Solves	Assumption	Typical RF use
`op`, dc sweep	$\mathbf{f}(\mathbf{x}_0) = \mathbf{u}_{\mathrm{DC}}$	static	bias point, dc transfer curves
`ac`	Equation 72	small signal, LTI	gain, bandwidth, impedances, loop gain
`acsp`	Equation 72 with ports	small signal, LTI	matching, gain, isolation, stability, passives
`noise`	Equation 74	small signal, stationary noise	NF of LNAs and baseband, noise contributions
`tran`	Equation 75	none	start-up, settling, switching, large-signal reference

10.2 Advanced RF Analyses

The fundamental analyses either linearize around a constant operating point (ac, noise) or integrate the full nonlinear equations in time (tran). Many RF circuits sit in between: a **large periodic signal**, such as the LO of a mixer, the clock of a switched-capacitor (SC) filter or sampler, the carrier of a PA, or the oscillation itself in an oscillator, sets a periodically time-varying operating point, while the information-bearing signal and the noise are small. The advanced analyses exploit this structure in two steps:

1. **Large-signal step.** The periodic steady state is computed directly, either in the time domain (shooting PSS, Section 10.2.2) or in the frequency domain (HB, Section 10.2.3), without simulating the settling transient.
2. **Small-signal step.** The circuit is linearized around this periodic steady state. The resulting LPTV system is analyzed for frequency-translating transfer functions (PAC, HBAC, Section 10.2.4) and for cyclostationary noise (PNOISE, HBNOISE, Section 10.2.5).

Transient noise (Section 10.2.1) is the exception: it is a straightforward extension of the transient analysis, without any linearization.

10.2.1 Transient Noise

In a **transient noise** simulation, every noise source of the circuit is replaced by a random time-domain waveform with the correct PSD within a band from $f_{\min}$ to $f_{\max}$, and injected into an ordinary transient analysis. White noise is generated from independent random samples, and flicker noise is approximated by a sum of filtered (Lorentzian) processes whose corner frequencies cover the band from $f_{\min}$ to $f_{\max}$. The samples are either held constant on a fixed time grid (zero-order hold) or integrated as stochastic differential equations along the actual time steps. To resolve the noise up to $f_{\max}$, the time step is limited to a fraction of $1/(2f_{\max})$.

Because the circuit equations are integrated without linearization, transient noise includes in principle everything: switching, strong nonlinearity, sampling and noise folding, the conversion of noise into jitter, and even noise-induced decisions such as metastability and bit errors

in comparators. In practice, it is also the numerically most fragile of all analyses, and the price is high:

- **Statistics.** Each run produces one random realization. A PSD, a jitter value, or an error probability requires long records, averaging, or many runs with different random seeds, and the frequency resolution is $1/T_{\mathrm{sim}}$. A fixed seed reproduces a run only for an unchanged netlist, so after a circuit change, a difference within the scatter of the seeds is not a result.
- **Run time.** The step size is limited by $f_{\max}$, even in parts of the waveform where the signal itself would allow large steps.
- **Band limits.** Flicker noise down to $f_{\min}$ requires $T_{\mathrm{sim}} \gg 1/f_{\min}$, and a default lower band edge derived from $f_{\max}$ (in VACASK, $f_{\max}/1000$) can silently remove all flicker noise at the offsets of interest. When the noise is integrated along the actual time steps, white noise extends up to the Nyquist frequency of these steps. The waveform must then be resampled at least as finely as the simulator's own steps before a PSD is computed, or out-of-band noise aliases into the band.
- **Numerical artifacts.** Device noise is tiny compared to the signals, and on stiff, strongly nonlinear circuits the random excitation itself can cause numerical errors that a noiseless run does not show. Run the analysis at two or more reduced noise amplitudes (a scaling factor such as `noisescale` in VACASK makes this easy), and use the part of the output noise power that scales with the square of the amplitude. Accept a higher-order part only if its size and order match a physical mechanism. A run at zero noise amplitude with otherwise identical settings, which uses the same time grid, shows the numerical floor.
- **Clocked circuits.** For comparators and sampled-data circuits, make sure that the run completes for the full number of decisions or cycles the statistics need, and validate the result against an independent estimate, such as an analytic kT/C budget or a periodic noise analysis.

Typical applications are comparators (input-referred noise from the decision statistics: the dc input at which the output is zero in about 16 % of the decisions equals the rms input noise [56]), sampled-data circuits and ADCs (kT/C noise and the SNR including all effects), the jitter of oscillators, clock buffers, and dividers, and, increasingly important, an independent cross-check of PNOISE and HBNOISE results. Transient noise shares no solver code with the periodic analyses, but it uses the same device noise models. Their agreement therefore validates the implementation of the analyses rather than the noise models, and its statistics limit it to about a decibel.

10.2.2 Periodic Steady State by Shooting (PSS)

Consider a circuit driven by large-signal sources with period $T = 1/f_0$. After the start-up transient has died out, the circuit is in its **periodic steady state** $\mathbf{x}_{\mathrm{s}}(t+T) = \mathbf{x}_{\mathrm{s}}(t)$. Let $\boldsymbol{\phi}_T(\mathbf{x}_0)$ denote the state reached after integrating the circuit equations over one period, starting from the state $\mathbf{x}_0$. The **shooting method** searches for the initial state that reproduces itself after one period [57]:

$$\boldsymbol{\phi}_T(\mathbf{x}_0) - \mathbf{x}_0 = \mathbf{0}. \tag{76}$$

This is again solved with Newton's method,

$$(\mathbf{\Phi}_T - \mathbf{I})\Delta\mathbf{x}_0 = \mathbf{x}_0 - \phi_T(\mathbf{x}_0), \qquad \mathbf{\Phi}_T = \frac{\partial \phi_T}{\partial \mathbf{x}_0}, \tag{77}$$

where the sensitivity matrix $\mathbf{\Phi}_T$ (the *monodromy matrix*) is computed alongside the transient integration over one period. Every Newton iteration therefore costs one period of transient simulation. For a linear circuit, the method converges in a single iteration, and a mildly nonlinear circuit needs a few. Usually, a short stabilization transient precedes the Newton loop to provide a good initial guess. In large circuits, the dense matrix $\mathbf{\Phi}_T$ would be prohibitive, and matrix-free iterative (Krylov subspace) solvers made shooting practical for circuits with thousands of nodes [58]. VACASK currently integrates the full matrix $\mathbf{\Phi}_T$, which is fine for circuit blocks but becomes expensive for large (e.g., post-layout) netlists.

The main advantage is illustrated in Figure 107: a high-Q circuit that needs many periods to settle in a transient analysis (about 110 for $Q = 50$) is solved with one or a few shooting iterations of one period each, because Newton's method jumps directly to the steady state.

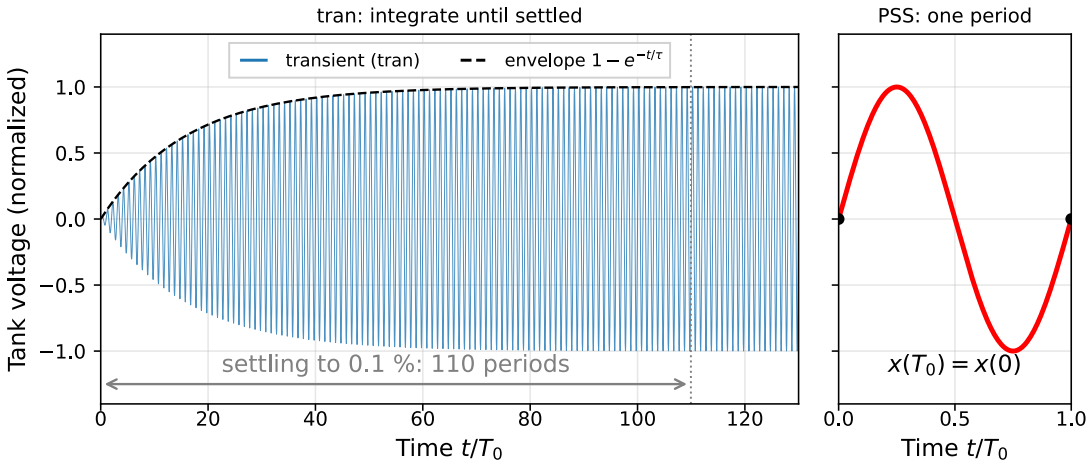


Figure 107: Transient response of a resonator with $Q = 50$ that is driven at its resonance frequency from $t = 0$. A transient analysis has to integrate about 110 periods until the amplitude is within 0.1 % of its final value. A PSS analysis solves for the single steady-state period (red) directly, and for this linear tank a single shooting iteration over one period suffices.

Since the integration happens in the time domain with adaptive steps, PSS handles sharp edges and strongly nonlinear behavior just as well as a transient analysis: switching mixers, SC circuits, frequency dividers, rectifiers, and switching PAs (class D, E, F). Its main limitation is that it needs a **single period**: all large-signal sources must be periodic with a common period T. A two-tone test with tones at 2.400 GHz and 2.401 GHz, for example, has a common period of 1 µs, i.e., 2400 carrier periods, which makes shooting inefficient. Note that only the *large* signals count: a small RF signal handled by PAC (Section 10.2.4) does not enter the period.

For **oscillators** (autonomous circuits), the period is unknown and becomes an additional unknown of the Newton iteration. As any time-shifted copy of a periodic solution is also a solution, a phase condition is added to make the problem well-defined. An autonomous PSS directly delivers the oscillation frequency and the steady-state waveform, without simulating the start-up. The circuit has to be kicked away from its (unstable) dc operating point by an initial condition, otherwise the analysis converges to the trivial dc solution.

> **!** PSS Does Not Prove Start-Up
>
> PSS finds a periodic solution, but it does not tell whether the circuit actually reaches it after power-up. Whether an oscillator starts reliably over all corners (or whether a divider locks into the intended mode) must still be checked with a transient analysis.

From the periodic waveforms, the Fourier coefficients follow directly: harmonic distortion, the dc shift of a rectifier or power detector, the duty cycle of an LO signal, the efficiency of a PA (from the output power at the fundamental and the averaged supply power), and, for oscillators, frequency and amplitude versus a tuning voltage (i.e., the tuning curve and K_{VCO}, see Section 6.6).

10.2.3 Harmonic Balance (HB)

Harmonic balance describes every unknown by a truncated Fourier series with K harmonics (not to be confused with the stability factor K of Section 4.2),

$$\mathbf{x}(t) = \sum_{k=-K}^{K} \mathbf{X}_k \, e^{jk\omega_0 t}, \qquad \mathbf{X}_{-k} = \mathbf{X}_k^{*}, \tag{78}$$

and solves for the Fourier coefficients $\mathbf{X}_k$ directly [59]. Inserted into Equation 70, the time derivative becomes an exact multiplication by $jk\omega_0$, and the circuit equations must hold ("balance") at every harmonic:

$$\mathbf{F}_k(\mathbf{X}) + jk\omega_0 \, \mathbf{Q}_k(\mathbf{X}) = \mathbf{U}_k, \qquad k = -K, \ldots, K. \tag{79}$$

Here, $\mathbf{F}_k$ and $\mathbf{Q}_k$ are the Fourier coefficients of $\mathbf{f}(\mathbf{x}(t))$ and $\mathbf{q}(\mathbf{x}(t))$. Nonlinear device equations cannot be evaluated in the frequency domain, so HB transforms the present estimate of the spectrum into the time domain (inverse FFT), evaluates the devices at the sampled time points, and transforms the results back (FFT). All harmonics of all nodes are solved for jointly with Newton's method, so the linear system has $(2K+1)$ times as many unknowns as the circuit.

With several large tones (e.g., a two-tone test, or LO and blocker), the solution is *quasi-periodic*, and the spectrum contains all mixing products $k_1\omega_1 + k_2\omega_2$. HB handles this naturally, without the long simulated time that a transient analysis needs for closely spaced tones, and the spectrum is truncated, e.g., to $|k_1| + |k_2| \le K$ ("diamond" truncation). The truncation then limits the accuracy of small intermodulation products, so check that the IM3 products rise with a slope of 3 before extrapolating an intercept point.

Strengths and weaknesses of HB are complementary to those of shooting:

- **Linear and distributed elements** are described exactly in the frequency domain, including transmission lines and, in simulators that accept them, frequency-domain data like S-parameter files from EM simulations. Such data, or a lumped model fitted to them, must be valid and passive from dc up to the highest harmonic and mixing product, not only in the band of interest.
- **High-Q circuits** pose no settling problem, as no transient is simulated.

- **Mildly nonlinear circuits** (LNAs, linear PAs, baseband amplifiers) are described accurately with a few harmonics, which makes HB very efficient for compression, IP3, and blocker simulations.
- **Strongly nonlinear circuits with sharp edges** need many harmonics, as the spectrum of a square wave decays only as $1/k$ (see Figure 108). Every additional harmonic enlarges the nonlinear system, which also becomes increasingly ill-conditioned, so accuracy has to be traded against convergence and run time, and a truncated spectrum shows Gibbs ringing. For hard-switching circuits (SC circuits, switching mixers with square-wave LO, dividers), HB may not converge at all, and PSS is the better choice.
- **Convergence** can be difficult at high drive levels. Simulators ramp the source amplitudes (source stepping), and it often helps to sweep the drive level up and use each solution as the starting point of the next one.

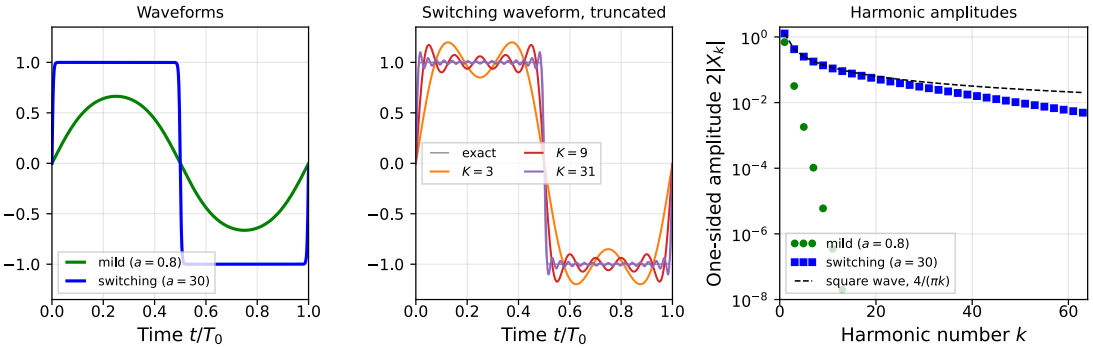


Figure 108: Harmonic content of a mildly nonlinear and of a hard-switching waveform, both modeled as $\tanh(a \sin \omega_0 t)$ (left: $a = 0.8$ and $a = 30$). The harmonics of the mildly nonlinear waveform fall off very quickly, so a few harmonics describe it well. Those of the switching waveform decay only roughly as $1/k$ (right, one-sided amplitudes $2|X_k|$), and a harmonic-balance spectrum truncated at K harmonics shows Gibbs ringing at the edges (center).

The harmonic count K is the key accuracy parameter of HB, just as the step size is for a transient analysis. Too small a K does not show up as a convergence failure: Newton's method converges, but to the solution of the truncated problem. The convergence in K need not be monotonic either, so two neighboring values that agree prove little. Compare with a much larger K or, better, with an independent engine (PSS), and repeat the check over the whole drive sweep, as the required K grows with the drive level: a value that is sufficient at the nominal level can be several dB off beyond the 1 dB compression point. Oscillators can also be simulated with HB, using an autonomous variant that adds the unknown frequency and a phase condition, similar to the autonomous PSS. Check whether your simulator supports it (in VACASK, oscillators are solved with the autonomous `pss`).

Table 9: Comparison of shooting (PSS) and harmonic balance (HB)

Property	PSS (shooting)	HB
Domain	time	frequency (devices evaluated in time)
Unknowns	initial state $\mathbf{x}_0$	Fourier coefficients $\mathbf{X}_k$
Sharp edges, switching	well suited	many harmonics, may not converge
Mildly nonlinear	well suited	very efficient
Distributed elements, S-parameter data	need a time-domain model	exact (simulator-dependent)
Several incommensurate large tones	not possible	well suited
High-Q, long settling	efficient	efficient
Oscillators	autonomous PSS	autonomous HB (simulator-dependent)
Small-signal and noise analyses	PAC, PNOISE	HBAC, HBNOISE

10.2.4 Periodic Small-Signal Analysis (PAC, HBAC)

We now apply a small signal $\Delta\mathbf{u}(t)$ to a circuit in its periodic steady state $\mathbf{x}_\mathrm{s}(t)$ and linearize Equation 70 around $\mathbf{x}_\mathrm{s}(t)$:

$$\mathbf{G}(t)\,\Delta\mathbf{x}(t) + \frac{\mathrm{d}}{\mathrm{d}t}[\mathbf{C}(t)\,\Delta\mathbf{x}(t)] = \Delta\mathbf{u}(t). \tag{80}$$

In contrast to Equation 72, the Jacobians $\mathbf{G}(t) = \partial\mathbf{f}/\partial\mathbf{x}$ and $\mathbf{C}(t) = \partial\mathbf{q}/\partial\mathbf{x}$ are now evaluated along $\mathbf{x}_\mathrm{s}(t)$ and are periodic in time. The small-signal circuit is a **linear periodically time-varying (LPTV)** system. Exactly like the time-variant mixer of Section 5.2, such a system converts an input at the offset frequency f into outputs at all **sidebands** $f + kf_0$. We therefore expand the Jacobians into Fourier series, $\mathbf{G}(t) = \sum_p \mathbf{G}_p e^{jp\omega_0 t}$ (same for $\mathbf{C}$), and the excitation and the response into sideband phasors, $\Delta\mathbf{u}(t) = \sum_k \Delta\mathbf{U}_k e^{j(\omega+k\omega_0)t}$ and $\Delta\mathbf{x}(t) = \sum_k \Delta\mathbf{X}_k e^{j(\omega+k\omega_0)t}$. Collecting the terms at each sideband yields the **conversion matrix** equation [53]

$$\sum_{l=-K}^{K} [\mathbf{G}_{k-l} + j(\omega + k\omega_0)\,\mathbf{C}_{k-l}]\Delta\mathbf{X}_l = \Delta\mathbf{U}_k, \qquad k = -K, \ldots, K, \tag{81}$$

which couples the sidebands l to the sidebands k through the harmonics $\mathbf{G}_{k-l}$ and $\mathbf{C}_{k-l}$ of the time-varying operating point. Without time variation, only $\mathbf{G}_0$ and $\mathbf{C}_0$ are nonzero, and Equation 81 falls apart into independent copies of the ac analysis of Equation 72. Solving Equation 81 for a source at sideband l and observing sideband k gives the **harmonic transfer function** $H_{k,l}(f)$. The offset frequency f is swept, and each point requires one linear solve of a system with $(2K+1)$ times as many unknowns as the circuit. This is far cheaper than a transient simulation, but the cost grows quickly with K, as discussed below.

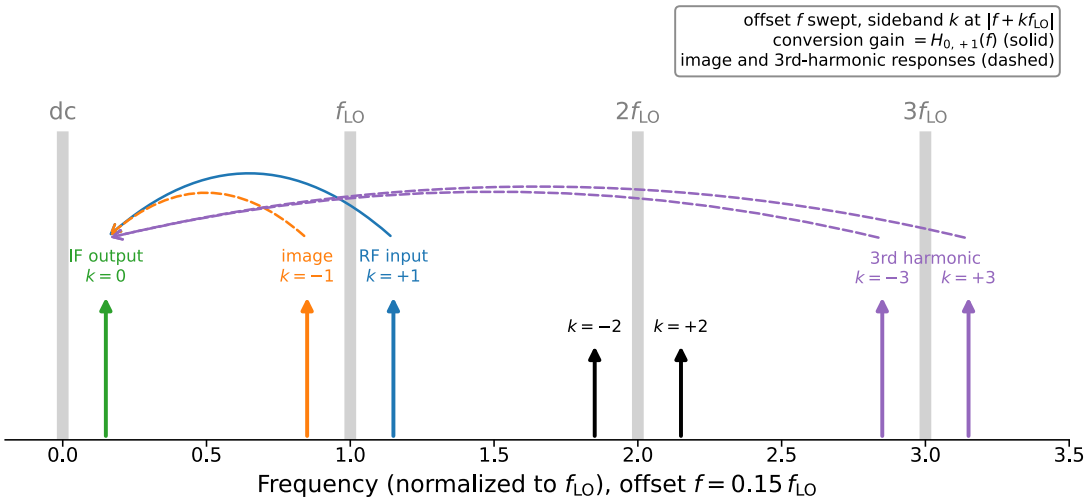


Figure 109: Frequency grid of a periodic small-signal analysis (PAC or HBAC) of a downconversion mixer. The large-signal LO and its harmonics kf_{LO} (gray) are computed by PSS or HB. The small signal lives on the sidebands $|f + kf_{\mathrm{LO}}|$ of the swept offset frequency f. The conversion gain is the transfer from the RF input at sideband $k = +1$ to the IF output at sideband $k = 0$. The image ($k = -1$) and the third-harmonic sidebands ($k = \pm 3$) convert to the same IF.

Figure 109 shows the frequency grid for a downconversion mixer. The LO is the large signal, the RF and IF are small signals:

- The **conversion gain** is the transfer from the RF at $f_{\mathrm{LO}} + f$ (input sideband $+1$) to the IF at f (output sideband 0), i.e., $H_{0,+1}(f)$, swept over the IF band in a single analysis.
- The **image response** is $H_{0,-1}(f)$ from $f_{\mathrm{LO}} - f$. A single mixer converts both sidebands with equal magnitude. In an I/Q (image-reject) receiver, the ratio of the two, evaluated after the I and Q paths are combined, gives the image rejection.
- The **harmonic responses** $H_{0,\pm 3}(f)$ show how much of an interferer around $3f_{\mathrm{LO}}$ appears at the IF, which matters for switching mixers in wideband receivers (compare the Fourier series of the switching function in Section 5.2).
- For an **upconversion mixer**, the input is at the baseband sideband 0 and the output at sideband ± 1.

> ! Why an AC Analysis of a Mixer Fails
>
> An ac analysis linearizes around the dc operating point, where the LO source sits at its dc value. The mixer is "frozen" in one LO state, the circuit is LTI, and no frequency conversion takes place: the ac analysis shows at most a feedthrough at the RF frequency, but no IF output. Frequency-converting circuits need PAC or HBAC.

Further applications of the periodic small-signal analyses are:

- **SC filters and sampling circuits**, where the clock is the large signal. The transfer from input sideband k to output sideband 0 quantifies aliasing, i.e., how much of an input at $f + kf_{\mathrm{clk}}$ appears at f.

- **N-path filters and passive mixers**, where the input impedance around f_{LO} (see Equation 53) is obtained from the transfer at the same sideband.
- **Power detectors**, whose video response (the detected baseband output for an amplitude-modulated input) is the transfer from sideband ± 1 around the carrier to sideband 0.
- **Amplifiers under a large blocker**, where the blocker is the large signal and the wanted signal the small one. This directly shows the gain reduction by desensitization (see Equation 11).
- **Chopper amplifiers**, where the chopping clock is the large signal. The transfer from input sideband 0 to output sideband 0 is the gain after demodulation, while the transfers to the sidebands around the chopping harmonics show how much of the signal appears as ripple.

PAC and **HBAC** solve the same Equation 81. They differ only in the large-signal solution underneath: PAC builds on the shooting PSS (single fundamental, any waveform shape), HBAC on harmonic balance (also quasi-periodic, e.g., with an LO and a large blocker as two tones, but limited by the harmonic count of the HB solution). The naming follows the engine: the analyses on top of shooting carry a "P" (PAC, PNOISE, and in some simulators PXF and PSP for transfer functions and S-parameters), those on top of harmonic balance carry "HB" (HBAC, HBNOISE, HBXF, HBSP).

The truncation K of Equation 81 must cover the harmonic content of $\mathbf{G}(t)$ and $\mathbf{C}(t)$: a few harmonics are enough for smooth waveforms, while hard-switching circuits like SC filters with steep clock edges can require tens to hundreds of harmonics. For the large signal, the PSS-based analyses have a clear advantage: the periodic steady state is computed in the time domain and is not truncated at all, while HB has to carry all these harmonics in its nonlinear system. In the Gilbert-cell example of VACASK, for instance, a large trapezoidal LO drives the switching quad: HB, and with it HBAC, currently does not converge, while PAC delivers the sidebands with 20 harmonics.

The small-signal step, however, is truncated in both families when it is solved with Equation 81, as in VACASK. Every nonzero Jacobian entry becomes a dense $(2K+1) \times (2K+1)$ block, so the memory grows with K^2. With hard switching, the truncation itself becomes the limit: the truncated Fourier series of a switch conductance does not vanish while the switch is off, so the switch leaks in the small-signal model. On high-impedance nodes, such as the virtual ground of an SC integrator, this spurious leak can falsify the low-frequency gain and noise until K reaches several hundred, which may be impractical. Check the convergence in K (keeping in mind that it need not be monotonic, see Section 10.2.3), and fall back to transient simulations where it is not reached.

10.2.5 Periodic Noise Analysis (PNOISE, HBNOISE)

The noise analysis of Equation 74 misses two effects that a large periodic signal introduces:

1. **Modulation.** The noise sources depend on the time-varying operating point: channel thermal noise follows the channel conductance (see Section 2.3.1), shot noise follows the current $I(t)$, and flicker noise depends on the bias as well. Such a source can be written as $n_i(t) = m_i(t)\, w_i(t)$, with a stationary noise process $w_i(t)$ and a periodic modulation $m_i(t)$. Its statistics vary periodically with time, so the noise is **cyclostationary**.
2. **Frequency translation.** The LPTV transfer of Equation 81 moves noise from all sidebands $f + l f_0$ to the output frequency f, which is called **noise folding**.

Combining both effects, with $M_{i,m}$ the Fourier coefficients of $m_i(t)$, S_i the PSD of the stationary process $w_i(t)$, and $H_{i,k}(f)$ the transfer function from source i at sideband k to the output, gives the output PSD [60], [61]

$$S_{\mathrm{out}}(f) = \sum_i \sum_l \left| \sum_k H_{i,k}(f)\, M_{i,k-l} \right|^2 S_i(f + l f_0), \tag{82}$$

with all spectral densities written two-sided. For a constant operating point ($M_{i,0} = 1$, all other coefficients zero, and $H_{i,k} = 0$ for $k \neq 0$), Equation 82 reduces to Equation 74. A single adjoint solve per offset frequency delivers the transfer functions of all sources, and the results are again broken down into contributions per device and per noise mechanism. **PNOISE** builds on the shooting PSS, **HBNOISE** on harmonic balance, exactly as PAC and HBAC. HBNOISE can also start from a multi-tone (quasi-periodic) HB solution, e.g., to obtain the noise figure of a mixer with an LO and a large blocker applied at the same time.

i Note 10: Noise Folding in an Ideal Switching Mixer

An ideal double-balanced switching mixer multiplies the RF input by a square wave toggling between $+1$ and -1 with 50% duty cycle (see Section 5.2), whose Fourier coefficients have the magnitude $|c_k| = 2/(\pi|k|)$ for odd k and vanish for even k. The switches are noiseless, and the only noise is the white thermal noise of the source resistance, with the PSD S_{n} at every frequency. The noise from each sideband k folds to the IF with the weight $|c_k|^2$. By Parseval's theorem, the sum of all weights equals the time average of the squared switching function, which is one:

$$S_{\mathrm{out}} = S_{\mathrm{n}} \sum_k |c_k|^2 = S_{\mathrm{n}}.$$

The wanted RF signal at sideband $+1$ is converted with the power gain $|c_1|^2 = 4/\pi^2$. In the single-sideband (SSB) noise factor, only the source noise at the signal frequency counts as input noise, hence

$$F_{\mathrm{SSB}} = \frac{S_{\mathrm{out}}}{|c_1|^2 S_{\mathrm{n}}} = \frac{\pi^2}{4} \approx 3.9\,\mathrm{dB}\ .$$

If the signal occupies both sidebands ± 1 (as in a direct-conversion receiver), the source noise of both counts as input noise, and the double-sideband (DSB) noise factor is $F_{\mathrm{DSB}} = \pi^2/8 \approx 0.9\,\mathrm{dB}$, i.e., 3 dB lower. An analysis that keeps only the sidebands ± 1 ($K = 1$) would miss the 19 % of the noise folded from around $3f_{\mathrm{LO}}$, $5f_{\mathrm{LO}}$, and so on, and underestimate the SSB noise figure by 0.9 dB (3.0 dB instead of 3.9 dB, see Figure 110).

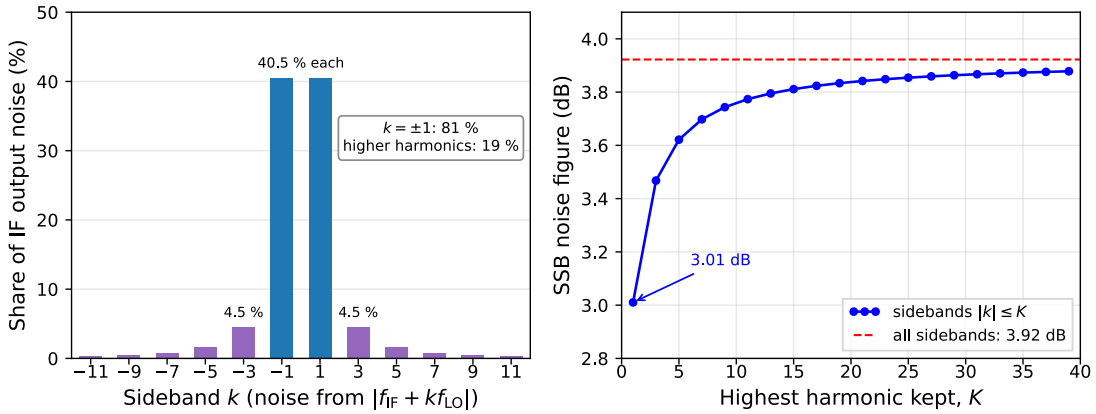


Figure 110: Noise folding in an ideal double-balanced switching mixer driven by a square-wave LO with 50 % duty cycle. Left: share of the white source noise that each sideband k contributes to the IF output, $|c_k|^2 = (2/(\pi k))^2$ for odd k, and the shares add up to one. Right: single-sideband noise figure obtained when the analysis keeps only the sidebands up to $|k| \le K$. Keeping only the first harmonic ($K = 1$) underestimates the noise figure by 0.9 dB.

The periodic noise analyses are required wherever a large periodic signal modulates the noise or translates it in frequency:

- **Mixers**: SSB and DSB noise figure, including the noise folded from the image and from the sidebands of the LO harmonics, and the flicker noise of the switches that the LO translates to the IF (compare the discussion of passive and active mixers in Section 5.2).
- **Sampling and SC circuits**: wideband switch noise folds into the baseband (kT/C noise). Many harmonics are needed, and with steep clock edges the conversion matrix may not converge in practice (see Section 10.2.4), so transient noise or an analytic kT/C budget are the alternatives.
- **Power detectors**: the noise with the LO (or the input carrier) applied, which can differ substantially from the quiescent noise analysis.
- **Amplifiers under a blocker**: the noise figure in the presence of a large interferer.
- **Chopper amplifiers**: the flicker noise that remains at baseband after chopping, and the white noise folded back from the chopping harmonics.
- **Oscillators**: the phase noise. An oscillator's noise is cyclostationary by nature, and its noise sources are converted into phase noise in a time-dependent way, which the impulse sensitivity function describes [62]. A rigorous nonlinear analysis shows that the phase of an oscillator diffuses like a random walk, so that the output spectrum is a Lorentzian that stays finite at the carrier, whereas a linear analysis predicts phase noise that grows without bound towards the carrier [63]. RF simulators compute the phase noise with a periodic noise analysis around an *autonomous* steady state, including the upconverted flicker noise of the $1/f^3$ region, which Leeson's equation (Equation 56) does not cover. Such an analysis relies on the time-shift invariance of the oscillator: a time-shifted copy of the periodic solution is a solution, too, so the phase of the linearized circuit is free, and the noise accumulates in it as phase noise that rises with $1/f^2$ towards the carrier. Every numerical error that breaks this invariance, the harmonic truncation of HB as well as the time discretization of shooting, pins the phase below some offset frequency. A phase noise that flattens towards the carrier instead of rising with $1/f^2$ is therefore a symptom, not a result, and the analysis has to

be refined until this corner lies below the offsets of interest (Note 11). For oscillators with logic-like edges, such as ring oscillators, the periodic noise analysis needs more harmonics than are affordable, and transient noise is the method of choice. A plain noise analysis of an oscillator is meaningless, as it linearizes around the unstable dc operating point.

> **Note 11: Example: Phase Noise of an LC Oscillator in VACASK**
>
> A parallel LC tank at $f_0 = 1$ GHz ($R_\mathrm{p} = 500\,\Omega$, $C = 10$ pF, $Q = 31$), whose loss is compensated by a noiseless negative conductance $i = -g_1 v + g_3 v^3$ (Van der Pol), oscillates with $V_\mathrm{p} = 1$ V. The tank resistance is then the only noise source, and the phase noise follows Equation 57 with $F = 1$, e.g., -126.8 dBc/Hz at 100 kHz offset. VACASK's `pnoise` around the autonomous PSS (`pss` with `oscillator=1`) reproduces it within 0.02 dB from 10 kHz to 1 MHz, but only with a fine enough time step. With the default settings, the result flattens below about 20 kHz ($f_0/50\,000$) and is already 7.5 dB too low at 10 kHz. The corner depends neither on the number of harmonics (`truncharm` from 5 to 80 gives the same result) nor on the tolerances, but on the time step with which `pnoise` integrates the period that it samples for the conversion matrix. Four times as many points per period (`pss_minpts`) move it four times lower, and limiting the time step between the sampling points (`options tran_fbr=0.02` before the `pnoise` analysis) moved it below 1 kHz, i.e., $f_0/10^6$. All numbers are from the VACASK version of the IIC-OSIC-TOOLS container 2026.10.

For the sweep of the offset frequency, keep in mind that the upconverted flicker noise forms narrow skirts around every harmonic of the large signal. A coarse logarithmic sweep easily steps over them, and a point exactly at a harmonic is undefined for $1/f$ noise. A dense grid near (but not at) the harmonics resolves them.

10.3 Choosing the Right Analysis

Table 10 summarizes the advanced analyses, and Figure 111 condenses the selection into two questions: is there a large periodic signal that sets the operating point, and how sharp are the resulting waveforms?

Table 10: Comparison of the advanced RF analyses

Analysis	Builds on	Linearized	Captures	Main limitation
transient noise	tran	no	in principle everything: switching, folding, jitter, decisions	statistics, run time, numerical artifacts
PSS	shooting (time)	no	periodic steady state, harmonics, oscillation frequency	one common period
HB	Fourier series	no	(quasi-)periodic steady state, IM products	many harmonics for sharp edges
PAC, HBAC	PSS, HB	yes (LPTV)	conversion gain, sideband transfer, aliasing	small signals only
PNOISE, HBNOISE	PSS, HB	yes (LPTV)	cyclostationary noise, folding, mixer NF	small noise only

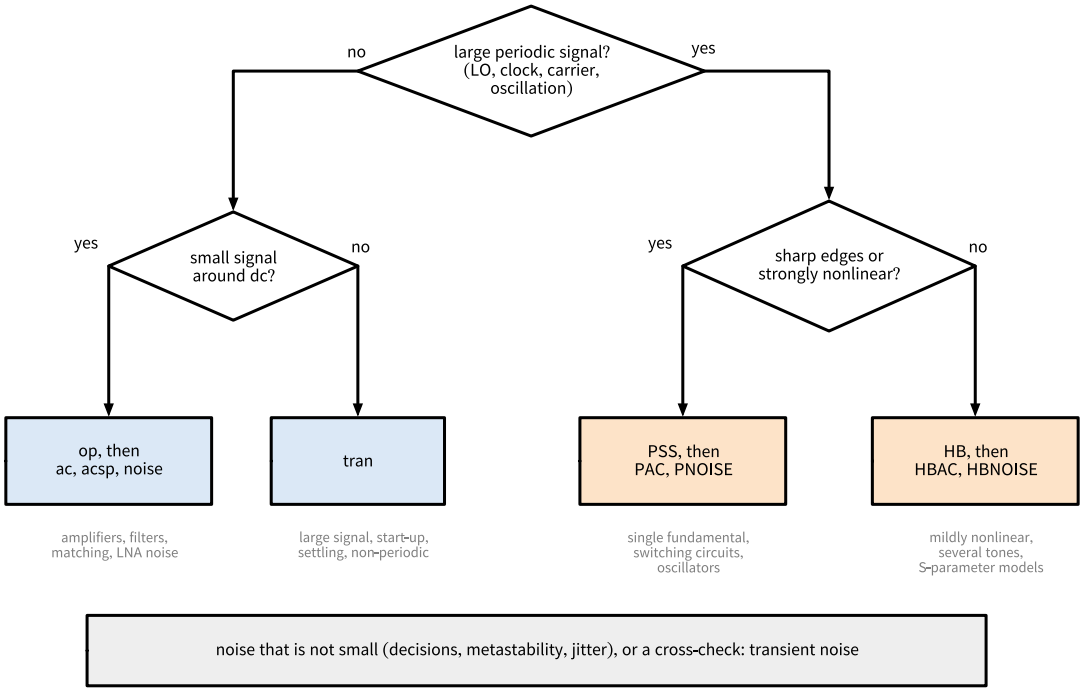


Figure 111: Choosing an analysis. The first question is whether a large periodic signal (LO, clock, carrier, or the oscillation itself) sets the operating point, the second is how sharp the resulting waveforms are. Transient (noise) simulation remains the reference that every other analysis should be cross-checked against.

For typical RF building blocks, Table 11 lists the key metrics and the recommended analyses. The first analysis in each row is usually the workhorse, and the others refine or cross-check it.

Table 11: Recommended analyses for typical RF building blocks.

Circuit	Key metrics	Recommended analyses
LNA	match, gain, NF, stability, $P_{1\mathrm{dB}}$, IIP3, blocker NF	op, acsp and noise, HB for $P_{1\mathrm{dB}}$ and two-tone IIP3, HBAC and HBNOISE with a blocker
Mixer	conversion gain, SSB/DSB NF, IIP3, image and harmonic rejection	PSS (hard-switching LO) or HB, PAC or HBAC, PNOISE or HBNOISE, two-tone HB for IIP3
Oscillator, VCO	start-up, frequency, tuning range, phase noise	tran for start-up, autonomous PSS, transient noise or periodic noise around the autonomous PSS (simulator-dependent)
SC circuits, sampling, N-path	transfer incl. aliasing, settling, kT/C noise	tran, PSS, PAC and PNOISE if they converge in K (not HB), transient noise
Power detector	responsivity, compression, video bandwidth, NEP, NF	HB or PSS power sweep, HBAC or PAC, noise with the LO off, HBNOISE or PNOISE with the LO on, transient noise
Power amplifier	output power, efficiency, AM/AM, AM/PM, harmonics	HB power sweep (with EM models of the matching network), PSS for switching PAs, tran
Divider, LO generation	operating range, duty cycle, I/Q phase error, jitter	tran, PSS, PNOISE or transient noise for jitter
Chopper amplifier	gain, ripple, residual flicker noise	tran, PSS with PAC and PNOISE
Comparator, latch	offset, noise, metastability	tran, transient noise
Passives	Q, S-parameters	acsp with EM-derived models

A complete PLL contains time constants from the VCO period (ns) to the loop settling time (µs), which makes a full transistor-level steady-state simulation very expensive. In practice, the blocks (VCO, divider, phase detector, charge pump) are characterized individually with the analyses above, and the loop is evaluated with the linear phase-domain model of Section 7 and with behavioral transient simulations.

Whatever analysis is used, a result is only trustworthy after it has been checked:

- **Convergence parameters.** Increase the harmonic count (HB, PAC, PNOISE) well beyond the value that seems sufficient, tighten the maximum time step and the tolerances (tran), and lengthen the stabilization time (PSS) until the result no longer changes, and repeat these checks at the highest drive level.
- **Independent methods.** HB and PSS compute the same steady state with different algorithms, and where both converge, PAC must agree with HBAC and PNOISE with HBNOISE. PAC or HBAC conversion gain can be compared with a two-tone transient, and PNOISE or

HBNOISE with transient noise at reduced noise amplitude. With the large signal switched off, PNOISE and HBNOISE must reproduce the plain noise analysis.

- **Simulator messages.** A simulation that finishes is not necessarily a correct one. Check the log for aborted analyses, convergence warnings, and devices that were silently dropped from the netlist. Neither a zero exit status nor an existing result file proves that an analysis ran, so delete old results before rerunning.

> **i** Note 12: Example: A 160 GHz Power Detector
>
> The open-source SPARX receiver [55] contains a Schottky-diode power detector at 160 GHz, characterized with VACASK as a mixer with a 159 GHz LO. Its testbenches use most of the analyses of this appendix:
>
> - **HB** power sweep for the transfer curve (responsivity and compression), cross-checked with **PSS** at nine drive levels. The two agree within 1.2 % over an input range of 80 dB.
> - **HBAC** for the conversion gain of each sideband to the IF and the video bandwidth, cross-checked with **PAC** (within 0.03 dB).
> - **noise** (LO off) and **HBNOISE** (LO on) for the noise figure: the LO raises the output noise, so the noise analysis at the dc operating point is 1.2 dB to 2.6 dB optimistic over the IF band, and far more at high LO drive. **PNOISE** reproduces HBNOISE within 0.2 dB.
> - **Transient noise** with the LO on, run at reduced noise amplitudes and reduced to its linear part, agrees with HBNOISE within about 1 dB.

10.4 Simulator Support

Table 12 lists the analyses of this appendix in the two open-source simulators used in this course, in the ngspice release 47 and the VACASK version of the IIC-OSIC-TOOLS container 2026.10. The ngspice development branch already contains a driven PSS with PAC, PNOISE, PXF and PSP, and quasi-periodic analyses (`qpss`, `qpac`, `qpnoise`), which are not part of a release yet (November 2026). Among the commercial RF simulators, Cadence Spectre RF offers all of them, plus further variants like quasi-periodic shooting and periodic transfer functions and S-parameters (PXF, PSP), while Keysight ADS is built around harmonic balance with its small-signal and noise variants, and adds envelope simulation for modulated signals.

Table 12: Analyses in the open-source simulators VACASK and ngspice.

Analysis	VACASK (container 2026.10)	ngspice 47
dc operating point, dc sweep	`op`, `op` inside a `sweep`	`.op`, `.dc`
ac	`ac`	`.ac`
S-parameters	`acsp`	`.sp`
noise	`noise`	`.noise`
transient	`tran`	`.tran`
transient noise	`tran` with `noisefmax` (device noise)	`.tran` with `trnoise` sources
PSS	`pss` (driven and autonomous)	`.pss` (experimental, autonomous only)
HB	`hb` (single and multi-tone)	not available
PAC, PNOISE	`pac`, `pnoise`	not available
HBAC, HBNOISE	`hbac`, `hbnoise`	not available

The following VACASK excerpt shows the two-step flow of Section 10.2 for a downconversion mixer: the periodic steady state with the LO is computed and stored once, and the conversion gain and the noise figure are evaluated around it. The RF source carries a unit small-signal excitation at sideband $+1$, and the IF output (node `out`) is observed at sideband 0. Device models and the mixer itself are omitted, and the syntax is that of the VACASK version above (older builds treated `pss` as autonomous by default and used other parameter names). The noisy source resistance supplies the input noise. In the pumped circuit, its contribution at the output also contains its noise folded from the image and the harmonic sidebands. The SSB noise factor therefore divides the output noise by the source noise converted from the signal sideband alone, which is $4kTR_{\mathrm{s}}$ times the power gain `gain` that `pnoise` reports for the input sideband (as in Note 10), and not by the total contribution of the source resistance.

```
// LO: large-signal sine, RF: small-signal tone at sideband +1 (f_LO + f)
vlo (lo 0) vsource type="sine" sinedc=0 ampl=0.5 freq=2.4G
vrf (rfs 0) vsource dc=0 spur={1} smag=[1]
rs (rfs rf) resistor r=50

control
  var flo=2.4G
  analysis op1 op
  // large signal: periodic steady state with the LO, stored for reuse
  analysis pss1 pss tper=1/flo tstab=20/flo store="lo_ss"
  // conversion gain from the RF (sideband +1) to the IF (sideband 0)
  analysis pac1 pac ( ic="lo_ss" psssolve=0 truncharm=40 outharm=0
                      from=100k to=100M mode="dec" points=10 )
  // output noise at the IF, gain from the RF at sideband +1 (SSB NF)
```

```
  analysis pn1 pnoise ( ic="lo_ss" psssolve=0 truncharm=40
                        in="vrf" inharm=1 out="out" outharm=0
                        from=100k to=100M mode="dec" points=10 )
endc
```

The harmonic-balance flow with `hb`, `hbac`, and `hbnoise` is analogous, with three differences: the large signal is described by its fundamental frequencies and harmonic count (e.g., `freq=[2.4G] nharm=9`) instead of the period, a stored solution is reused with `nodeset="lo_ss" hbsolve=0`, and spurs are given as tone-weight vectors (e.g., `spur={[1]}`, `outspur=[0]`, `inspur=[1]`). VACASK ships worked examples for a Gilbert-cell mixer in multiplier mode (where PAC is compared with HBAC and PNOISE with HBNOISE) and in switching mode (where VACASK's HB currently does not converge, but PAC and PNOISE do), for the PAC analysis of an SC filter, and for the phase noise of an oscillator from transient noise. Further examples at circuit level, including the 160 GHz power detector, are found in the testbenches of the SPARX repository [55].

10.5 Summary

! Key Takeaways

- **op, ac, acsp, noise** linearize around a constant operating point: fast and insightful, but blind to distortion, frequency conversion, and noise folding.
- **tran** is the general reference, but RF time scales make it expensive, and a maximum time step is mandatory.
- **PSS** (time domain) and **HB** (frequency domain) find the periodic steady state directly. PSS suits switching circuits and oscillators, HB mildly nonlinear and multi-tone problems and circuits with distributed elements.
- **PAC, HBAC** and **PNOISE, HBNOISE** linearize around this steady state (LPTV) and give conversion gain, aliasing, and cyclostationary noise including folding.
- **Transient noise** captures everything in principle, at a high cost and with numerical pitfalls. Run at reduced noise amplitude, it is an independent cross-check.
- Increase harmonics, tighten time steps, and compare independent methods before trusting a result.

11 Appendix: Examination Questions

This chapter contains a collection of potential questions covering the topics discussed in this course. The questions are organized by topic and are intended to help students prepare for examinations.

11.1 Introduction and Wireless Basics

1. What is the basic function of a radio-frequency transceiver? Describe the main components and their roles.

2. Explain Friis' transmission formula (Equation 1). What factors determine the received power in a wireless link? How does the free-space path loss (FSPL) change with distance and frequency?

3. What are the main challenges of wireless communication? Name at least four properties of a wireless channel that make RF design difficult.
4. What is the definition of "dBm"? Why are logarithmic metrics commonly used in RF systems?
5. What is the Shannon-Hartley theorem (Equation 5)? Calculate the channel capacity for a system with 10 MHz bandwidth and an SNR of 20 dB. What factors can be changed to increase the channel capacity?
6. Name at least five different wireless standards. For each, state the approximate frequency range and typical application.

11.2 Linearity

7. Describe the nonlinear model used in RF systems (Equation 6). What happens when a single sinusoidal signal passes through a weakly nonlinear system? What frequency components are generated?
8. What is gain compression? Draw a simple transfer characteristic and show the 1 dB compression point ($P_{1\text{dB}}$). Explain the difference between compressive and expansive behavior.
9. Explain the concept of intermodulation. Using a two-tone test and assuming third-order nonlinearity, which intermodulation products are created? Why are the IM3 products at $2\omega_1 - \omega_2$ and $2\omega_2 - \omega_1$ particularly problematic?
10. What is the third-order intercept point (IP3)? Draw a diagram showing the relationship between input power, fundamental output power, and IM3 products. How can IIP3 be calculated from measurement data using Equation 15?
11. How can nonlinearities be cascaded in a calculation? Given an LNA with IIP3 = −10 dBm and gain = 20 dB followed by a mixer with IIP3 = 5 dBm, what is the overall approximate IIP3 of the cascade? Which block dominates?
12. What is the approximate relationship between $P_{1\text{dB}}$ and IIP3 for mildly nonlinear systems?

11.3 Noise

13. What are the main types of noise relevant for electronic circuits? Describe thermal noise, shot noise, and flicker noise with their governing equations.
14. What is the thermal noise voltage of a resistor with value R? Calculate the noise voltage spectral density of a 1 kΩ resistor at room temperature.
15. What is the maximum noise power that can be extracted from a resistor? Derive the result using the maximum power transfer theorem and explain why it equals kT per Hz.
16. What are the main noise sources in a MOSFET? How can the input-referred noise voltage be minimized? How does flicker noise depend on transistor dimensions?
17. What is the noise figure (NF)? State the definition using SNR at input and output (Equation 21). What is the noise figure of an ideal noiseless amplifier? What is the noise figure of a passive lossy element with loss L?

18. What is the Friis equation (Equation 22)? For a system with two cascaded blocks (Block 1: NF_1 = 2 dB, G_1 = 15 dB; Block 2: NF_2 = 10 dB), calculate the total noise figure. Why is the noise figure of the first block so important?

19. What is the sensitivity of a wireless receiver? A receiver requires an SNR of 12 dB, has a channel bandwidth of 200 kHz, and a total noise figure of 5 dB. Calculate the sensitivity.

11.4 Modulation and Communication Concepts

20. What is modulation? Describe the basic types: amplitude modulation (AM/ASK), frequency modulation (FM/FSK), and phase modulation (PM/PSK). What is QAM?

21. Draw the constellation diagram of a QPSK or 16-QAM modulated signal. What impairments can affect the constellation, and how would they manifest in the diagram? What is Error Vector Magnitude (EVM)?

22. Explain the concept of pulse shaping. Why is the sinc pulse important, and what is its relationship to the rect function in the frequency domain?

23. What is Orthogonal Frequency Division Multiplexing (OFDM)? How are symbol period and sub-carrier spacing related? What are the advantages of OFDM?

24. What is the difference between Frequency Division Duplex (FDD) and Time Division Duplex (TDD)? What are the advantages and disadvantages of each? How does this affect transceiver design?

25. How can multiple users access a radio channel simultaneously? Describe FDMA, TDMA, and CDMA.

11.5 Transceiver Architectures

26. Draw a block diagram of a direct-conversion (zero-IF) receiver. Explain the function of each block. What are the advantages and disadvantages of this architecture?

27. What is the "image frequency" problem in a heterodyne receiver? How can it be addressed?

28. Describe the differences between direct-conversion and heterodyne receiver architectures. What are LO leakage, dc offsets, and flicker noise problems in direct-conversion receivers?

29. Draw a block diagram of a direct-conversion transmitter. What is oscillator pulling, and how can it be minimized?

30. What is a duplexer? When is it needed? Compare the requirements for duplexers vs. TX/RX switches.

11.6 Low Noise Amplifiers

31. What are the key performance parameters of an LNA? List at least five important specifications and describe their typical values.

32. What is impedance matching? What is the return loss, and how is it related to the reflection coefficient Γ? What return loss is typically considered acceptable?

33. What is amplifier stability? Explain the μ stability factor. What conditions must be satisfied for unconditional stability?

34. Draw the schematic of a resistively matched common-source LNA. Derive its noise figure and explain why this topology is limited to a minimum NF of 3 dB.

35. Draw the schematic of a common-gate LNA. What is its input impedance? Explain why the noise factor is limited by $F = 1 + \gamma$.

36. Draw the schematic of an inductively-degenerated common-source LNA. Derive the input impedance (Equation 44). How can a purely resistive input impedance be achieved?

37. What is the advantage of the inductively-degenerated LNA over the common-gate LNA in terms of noise figure? How can the noise figure be minimized?

38. Describe the shunt-feedback LNA topology. What are its advantages, and how does the input impedance depend on the feedback and load impedances?

11.7 Mixers

39. What are the two fundamental approaches to implementing frequency conversion in a mixer? Explain the difference between nonlinear and time-variant mixers.

40. Draw the block diagram of an unbalanced passive mixer using a single MOSFET switch. Derive the output signal expression and identify the conversion loss.

41. What is a double-balanced mixer? Draw its circuit diagram. What are the advantages over an unbalanced and over a single-balanced mixer, respectively? What is the conversion gain/loss of an ideal double-balanced mixer, and where do the 6 dB compared to the unbalanced mixer come from?

42. Draw and explain the Gilbert cell mixer. How does it work? What is the advantage of using a bipolar or CMOS Gilbert cell?

43. What is an N-path filter? Explain how it can create a high-Q bandpass filter centered at the LO frequency. Why is this useful for RF receivers?

44. Describe three different methods to generate quadrature (I/Q) LO signals. What are the advantages and disadvantages of each method?

45. Explain the flip-flop based I/Q generation method. How can 25% duty cycle LO signals be generated?

11.8 Oscillators

46. What is the Barkhausen criterion for oscillation? Describe the conditions for sustained oscillation in a feedback system.

47. Draw an LC tank circuit with loss. What is the resonance frequency? How is the quality factor Q defined? What is the relationship between Q and the energy stored vs. energy dissipated per cycle?

48. Describe the concept of phase noise. What is its physical meaning? In what unit is it typically specified? Calculate the phase noise for an oscillator with 10 dBm output power and −80 dBm noise at 1 MHz offset measured with 1 kHz bandwidth.

49. What is Leeson's equation (Equation 56)? Which factors influence the phase noise of an oscillator? How can phase noise be minimized?

50. What is reciprocal mixing? Why is it problematic in wireless receivers? Derive the maximum allowable phase noise given a blocker power and sensitivity requirement.
51. Draw a cross-coupled LC oscillator (NMOS, PMOS, or CMOS topology). What is the negative resistance provided by the cross-coupled pair? How does oscillation start and stabilize?
52. How can frequency tuning of an LC oscillator be achieved? Describe different varactor implementations (MOS varactor, switched capacitor, PN junction).

11.9 Phase-Locked Loops

53. Draw a simple block diagram of a Type-I phase-locked loop and explain the function of each component (PD, LPF, VCO, divider). What are the steady-state phase and frequency errors?
54. What is a phase-frequency detector (PFD)? How does it differ from a simple XOR phase detector? Draw its implementation and timing diagrams for different input conditions.
55. Draw a simple charge-pump circuit. How does it work? Why is the resistor in the loop filter necessary?
56. What is the difference between a Type-I and a Type-II PLL? What are the steady-state errors for phase and frequency in each case?
57. Write the closed-loop transfer function of a charge-pump PLL with a simple RC loop filter. Identify the natural frequency ω_n and damping factor ζ. What is a reasonable value for ζ?
58. Describe the phase noise profile of a PLL output. How do the reference oscillator phase noise and VCO phase noise contribute at different offset frequencies? What determines the crossover frequency?
59. What is an all-digital PLL (ADPLL)? Draw its block diagram. What are the advantages and challenges compared to analog PLLs?
60. Describe the operation of a time-to-digital converter (TDC). How does quantization noise affect the PLL phase noise?
61. What is a fractional-N PLL? Why is it useful? How does delta-sigma modulation help reduce spurious tones?
62. What is the noise transfer function (NTF) of a first-order delta-sigma modulator? How does it shape quantization noise?

11.10 Power Amplifiers

63. Draw a simple single-ended power amplifier. What is the trade-off between supply voltage, load impedance, and output power? Calculate the maximum output power for $V_{\mathrm{DD}} = 3$ V and $R_{\mathrm{L}} = 50\ \Omega$.
64. What is the definition of drain efficiency and power-added efficiency (PAE) of a PA?
65. What is the AM/AM and AM/PM distortion in a power amplifier? How can the Saleh model be used to characterize PA nonlinearity?
66. Describe load-pull measurements. Why are they important for PA design?

67. What are the different classes of operation for power amplifiers (Class A, B, AB, C)? Describe the conduction angle and theoretical maximum efficiency for each class.

68. What are Class D, E, and F amplifiers? How do they achieve high efficiency?

69. What is envelope tracking (Class G/H operation)? Why is it beneficial for power amplifiers used with non-constant envelope modulation schemes?

70. What is neutralization in power amplifier design? Why is it needed, and how is it implemented in a differential PA?

11.11 Calculation Problems

71. **FSPL Calculation**: Calculate the free-space path loss for a 5.8 GHz Wi-Fi link over a distance of 100 m. Express your answer in dB.

72. **Channel Capacity**: A mobile communication system operates with a 20 MHz channel bandwidth. The SNR at the receiver is 15 dB. Calculate the theoretical maximum data rate according to the Shannon-Hartley theorem. If the actual data rate is 75 Mbps, what is the spectral efficiency in bits/s/Hz?

73. **IIP3 Cascade**: Calculate the overall approximate IIP3 for the following cascade: LNA (IIP3 = −5 dBm, gain = 15 dB), Mixer (IIP3 = 8 dBm, gain = 10 dB), IF amplifier (IIP3 = 15 dBm, gain = 40 dB).

74. **Cascade Noise Figure**: An RF front-end consists of: RF filter (loss 2 dB), LNA (NF = 1.5 dB, gain = 18 dB), mixer (NF = 8 dB, gain = 6 dB). Calculate the total noise figure.

75. **Sensitivity Calculation**: A Bluetooth LE receiver has a noise figure of 8 dB, channel bandwidth of 2 MHz, and requires an SNR of 15 dB. Calculate the receiver sensitivity.

76. **LNA Input Impedance**: An inductively-degenerated common-source LNA has C_{gs} = 200 fF, g_m = 40 mS, and L_{deg} = 1 nH. Calculate the real part of the input impedance. What additional inductance is needed to resonate with C_{gs} at 2.4 GHz?

77. **Reciprocal Mixing**: A receiver has a desired sensitivity of −90 dBm with an adjacent channel blocker at −30 dBm located 3 MHz from the carrier. The channel bandwidth is 100 kHz, and the demodulator requires an SNR of 10 dB. What is the maximum allowable LO phase noise at 3 MHz offset, assuming the reciprocal mixing noise may consume the entire noise budget?

78. **Phase Noise**: An oscillator operates at 2.4 GHz with a tank Q of 15, tank resistance R_p = 500 Ω, oscillation amplitude of 0.8 V peak, and noise factor F = 2. Calculate the phase noise at 1 MHz offset using Leeson's equation.

79. **PLL Loop Parameters**: A charge-pump PLL has I_{CP} = 100 µA, K_{VCO} = 100 MHz/V, C_{int} = 10 nF, and N = 100. Calculate the natural frequency ω_n and the required R_{int} for a damping factor ζ = 0.707. Watch the units of K_{VCO}!

80. **PA Efficiency**: A power amplifier delivers 27 dBm output power and draws 400 mA from a 3.3 V supply. Calculate the drain efficiency and the PAE if the input power is 0 dBm.

11.12 RF Circuit Simulation (Optional)

These questions refer to the appendix on RF circuit simulation (Section 10) and are optional.

81. Which assumptions about the circuit waveforms do the analyses op, ac, noise, tran, PSS, and HB make? Sketch how the analyses build on each other (Figure 106).

82. Why does a small-signal ac analysis of a mixer show no conversion gain? Which analysis do you use instead, and why?

83. Explain the shooting method (Equation 76). Why does a PSS analysis of a high-Q circuit converge much faster than a transient analysis? What are the limitations of PSS, and why does a converged PSS not prove that an oscillator starts up?

84. Explain the principle of harmonic balance (Equation 79). For which circuits is HB efficient, and for which circuits is PSS the better choice? Why can HB fail to converge for a hard-switching mixer, while a PSS-based PAC analysis of the same circuit works?

85. What is a linear periodically time-varying (LPTV) system? Explain the conversion matrix (Equation 81) and the meaning of the sidebands. How do you obtain the conversion gain of a mixer and the image rejection of an I/Q receiver with PAC or HBAC?

86. What is cyclostationary noise? Which two effects does a periodic noise analysis (PNOISE, HBNOISE) capture that a plain noise analysis misses? Name three circuits where this matters.

87. How does a transient noise simulation work? List its advantages and disadvantages compared to PNOISE and HBNOISE. How can you check that a transient noise result is not dominated by numerical errors?

88. Which analyses would you use to characterize (a) an LNA, (b) a switched-capacitor filter, (c) an LC VCO, (d) a passive mixer, and (e) a power detector? Justify your choice.

89. **Transient Simulation Effort**: A downconversion mixer with f_{LO} = 5.8 GHz is simulated in the time domain with 40 time points per LO period. The IF spectrum shall be computed with a frequency resolution of 50 kHz. How many time points are needed, ignoring the settling time? How does a PSS analysis followed by a PAC analysis avoid this effort?

90. **Noise Folding**: An ideal double-balanced switching mixer multiplies the RF input by a square wave toggling between $+1$ and -1 with 50 % duty cycle, and the only noise source is the source resistance. Calculate the SSB and the DSB noise figure. By how much is the noise figure underestimated if a periodic noise analysis keeps only the sidebands ± 1 ($K = 1$)?

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